

The Japanese DP and the Structure of its Modifiers

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Preface

This book is a revised and shortened version of my PhD dissertation which I defended on June 28th 2024 at Goethe University Frankfurt. It aims at an analysis of the Japanese DP in line with the cartographic research program. On the one hand, the study presented here emphasizes the validity of the cartographic assumptions and their applicability to languages outside the Indo-European cosmos, but at the same time it shows that Japanese forces us to rethink many previously made claims and allows for refinements of these assumptions and a better understanding of the DP in cross-linguistic perspective.

The audience I would like to address are linguists interested in the DP and nominal modification as well as scholars in the field of Japanese studies who want to take a closer look at the different types of modification and the internal structure of modifiers in Japanese and to see how these language-specific phenomena can be transferred into a cross-linguistic framework. I strongly believe that both parties can equally benefit from reading this book.

The book consists of 6 chapters. Chapter 1 introduces the scope of this book and the research questions to be answered and gives an overview over the main hypotheses of this book. It also presents an overview over the Japanese nominal domain and a note on data collection. Chapter 2 is concerned with the theoretical background, the research program *cartography*. Chapter 3 addresses the main hypotheses defended in this book. It presents in detail the assumed DP structure and discusses the implications and necessary amendments to be made in line with the Japanese data, compared to earlier studies. The second part of this chapter is dedicated to the monomoraic elements co-occurring with non-verbal modifiers, the widely-distributed element NO, as well as the elements I and NA, which solely occur with the two Japanese adjective groups. I propose that those elements are instances of a syntactic head MOD which licenses elements in attributive position.

Chapters 4 and 5 are dedicated to the direct and indirect domain respectively. In chapter 4, I will outline each group of direct modifiers based on criteria for direct modification, both cross-linguistic and language-specific. Following is a discussion on the lack of ordering restrictions, otherwise a prerequisite for direct modification. Chapter 5 starts with an account of the complex elements co-occurring with Japanese

modifiers. Then the size of indirect modifiers will be evaluated and a uniform CP status will be argued for. This chapter provides similar findings to the direct domain and closes with the assumption that more complex relative clauses are more prone to move to a higher position within the indirect domain.

The book closes with a summary and an outlook. Following are two appendices which contain information on the group of *extended modifiers* and a detailed taxonomy of the group of *no*-modifiers.

All Japanese examples throughout this book, if not quoted otherwise, were provided to me by native speakers of Japanese. Most examples were kindly provided to me with additional discussion by Ken Hiraiwa. Other native speakers who helped me with examples are: Jiro Inaba, Emiko Hayatsu, Chikako Takeuchi, Noboru Miyazaki.

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List of Abbreviations

A	Adjective	Ins	Instrumental
Acc	Accusative	Ipfv	Imperfective
Adj	Adjectivizer	Ln	Light Noun
Adjez	Adjectival Ezāfe	Loc	Locative
Adv	Adverb/Adverbial	M	Masculine
AgrP	Agreement Projection	Mod	Modification (Syntactic Head)
All	Allative	N	Neuter
Ant	Anterior	Neg	Negation/Negative Element
Att.Part	Attributive Particle	Nom	Nominative
Aux	Auxiliary	Num	Numeral
Cl	Classifier	Obj	Object
Com	Comitative	Obl	Oblique Case
Comp	Complementizer	Op	Operator
Cop	(Dummy) Copula	Pass	Passive
Dat	Dative	Perf	Perfective
Decl	Declarative Sentence Ending	Pl	Plural
Dem	Demonstrative	Pol	Politeness/Polite Element
Det	Determiner	Poss	Possessive
Dir	Direct Case	Pot	Potential
Emp	Emphatic Marker/Particle	Pred	Predicative Copula
Epis	Epistemic Element	Prs	Present/Non-Past
Ez	Ezāfe	Prs.Ptcp	Present Participle
F	Feminine	Pst	Past
Foc	Focus	Pst.Ptcp	Past Participle
Fp	Functional Projection	Ptcp	Participle
Gen	Genitive	Q	Question marker
Ger	Gerund	Qa	Qualitative Adjective
Hon	Honorative	Qm	Qualitative Modifier
Ill	Illative	Ra	Relational Adjective

Rc	Relative Clause
Rel	Relative Marker/Particle
Rm	Relational Modifier
Rp	Relator Phrase
Rrc	Reduced Relative Clause
Rtk	<i>rentaikei</i> (Attributive Form)
Rtro	Retrospective Mood
Ryk	<i>renyōkei</i> (Nominalization Form)
Sfp	Sentence Final Particle
Sg	Singular
Shk	<i>shūshikei</i> (Sentence-final Form)
Sub	Subject
Top	Topic
UG	Universal Grammar
UoL	Unit of Language
V	Verb
Vol	Volitional

Notational Conventions

When I refer to *Japanese*, what I mean is *Modern Standard Japanese*, the official version of the language spoken predominantly in the areas in and surrounding Tokyo. *Pre-modern Japanese* is a broad term for any stage of the language prior to the Meiji-period (1868-1912). *Classical Japanese* (or *bungo*), refers to the premodern (mostly) written stage of the language based on the literary style of the Heian-period (794–1185) and in use until the language reform carried out in the Meiji-period.

In the context of nominal modification, I will often prefer the term *modifier* over *adjective* with regard to Japanese. This is because adjectival status cannot always be unequivocally assigned to the modifiers under question. I will only use the term *adjective* when I clearly refer to *i*- and *na*-adjectives respectively. With regard to other languages, I will in most cases stick to the terminology used by the respective authors.

I will represent syntactic trees for Japanese as head-final with the head to the right and the complement to the left. This means that in many cases no movement steps need to be assumed for Japanese that would be necessary if one follows to the bone the traditional cartographic system where everything is derived from a head-initial structure. Note, however, that this does not affect in any way the hierarchies and the structural precedences depicted here for the Japanese DP.

In the glosses, I will display copular elements, tense markers as well as the attributive MOD-head, all of which are heads, connected to their preceding elements with a hyphen. This is done for ease of notation and readability. The tree structures will display them correctly as heads.

All Japanese examples are transliterated following the Modern Hepburn Romanization System. I have chosen this system due to the fact that 1), pronunciation of Japanese material is depicted in a straightforward manner and 2), because the Modern Hepburn-Romanization System is a unified and standardized system that does not allow room for ambiguity. This means that lengthened vowels, for instance [a:], will be displayed via a macron over the vowel (<*ā*>). The only exception is [e:], which will be transliterated as <*ei*> sticking close to the original script. The semi-consonant /y/ will only be transliterated when it affects pronunciation, that is when it causes palatalization of the preceding consonant, not when it serves the function

in the original Japanese script to display certain consonant-vowel combinations. For instance, the mora しゃ, pronounced [ɕa], will be transliterated as /sha/, which is very close to the original pronunciation, but never as <sya> or <shya>. Transliterations of Persian follow the scientific DMG-system (DIN 31635), transliterations of languages with Cyrillic alphabets follow the scientific DIN-system as well.

Chapter 1

Introduction

1.1 Aim of this book

This book presents an analysis of the Japanese DP as part of the research program *cartography* (Pollock 1989; Cinque 1993; Rizzi 1997; Cinque 1999, 2006, 2010, 2017, 2020, 2023 among others). Its main goal is to derive a structure of the DP and all its domains in which all types of modifiers can be incorporated. This DP structure to be proposed is built on previous cartographic analyses on the DP (Cinque, 1993, 2010, 2020; Svenonius, 2008; Laenzlinger, 2005, 2011; Kim, 2019). Importantly, it does not only *accommodate* the Japanese data, but, rather, is *enriched* by it, shaping our understanding of the DP and nominal modification as a research topic.

Specifically, cartography provides the necessary tool for an analysis of the Japanese DP, highlighting Japanese as a language which does not behave entirely differently from other languages analyzed as part of this research field. There are, however, significant points of deviation, which, I argue to be even more important for future analyses on the DP. With the in-depth investigation performed in this book, I hope to achieve two things: On the one hand, to reveal new insights for the Japanese nominal domain, and, on the other hand, to allow for cross-linguistic generalizations from this language, which enhance our linguistic knowledge and the inventory of our linguistic tools and categories.

In this introductory chapter, I will present the scope and the main research questions of this book, as well as the main motivation for writing it, as well as the most important claims. In section 1.4, I present an overview over the Japanese nominal modification system. In the final section, I give an overview over data collection.

1.2 Scope of this book and questions to be answered

It is a characteristic of Japanese that the syntactic role a modifier performs within the DP is not visible from the surface structure, and neither are its syntactic function or its syntactic position. This can be attributed to two main morphosyntactic peculiarities. First, in most cases, there is no morphological difference between attributively used lexemes and their predicative counterparts. Second, and perhaps more importantly, there is *only one* surface construction, only one mechanism, available for lexemes partaking in nominal modification. Consider the minimal pairs below which present an *i*-adjective (1) and a verb (2) in predicative and in attributive position.¹ Essentially, it seems as if the word order has been shifted around and we only know that the b) examples involve a modification structure from the very strict SOV order present in Japanese.

- (1) a. Ano doresu-wa taka-i.
that dress-TOP expensive-I
'That dress is expensive.'
- b. ano taka-i doresu
that expensive-I dress
'that expensive dress' *i*-adjective, non-past
- (2) a. Kare-ga [hon-wo kat-ta].
he-NOM book-ACC buy-PST
'He bought a book.'
- b. [kare-ga kat-ta] hon
he-NOM buy-PST book
'the book (which) he bought' verb, past

Now, compare (1) to the different corresponding English DPs depicted below.

- (3) a. the expensive dress
b. the dress which was expensive
c. a dress, expensive but beautiful

The English adjective *expensive*, besides performing regular adjectival modification in (3-a), can appear inside a relative clause (3-b). Such relative clauses are explicitly marked via relative pronouns or complementizers, elements which are absent in Japanese (Kuno, 1973; Comrie, 1998). Furthermore, the adjective can occur postnominally as depicted in (3-c). Postnominal adjectives are in English typically analyzed as *reduced relative clauses* (RRCs; Sadler and Arnold 1994; Larson 1998;

¹The verb is inflected for past, highlighted via the past ending -TA, or the allomorph /da/. See Chapter 1.4.3 for a first discussion on the morpheme -I.

Cinque 2010). They are reduced since they lack a relative pronoun/a complementizer and a copula/a verb (Douglas, 2016; Harwood, 2018). This reduction can be depicted as follows.

- (4) a. a dress ~~which/that~~ is expensive but beautiful
 b. the stars ~~which/that~~ are visible

In many languages, adjectives agree with the noun in case, number and gender. In some languages, including German and Dutch, they do this only in attributive (5-a), but not in predicative position (5-b). This predicts agreement to be also absent when the adjective appears inside a relative clause (5-c) (Roehrs 2008; Belk 2017; Zeijlstra 2020 a.o.).

- (5) a. der rot-e Tisch (German)
 the red-NOM.SG.M table.NOM.SG.M
 ‘the red table’
 b. Der Tisch ist rot.
 the table.NOM.SG.M is red.∅
 ‘The table is red.’
 c. der Tisch, der rot ist
 the table.NOM.SG.M the red.∅ is
 ‘the table which is red’

Nominal modifiers with a more predicative structure are called *indirect* or *predicative* modifiers, contrasting them with *direct* or *attributive* modifiers, those not related to predicativity (Sproat and Shih, 1991; Alexiadou *et al.*, 2007; Cinque, 2010). Another way in which languages can cue different types of modification is through the optional appearance of additional morphosyntactic elements. For example, the element DE in Mandarin Chinese, which is optional with adjectival modifiers, but mandatory with verbal modifiers, has been analyzed as a clear indicator of indirect modification (Sproat and Shih, 1991, 571).

- (6) hǎo (de) tiānqì
 good DE weather
 ‘good weather’ (Sproat and Shih, 1991, 573)

In Japanese, however, where no agreement between head and modifier is observable, where modifiers can only appear prenominal, and where only one modification structure exists, a simple modifier in non-past as given above in (1), is structurally ambiguous. These facts have led to the prevailing assumption that this language simply lacks direct modification, with all modifiers being interpreted as relative clauses (for example Kuno 1973; Hinds 1988; Whitman 1981; Kaplan and Whitman 1995).

Admittedly, structural ambiguity also holds for cases in English in which a single adjectival modifier modifies a noun, such as *red book*. However, by relating the two types of modifiers to two different structural positions, besides their two different internal structures, cartography offers another diagnostic, namely ordering. Observe (7).

- (7) a. big red book
b. ??red big book

One order of the adjectives in (7) is clearly more natural than the other, given an out-of-the-blue context. Rather than assuming cognitive principles, cartography explains this behavior through selection. Simply said, the projection hosting direct modifiers of SIZE structurally dominates the one hosting direct modifiers of COLOR. Importantly, in the indirect domain, where several functional projections of the same category can co-exist, no structural dominance exists, predicting the absence of strict order between indirect modifiers of the same type (Sproat and Shih, 1991; Baker, 2003b).

- (8) a. book that is big that is red
b. book that is red that is big

Crucially, the claim that modification in Japanese involves only relative clauses extends to ordering restrictions, or rather the absence thereof. Different from the minimal pair in English given in (7), no ordering restrictions, or even tendencies, exist in the corresponding Japanese example (9).

- (9) a. ōki-i aka-i hon
big-I red-I book
b. aka-i ōki-i hon
red-I big-I book
'big red book'

Even if we were to maintain the null hypothesis that lexical items in every language, unless proven otherwise, can appear as both types of modifiers, direct or indirect, these facts named above make it hard to argue for Japanese as a language in which both types exist. This claim also persists in the few theoretical works that discuss this issue with regard to this language (Sproat and Shih, 1991; Baker, 2003b; Laenzlinger, 2011).

Apart from the fact that such analyses leave open how exactly indirect modifiers are structurally integrated in the DP, Japanese does exhibit direct modifiers when applying another criterion: that of *non-predicativity*. Taking predicate nature as a characteristic of indirect modifiers (Bolinger, 1967; Alexiadou *et al.*, 2007; Cinque,

2010; Panayidou, 2013), the lack of predicativity is a characteristic of unambiguously direct, or *direct-only modifiers*. Such modifiers, which have sometimes been noted in the literature (see overview in Watanabe 2017), include non-intersective modifiers such as *shōrai-no* ‘future’ depicted in (10) and modifiers of NATIONALITY and MATERIAL as depicted in (11)². Intriguingly, most direct-only modifiers co-occur with the element NO.

- (10) a. *shōrai-no daitōryō*
future-NO prime.minister
‘future prime minister’
b. **Kono daitōryō-wa shōrai-da.*
this prime.minister-TOP future-COP
(‘This prime minister is future.’)
direct modifier, non-intersective

- (11) a. *Denmāku-no / chīku-no isu*
Denmark-NO / teak-NO chair
‘a Danish chair / a teak chair’
b. **Ano isu-wa Denmāku-da / chīku-da.*
that chair-TOP Danish-COP / teak-COP
(‘This chair is Danish / teak.’)
direct modifier, NATIONALITY/MATERIAL

Crucially, even among direct-only modifiers no ordering restrictions can be detected in Japanese, contrary to, for instance, English.

- (12) a. *Denmāku-no chīku-no isu*
Denmark-NO teak-NO chair
‘Danish teak chair’
b. *chīku-no Denmāku-no isu*
teak-NO Denmark-NO chair
‘teak Danish chair’
(13) a. Danish teak chair
b. ??teak Danish chair

The first main goal of this book is therefore to derive a DP-structure in which both types of modifiers, direct and indirect, can be integrated and which can account for this freedom of ordering in Japanese, which is rather surprising for cartographic

²In such examples, I depict possible usage in context-neutral sentences, abstracting away from contexts which ameliorate these modifiers in predicative position. These include contrastive contexts (Hoshi, 2002), contexts in which the head noun is elided in the second instance, thus essentially a case of modification (Nagano, 2018), or contexts in which the copula acts as a placeholder for another verb, so-called eel-constructions (Okutsu, 1978). See chapters 4.2 and 4.3.1.

Finally, it is a desideratum to analyze the internal structure of both types of modifiers. First, this concerns morphologically complex elements, for example, KATTA marking past inflections of *i*-adjectives.

- (14) ano taka-katta doresu
that expensive-KATTA dress
'that dress which was expensive' *i*-adjective, past

Second, this concerns the elements consisting only of one mora, namely -I following the *i*-adjective *taka-i* ‘high, expensive’ in (1), -NA following *na*-adjectives, and the most versatile -NO occurring in (10).³ The internal structure of modifiers is often treated orthogonally to their role inside the DP, but by bringing these issues together, I derive a complete structure of the Japanese DP, also including the micro-level.

This section presents the main hypotheses to be defended in this book. These hypotheses concern the structure of the DP, the absence of specific functional projections and the internal structure of modifiers.

I propose that the Japanese DP looks as depicted in figure 1.1.

6

I adopt a split-DP approach (Laenzlinger, 2005, 2011; Kim, 2019) and propose that the Japanese DP consists of a DP-external domain, the left periphery, and a DP-internal domain. The DP-internal domain consists furthermore of a domain for indirect (predicative) and a lower domain for direct (attributive) modification. Those two domains are separated by the indefinite head *d*.⁴ Note that the DP-external domain is represented by the left part of the tree, and the DP-internal domain is represented on the right part. The thresholds between the three domains are depicted in boldface.

This book focuses on the DP-internal domain, but a discussion on the left periphery can be found in Chapter 3.1.3 where I show that this area must encompass landing sites for quantificational modifiers, non-restrictive relative clauses and focused modifiers, but that essentially all types of modifiers can move there.

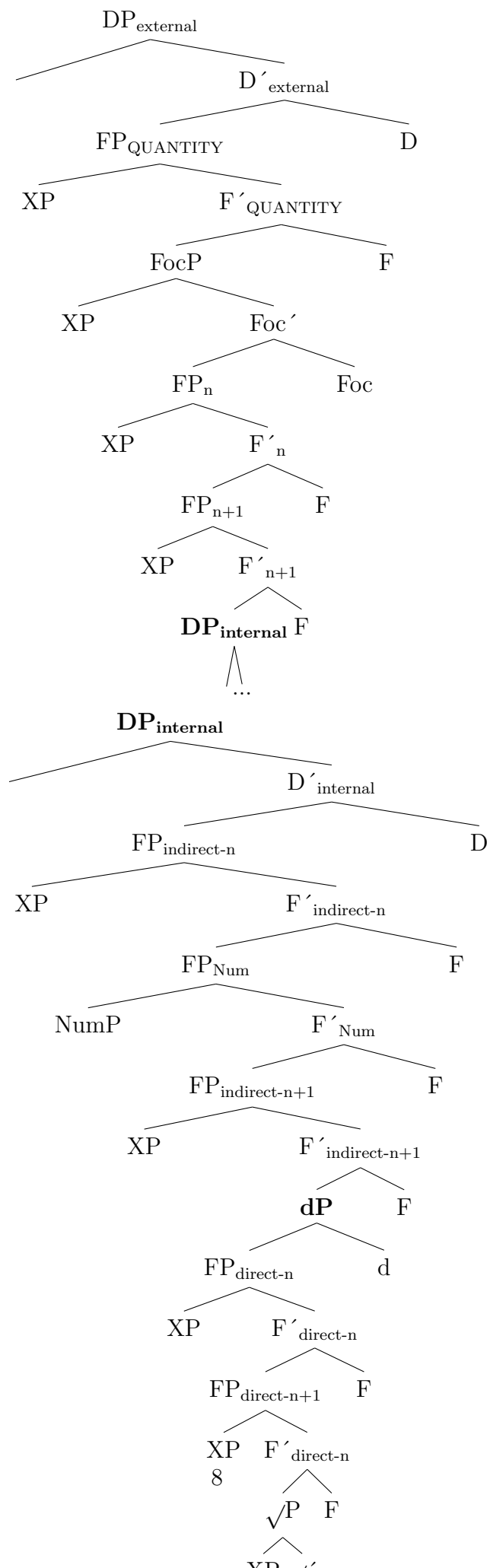
Modifiers, unequivocally depicted as XPs in figure 1.1, merge in the specifier position of functional projections (FPs) within the DP. A crucial assumption that differentiates the system defended here from accounts such as Cinque (2010, 2020) is that functional projections (with one exception) are not dedicated to specific semantic categories in the direct domain, and neither to different sizes of relative clauses in the indirect domain. In other words, these functional projections are unspecific. The labels n , $n+1$, are meant to depict that there is potentially an infinite number of functional projections inside every domain, but that there is a hierarchy, depicted by the numbers and the “plus” symbol.

1.3.2 Unspecified functional projections predict free order

I propose that this is the general DP spine valid in every language, but that cross-linguistic variation should be permitted concerning the nature of the functional projections present in that spine. Some languages, including English, have the relevant language-specific tools, called *Units of Language* in Wiltschko (2014), available to create functional projections dedicated to fine-grained semantic categories such as SHAPE or COLOR. In turn, this lead to the familiar ordering restrictions within the direct domain.

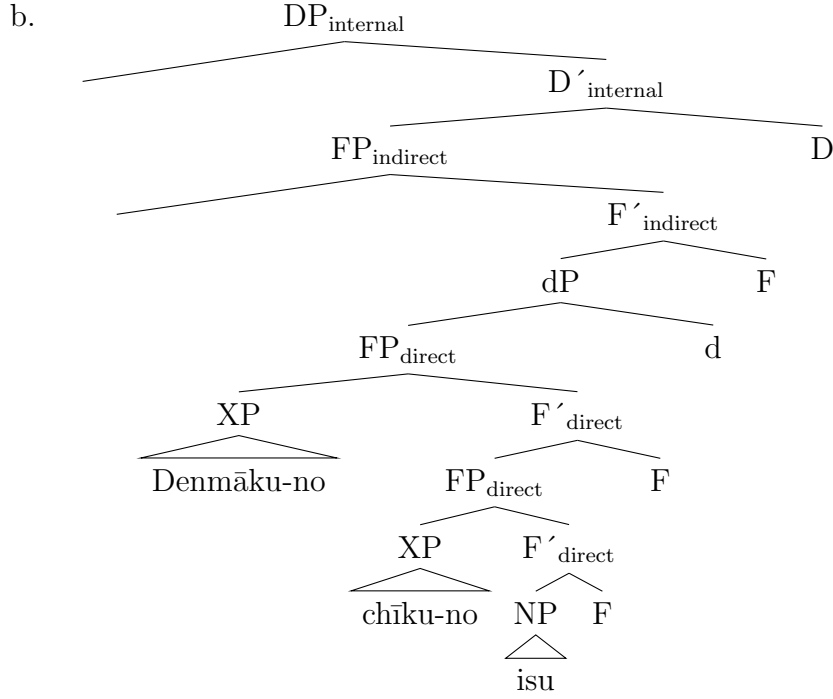
In turn, Japanese modifiers are licensed by functional heads in the respective domains they belong to, but this language lacks the language-specific tools to form *dedicated* functional projections. Since modifiers are not merged in such dedicated, and rigidly ordered, functional projections, their merging position is not predetermined and hence they can occur in any order. As a result, even for a pair of non-predicative modifiers, which should exhibit ordering restrictions given what we know from other languages, no such ordering restrictions are observable.

⁴The head *d* is the small indefinite head of a relative clause if there is one in the structure (Cinque, 2010, 2020). See chapter 2.4.1.3 for discussion.

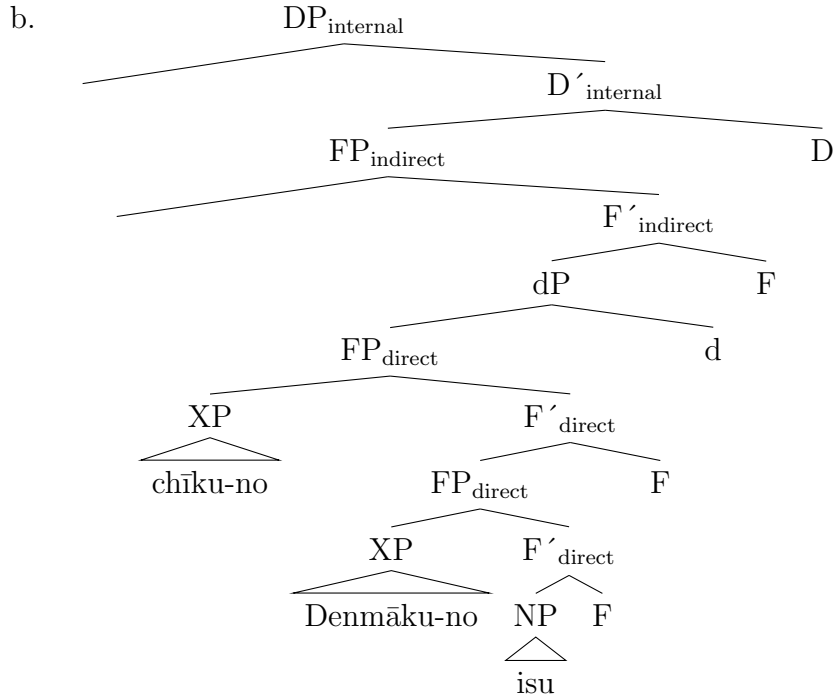


This is illustrated below for the two modifiers of NATIONALITY *Denmāku-no* ‘Danish’ and of MATERIAL *chīku-no* ‘teak’. Their two possible surface orders are simply the result of different merging positions within the DP. See Chapter 4.4 for discussion on ordering in the direct domain.

- (15) a. Denmāku-no chīku-no isu
Denmark-NO teak-NO chair (= (12-a))



- (16) a. chīku-no Denmāku-no isu
teak-NO Denmark-NO chair (= (12-b))



Concerning indirect modifiers, a similar situation holds. They only come in one size in Japanese, namely CP, which means Japanese only has one type of (head-external) relative clause. Again, this predicts the absence of ordering restrictions, also within the indirect domain. However, the indirect domain also encompasses a projection

for numerals and in principle, indirect modifiers can occur in a position above or below the relevant projection hosting numerals.

- (17) a. [pātī-de at-ta [futari-no josei]]
 party-LOC meet-PST two-NO woman
 b. [futari-no [pātī-de at-ta josei]]
 two-NO party-LOC meet-PST woman
 ‘the two women who I met at a party’

Furthermore, indirect modifiers of higher complexity, for instance if they contain an overt subjects or adverbials, are more natural in a higher position.

- (18) a. [doresu-ga utsukushi-k-at-ta [futari-no josei]]
 dress-NOM beautiful-PRED-COP-PST two-NO woman
 b. #[futari-no [doresu-ga utsukushi-k-at-ta josei]]
 two-NO dress-NOM beautiful-PRED-COP-PST woman
 ‘the two women whose dresses were beautiful’

This suggests that indirect modifiers originate low, in a position below FP_{Num} , but can move up. This discussion will be picked up in Chapter 5.4.

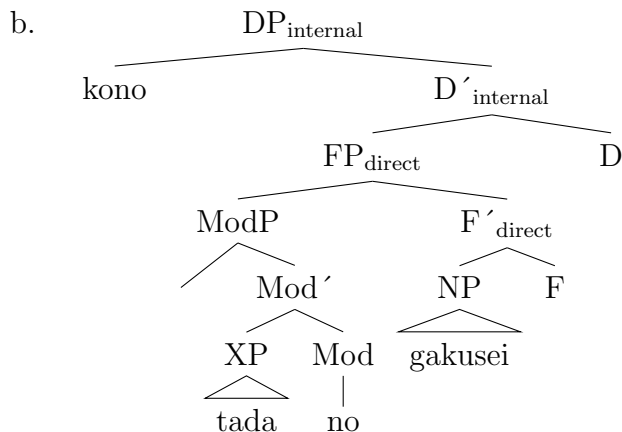
1.3.3 Internal structure of modifiers

The final hypothesis concerns the internal structure of modifiers. Following Nishiyama (1999), I analyze complex elements such as *katta*, given in (14) as combinations of two copulas and a tense marker along the following lines. This analysis will be outlined in Chapter 5.2.

- (19) ano taka-k-at-ta doresu
 that expensive-PRED-COP-PST dress
 ‘that dress which was expensive’ *i*-adjective, past

On the other hand, the monomoraic elements co-occurring with Japanese non-verbal modifiers in attributive position are not manifestations of copulas and tense markers. Rather, I argue that these elements, exemplified below with NO, are overt manifestations of a syntactic head labeled MOD (Rubin, 1994). This proposal is similar to the Mod-Insertion hypothesis of Kitagawa and Ross (1982), but crucially does not assume post-syntactic insertion. Instead, MOD has a syntactic function and licenses the preceding modifier in attributive position along the following lines illustrated with a direct modifier.

- (20) a. kono tada-no gakusei
 this mere-MOD student
 ‘this mere student’



I argue for a unified analysis of NO which is not contingent on the nature of the different elements that precede it. Furthermore, also the elements following adjectives are instances of such a syntactic head, which means its spell-out form is contingent on the word class of the preceding modifier. Readers interested in this discussion are referred to Chapter 3.2.

1.4 The Japanese nominal domain

In this section, I will present the characteristics of the Japanese language that are relevant for this book. I focus on the nominal domain and nominal modifiers.

1.4.1 General information and word classes

Japanese is the first language of approximately 125 million people and the official language in Japan and certain regions of Palau. It is a strictly head-final SOV language in which the verb cannot be moved from sentence-final position (Kuno 1973; Martin 1975; Iwasaki 2013 among others). Another key feature is its agglutinative character. Grammatical information is encoded via affixes, predominantly suffixes.

There is a classical divide between inflecting and uninflecting, as well as dependent and independent word classes,⁵ making *inflection* a key characteristic for differentiating between word classes. What is meant by this term is not the familiar morphological alternation dependent on categories such as number or person, rather inflection (Japanese: *katsuyō* 活用) denotes the conjugation of verbal categories in different inflectional forms. These inflectional forms, or *katsuyōkei* (活用形), not

⁵Following the terminology of Spencer (2020), *inflecting* denotes all those word classes in a specific language whose members generally do inflect. *Uninflecting* denotes word classes whose members never inflect, while *uninflectable* denotes specific members of an otherwise inflecting word class that for various reasons do not inflect.

only encode different functions, they are also encoded for which functional, or lexical, elements they host.⁶ Inflecting word classes include verbs, the two adjective classes and the functional category of so-called *auxiliary verbs* (Japanese: *jodōshi* 助動詞).⁷

1.4.2 Modifiers and modification

The strict head-finality is also apparent in the nominal domain. Modifiers exclusively precede the noun and include numerals, verbs, nominals, two adjective groups, and demonstratives, but not articles. The two adjective groups, in Japanese called *keiyōshi* 形容詞, lit. ‘qualifying words’, will consistently be referred to as *i-adjectives* and *na-adjectives* throughout this book reflecting their systematic morphological differences. *i*-adjectives such as *taka-i* ‘high, expensive’ end in -I in attributive position, but also in predicative position to mark non-past, whereas NA-adjectives such as *shikku-na* ‘chic’ end in -NA in attributive position in non-past. In predicative position, they exhibit the form -DA, and this difference essentially reflects the difference between *shūshikei*, the sentence-final form, and *rentaikei*, the attributive form, of Classical Japanese.⁸

⁶Six inflectional forms are recognized on the basis of Classical Japanese: I *mizenkei* 未然形 (negative form/irrealis), II *renyōkei* 連用形 (nominalization/adverbial form), III *shūshikei* 終止形 (sentence-final form, also conclusive), IV *rentaikei* 連体形 (attributive form), V *izenkei* 已然形 (concessive form) and VI *meireikei* 命令形 (imperative form). These forms and the respective terminology are still referred to with regard to Modern Japanese, but the amount and nature of these inflectional forms is subject to debate. This is because for many inflectional paradigms, the different forms have become syncretic and hard to distinguish (see Köhlich to appear for an overview). Furthermore, strong phonological processes have blurred the borders of inflectional form and suffix making inflection appear almost fusional. Some clear instances of inflectional forms can be observed for verbs of the *consonant inflectional paradigm*, for example *nom* ‘to drink’. Compare the following stem vowel alternations.

- | | | | |
|------|----|----------------------------|----------------|
| (21) | a. | nom- u | |
| | | drink-PRS | |
| | | ‘I/you/etc. drink’ | sentence-final |
| | b. | nom i -ta-i | |
| | | drink-VOL-PRS | |
| | | ‘I/you/etc. want to drink’ | nominalization |
| | c. | nom e -ba | |
| | | drink-if | |
| | | ‘if I/you/etc. drink’ | concessive |

Note also that I use the glosses SHK (*shūshikei*), RTK (*rentaikei*) and RYK (*renyōkei*) mostly with reference to Classical Japanese and only when relevant with reference to Modern Japanese.

⁷As will become obvious on multiple occasions, it is a matter of debate whether *na*-adjectives do inflect (see Uehara 1998; Kishimoto and Uehara 2016).

⁸This difference has been reduced in all other inflecting word classes since around Late Middle Japanese (12–16 AD), when the attributive form started to appear in sentence-final environments. The dichotomy is still observable for inflecting word classes in the different varieties of Okinawan (Ryūkyūan), which go back to the same origin as Japanese, but in many instances have preserved features of Proto-Japanese.

Thus *na*-adjectives, in non-past, alongside nouns to be reviewed below, provide one exception to the observation that Japanese lexemes always appear in the same form in attributive and in predicative position.

- (22) a. Ano doresu-wa shikku-da.
 that dress-TOP chic-DA
 ‘That dress is chic.’
 b. ano shikku-na doresu
 that chic-NA dress
 ‘that chic dress’ *na*-adjective, non-past

1.4.3 Adjectives, I and NA

The two adjective groups exhibit the typical differences observable in languages with two adjective groups (Dixon, 2004). *i*-adjectives are the older group and most lexemes belong to the native Japanese stratum, whereas *na*-adjectives entered the language in Late Middle Japanese and have only received the status as an independent word class in the middle of the 20th century (Katō 2003; Lewin 2003; Frellesvig 2010 a.o.). Most lexemes of this adjective class belong to non-native, foreign strata, mostly Sino-Japanese, but also English, but there are also some native lexemes, then often with typical endings *-ka*, *-raka* or *-yaka* (see for instance Uehara 1998, 210–213 for diachronic differences). This also means that *na*-adjectives present an open class, accepting new loanwords, while *i*-adjectives are a closed class.

As for their semantics, *i*-adjectives typically express basic notions. More detailed meanings are denoted via *na*-adjectives. However, both groups can in principle express all semantic categories of various well-known hierarchies (Scott 2002 among others), although some categories are biased towards one group. More basic categories such as DIMENSION show a higher percentage of *i*-adjectives, the more versatile category QUALITY incorporates more *na*-adjectives (Uehara, 1998).

There exist various analyses of the monomoraic elements I and NA. Until the 2000’s, I was analyzed either as an inflecting tense morpheme (Teramura, 1982; Murasugi, 1991; Urushibara, 1993; Uehara, 1998), or (part of) a copula (Kubo, 1992). NA on the other hand was perceived less controversially and predominantly analyzed as the copula, alongside its predicative equivalent DA, which generally received unequivocal copula-status (for example Urushibara 1993; Nishiyama 1999). One reason for this is that not only DA, but also the attributive NA, can be paraphrased by the complex element DEARU, which has been identified as the canonical form of the copula and is historically related to DA (see for example Urushibara

1993; Nishiyama 1999; Narahara 2002; see Miyama 2011 for historical evidence). This is shown below.⁹

- (23) a. Ano doresu-wa shikku-dearu.
that dress-TOP chic-COP
'That dress is chic.'
- b. ano shikku-dearu doresu
that chic-COP dress
'that dress which/that is chic'
- na*-adjective, non-past with copula DEARU

Furthermore, both adjective groups can express past, and when they do, they exhibit the same complex morphological forms in attributive as in predicative position, meaning the formerly observed difference in the case of *na*-adjectives vanishes. *na*-adjectives co-occur with the complex element DEATTA, which corresponds to DEARU in non-past, or the shorter form DATTA, where the syllable /e/ has been reduced and which corresponds to DA.¹⁰

i-adjectives on the other hand seem to truly inflect for past. The complex element KATTA occurs instead of I, again in both attributive and predicative position. At first glance, these two elements seem to not bear any resemblance to one another.

- (24) a. Ano doresu-wa shikku-d(e)atta.
that dress-TOP chic-COP.PST
'That dress was chic.'
- b. ano shikku-d(e)atta doresu
that chic-COP.PST dress
'that dress which was chic'
- na*-adjective, past
- (25) a. Ano doresu-wa taka-katta.
that dress-TOP expensive-PST,
'That dress was expensive.'
- b. ano taka-katta doresu
that expensive-PST, dress
'that dress which was expensive'
- i*-adjective, past

A landmark analysis to be adopted in this study comes from Nishiyama (1999), who aimed at a unifying analysis of I and NA as combinations of two copulas and a tense marker. This analysis, although its adoption at face value brings about some problems, is important in the sense that it offers a parallel analysis of the two

⁹The two elements do not display a significant difference in meaning, but DEARU is mostly observed in written register and rarely in spoken language. It is also degraded in attributive position, although not impossible. I will gloss both elements simply as COP for now. For the relevant analysis see Chapter 5.2.

¹⁰Same as DEARU and DA, these two forms display a difference in register, but are otherwise interchangeable.

elements. In earlier studies, often, but not always (Backhouse, 1984), the dichotomy between *inflecting tense marker* (I) vs. *copula* (NA) reflected the word class status of relevant lexemes, *i*-adjectives were analyzed as (verbal) inflecting predicates, and *na*-adjectives as non-inflecting nouns (see most of all Uehara 1998).

What should be pointed out, however, is that the modifiers that co-occur with I and NA are adjectives.¹¹ First, note that, abstracting away from semantic reasons, such lexemes are typically gradable as exemplified for a *na*-adjective below. Gradability is a typical feature of the adjectival category (Baker, 2003a; Dixon, 2010) and in Japanese expressed via comparative-formation with the particle *yor*i(-*mo*) ‘than’ and the optional adverb *motto* ‘more’ (26-a), degree adverbs (26-b), or superlative-formation via the adverb *ichi-ban* ‘most’ (Lit. ‘one-number’) (26-c), see Bhatt and Takahashi (2011).

- (26) a. Kono machi-wa ano machi-yori (motto) nigiyaka-da
 this city-TOP that city-than (more) lively-COP
 ‘This city is more lively than that city.’
 b. totemo nigiyaka-na machi
 very lively-NA city
 ‘very lively city’
 c. ichi-ban nigiyaka-na machi
 number-one lively-NA city
 ‘the most lively city’

Another, more language-specific, feature, of adjectivehood is nominalization with the suffix -SA ‘-ness’, which only nominalizes adjectives, but never nouns, since they are already nominal. As shown below, *i*-adjectives (*samu-i* ‘cold’) and *na*-adjectives (*shizuka-na* ‘quiet’) can be nominalized, but nouns (*sensei* ‘teacher’) or verbs (*tabe* ‘eat’) cannot.¹²

- (27) a. samu-sa
 cold-ness
 b. shizuka-sa
 quiet-ness
 c. *sensei-sa
 teacher-ness

¹¹The fact that *i*-adjectives seem rather “verby” and *na*-adjectives rather “nouny” has prompted several researchers to subsume them under the headlines of *Verbals* and *Nominals* (Uehara, 1998), sometimes denying the adjective category in Japanese in general (Wetzer, 1996). In fact, the analysis of Nishiyama (1999) and its subsequent incorporation in the study of Baker (2003a) were important steps towards the assumption of two adjective groups in Japanese.

¹²See Teramura (1982); Wenck (1974); Urushibara (1993); Rickmeyer (1995); Katō (2003); Muraki (2012) for a similar application of this characteristic. Other, more specific suffixes which only attach to adjectives are -*garu*, which derives verbs from adjectives that express (often negative) physiological or psychological states (Urushibara, 1993), and the verbal suffix -*sugiru* ‘to exceed; too’.

- d. *taberu-sa
eat-ness

Another characteristic differentiating adjectives from nouns is the inability to appear with particles, especially -GA for nominative and -WO for accusative case (Suzuki, 1972; Martin, 1975; Teramura, 1982; Nakayama, 2004; Muraki, 2012), in other words the lack of *referential use* (Oshima *et al.*, 2019), as shown below.

- (28) a. *Samu-(i-)ga ku-ru.
cold-NOM come-PRS
(‘(The/An) cold comes’.)
b. *Samu-(i-)wo mi-ru.
unknown-ACC see-PRS
(‘to see (the/a) cold’) *i*-adjective
- (29) a. *Shizuka-ga ku-ru.
quiet-NOM come-PRS
(‘(The/An) quiet comes.’)
b. *Shizuka-wo mi-ru.
quiet-ACC see-PRS
(‘to see (the/a) quiet’) *na*-adjective

1.4.4 *no*-Modifiers and NO

The discussion above was meant to show that I and NA co-occur only with adjectives. This subsection discusses the monomoraic element NO, whose distribution is not restricted to a specific word class.

1.4.4.1 Nouns

In most grammars of Japanese, NO is strongly, often exclusively, connected to the co-occurrence with nouns. In fact, nouns share their entire inflectional paradigms with *na*-adjectives, except for this element which they co-occur with in attributive position, contrary to the element NA that co-occurs with *na*-adjectives. The prototypical relationship between two nouns connected via NO is arguably possession, which explains why this element is predominantly glossed as GEN.¹³

- (30) a. sensei-no hon
teacher-NO book
‘book of the teacher’
b. Tarō-no jitensha
Taro-NO bicycle
‘Taro’s bicycle’ noun, possessive

¹³For discussion and different opinions see Murasugi (1991); Uehara (1998); Backhouse (2004). I will gloss it as NO until I present the proper analysis as a syntactic head in Chapter 3.2.

The lexemes combined via NO in (30) are clearly nouns. Syntactically, they can appear with case particles as visible in (31), from a semantic and pragmatic point of view, they establish a *reference* to some entity, most obvious with the proper name *Tarō*, alongside common ground between speaker and hearer.

- (31) sensei / Tarō / jitensha / hon-wo mi-ta
 teacher / Taro / bicycle / book-ACC see-PST
 ‘saw the teacher / Tarō / the/a bicycle / the/a book’

Several other relationships can be invoked when two nouns are connected via NO, prominently argument roles (Murasugi, 1991). The following example is ambiguous between a subject and an object reading.

- (32) Itaria-no hakai
 Italy-NO destruction
 1. ‘Italian destruction’ (= destruction by Italy, *subject*)
 2. ‘Italian destruction’ (= destruction of Italy, *object*)

A subject and an object can even appear side by side within the same DP, both times followed by NO. Usually (abstracting away from meaning), the reading where the first noun is interpreted as the subject is more accessible, but permutation is naturally possible.¹⁴

- (33) yabanjin-no toshi-no hakai
 barbarian-NO city-NO destruction
 ‘the destruction of the city by the barbarians’ (Murasugi, 1991, 7)

These DPs can be paraphrased as sentences where the modifying nouns bear nominative and accusative case particles and the modified noun forms a predicate with the light verb *su*, also showing that *no* is mutually exclusive with case particles.

- (34) Yabanjin-ga toshi-wo hakai-shi-ta.
 barbarian-NOM city-ACC destruction-do-PST
 ‘The barbarians destroyed the city.’

Furthermore, NO can license the attributive modification of nouns that bear other diverse argument roles. In this case, those nouns typically appear with postpositions in order to make the role they bear clearer. These postpositions include DE (instrumental, locative), NI (dative, locative, allative, illative) and KARA (ablative)

¹⁴Examples of multiple occurrences of NO also show that it is recursive in the sense that in theory an infinite number of elements can be connected via NO in the same DP. This is similar to English *of* or genitive *-s*, with the difference that two or more argument genitives are not allowed in one DP as also visible in the translations of the relevant example (Murasugi, 1991, 86).

(Murasugi, 1991; Saito *et al.*, 2008).¹⁵ Also, note that these examples give a first indication that not everything that co-occurs with NO is a noun.¹⁶

- (35) a. ishi-de-no kōgeki
stone-INS-NO attack
‘an attack with stones’ instrumental (Saito *et al.*, 2008, 248)
- b. Tōkyō-de-no kaigi
Tokyo-LOC-NO meeting
‘the meeting in Tokyo’ locative
- c. Tōkyō-kara-no densha
Tokyo-ABL-NO train
‘the train from Tokyo’ ablative (Murasugi, 1991, 50)

Note that NO always follows the respective postposition and must be closest to the noun, highlighting its connecting function.

- (36) *ishi-no-de kōgeki
stone-NO-INS attack

Nouns performing modification duties do not necessarily have to bear an argument status. Sometimes they are adjuncts (37). This also includes cases of possessor-possessee relationship where the possessor appears as the second noun (38).

- (37) a. ame-no hi
rain-NO day
‘rainy day’ (Saito *et al.*, 2008, 251)
- b. otoko-no ko
man-NO child
‘male child (=boy)’
- c. kuruma-no zasshi
car-NO magazine
‘magazine about cars’ adjuncts
- (38) a. megane-no gakusei
glasses-NO student
‘the student with glasses’
- b. shiraga-no gakusei
white.hair-NO student
‘the student with white hair’
- (Katō, 2003, 96)

¹⁵When the context is sufficiently clear, the postpositions preceding NO can also be omitted.

¹⁶In fact, this is one argument for the often drawn distinction between case particles and postpositions. Case particles do not co-occur with NO. Note furthermore that PPs can also co-occur with case particles, which modifiers belonging to the adjective groups can never do (Murasugi, 1991; Urushibara, 1993).

1.4.4.2 Adjectival Modification

Although most Japanese grammars (Iwasaki 2013; Tsujimura 2014; Hasegawa 2015 etc.), including the so-called *school grammar* (Hashimoto 1948 (1934)), treat lexemes preceding NO as nouns, NO is not restricted to the lexical class of nominals, but is also involved in (qualitative) adjectival modification (Muraki, 2012; Oshima *et al.*, 2019). A good example is offered with the lexeme *mumei-no* ‘unknown’, which appears (almost) exclusively with NO, but exhibits clear characteristics of adjectivehood. These include gradability, at least with adverbs such as *kanzenni* ‘completely’ (39-b) and inside comparative clauses (39-c), the possibility to be nominalized via the suffix -SA ‘-ness’ (39-d), and the impossibility to appear with nominative and accusative case particles, (39-e) and (39-f). For these reasons, such lexemes are sometimes referred to as *no-adjectives* (Mio, 1942, 1958; Muraki, 2012), making them a third morphological subgroup of the adjectival category in Japanese (for different analyses of *mumei-no* see Teramura 1982; Katō 2003; Morita 2013, for an overview see Köhlich 2023).

- (39) a. *mumei-no haiyū*
 unknown-NO actor
 ‘unknown actor’
- b. *kanzenni mumei-no haiyū*
 completely unknown-NO actor
 ‘completely unknown actor’
- c. *Tanaka-san-yori (motto) mumei-no haiyū-to at-ta.*
 Tanaka-Mr.-than more unknown-NO actor-COM meet-PST
 ‘I met with a more unknown actor than Mr. Tanaka.’
- d. *mumei-sa*
 unknown-ness
 ‘unknownness; anonymity’
- e. **Mumei-ga ku-ru.*
 unknown-NOM come-PRS
 (‘(The/An) Unknown comes.’)
- f. **Mumei-wo mi-ru.*
 unknown-ACC see-PRS
 (‘I see (the/an) unknown.’) *no-adjective*

Importantly, lexical modifiers co-occurring with NO are not only clear nouns or clear adjectives, but include many in-between cases. Some modifiers can occur with NA in addition to NO, some can occur with both and exhibit nominal use, there are also some with only occur with NA, but are simultaneously nominals.¹⁷ I will refer to all modifiers that appear with NO as *no-modifiers* adopting a uniform non-committal label for relevant elements.

¹⁷For more on this see Chapter 1.5 below and Köhlich (2023).

Such lexical *no*-modifiers truly share the rest of their inflectional paradigm with *na*-adjectives, meaning they also appear predicatively with the copula *DA*. *NO* as well as *NO* can be paraphrased by the canonical copula *DEARU*.

- (40) a. Ano haiyū-wa mumei-da.
that actor-TOP unknown-COP
‘That actor is unknown.’
b. Ano haiyū-wa mumei-dearu.
that actor-TOP unknown-COP
‘That actor is unknown.’
c. ano mumei-dearu haiyū
that unknown-COP actor
‘that unknown actor’

1.4.4.3 Functional Elements

Besides lexical elements, all functional elements which serve as nominal modifiers, are connected to the head noun via *NO*. This holds for most quantifiers, perhaps an in-between case between lexical and functional modifiers.¹⁸

- (41) takusan-no / subete-no gakusei
a.lot-NO / all-NO student
‘many / all students’ quantifiers

Clear functional elements that can appear as attributive modifiers in Japanese are *wh*-elements (42) (see for instance Watanabe 2001) and classifiers, which follow numerals (43) (Saito *et al.* 2008).

- (42) Doko-no ATM-de shukkin deki-mas-u ka.
where-NO ATM-LOC withdraw can-AUX.POL-PRS Q
‘At which ATMs can you withdraw?’ (Lit. ‘At ATMs where...’) ¹⁹
wh-elements
- (43) a. san-bon-no bīru
three-CL-NO beer
‘three bottles of beer’
b. go-nin-no gakusei
five-CL-NO student
‘five students’ numeral classifiers

NO is also the final mora of demonstratives of the type *kono* ‘this’, but here has a bound character in Modern Japanese.

¹⁸Eventually, I will treat quantifiers as lexical elements. See Chapter 3.1.3.2.

¹⁹<https://help.jibunbank.co.jp/faq-detail.html?id=182,2023/02/05>.

Compared to I and NA, the status of NO is more versatile and harder to pinpoint. The question is whether every instance of NO can be reduced to a single function, at least DP-internally, or whether different instances of NO should be postulated. Possible analyses of NO are outlined in detail in Chapter 3.2.6, but this book attempts at a unified analysis of NO, namely as a syntactic head that licenses modification. The analysis depicted in (20) can therefore be applied to all examples mentioned in this section.²⁰ For instance a numeral (containing a classifier) would be integrated as follows.

- b.
-
- ```
graph TD
 DP_internal[DPinternal] --- Node1[]
 DP_internal --- D_prime_internal[D'internal]
 Node1 --- Empty1[]
 D_prime_internal --- FP_Num[FPNum]
 D_prime_internal --- D[D]
 FP_Num --- ModP[ModP]
 FP_Num --- F_prime_Num[F'Num]
 ModP --- Empty2[]
 ModP --- Mod_prime[Mod']
 Mod_prime --- ClP[ClP]
 Mod_prime --- Mod[Mod]
 ClP --- Empty3[]
 ClP --- Cl_prime[Cl']
 Cl_prime --- NumP[NumP]
 Cl_prime --- Cl[Cl]
 NumP --- Empty4[]
 NumP --- san[san]
 Cl --- bon[bon]
 F_prime_Num --- NP[NP]
 F_prime_Num --- F[F]
 NP --- Empty5[]
 NP --- biru[biru]
 F --- no[no]
```

It seems that NO appears in all instances of (non-adjectival) modification, which is why it is often treated similar to Mandarin DE (see Saito *et al.* 2008). However, it is illicit after verbal relative clauses including a finite (past or non-past) verb (see for instance Murasugi 1991; Saito *et al.* 2008).

22

- (46) a. kare-ga kat-ta (\*no) hon  
 he-NOM buy-PST NO book  
 ‘the book that he bought’  
 b. hon-wo yon-de-i-ru (\*no) gakusei  
 book-ACC read-GER-be-PRS NO student  
 ‘the student who is reading a book’

Similarly, NO never appears after complex morphological past forms, which also contain a tensed form.

- (47) ano shizuka-datta (\*no) yoru  
 that quiet-COP.PST NO night  
 ‘that night which was quiet’

Furthermore, it is never the case that any of the elements NO, NA and I co-occur.

- (48) aka-i (\*no) hana  
 red-I NO flower  
 ‘red flower’  
 (49) kanzen-na (\*no) baka  
 complete-NA NO idiot  
 ‘complete idiot’

Finally, NO is never possible in predicative position.<sup>21</sup>

- (50) Ano haiyū-wa mumei (\*no) da.  
 that actor-TOP unknown NO COP  
 ‘That actor is unknown.’

## 1.5 Data Collection

The theoretical insights of this book are mainly based on an empirical analysis in the Balanced Corpus of Contemporary Written Japanese (BCCWJ).<sup>22</sup> All relevant materials are saved at the following repository: <https://osf.io/u9txp/>. I have analyzed approximately 2300 lexemes which were predominantly collected from the scientific literature, in total from a body of 23 different studies. The categories according to which I incorporated these lexemes were either (*alleged*) *inability to be used predicatively* or *use with -no, but no clear nominal status*. This list of lexemes will be referred to as the *data set*. The repository contains two files with different annotations of this data set. Data set 1 contains a list of all relevant lexemes annotated based on the judgment of each respective author for cross-referencing

<sup>21</sup>The form *no-da* exists, however in this case *no* is an emphatic element whose usage domain is pragmatically restricted.

<sup>22</sup><https://chunagon.ninjal.ac.jp/auth/login> (Maekawa *et al.*, 2014).



purposes. Data set 2 displays the frequency of each relevant variable tested for all lexemes in the corpus.<sup>23</sup>

The repository also contains a data paper, which in detail outlines the design and structure of the corpus and its analysis tools and depicts all necessary steps and search queries that were performed (Köhlich, 2023). It also gives a more detailed overview of the results of the analysis that can be granted here. Readers are welcome to replicate this study, verify the results, implement refinements and provide feedback.

My goal was to establish an overview of how big the inventory of direct, that is non-predicative, modifiers in Japanese can actually be. In this respect, one must be aware that the existing literature on this matter can only be a reference point. Approximately 75% of the lexemes in the data set are taken from the paper collection published as a monograph by MURAKI Shinjirō (Muraki, 2012). Muraki presents long (yet non-exhaustive) lists of different kinds of *no*-modifiers drawn from various Japanese dictionaries, among those an important list of non-predicative lexemes (Muraki, 2012, 96–98), and analyzes them according to his own introspective judgment.<sup>24</sup> The present corpus analysis had the goal of verifying such judgments.

I therefore tested for the variable *predicative use with copula* for all modifiers in the data set. A modifier was classified as direct (group 5) if it was either completely lacking predicative use, or if it yielded no more than 5 predicative occurrences vs. at least 20 attributive ones, or alternatively a ratio of  $5 > 1$  towards attributive use.<sup>25</sup> The relevant numbers are given below.

- (51)    a. Direct modifiers in total: 511  
          b. Out of which direct modifiers with 0 hits in predicative position: 110  
          c. Out of which direct modifiers with no more than 5 hits in predicative position: 190

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<sup>23</sup>Compilation of these lexemes finished at the beginning of 2022. Subsequent analysis began in March 2022 and was finished in August 2023. Readers are referred to a research report published by the Tokyo Gakugei University at the beginning of 2022 (Abe *et al.*, 2022) in which all lexemes given in Muraki (2012) are analyzed for the variables *attributive use*, *predicative use*, *adverbial use* and *referentiality*. This is therefore a major contribution to the field and many aspects are shared between this research report and my own study although sometimes numbers diverge slightly. See the data paper for relevant details.

<sup>24</sup>The validity of introspective judgments, even with regard to seemingly well-known examples and languages, has been put into question in recent years (Gibson and Fedorenko, 2010), and this also applies to Japanese.

<sup>25</sup>As a rule of thumb, only lexemes with at least 5 occurrences are reported in the data set, and only lexemes with at least 25 occurrences were categorized into one of the different groups. I am aware of the fact that fewer predicative occurrences do not automatically mean that a modifier is non-predicative, in other words direct-only. Nevertheless, a skew towards attributive use signals that this is the prototypical usage domain of a given modifier, which is why I classified relevant lexemes into this group even though they might be possible in predicative position.

- d. Prototypical direct modifiers with a ratio of 5>1 towards attributive use: 211

Besides the goal of determining which lexemes are candidates for direct modifiers, another important goal was to point out the heterogeneous word class status of *no*-modifiers. For this the following variables were investigated for each lexeme. Consult Appendix B for more information and numbers on all relevant groups.

- (52)
  - a. occurrences in attributive use
  - b. element in attributive position: NA or NO
  - c. referential use with case particles for subject and object
  - d. nominalization

One might object that corpus data do not tell the whole story for a given phenomenon and indeed, corpora depict a representative sample of real-life language and only show what is *attested*, not what is *possible* or *impossible*. When basing our analyses on corpus queries alone we must keep in mind that certain occurrences might be missed or even overlooked. Nevertheless, a corpus is a powerful tool to give us an overview of a language and its patterns, which in turn allows us to draw important generalizations – if we know how to look for them. As noted in Desagulier (2017, 7), “even if no corpus can provide access to the true, unknown law of a language, a corpus can be considered a sample drawn from this law.”

In order to further balance out these aspects, I have also performed qualitative interviews with native speakers on several occasions and had discussions with many consultants. The discussion on direct modifiers in Chapter 4 is based on a qualitative study with five native speakers to inquire their judgments on grammaticality and acceptability. The study on the position of relative clauses in Chapter 5 is also based on interviews with, three different, native speakers. All these measures were undertaken to ensure the empirical adequacy of the data and the correct theoretical implications to be drawn from them.

# Chapter 2

## Theoretical Background: Cartography

This chapter gives a general introduction to the theoretical background and serves as a reference to what my assumptions are based on and which amendments need to be made in Chapter 3. The framework this book is written in is *Cartography*, although it might be better to speak of a *research field* or a “*research topic*” (Cinque and Rizzi, 2010, 65). Most assumptions of this book come from works by Guglielmo Cinque (Cinque, 1993, 2010, 2017, 2020, 2021, 2023), but I also incorporate ideas from Laenzlinger (2005, 2011, 2015); Svenonius (2008) and Kim (2019). I start by outlining the core ideas of cartography, before tending to direct and indirect modification as given by Cinque (2010). I then discuss the additional approaches and close with an overview over previous studies of Japanese in the cartographic program.

### 2.1 Basic ideas and concepts

What *Cartography* offers is an elaborately articulated structure, a *map* staying true to its name, of the noun phrase and the clause valid in every language. This research program is an offspring of X'-modeling and its adaptation to functional heads (Chomsky, 1986) and developed parallel to Minimalism. Following Pollock's fine-grained universal structure of the TP area (published in Pollock 1989), the clausal spine was constantly refined, most noteworthy through Rizzi's split-CP-hypothesis (Rizzi, 1997) and the highly articulated hierarchy of adverbials across different levels of the clause in Cinque (1999). Following the doctoral thesis of Steven Abney (1987), the noun phrase was in the late 90's and early 2000's extended with several projections as well, including projections for demonstratives (Brugé, 2002), numerals (Ritter, 1993a,b) as well as left-peripheral focus and topic projections (Dimitrova-

Vulchanova and Giusti, 1998; Ihsane and Puskás, 2001; Aboh, 2004, 2006) and of course projections for (adjectival) modifiers, the focal point of this book.<sup>1</sup> One goal of this endeavor, and cartography in general, is to argue for a parallel structure of the clause and the noun phrase.<sup>2</sup>

The essential idea of Cartography is that the extended projection of every lexical head, N and V, contains several functional heads (F) where features are encoded and which give merging and landing sites in their specifier position for lexical and functional elements. Crucially, every functional projection is specified for which element(s) it hosts. Furthermore, these functional heads are ordered, are mapped, in a hierarchy that is based on structural selection and this hierarchy, this map, is universal, thus uniform in every language. This hypothesis also entails that every functional category attested in a given language is present in the structural make-up of all languages. As of now, we have no way of knowing how many functional projections there actually are in the clause and the noun phrase (Rizzi and Cinque, 2016, 154–157).

Languages thus essentially show parametric differences only in the lexicon, not in the syntax. They differ in how (and if) they realize each functional head and its specifier in the surface structure, as well as the type of movement they admit, if any. Essentially, this has the consequence that the structure offered by cartography, although rigid at first glance and in a way the “strongest position linguists can take” (Bortolotto, 2016, 9), is constantly the subject of development and refinement, with every new language and micro-parameter investigated, as well as every new approach and perspective taken. This makes this research field not only highly relevant and powerful, but also extremely versatile and fluctuating.<sup>3</sup>

### 2.1.1 Ordering

The rigid order of functional projections (correctly) predicts the existence of ordering restrictions of the type we saw in English in the introduction. A well-established ordering hierarchy of adjectives attested in several unrelated languages (Sussex, 1974; Dixon, 1982; Hetzron, 1978) and partially attributed to cognitive factors (Sproat and Shih, 1987, 1991) reads as follows, where the symbol ‘ $\succ$ ’ means ‘*precedes in linear order*’.

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<sup>1</sup>See Alexiadou *et al.* (2007) for a large overview on the noun phrase, evidence for these projections and a historical overview of the literature.

<sup>2</sup>See Cinque (1993, 1999, 2010, 2020); Rizzi (1997, 2004); Cinque and Rizzi (2010) and for a list of publications until 2015 Rizzi and Cinque (2016, 144–151). As also becomes obvious there, the clause has been the locus of much more attention, and far fewer controversy.

<sup>3</sup>See Cinque (1993, 1999, 2010, 2020); Rizzi (1997, 2004); Shlonsky (2004); Poletto (2000); Scott (2002); Laenzlinger (2011) and many more. For an overview see Cinque and Rizzi (2010); Rizzi and Cinque (2016).

- (1) QUALITY  $\succ$  SIZE  $\succ$  SHAPE  $\succ$  COLOR  $\succ$  PROVENANCE  $\succ$  N  
(Sproat and Shih, 1991)

It was a central achievement of the cartographic research field (most notably Cinque 1993; Scott 2002; Laenzlinger 2005; Panayidou 2013 and Cinque 2010) to attribute these ordering principles to structural selection, getting rid of adjunction, which cannot explain ordering preferences.

In other words, each of these semantic categories corresponds to a functional head and adjectives are merged as specifiers of the head that matches their semantic value. An adjective denoting SIZE would merge in FPSIZE and so on.

The hierarchical order of these heads can be represented as functional selection in a bracketing format (2) or as in (3) where the symbol ‘ $\succ$ ’ means ‘*is structurally/hierarchically higher*’.<sup>4</sup>

- (2) [QUALITY [SIZE [SHAPE [COLOR [PROVENANCE [N ]]]]]]  
(3) QUALITY  $\succ$  SIZE  $\succ$  SHAPE  $\succ$  COLOR  $\succ$  PROVENANCE  $\succ$  N

This functional hierarchy is expected to be cross-linguistically valid and was verified among others for Romance, Greek, Romanian, Welsh, Mandarin (see for example Scott 2002, the appendix of Cinque 2010 and many examples in Alexiadou *et al.* 2007).

### 2.1.2 Left-right asymmetries

A fundamental contribution of Cartography, first systematized in Cinque (2005, 2009) and extended in Cinque (2023, 2025), was to shed light on so-called *Left-right asymmetries*. It has been a central cross-linguistic observation (see for example Rijkhoff 2002), and Greenberg’s Universal 20 (Greenberg, 1963), that there is a restriction on possible orders of modifiers around a lexical head when they are all on the same side of the head: When they appear on the left side of the lexical head, they are ordered in only one specific way, but when they follow the lexical head, two orders are possible: Either the identical order maintaining linear precedence, or the mirror-image, in which the order has been reversed compared to the pre-head order. This observation was verified for the order of adjectives (Cinque, 2009; Shlonsky, 2004), but for many other domains as well, including the order of adverbs as well as circumstantial and directional/locative prepositions around the verb (Cinque, 2009, 167–168).

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<sup>4</sup>When an element A is hierarchically higher than an element B, it precedes it in structural, but not necessarily in linear order since movement operations might yet take place. The symbol ‘ $\succ$ ’ means ‘precedes in linear order’. The same technique is applied in Panayidou (2013).

A very illustrative case, and perhaps the central case of application in the nominal domain, is the order of demonstratives, numerals and adjectives with respect to the noun. Whereas languages such as English or German reflect the underlying order *demonstrative*  $\succ$  *numeral*  $\succ$  *adjective*  $\succ$  *noun*, in languages in which these constituents appear postnominally (in canonical order), two possibilities are found: The relative order of modifiers as they appear before the noun in English, or the mirror-image of the English order. Crucially however, the mirror-image order in prenominal position is not attested as the natural order of constituents in any language. This is visualized below.

- (4) Order of demonstratives, numerals and adjectives (Cinque, 2009, 166)
- a. Dem  $\succ$  Num  $\succ$  A  $\succ$  N (English, German...)
  - b. N  $\succ$  Dem  $\succ$  Num  $\succ$  A (Abu', Kikuyu...)
  - c. N  $\succ$  A  $\succ$  Num  $\succ$  Dem (Gungbe, Thai...)
  - d. \*A  $\succ$  Num  $\succ$  Dem  $\succ$  N (not attested)

Although humans do have a strong bias towards homomorphism, that is placing all elements on the same side of a lexical head (cf. Martin *et al.* 2020, references therein and further work by the same authors), in many different languages, modifiers appear on both sides of a lexical head in linear order. The crucial point is however that not all mathematically possible 24 (factorial 4) orders are attested as base orders across the world's languages, but only 14 out of 24 (Cinque 2005, 2023, and see already Greenberg 1963; Hawkins 1983). Besides the order A  $\succ$  Num  $\succ$  Dem  $\succ$  N, neither are Num  $\succ$  Dem  $\succ$  N  $\succ$  A or Dem  $\succ$  A  $\succ$  Num  $\succ$  N attested (Cinque, 2005, 319–320).

Under the anti-symmetric approach to linearization of Kayne's 1994 Linear Correspondence Axiom (LCA), according to which there is only one structural hierarchy of elements,<sup>5</sup> and the relation between these elements is governed via *asymmetric c-command*,<sup>6</sup> those left-right asymmetries have been argued to fall naturally, given the assumption that only one underlying hierarchy exists and all other orders arise via movement. The purpose of movement in Cartography is thus primarily the alteration of linear precedence, not feature checking as in minimalist frameworks.

Second, one needs to take into account that movement is restricted, otherwise we would see all of the 24 possible orders attested cross-linguistically. It is only the lexical head that moves and only via phrasal movement to the left, but in different fashions. It moves either alone, crossing each higher functional projection it is c-

<sup>5</sup>“ $d(A)$  is a linear ordering of  $T$ ”, (Kayne, 1994, 6).

<sup>6</sup>“Let  $X, Y$  be nonterminals and  $x, y$  terminals such that  $X$  dominates  $x$  and  $Y$  dominates  $y$ . Then if  $X$  asymmetrically  $c$ -commands  $Y$ ,  $x$  precedes  $y$ .”, (Kayne, 1994, 31); see also Moro (1997, 2000).

commanded by (and possibly halting in-between causing non-homomorphic orders), or in a pied-piping fashion (also *roll-up* or *snowballing*, Ross 1967) where it takes each constituent present in each functional projection it crosses along and reverses their order (Cinque, 2017, 2023). When the noun has reached its destination, the order of all element it has taken along has been reversed accounting, for instance, for the order  $N \succ A \succ \text{Num} \succ \text{Dem}$  (Cinque, 2005, 318). Pied-piping can also occur partially, deriving for example the order  $\text{Dem} \succ \text{Num} \succ N \succ A$  in which the NP has moved to an agreement position above the adjective. Orders such as  $^*\text{Num} \succ \text{Dem} \succ N \succ A$  can, however, never be achieved since the order  $\text{Num} \succ \text{Dem}$  can only be achieved if the noun moves across these categories, which it has not done (Cinque, 2005, 317, 318).<sup>7</sup>

These mechanics can be extended to all other domains mentioned above as well, that is the order of adjectives, adverbs and prepositions, among others. As recently argued in Cinque (2025), it is exactly this point, the applicability across different domains and the prediction on which orders are attestable, that constitutes a big advantage of the movement approach over other approaches. I will include a brief discussion on how the account proposed in this book fares with these observations in Chapter 6.1.3.

### 2.1.3 The Japanese DP: No NP-movement

Admittedly, there is a skew in cartographic research towards languages of the Indo-European family. Japanese is considered in Cinque (2017, 3–4) a “closest to the ideal” head-final language. This is because modifiers (demonstratives, relative clauses, numerals, adjectives) systematically precede the noun, but most functional morphemes (suffixes for plurality or case) follow it.<sup>8</sup> An example of a Japanese DP containing an adjective and a plural suffix is illustrated in (5).

- (5) [Waka-i gakusei-tachi-ga] ki-ta.  
 young-I student-PL-NOM come-PST  
 ‘The young students came.’

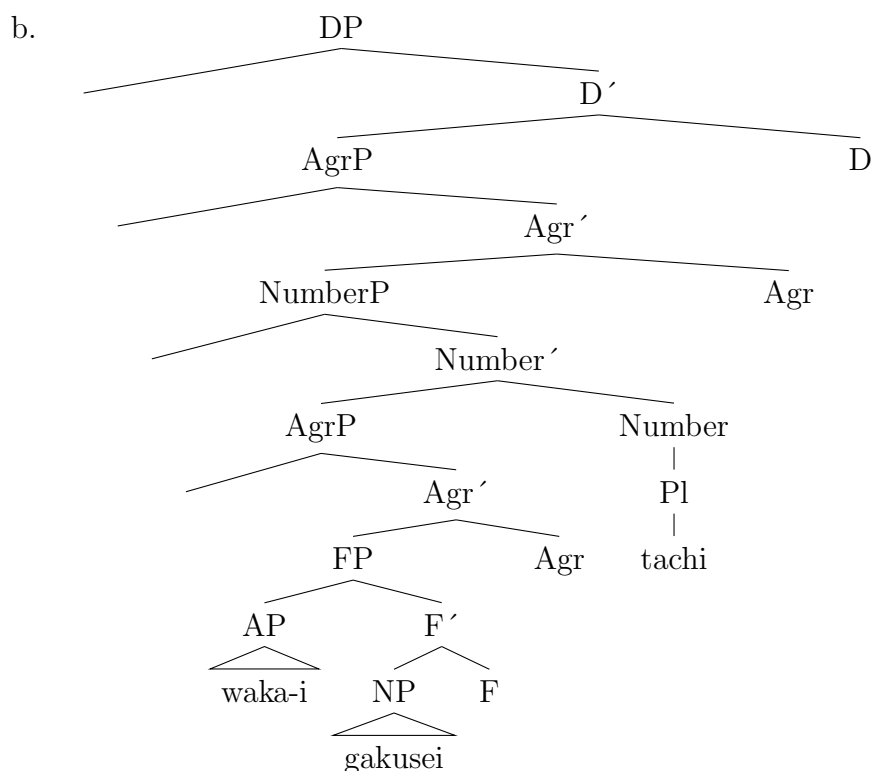
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<sup>7</sup>Matters are more complex. There is another type of pied-piping, namely the so-called *pictures-of-whom type* or *regressive* pied-piping, in which the head moves and takes its constituents with it, however *without* reversing the order. Furthermore, several movement operations of different types can combine in one derivation. An important argument for the complexity of these movement operations is the derivation of certain rare constituent orders across languages. For instance, if we take  $\text{Dem} \succ \text{Num} \succ A \succ N$  as the base order, the order  $\text{Num} \succ A \succ N \succ \text{Dem}$ , for which Cinque (2005, 2023) finds 51 languages, can only be derived if the NP moves above the functional projection hosting NumP in a pictures-of-whom pied-piping fashion, not reversing the order, and then continues moving above DemP in a whose-pictures type fashion, this time reversing the order (Cinque, 2023, 36). See Cinque (2025, 9–11) for an application to adjective order. For a recent contribution on U20-effects in a nanosyntactic framework see Pinzin and Mattiuzzi (2025).

<sup>8</sup>Exceptions are honorific morphemes, which are realized as prefixes.

As modifiers always precede the noun, no NP-movement needs to be assumed in the strict sense. Under the assumption that adjectives such as *waka-i* ‘young’ are embedded in specifier positions of functional projections, but number suffixes are heads, order falls out naturally as depicted in the structural representation of (5) in (6). Note that I deviate here from the strict application on Kayne’s LCA in depicting Japanese trees as head-final. Movement does become necessary if we were to depict Japanese structures as head-initial.<sup>9</sup>

- (6) a. *waka-i gakusei-tachi*  
       young-I student-PL



## 2.2 Direct and indirect modification

In this section, I review the differences between direct and indirect modifiers, first, in order to derive reliable diagnostics for the type of modification, second, to relate their different structural positions to these diagnostics. In the following two sections, I elaborate on direct-only modifiers and the derivation of indirect modifiers. Discussions will be kept short here as they will be picked up in the following chapters.

<sup>9</sup>This point also extends to other languages and phenomena. See Buring and Hartmann (1997) for a discussion on German extraposed relative clauses, which are hard to analyze with the rigid assumptions of the LCA. See Abels and Neeleman (2012) for general discussion of the problems this axiom entails.



### 2.2.1 Introduction

The notions *indirect* and *direct modification* were established by Sproat and Shih (1991). The authors used them as an explanation adjectives in Mandarin Chinese are ordered rigidly in some instances, but not in others. Precisely, they noted that an overt exponent of indirect modification is the element DE 的 (Sproat and Shih, 1991, sec. 2).<sup>10</sup> This element is optional in adjectival modification, but when it appears, it causes the absence of otherwise attested ordering restrictions, which are otherwise similar to the English case (Sproat and Shih, 1991, 579). This is shown below.

(7) SIZE  $\succ$  COLOR without DE

a. xiǎo lǜ huāpíng  
small green vase

b. ?lǜ xiǎo huāpíng  
green small vase  
'small green vase'

(Sproat and Shih, 1991, 566)

(8) SIZE vs. COLOR with DE

a. xiǎo de lǜ de huāpíng  
small DE green DE vase

b. lǜ de xiǎo de huāpíng  
green DE small DE vase

Furthermore, this element is mandatory in verbal modification, which arguably always involves a relative clause structure.

(9) [nèizhī jiào \*(de)] gǒu  
that bark DE dog

'the dog which is barking'

(Sproat and Shih, 1991, 573)

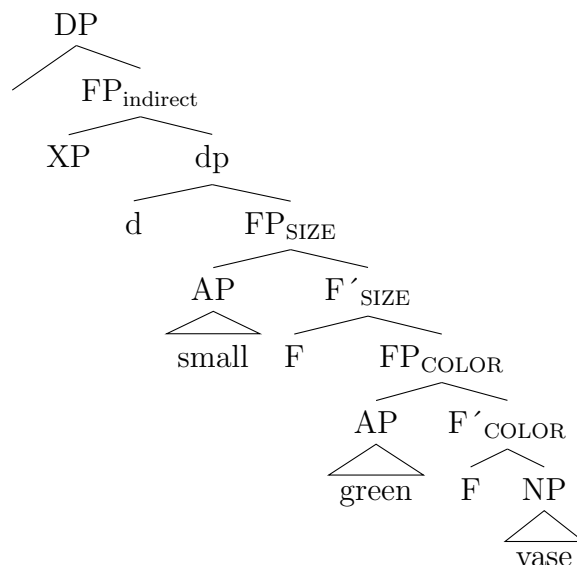
As mentioned above, cartography predicts that direct adjectives are merged in the specifier positions of functional heads that match their semantics and which are rigidly ordered. And since  $FP_{\text{SIZE}}$  structurally dominates  $FP_{\text{COLOR}}$ , linear adjacency is expected in context-neutral utterances. The hierarchical structure is depicted below for English (Cinque, 2010, 28–30).<sup>11</sup>

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<sup>10</sup>Analyzing the precise nature of DE has been the task for a long time in syntactic research. Sproat and Shih (1991) analyze it as a complementizer (see also del Gobbo 2010), but authors including Cinque (2010) do not share this proposal and have shown that some direct modifiers in fact have to appear with this element, (cf. Paul 2005, 2010, 2012; Yang 2005) and analyze it as 'phrasal modifier'. Other analyses include 'case assigner' (Larson, 2009) or 'linker' (den Dikken and Singhapreecha, 2004).

<sup>11</sup>The threshold between the direct and the indirect domain is the indefinite head d. See Chapter 2.4.1.3.

- (10) a. small green vase  
 b. ??green small vase  
 c.



Indirect modifiers, on the other hand, are merged in the projection of relative clauses and since these functional projections are unordered with respect to semantic category, it is expected that indirect modifiers exhibit free order on the surface. This is what is at stake in Mandarin in (8), claimed Sproat and Shih (1991), and this behavior is also observable in English (Baker 2003b).<sup>12</sup>

- (11) a. a vase that is small that is green  
 b. a vase that is green that is small

### 2.2.2 Reduced relative clauses and ambiguities

Now, as mentioned in the introduction, most adjectives in English are actually ambiguous between a direct and an indirect status. Direct modifiers, here called *direct-only*, are a very rare species. In fact, all adjectives that can have access to predicative use can have an indirect status when they appear in a simple modification structure. They are then not embedded under a full (also: finite) relative clause, but are instances of so-called *reduced relative clause* (RRC) (Cinque 2010, 54–55, Douglas 2016; Harwood 2018), relative clauses which lack a relative pronoun/a complementizer and a verb/a copula. This fact has also been put forth to explain deviations from expected ordering restrictions. If two indirect adjectives modify a noun, no fixed order is predicted, and languages where adjectives never fol-

<sup>12</sup>Speakers might disagree on the possibility of relative clause stacking. And while (11) does sound a bit stilted and was chosen mostly for the parallelism to the Mandarin example, same as the German example in (28), examples of stacked relative clauses can be found, although preferably with longer relative clauses such as “a book [that I wanted to read for a long time] [that was recommended to me by my teacher]”.

low ordering restrictions are predicted to only have indirect modification adjectives (cf. Cinque 2010, ch. 3.3.2).

Verbal reduced relative clauses are easily recognizable in English, namely by virtue of lacking relative pronouns and copula, as for instance is the case with participle and infinitival relative clauses (Kayne, 1994; Cinque, 2020), but there are also clear cases of indirect adjectives embedded in reduced relative clauses. Prototypical examples are postnominal adjectives of the type *the man proud of his children* (Williams, 1982; Kayne, 1994; Sadler and Arnold, 1994; Larson, 2000; Larson and Marušič, 2004). While most adjectives in English are only licit in this position when they occur with complements or adjuncts,<sup>13</sup> there are some adjectives that can appear independently in this position, in addition to the prenominal position. Those include adjectives with the ending *-ible/able* as in the following famous example from Bolinger (1967, 3–4).<sup>14</sup>

- (12)    a.    the (only) visible stars  
           b.    the (only) stars visible

It is well-known that such minimal pairs in English exhibit a meaning difference. In postnominal position, the adjectives denote a temporary, passing property, dubbed *stage-level* reading. In prenominal position, they are ambiguous and can have this reading but additionally can also be understood as denoting an enduring property dubbed *individual-level* reading (see Sadler and Arnold 1994; Larson 1998; Larson and Marušič 2004 and especially Cinque 2010, Chapter 2.1). This allows the following paraphrases.

- (13)    a.    the (only) stars visible  $\approx$  ‘the only stars visible at the moment/that happen to be visible’ (stage-level)  
           b.    the (only) visible stars  $\approx$  ‘the only stars visible at the moment/that happen to be visible’ (stage-level) OR ‘the only stars visible in general’ (individual-level)

There are many such minimal pairs in English. What is common to each of them is that it is always the prenominal variant that is ambiguous, while the postnominal variant is restricted to a certain reading as shown below for the famous case of *beautiful dancer* which can either have an intersective or a non-intersective reading (Larson 2000; Vendler 1957; Siegel 1976; Larson 1995).

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<sup>13</sup>The *Head-Final Filter* of Williams (1982), cf. Alexeyenko and Zeijlstra (2021); Richards (2022, 2023).

<sup>14</sup>Adjectives with the initial prefix *a-* such as *afraid* or *asleep* predominantly occur in postnominal position (Bolinger 1967; Larson 1998; Larson and Marušič 2004; Alexiadou *et al.* 2007, 295; Belk 2017, ch. 2.3).

- (14) Non-Intersective vs. Intersective (Larson, 1995, 145)
  - a. Olga is a beautiful dancer. (ambiguous)
  - b.  $\approx$  ‘Olga is a dancer that dances beautifully.’ (non-intersective reading)
  - c.  $\approx$  ‘Olga is a dancer and Olga is beautiful.’ (intersective reading)
- (15) Non-Intersective vs. Intersective (Cinque, 2010, 9)
  - a. Olga is a dancer more beautiful than her instructor. (only intersective)
  - b.  $\neq$  ‘Olga is a dancer that dances beautifully.’ (non-intersective reading)
  - c.  $\approx$  ‘Olga is a dancer and Olga is beautiful.’ (intersective reading)

Cartography offers an explanation for this: The ambiguity in interpretation observed in prenominal position arises since the adjective is syntactically ambiguous - it could be either direct or indirect. On the other hand, in postnominal position, the adjective can only be indirect. This can be verified when embedding it in a full relative clause. Again, only one reading is available: The same as in the postnominal use. This confirms that postnominal adjectives in English are predicative in nature (Cinque, 2010, 18–19).

- (16) Non-Intersective vs. Intersective, full RC (Cinque, 2010, 9)
  - a. Olga is a dancer who is beautiful. (only intersective)
  - b.  $\neq$  ‘Olga is a dancer that dances beautifully.’ (non-intersective reading)
  - c.  $\approx$  ‘Olga is a dancer and Olga is beautiful.’ (intersective reading)

As a consequence, the reading not accessible in postnominal position, and not derived from predicative use, must be the direct reading and when adjectives such as *beautiful* exhibit that reading, they must be interpreted as direct.

### 2.2.3 Direct-only modifiers

There is a small group of adjectives that are unambiguously direct. Those are called *direct-only* in this book. According to Cinque (2010, ch. 4.1), these modifiers are *functional* and *phrasal*. They are functional because they arguably form a closed class, in most languages only with a couple of members (Cinque, 2010, 43–44), explaining many cross-linguistic intersections. They are phrasal because they can appear with complements and have a phrasal character in some languages (Cinque, 2010, 44–49). Below, I will review some well-known members as well as characteristics, but it should be kept in mind that both are, most of all, prototypical.

A well-known group of direct-only modifiers are *non-intersective*, also called *intensional*, adjectives (Kamp and Partee, 1995; Panayidou, 2013), such as *former*. Their inability to appear in predicative position, and thus also inside a relative



- (22) The visible visible stars include Capella. (Larson, 1998, 155)  
the stage-level individual-level stars

The two modifiers can also be spread out on both sides of the noun. Then, as expected, the postnominal modifier must bear the indirect modification, the stage-level reading (Cinque, 2010, 19–20).

- (23) The visible stars visible include Capella. (Larson, 1998, 155)  
the individual-level stars stage-level

These observations hold for the other dichotomies, for example the intersective/non-intersective case.

- (24) a. Olga is a beautiful beautiful dancer.  
Olga is a intersective non-intersective dancer  
(Larson and Marušič, 2004, 281)
- b. Olga is a beautiful dancer more beautiful than her  
Olga is a non-intersective dancer more intersective  
instructor.

(Cinque, 2010, 19)

### 2.2.5 More on ordering restrictions

Standard cartography explains ordering restrictions among direct modifiers as a result of the distinct positions of the respective functional heads in the direct domain. Several hierarchies of functional heads for direct modification have been proposed, but a standard hierarchy for object nouns could look as follows. I have chosen the label DIMENSION as a more broad term than SIZE. It also includes the categories LENGTH and WIDTH from Scott (2002, 114).<sup>15</sup>

- (25) QUALITY > DIMENSION > SHAPE > COLOR > NATIONALITY > MATERIAL  
> N

Importantly, ordering restrictions are only maintained in *hierarchical* modification. In *parallel* modification, achieved via syndetic or asyndetic coordination, no ordering restrictions are expected (Sproat and Shih, 1991, 578–580).

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<sup>15</sup>Modifiers of *action* or *event* nouns (Grimshaw, 1990), correspond to adverbs in the verbal domain (Cinque, 1993, 1999; Alexiadou, 1997) and are often given in the following hierarchy: SPEAKER-ORIENTED > SUBJECT-ORIENTED > MANNER > THEMATIC > N (Alexiadou *et al.*, 2007, 324).

## (26) Parallel Modification (English)

- |    |                    |           |
|----|--------------------|-----------|
| a. | a red and big book | syndetic  |
| b. | a red, big book    | asyndetic |

**2.2.6 Summary: Direct vs. indirect modifiers**

The defining characteristics that differentiate direct from indirect modifiers which offer valid diagnostics and tests to determine the status of a given modifier are summarized in the following table (cf. Cinque 2010, 33 for a more detailed version).

| Direct                                                         | Indirect                                                      |
|----------------------------------------------------------------|---------------------------------------------------------------|
| Exhibit special readings, such as non-intersective, modal etc. | Exhibit the opposite readings: Intersective, implicit-RC etc. |
| Take part in idiomatic constructions                           | Only give literal interpretation                              |
| Impossible as predicates                                       | Possible as predicates                                        |
| Closer to the head noun                                        | Further away from the head noun                               |
| Ordered rigidly                                                | Show free/arbitrary ordering                                  |
| Tend to be more simple                                         | Are usually complex                                           |

Table 2.1: Differences Between Direct and Indirect Modifiers

**2.2.7 Cross-linguistic perspective**

It is an important enterprise of the cartographic research programs to verify those facts in cross-linguistic perspective. And indeed, they can, and often do, serve as valid diagnostics for modification status in other languages as well. A cartographic account of the DP has been proposed for French (Laenzlinger, 2005, 2011), Greek (Cinque 2010, appendix; Panayidou 2013), Russian (Siegel 1976; Cinque 2010, appendix), Dutch (Belk, 2017), Korean (Kang, 2005, 2006; Kim, 2019), Javanese (van der Klock, 2009) and Kipsigis (Kouneli, 2019).<sup>16</sup> That ordering restrictions, or at

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<sup>16</sup>See also the appendix of Cinque (2010) for Romanian, Maltese and German and the typological sample in Laenzlinger (2010, 2011).

least tendencies, have cross-linguistic validity is exemplified for German in (27). Again, once the modifiers are embedded in corresponding relative clauses, their order becomes free (28).

(27) SIZE vs. COLOR (German)

- a. ein großes rotes Buch  
a big red book
- b. ??ein rotes großes Buch  
a red big book  
'a big red book'

(28) SIZE vs. COLOR indirect (German)

- a. ein Buch, das groß ist, das rot ist  
a book that big is that red is
- b. ein Buch, das rot ist, das groß ist  
a book that red is that big is  
'a book which is big which is red'

The resolution of ambiguities also holds in Italian (Romance), where adjectives are freer in their distribution to the left and to the right of the noun. Ambiguous adjectives of the *beautiful*-type show this ambiguity in their prototypical position as well, but this is the postnominal position (29). On the contrary, in the, more marked, prenominal position, the relevant adjectives only have one interpretation. This interpretation is the opposite of the English postnominal cases, thus corresponds to the direct reading (30). From this, Cinque (2010, ch. 2.4) concludes that prenominal adjectives in Italian must be direct (examples taken from Cinque 2010, 10).

- (29) Un **attaccante buono** non farebbe mai una cosa del genere.  
a forward good NEG would.do never a thing of.DET sort  
1..  $\approx$  'A forward good at playing forward would never do such a thing.'  
(non-intersective)  
2.  $\approx$  'A good-hearted forward would never do such a thing.' (intersective)
- (30) Un **buono attaccante** non farebbe mai una cosa del genere.  
a good forward NEG would.do never thing of.DET sort  
1.  $\approx$  'A forward good at playing forward would never do such a thing.'  
(non-intersective)  
2.  $\neq$  'A good-hearted forward would never do such a thing.' (intersective)

The same ambiguous adjective with two different readings is only possible postnominally. Nevertheless, the relative closeness is preserved: The adjective with the direct modification interpretation must be closer to the noun than the adjective with the indirect modification reading (Cinque, 2010, 20–22).



- (31) Un attaccante **buono** buono  
 a forward good(non-intersective) good-hearted(intersective)  
 ‘a good-hearted good forward’ (Cinque, 2010, 21)

## 2.3 Groups of direct-only modifiers

This section presents the two biggest cross-linguistic groups of direct-only modifiers. Besides non-intersective modifiers, which were already looked at, I will discuss *relational* modifiers. Both these groups are relevant for the discussion of direct modifiers in Japanese in Chapter 4.

### 2.3.1 Non-intersective modifiers

#### 2.3.1.1 Truly non-intersective modifiers

Non-intersective modifiers of the type *former* or *alleged* are characterized by the fact that they do not denote a set intersection with the noun they modify (Vendler, 1957; Siegel, 1976; Bolinger, 1967; Kamp and Partee, 1995; Larson, 1998). This is illustrated with the English adjective *former* below where there is no set intersection between the set of presidents and the set of former entities pertaining to Obama.

- (32) a. Obama is the/a former president.  
 b. [former president]  $\neq$  [former]  $\cap$  [president]

As predicative use enforces the intersective reading, it is only logical that these modifiers cannot be used predicatively.

- (33) \*This president is former.

Such *truly non-intersective* modifiers often refer to a time (former, future, present), a place (*dortig*; German: ‘over there’) or a manner (possible, main), highlighting their connection to adverbs. This group has also been recognized in Japanese (Yamakido, 2005; Ayano, 2010) and behaves alike. Non-intersective modifiers in Japanese cannot be used predicatively and do not denote intersections with the noun.

- (34) a. Tarō-ga shōrai-no purosakkāsenshu-da.  
 Taro-NOM future-NO pro.soccer.player-COP  
 ‘Taro is a future professional soccer player.’  
 b. \*Ano purosakkāsenshu-ga shōrai-da.  
 that pro.soccer.player-NOM future-COP  
 (‘That professional soccer player is future.’)  
 c. [shōrai-no purosakkāsenshu]  $\neq$  [shōrai]  $\cap$  [purosakkāsenshu]

### 2.3.1.2 Subjective modifiers

Notably, there is an in-between-class of modifiers between intersective and non-intersective modifiers, usually called *subjective*, where modifiers co-occur with nouns depicting an event or a profession and only modify part of the extension of those nouns. A well-known example is the following (Kamp and Partee, 1995, 137–138).

(35) Mary is a skillful surgeon.

The most salient interpretation of this phrase is that Mary is skillful as a surgeon, so she is experienced in her profession of operating people. The interpretation ‘Mary is skillful.’ again seems not really available, because we do not know what she could be skillful at. Subjective modifiers are able to appear in predicative position and thus also in relative clauses, but then, only one interpretation remains, namely the intersective one. The subjective interpretation, on the other hand, is inaccessible.

- (36)
- a. Mary is a surgeon who is skillful. (only intersective)
  - b.  $\neq$  ‘Mary is skillful as a surgeon.’ (subjective)
  - c.  $\approx$  ‘Mary is a surgeon and Mary is skillful.’ (intersective)

Subjectivity is the source of most ambiguities mentioned above, including the famous *beautiful dancer* example (Larson 1995, 145; Vendler 1957; Siegel 1976). The paraphrase ‘beautiful as a dancer’ is in fact the subjective, while the paraphrase ‘beautiful and a dancer’ is the intersective reading.

- (37)
- a. Olga is a beautiful dancer.
  - b.  $[\text{beautiful dancer}] \subseteq [\text{dancer}]$  (subjective)
  - c.  $[\text{beautiful dancer}] = [\text{beautiful}] \cap [\text{dancer}]$  (intersective)

Another well-known ambiguous example is the following.

- (38)
- a. Peter is an old friend. (Larson, 1998, 146)
  - b.  $[\text{old friend}] \subseteq [\text{friend}]$  (subjective: ‘long-time friend’)
  - c.  $[\text{old friend}] = [\text{old}] \cap [\text{friend}]$  (intersective: ‘aged friend’)

In Chapter 4.3.1, it will be shown that Japanese nominal modification does not readily allow such ambiguities, one reason why direct modification has been denied. Instead, ambiguities are resolved via different lexemes, each exhibiting an intersective and a non-intersective reading.

**2.3.1.3 Privative modifiers**

Another group of non-intersective modifiers are privative modifiers. Such modifiers, often similar in spirit to the meaning ‘fake’, denote a zero intersection with their head noun.

- (39) a. a fake gun  
 b.  $[\text{fake gun}] \cap [\text{gun}] = \emptyset$  (Morzycki, 2016, 25)

Interestingly, such modifiers can appear in predicative position, which is why Par-tee (2009) has proposed that privative adjectives in English and Russian might be subsective after all, therefore not direct-only.

- (40) This gun is fake / pretend / artificial. (Morzycki, 2016, 25)

This is also the case in Japanese where predicative use of the privative modifier *nise-no* ‘fake’ is degraded, but not entirely ungrammatical.

- (41) ?Kono pisutoru-wa nise-da.  
 this gun-TOP fake-COP  
 ‘This gun is fake.’

As shown in Cinque (2014, 21–23) however, there is reason to analyze privative modifiers as direct-only. Consider the following minimal pair with the Italian equivalent *falso* ‘false’.

- (42) a. le false banconote  
 DET fake bank.note.PL  
 only: ‘Monopoly money’  
 b. le banconote false  
 DET bank.note.PL fake  
 only: ‘counterfeited money’ (Cinque, 2014, 21)

In prenominal position – remember the position only having a direct modifier reading in Italian – *falso* has a reading as ‘*Monopoly money*’, whereas in postnominal position it only has a reading as ‘*forged/counterfeited money*’. As expected, this reading corresponds to the reading in predicative position.

- (43) Quelle banconote sono false.  
 this bank.note.PL are fake  
 1.  $\approx$  ‘This money is counterfeited.’  
 2.  $\neq$  ‘This money is Monopoly money.’ (Cinque, 2014, 21)

### 2.3.1.4 Adnominal degree modifiers

A final group is what Morzycki (2016) calls *adnominal degree modifiers*.

- (44) He is a complete / total / utter / absolute / true idiot.  
(Morzycki, 2016, 56–57)

As the name suggests, Morzycki (2014, 2016) does not see these elements as lexical adjectives, but rather as functional modifiers that denote a (often, but not always extreme) point on a scale for certain nouns. Nevertheless, they are relevant for this study since they cannot be used predicatively.

- (45) \*This idiot is complete / total / utter / absolute / true.

## 2.3.2 Relational adjectives

### 2.3.2.1 Nominal derivation and syntactic constraints

*Relational adjectives*, such as *environmental* or *technical*, make up a relatively rich research field. They have been discussed predominantly with regard to Romance languages and English, but also in Japanese (Bisetto, 2010; Nagano and Shimada, 2015; Nagano, 2016, 2018; Shimada and Nagano, 2018; Morita, 2010, 2011, 2013) and will be picked up in Chapter 4.3.2. Morphologically, they are characterized via their denominal appearance, exhibiting typical suffixes such as *-al*, *ic(al)* in English or *-ar*, *-lich* or *-isch* in German. This oftentimes gives relational adjectives a scientific flavor (cf. Gunkel and Zifonun 2008).<sup>17</sup> Semantically, they denote a specific relationship to the nouns they modify and do not just simply add a quality, as would ordinary *qualitative* adjectives. Syntactically, the following cross-linguistic characteristics of relational adjectives have been established.<sup>18</sup>

- (46) Characteristics of relational adjectives
- a. They cannot be used predicatively. (\*this crisis is environmental)
  - b. They are not gradable. (\*more/most environmental crisis; \*very environmental crisis)

<sup>17</sup>They are therefore also sometimes called “denominal adjectives” or “(denominal) pseudo-adjectives” (Postal, 1969; Bartning, 1980). See Bortolotto (2016, 52–56) for an overview.

<sup>18</sup>For English and Romance see Postal (1969); Downing (1977); Levi (1978); Bartning (1980, 1984, 1986); Warren (1984); Bosque (1993); Bosque and Picallo (1996); McNally and Boleda (2004); Fábregas (2007, 2014); Gunkel and Zifonun (2008); Roy (2010); Zifonun (2011); Rainer (2013); Bortolotto (2016), and for an overview Alexiadou *et al.* (2007, Chapters 3.3 and 9.2). They have also been discussed in German (Hotzenköcherle, 1967; Schäublin, 1972; Eichinger, 1979; Frevel and Knobloch, 2005; Trost, 2006a), Russian (Mezhevich, 2002), Polish (Cetnarowska, 2015), Hungarian (Kenesei, 2014) and Selkup (Spencer and Nikolaeva, 2017; Nikolaeva and Spencer, 2020). See Nagano (2016, 44) or Bortolotto (2016, ch. 2.3) for similar overviews.

- c. They have to be strictly adjacent to their head noun and cannot be combined with regular qualitative adjectives. (\*environmental huge crisis; \*huge and environmental crisis; \*environmental and huge crisis)
- d. They appear in certain positions from which they cannot move, typically they occur in postnominal position in Romance languages. (\**chimico processo*; *processo chimico* ‘chemical process’, Italian; Bortolotto 2016, 59)
- e. They cannot be nominalized. (\*environmental-ity)
- f. They lack polarity/do not come in antonym pairs. (\*un-environmental; \*non-environmental)

The fact that they cannot be used predicatively defines them as direct modifiers as tested in many different languages.

- (47) a. \*My appointment this afternoon is dental. (English) (Rae, 2010, 49)  
b. \*This linguist is theoretical.

- (48) \*Dieser Linguist ist theoretisch. (German)  
this linguist is theoretical

- (49) \*L’élection est présidentielle. (French)  
DET.election is presidential

(Bortolotto, 2016, 68)

- (50) \*Magazin byl kniž-n-yj. (Russian)  
store was book-RA-NOM.SG.M

(Mezhevich, 2002, 100)

Importantly, relational adjectives can occur in predicative position in certain circumstances. One involves the class shift into a qualitative adjective (cf. Nagano 2018).

- (51) The way he approached this problem was very theoretical.

This shift can be facilitated by degree adverbs such as *very* as in (52). The *ethnic* adjective *Italian* (Alexiadou and Stavrou 2011, see below) has acquired a qualitative meaning as shown by the available paraphrases.

- (52) a. The party yesterday was very Italian.  
b. ≠ ‘The party took place in Italy, was organized by Italians, etc.’ (relational reading)  
c. ≈ ‘The party featured many Italian dishes, music, etc.’ (qualitative reading)

Another way in which relational adjectives can be used predicatively is when they have a *kind-reading* (McNally and Boleda, 2004) as in the following example from Catalan. This example can essentially be paraphrased by *this tuberculosis can be of the pulmonary kind/type*, and from a syntactic point of view allows the recovery of a noun *kind* or *type*.

- (53) La tuberculosi pot ser pulmonar. (Catalan)  
 DET tuberculosis can be pulmonary  
 ‘The tuberculosis could be pulmonary.’

(McNally and Boleda, 2004, 189)

Relational adjectives also resist gradability, at least with *indefinite degree adverbs* (Fábregas, 2014) of the type *very*. Many are however compatible with so-called *modifiers of extension* (Fábregas 2014; called ‘*endpoint-oriented modifiers*’ in Kennedy and McNally 2005) such as *completely* or *100 percent*.

- (54) This is a \*very / completely / 100 percent environmental problem.

Relational modifiers, as direct modifiers, are not only very low in the DP, they are also lower than many other direct modifiers, predicting strict adjacency to the noun. This holds cross-linguistically as shown for French below. Relational adjectives, such as *industriel* ‘industrial’, obligatorily appears postnominally, and must be closer to the noun than qualitative adjectives, whatever status one assumes for them.

- (55) a. un développement industriel rapide (French)  
 a development industrial rapid  
 ‘a fast industrial development’  
 b. \*un développement rapide industriel  
 a development rapid industrial (Bortolotto, 2016, 61)

In Japanese, relational adjectives have been identified amongst the group of *no*-modifiers, free of morphological processes (Bisetto 2010; Nagano 2016 a.o.). Importantly they do fulfill the relevant characteristics, a restriction on predicativity and gradability.

- (56) a. kankyō-no kiki  
 environment-NO crisis  
 ‘environmental crisis’  
 b. \*Kono kiki-ga kankyō-da.  
 this crisis-NOM environment-COP  
 (‘This crisis is environmental.’)  
 c. \*totemo kankyō-no kiki  
 very environment-NO crisis  
 (‘very environmental crisis’)

This suggests that those lexemes are simply nouns, since no obvious derivational patterns are detectable. Furthermore, relational adjectives in this language do not compete with other constructions, such as compounds or PPs with the same meaning, which is very well observed in European languages (for example *nominal modification* vs. *noun modification* in English, ten Hacken 2013; Rainer 2013; Nikolaeva and Spencer 2020). Again, Japanese has only one option of modification at its disposal where other languages can show variation.

### 2.3.2.2 Classificatory, thematic and ethnic adjectives

While in many cases, relational adjectives, classifies the type of noun they modify, they can also express argument status. This happens when they modify event or action nouns (Grimshaw, 1990).

(57) environmental protection

Here, *environmental* expresses the role of the internal argument of the verb *to protect* underlying the action noun.<sup>19</sup> Relational adjectives used this way have been called *thematic* since they “saturate semantic roles“ (Bosque and Picallo, 1996, 360) contrasting them with *classificatory relational adjectives*. Thematic relational adjectives are interesting since cross-linguistic variation in the type of argument they can express has been detected. For instance, in German, object readings of thematic adjectives are categorically impossible (Gunkel and Zifonun, 2008).

(58) städt-isch-e            Reinigung            (German)  
city-RA-NOM.F.SG cleaning.NOM.F.SG  
1.  $\approx$  ‘cleaning by the city’ (subject)  
2.  $\neq$  ‘cleaning of the city’ (object)

(Gunkel and Zifonun, 2008, 294)

Another relevant subgroup are so-called *ethnic adjectives*. Ethnic adjectives are toponymic modifiers which denote a geographical origin and sometimes even titles and can be thematic or classificatory (Giorgi and Longobardi, 1991; Alexiadou and Stavrou, 2011; Arsenijević *et al.*, 2014; Kotowski, 2016). Arguably, they also show restrictions in their use as thematic modifiers, resisting the role of direct objects. This goes back to the following classic example of Kayne (1981, 111).

(59) a. Everyone deplored the Russian destruction of China. (Subject reading)  
b. \*Everyone deplored the Chinese destruction by Russia. (Object reading)

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<sup>19</sup>Theoretically, it could express a subject role, but our world knowledge tells us that the environment needs to be protected, rather than is the protector.

On the other hand, the THEME reading is possible in English with genitive modifiers (Kayne, 1981).

- (60) Everyone deplored China’s destruction by Russia. (Object reading)

Classificatory ethnic adjectives exist as well. As they correspond to the categories NATIONALITY/ORIGIN/PROVENANCE put forth in Sproat and Shih (1991); Scott (2002); Laenzlinger (2005); Cinque (2010), they are often not treated as true ethnic adjectives. Importantly, contrary to relational adjectives, these adjectives can, at least in some languages, appear in predicative position, including English (61) and Greek (62).

- (61) These bed sheets are Italian.

- (62) I tsanta tis ine italiki. (Greek)  
 DET bag POSS.3.SG COP Italian  
 ‘Her bag is Italian.’ (Alexiadou and Stavrou, 2011, 138)

In Japanese, modifiers that express NATIONALITY only appear among the group of *no*-modifiers and can never be used predicatively, neither with object nor with event nouns. This highlights their status as direct-only modifiers.

- (63) a. \*Kono kuruma-wa Nihon-da.  
 this car-TOP Japan-COP  
 (‘This car is Japanese.’)  
 b. \*Kono kōgeki-wa Nihon-da.  
 this car-TOP Japan-COP  
 (‘This attack is Japanese.’)

### 2.3.2.3 Categories and ordering

Relational adjectives also show ordering restrictions among each other analyzed in Bortolotto (2016) for many Romance languages.

- (64) a. escursioni montane estive (Italian)  
 hike.PL mountain.RA.PL. summer.RA.PL  
 b. ??escursioni estive montane  
 hike.PL summer.RA.PL mountain.RA.PL  
 ‘mountain summer hikes’ (Bortolotto, 2016, 148)

The systematic hierarchy established in Bortolotto (2016) reads as follows.

- (65) *Hierarchy of Relational Adjectives (based on Bortolotto 2016, 180)*  
 TIME > LOCATION > AGENT > INSTRUMENT > MATTER (>) THEME > N



### 2.3.3 Summary

Both, non-intersective as well as relational modifiers are thus groups of modifiers, whose existence in Japanese immediately verifies the existence of direct modification.

## 2.4 Indirect modification

This section outlines the two types of indirect modifiers and briefly discusses the syntax of indirect modifiers focusing on the derivation adopted in this book. Indirect modification will be the topic of Chapter 5.

### 2.4.1 Two kinds of indirect modifiers: Full vs. reduced relative clauses

#### 2.4.1.1 General differences

As mentioned above, indirect modifiers come in two types: *Full relative clauses* and *reduced relative clauses* (RRCs) (Cinque, 2010, 2020; Krause, 2001; Douglas, 2016; Harwood, 2018). Traditionally, reduced relative clauses have been assumed to be derived from full relative clauses (Smith, 1964) and the following characteristics are often invoked that highlight their reduced/impooverished nature (Cinque 2020, 203; Bhatt 1999; Krause 2001; Douglas 2016; Harwood 2018).

- (66) Characteristics that differentiate reduced from full relative clauses
- a. They do not make use of relative pronouns or complementizers.
  - b. Fewer positions on the Accessibility Hierarchy (Keenan and Comrie, 1977) can be relativized/only the subject can be relativized.
  - c. Tense is reduced/No tense is involved.
  - d. No overt subject appears.
  - e. A restriction to certain types of verbs applies.

These characteristics seem accurate in languages such as English, although the problem arises to differentiate reduced relative clauses from adjectives, if they do not appear postnominally or are participial or infinitival. Testing the status of a relative clause in a prenominal-RC language,<sup>20</sup> however, is almost impossible to test based on these criteria alone. That is mainly because in such languages, usually only one surface form for relative clauses exists and relative pronouns/complementizers are typically absent. While these facts speak for a reduced status, the fact that overt

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<sup>20</sup>With this notion I refer to languages with only prenominal relative clauses in line with Wu (2011).

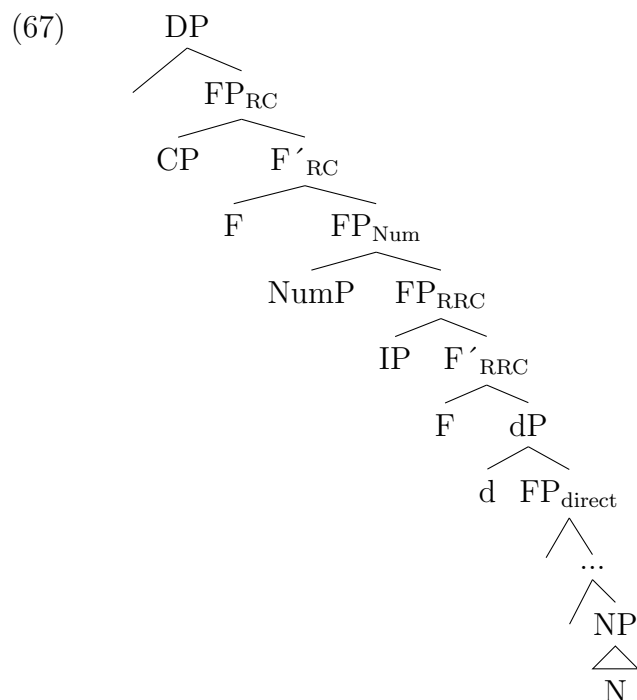
subjects typically appear freely and many positions on the Accessibility Hierarchy can be relativized speak for a larger status (cf. Wu 2011).

In the cartographic program, two structural differences have been identified for full and reduced relative clauses: 1. The internal make-up of reduced relative clauses contains less functional material. 2. They occupy different merging positions.

Regarding the first point, while full relative clauses are of the structural size CP, reduced relative clauses are (arguably) smaller, although their exact size is controversial. In seemingly well-known languages such as English, convincing evidence, mostly in the form of co-occurrence with high adverbials (Douglas, 2016; Harwood, 2018; Cinque, 2020) can be found that they must be at least of size IP. On the other hand, some authors have argued that reduced relatives in prenominal-RC languages are only vPs (Krause, 2001), while some assume a specific PrtP projection for participles (Bhatt, 1999).

#### 2.4.1.2 Ordering and the syntax of the indirect domain

Ordering restrictions, arguably, provide a better testing ground. According to Cinque (2010, 2020), the indirect domain (the area up to dP, see below) looks as depicted in the (simplified) tree below.



Within the indirect domain, the projections for full relative clauses are higher than the projections for reduced relative clauses. Recall that several projection for the same type have to be assumed, explaining cases of relative clause stacking. Furthermore, there is a projection for numerals intervening. This predicts the following overall hierarchy.

- (68) (DEMONSTRATIVE) (>) RELATIVE CLAUSE > NUMERAL > REDUCED RELATIVE CLAUSE (>) (dP) (>) (DIRECT ADJECTIVE) (>) (N)

In fact, the interaction with numerals serves as an important test bed as well: Full relative clauses cross-linguistically typically linearly precede numerals, while reduced relative clauses follow them, in languages where they are on the same side of the noun. That full relative clauses typically precede numerals has been identified already in Greenberg (1963) and Hawkins (1983) and is further discussed in Cinque (2005, 2023).<sup>21</sup> In those rigid head-final and head-initial languages which exhibit a more or less homomorphic order of constituents, this is what is observable (Cinque, 2020, ch. 3.5) as illustrated in the examples below, which provide evidence for the hierarchy *demonstrative* > *relative clause* > *numeral* (cf. Cinque 2020, 225).

- (69) [he taa-w kuttuwa ehida] iccasha adussa laagge-t-i  
 those I-DAT chicken having.brought five tall friend-PL-NOM  
 ‘those five tall friends [who brought me a chicken]’  
 Wolaytta, West Cushitic, head-final (Lamberti and Sottile 1997, 215)
- (70) gwóggî à dòṅò àryó-nì [ámê lócə ònèkò]-nì  
 dogs ATT.PART big two-this REL man 3.SG.kill.PERF-this  
 ‘these two big dogs [that the man killed]’  
 Lango, Eastern Nilotic, head-initial (Noonan 1992, 156)

On the other hand, the underlying order *numeral* > *RRC* can be verified in German ((71); cf. also Eichinger 1991; Trost 2006a) and English (72). Numerals and participle (reduced) relative clauses must both occur prenominally and are best in that order.

- (71) a. die drei [dort parkenden]<sub>RRC</sub> Autos (German)  
 the three there parking.PRS.PTCP cars  
 b. ??die [dort parkenden]<sub>RRC</sub> drei Autos  
 the there parking.PRS.PTCP three cars  
 ‘the three cars parking over there’
- (72) a. these (other) two [recently completed]<sub>RRC</sub> plays  
 b. ?these (other) [recently completed]<sub>RRC</sub> two plays (Cinque, 2020, 229)

### 2.4.1.3 A Note on the indefinite head dP

According to Cinque (2010, 2020), the threshold between the direct and the indirect domain is the indefinite head d, which is the “head of the relative clause itself” (Cinque, 2010, 34) and bears an identical constituent within the relative

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<sup>21</sup>Similarly, relative clauses precede adjectives, see Cinque (2005, 2023); Rijkhoff (2002, 298–299).

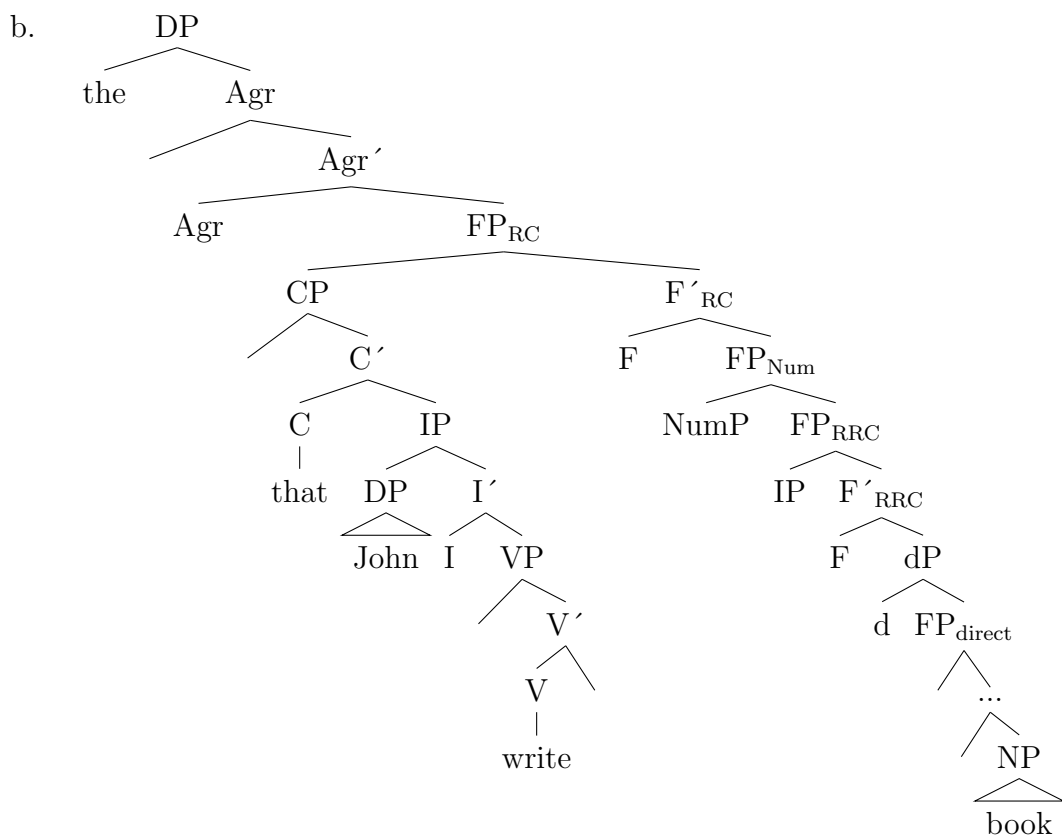
clause. The syntactic role of *d* as the *external head* and counterpart to the *internal head* is essentially relevant in Cinque’s theory on the double-headedness of relative clauses (Cinque, 2020). From a semantic point of view, *d* assigns “referential import” (Cinque, 2010, 34) meaning that modifiers above this head modify nouns with referentiality of their own. The author seems to have in mind a more referential status of relative clauses, which, for instance, matches the intersective interpretation, whereas direct modifiers not necessarily bear this feature of referentiality, which explains their non-intersective interpretation. In this book, *dP* will be displayed as the threshold between the two domains, but it does not serve a syntactic function beyond that, as the double-headed structure will not be adapted for Japanese.

## 2.4.2 The syntactic derivation of relative clauses

### 2.4.2.1 The underlying structure

A plethora of different analyses can be found for the structure and derivation of (the different types of) relative clauses in English alone (Schachter, 1973; Vergnaud, 1974, 1985; Kayne, 1994; Bianchi, 1999; Bhatt, 2002; Salzmann, 2019). What is common to them is that they must find a way to solve the *connectivity problem* (Salzmann, 2017), i.e. the question how the correct interpretation of the noun, both inside the relative clause and inside the matrix clause, can be ascertained. Prior to derivation, I assume the underlying structure of a full relative clause to look as follows following cartographic research until Cinque (2010).

(73) a. I read the book [that John wrote].



The relative clause is a full one, thus is of status CP. It merged in the specifier position of a corresponding projection for full relative clauses, which is higher than the projection for numerals and those for reduced relative clauses. The structure for a reduced relative clause looks similarly except for the fact that a CP-layer is lacking.

### 2.4.2.2 The derivation

Three major types of analyses have been invoked for head-external relative clauses, each of them ensuring in a different way that the noun can be interpreted correctly both inside the relative clause, where it is not present on the surface, and within the matrix clause:<sup>22</sup>

1. The Raising Analysis
2. The Matching Analysis
3. Operator Movement

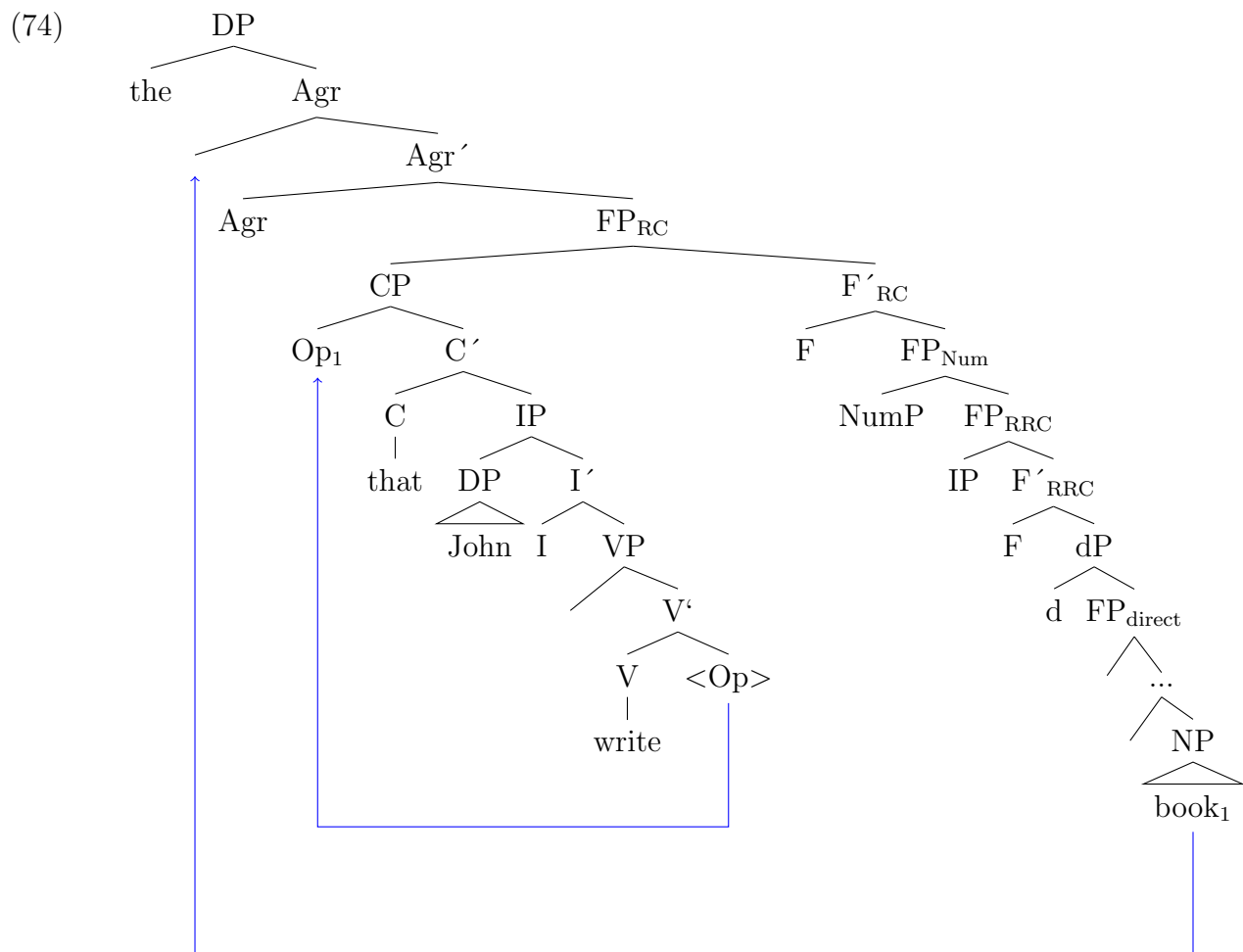
<sup>22</sup>Even among a single stream of analysis, several differences exist as to the fine details and technicalities of the derivation. I will simplify these details and differences in the following discussion. See Salzmann (2017, ch. 2) for an informative overview.

In the raising analysis, also known as *promotion* or *head-internal* analysis, the noun is structurally present inside the relative clause at the beginning of the derivation. It then *raises* out of the relative clause and leaves behind a trace. This guarantees the correct interpretation (Schachter, 1973; Vergnaud, 1974, 1985; Kayne, 1994; Bianchi, 1999; Bhatt, 2002).

According to the *matching* analysis, no movement happens. Instead, there are two instances of the noun, one inside and one outside the relative clause. The noun inside the relative clause (in some versions the complement of an operator) is deleted at PF under phonological identity with the noun in the matrix clause (Lees, 1961; Chomsky, 1965; Schachter, 1973; Sauerland, 1998, 2003; Salzmann, 2017, 2019).

Both analyses have been proven to be valid in different languages. Essentially, the difference boils down to the fact that in raising analyses, movement of the noun takes place. This predicts the availability of different syntactic reconstruction effects, such as idiomatic readings and island effects (see Cinque 2008a, 2020; Salzmann 2017, 2019 for overviews and discussions).

In this book, I will adopt the standard, older and not uncontroversial, *operator-movement* derivation for Japanese in Chapter 5.3.5, according to which a relative clause operator moves from within the relative clause, for instance from the complement position of the verb in object relative clauses, to Spec, CP. This operator is co-indexed with the head noun outside the relative clause and ensures it is interpreted correctly inside the relative clause as well (cf. Chomsky 1977, especially Miyamoto 2017 for Japanese). This derivation is also given as a possible derivation for Japanese, and other prenominal relative clause languages in Cinque (2020, 81), abstracting away from the double-headed structure. For English, the structure looks as follows, including movement of the noun to a higher functional projection for word order purposes.

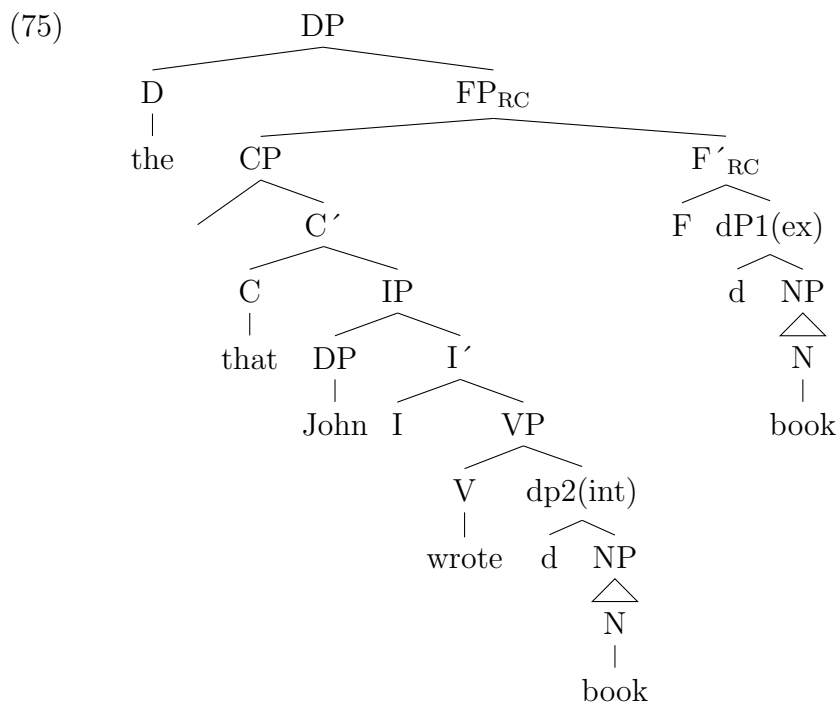


The difference between the raising analysis and the operator-movement analysis is thus the presence of an operator, which ensures correct interpretability, a commonality is the prediction of reconstruction effects, contrary to the matching derivation.<sup>23</sup>

### 2.4.2.3 A note on the double-headed structure

In recent times, Cinque (2020) has proposed a universal double-headed structure for relative clauses that looks as follows for a full relative clause.

<sup>23</sup>See Chapter 3.2 for why we need to assume this structure for seemingly simpler modifiers as well.



Crucial to this idea is that the noun is structurally present twice, once inside and once outside the relative clause,<sup>24</sup> and deletable in either a raising or a matching fashion. Either, the internal head, dP2, moves to Spec, CP, where it receives operator status and licenses subsequent deletion of the external head (raising), or the external head, dP1, stays in situ and licenses the deletion of the internal head dP2 (matching), cf. Cinque (2020, 25–45). This account is argued to be universal and both derivations are accessible in all languages at all times.

The problem with this account is that evidence for a double-headedness of the noun has only been proven in a small number of languages, including Kombai (Awyu, West Papua; cf. de Vries 2020) and Wenzhounese (Sinitic; cf. Hu *et al.* 2018). In most languages, including English and Japanese (although see Erlewine and Gould 2016 for possible exceptions), there is only one overt noun, or some version of it. Furthermore, as shown in Chapter 5.3.5, Japanese shows ample evidence for movement in the form of reconstruction effects, which in turn speaks in favor of either an operator movement or raising derivation. For these reasons, I will not adopt the double-headed structure here, but a simpler structure, and assume operator movement as the operation that derives relative clauses in Japanese.

### 2.4.3 Summary: Indirect modification

To summarize, indirect modifiers come in two types, full and reduced relative clauses. These modifiers differ in their structural make-up and syntactic height. In this book,

<sup>24</sup>In fact, what is structurally present twice, is not only the noun but any projection lower than the indefinite head d.



relative clauses will be assumed to be derived via movement of an operator from within the relative clause to a Spec, CP, which is co-indexed with the noun and ensures correct interpretability.

## 2.5 Additional accounts

In this section, I will present ideas from three additional proposals, that differ from the mainline approach in Cinque (2005, 2017, 2010, 2020) in different ways and will all enrich the system defended here. They all share the assumption of a large amount and universal character of functional projections within the DP, but add different ingredients to be adopted in the following. I will focus on what is relevant in these approaches for the purpose of this study.

### 2.5.1 Svenonius: Gradability and intersectivity

Larson (2021, 249–250), in a general critique on cartography (and see Truswell 2004, 2009; Svenonius 2008; Kotowski 2016 and Kim 2019 for different languages and categories), mentions the “problem of rigidity” according to which such a strict structural selection that we see in cartography should always be reflected in linear order (Kayne 1994; Cinque 2010 etc.) and thus deviations should never occur. However, as Svenonius (2008) points out, sometimes the ordering restrictions proposed in Scott (2002) and similar work seem rather “tailored” (Svenonius, 2008, 37). For instance, the order of the modifiers of DIMENSION and SHAPE in (76) is rather free. On the contrary, he does point out clear cases where ordering restrictions hold, for example between modifiers of DIMENSION and COLOR, illustrated in (77).

- (76) a. a long thick rope (DIMENSION  $\succ$  SHAPE)  
       b. a thick long rope (SHAPE  $\succ$  DIMENSION) (Svenonius, 2008, 37)
- (77) a. tiny red hats (DIMENSION  $\succ$  COLOR)  
       b. ??red tiny hats (COLOR  $\succ$  DIMENSION) (Svenonius, 2008, 36)

Svenonius dismisses ordering restrictions based on fine-grained semantic categories, and instead argues that such ordering restrictions only arise with modifiers with different levels of gradability (as well as intersectivity). For example, modifiers of COLOR, ORIGIN and MATERIAL are clearly non-gradable, while modifiers of DIMENSION, QUALITY and SHAPE are gradable (countable).

- (78) a. ??very red / Japanese / metal hat (COLOR, ORIGIN, MATERIAL)  
       b. very big / nice / round hat (DIMENSION, QUALITY, SHAPE)



Besides the idea that fine-grained functional projections might not (always) be at stake, what will also be adopted from Svenonius is a specific projection on the right edge of the DP for idiomatic modifiers. Arguably, such modifiers are embedded in category-less root phrases (Marantz, 1997). Such root phrases are embedded at the nP level. That they occupy the position right above the noun, becomes evident since nothing can intervene between idiomatic modifiers and the noun they modify without changing the idiomatic meaning (Svenonius, 2008, 36–38).

- (83) a. Minnesotan wild rice  
b. wild Minnesotan rice (only ‘uncultivated rice from Minnesota’)  
(Svenonius, 2008, 36–37)

Christopher Laenzlinger (1998; 2005; 2010; 2011; 2015) focuses on the parametric differences between Romance (French) and Germanic and extends his findings to a cross-linguistic perspective. He presents a study which discusses the CP and DP in 14 typologically diverse languages, one of these languages being Japanese. His approach and DP-structure are essentially similar to Cinque's modulo certain differences.

What is crucial for the account presented in this study is his adoption of a DP-external domain, a ‘*Vorfeld*’ or ‘ $\omega$ -domain’ as he calls it, above the ‘middle field’ and ‘low field’ (Laenzlinger 2011, 7; see Grohmann 2003 for a similar account). This area can be reached via individual movement of modifiers and is headed by a dedicated D-projections ‘DP<sub>determination</sub>’ above the *internal* ‘DP<sub>deixis</sub>’ (Laenzlinger, 2005, 668–669). The availability of two D-positions accounts for the overt occurrence of multiple D-elements in certain languages (including Lakota and Urama, see Becker 2018, 187–193) as well as well-known phenomena such as Romance superlatives (Brugé, 2002; Bernstein, 2001) or Greek determiner spreading (Androutsopoulou 1995, but see Lekakou and Szendrői 2012 for an alternative claim). Furthermore, the necessity of a left periphery in the nominal realm has been put forth by various authors to account for high topic and focus positions and movement to those (Giusti

<sup>26</sup>Manzini (2025) does not make the same reference to intersectivity, concretely since, as she argues, intersectivity and subsectivity are semantic concepts respectively and, according to her, are difficult to argue to have syntactic dimensions. Furthermore, she provides the data point in (82) where an intersective adjective precedes a subsective adjective in English, contrary to expectation. It should be noted, however, that different mechanisms might be at play here, such as scope dependencies, similar to cases addressed in Chapter 4.4.3.1 for Japanese.

- (82) It is not true that there are no beautiful good linguists. (Manzini, 2025, 10)

1996; Dimitrova-Vulchanova and Giusti 1998; Ihsane and Puskás 2001; Aboh 2004, 2006; Grohmann 2003; Grohmann and Panagiotidis 2015; cf. Laenzlinger 2011, 173–178). In DPs with only one determiner, this determiner always starts in the lower DP but can move up in case material has moved above it (Laenzlinger, 2005, 667).

The necessity of a left periphery as a landing site for modifiers serves mainly the purpose of accounting for otherwise inexplicable placement of modifiers. As Laenzlinger assumes that the canonical position of adjectives in French is postnominal, which is the result of NP-movement to satisfy agreement between adjective and noun, pied-piping the adjective in the process (Laenzlinger, 2005, 657–662), prenominal adjectives are only possible in the following circumstances which are all related to deictic or pragmatic contributions (Laenzlinger, 2005, 666). This echoes the arguments from the other authors mentioned above who proposed a left periphery to account for high modifiers with a topic or focus interpretation.

- (84) a. They have an emphatic or subjective interpretation such as *superbe* ‘superb’.  
 b. They are quantifiers.  
 c. They appear in weak forms (*gros* in the literal sense of ‘big’ as opposed to ‘grand’).

Such modifiers move to the specifier position of dedicated projections FocP/SubjectiveP, for focused/subjective adjectives, QuantP, for quantifiers, and WeakP for weak forms of French adjectives respectively (Laenzlinger, 2005, 666–668). In a DP such as (85), first the NP has moved cyclically to an agreement position in the DP-external domain, where it checks agreement with the adjectives *rouges* ‘red’, *magnifiques* ‘beautiful’ and *petites* ‘small’. The adjective *petites* subsequently moves to WeakP in the left periphery for pragmatic reasons (Laenzlinger, 2005, 667).

- (85) de petites voitures rouges magnifiques  
 DET small car red beautiful  
 ‘beautiful small red cars’ (Laenzlinger, 2005, 668)

The DP-structure, given in hierarchical format, proposed by Laenzlinger (2005, 2011) reads as follows.

- (86)  $DP_{\text{external/deictic}} > QuantP > FocP > WeakP > FP_{AgrNP} > DP_{\text{internal/determination}}$   
 $> FP_{PP} > FP_{AgrNP} > FP_{Pred} > FP_{Adj} > nP > NP$  (Laenzlinger, 2011, 35)

Notably, Laenzlinger does not present a systematic difference between direct and indirect modification and his projection PredP ( $FP_{Pred}$ ) is reserved for what he calls

“external adjectival predicates” (Laenzlinger, 2005, 670) explaining cases of three postnominal adjectives in French. This scenario is atypical, but possible, if the final one receives an emphatic reading such as in *une voiture rouge italienne MAGNIFIQUE* ‘a BEAUTIFUL red Italian car’ (Laenzlinger, 2005, 663–664, 671–672).<sup>27</sup>

The main point to be adapted from Laenzlinger is the left periphery within the DP and the possibility for individual movement. Specifically, Japanese also freely exhibits different kinds of modifiers in a position above demonstratives, among them quantifiers and direct modifiers, but no pragmatic factors are detectable.

- (87) a. takusan-no ano hon  
a.lot-NO that book  
‘those many books’  
b. izen-no kono rûru  
former-NO this rule  
‘this former rule’

### 2.5.3 Kim: A tripartite DP-structure based on Korean

Kim (2019) blends and amends the two preceding analyses of Svenonius (2008) and Laenzlinger (2005) and enriches them with her own ideas drawing from Korean data. Essential to her account as well is a tripartite DP-structure consisting of three layers: A *High Field* headed by  $DP_{d(eixis)/r(eferential)}$ , a *Middle Field* headed by  $DP_{q(uantification)}$  and a *Low Field* headed by  $DP_{p(redication)}$ . These DPs license [deictic, referential, definite], [quantificational] and [predicative] features respectively (Kim, 2019, 123).<sup>28</sup> Furthermore, she assumes independent movement of modifiers for reasons of feature-checking (and potentially at LF for semantic/pragmatic reasons) and even of whole DP domains and all material contained in them. Every functional projection, and this includes every DP projection, can multiply and “create” additional functional projections of the same kind in an adjunct fashion.<sup>29</sup>

What will be adopted here, besides the split-DP also argued for in Laenzlinger (2005, 2011), is her projection SpplP (SupplementaryP) situated above the highest DP and dedicated to *supplementary* relative clauses, which are not properly *inte-*

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<sup>27</sup>Also, the question remains why languages should exist that have agreement, but where prenominal adjectives are inflected, but postnominal adjectives uninflected, a point also raised in Bortolotto (2016, 20, Fn. 9). A good example is German, where inflected attributive adjectives in prenominal position are the standard case, but postnominal adjectives are possible, in which case they are not inflected (cf. Trost 2011) as in *Whisky pur* ‘pure whiskey’ (Trost, 2006b, 383).

<sup>28</sup>These DPs correspond to the layered structural mapping of satellites around a nominal or verbal nucleus proposed in Rijkhoff (2002, 218), and also to three different types of nominals, namely referential, quantificational and predicative (Kim, 2019, 114).

<sup>29</sup>“FPs of the same kind can be created in the left periphery of a DP structure as needed, forming an adjunction structure” (Kim, 2019, 121), cf. Koopman and Szabolcsi (2000).

*grated* in the DP (Kim, 2019, 116–119).<sup>30</sup> She gives the following example (from Huddleston and Pullum 2005).

- (88) The doctor, [with no name tag on], got into serious trouble.  
(Kim, 2019, 117)

A (reduced) relative clause such as *with no name tag on* is not essential to the DP, but *supplementary*. It is base-generated in the left periphery outside the scope of D. In Japanese, non-restrictive, or supplementary, relative clauses also obligatorily occur above demonstratives (example adapted from Ishizuka 2008, 7–8).

- (89) Ito-san-ni-wa musuko-ga hitori i-ru. [[Sakunen isha-ni  
Ito-Mr./Mrs.-LOC-TOP son-NOM one be-PRS last.year doctor-LOC  
nat-ta] sono musuko-ga] kekkon-shi-ta.  
become-PST that son-NOM marriage-do-PST  
'Mr./Mrs. Ito has one son. This son, who happened to become a doctor  
last year, got married.'  
non-restrictive relative clause  $\succ$  demonstrative  $\succ$  N

In the lower DP domain, Kim assumes the same projections as Svenonius (2008) and adds a projection for possessive modifiers and demonstratives called LocP (the satellite '*Location*' in Rijkhoff (2002)). She offers the following hierarchy.

- (90)  $RC_{\text{Sppl}} > DP_{\text{deictic/referential}} > DP_{\text{quantification}} > DP_{\text{predication}} > \text{FocP} > \text{LocP}$   
 $> \text{UnitP} > \text{PIP} > \text{SortP} > \text{nP} > \sqrt{\text{P}} > \text{NP}$  (Kim, 2019, 132)

Another important assumption to be adopted (see Chapter 5.3) is the unification of full and reduced relative clauses to one type of relative clause of the same size.

## 2.6 Further issues

In this section, I raise two additional issues. One anticipates potential problems that come with cartography, in the other, I briefly mention previous approaches to Japanese as part of this research program.

### 2.6.1 Potential problems of cartography

A system as rigid as cartography cannot be free of problems and criticism. Anticipating the discussion, I will discuss several potential problems below.<sup>31</sup>

<sup>30</sup>Note that *supplementary* is not equal to *non-restrictive*. While supplementary relative clauses are always non-restrictive, both, non-restrictive and restrictive, relative clauses can be integrated parts of the internal DP (Kim, 2019, 120).

<sup>31</sup>For this section, consider the recent discussions in Manzini (2025) and Cinque (2025).

One objection concerns the existence of movement operations for the sake of order and devoid of syntactic and semantic constraints, often dubbed “meaningless movement”. Several researchers have argued that this kind of movement should be kept apart from narrow syntax (most notably see Chomsky *et al.* 2019 and references cited therein). Recall from Chapter 2.1.3 that no NP-movement needs to be assumed in Japanese since the noun is always the final element in the DP.<sup>32</sup>

While this is a more ideological point, a more concrete point that begs further discussion in the future is the nature of the cross-linguistic restrictions on movement operations. How are they spread across different languages and language groups? Why do we only witness a particular type of movement, for instance progressive pied-piping, in a given language, while other languages allow both types (see footnote 7 in Chapter 2.1.2)? Furthermore, experimental approaches have cast doubt on universal syntactic principles and related ordering principles rather to cognitive factors, such as subjectivity and information gain. See Scontras *et al.* (2017); Dyer *et al.* (2023), the works cited therein and similar works by the same authors.

Sometimes, even specific subclasses of modifiers in a given language are restricted to specific positions and suggest a restriction to one particular type of movement. An interesting case are uninflectable adjectives of the type *blu* ‘blue’ in Italian, and perhaps in other Romance languages. Italian color adjectives, although restricted, can appear in prenominal and postnominal position, but this is never possible for uninflectable color adjectives such as *blu* illustrated below, same as for Relational Adjectives in this and other Romance languages.

- (91) a. *la (rossa) distesa (rossa) del mare* (Italian)  
           the red.F.SG expanse.F.SG red.F.SG of.the sea  
           ‘the red expanse of the sea’  
       b. *la (\*blu) distesa (blu) del mare*  
           the blue.Ø expanse.F.SG blue.Ø of.the sea  
           ‘the blue expanse of the sea’ (Mattiuzzi *et al.*, 2023, 110)

Such adjectives pose a problem for standard cartographic accounts, since no obvious relation between morphology and syntax can be drawn. Specifically, there is no obvious reasons why this specific type of modifiers must be subject to pied-piping. The problem also extends to the system of Laenzlinger (2005, 2011), which would predict only agreeing adjectives to surface postnominally. For recent discussion see also Mattiuzzi *et al.* (2023).

Equally problematic are data points, provided by several authors from different languages, that show that direct modifiers (adjectives) adhere to the strict ordering principles dictated by cartography cross-linguistically to different degrees (see for an

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<sup>32</sup>For meaningless movements also see Hall (2015).

overview Manzini 2025). I refer the reader to Chapter 4.4, where I discuss this point, specifically with regard to Japanese, but also to Persian and Modern Greek. Here, I would like to exemplify this point with recent data from Naasioi, a Papuan language. See Chapter 4.4.2.5 for discussion on this and another problematic language.

Brown (2024) provides very clear evidence that adjectives in Naasioi which are unequivocally direct do not have to adhere to ordering principles, as shown below (see the relevant paper for evidence that the provided adjectives are direct).

- (92) a. pangkaing mutaanu mosi (Naasioi)  
           big           black     dog  
       b. mutaanu pangkaing mosi  
           black     big           dog  
           ‘big black dog’ (Brown, 2024, 428)

As I will argue in this book, such cases are unproblematic if we assume that the nature of the functional projections available in the direct domain of the DP are subject to cross-linguistic variation.

Finally, a potential problem concerns the presumably infinitive amount of functional heads present in the nominal, and also the clausal, domain. Recall that, if we take the cartographic idea seriously, every new category found in a certain language must be present in the functional spine of all languages. This point is also raised in Larson (2021), who also reports the first of the two quotes below.

- (93) a. The cartographic approach follows this idea in assuming that all languages share the same principles of phrases and clause composition and the same functional make-up of the clause and its phrases. (Cinque and Rizzi, 2010, 68)  
       b. Comparison of many different languages may provide evidence for determining the precise relative order of the different projections by combining the partial orders overtly manifested by different languages into what, in principle, should be a unique consistent order/hierarchy, imposed by UG. (Cinque and Rizzi, 2010, 72)

A concrete problem arises if a modifier occurs that cannot readily be assigned to any of the existing categories of functional projections. An obvious solution would be to establish a new functional projection. But this would, at some point, potentially lead to an overabundance of categories, which is called the ‘*Problem of Plenitude*’ in Larson (2021, 248–249). This problem is circumvented in this study by assuming unspecific projections, an assumption that also solves the ‘*Problem of Rigidity*’ (Larson 2021, 249–250) which concerns the lack of expected ordering principles between two modifiers of different categories.



In sum, cartography faces the problems of meaningless movement, ordering derivations, plenitude and rigidity. As its counter-advocates argue, it overgenerates and undergenerates at the same time (Larson, 2021; Manzini, 2025). Still, this research topic is extremely powerful and offers the right tools and principles for an in-depth investigation of the Japanese DP. Especially important is the assumption of two different sources of modifiers and their different characteristics and derivations, furthermore theoretically predicting linear precedence. Since even though there might be more lenient cases, some ordering principles still do exist. All these issues will be further addressed in Chapter 4.4.

## 2.6.2 Japanese and cartography: Previous analyses and relevance

### 2.6.2.1 Laenzlinger on Japanese DP

Laenzlinger (2010, 2011) draws many implications for his universal DP structure from Japanese as a strict head-final language. His typological study is to my knowledge the only one that looks at the Japanese nominal domain in line with cartography. However, other than his detailed analysis of the Japanese clause, his discussion of the Japanese noun phrase falls out short. He gives the following example of three modifiers adhering to the expected order QUALITY  $\succ$  DIMENSION  $\succ$  COLOR.

- (94)    subarashi-i chiisa-na aka-i kuruma  
          wonderful-I small-NA red-I car  
          ‘wonderful small red car(s)’ (Laenzlinger, 2011, 189)

He claims that the expected order is also the preferred one, but that modifiers in general can be permuted without a change in meaning or “special focus effect(s)” (Laenzlinger, 2011, 190). He concludes that “at first sight Japanese (and also Tatar) pose a problem for previous theories to nominal modification and modifier ordering” (ibid.) and, as many researchers have done before him, solves this problem by analyzing all adjectives as stacked relative clauses predicting possible permutation. This finds support in his analysis of I (and NA) as tense morphemes (Laenzlinger 2011, 191, Fn. 106).

It was already shown that direct modification does exist in Japanese, so this analysis cannot be correct. It remains furthermore unclear why there should be a preferred order of modifiers in the first place, if permutation is always allowed, and why this order should be as in English. Another major problem is that his example (94) contains the attributive-only lexeme (*rentaishi*) *chiisa-na* ‘small’, which cannot function as a predicate, hence cannot be predicative and be embedded in a relative clause as shown in (95). This group of modifiers is the topic of Chapter 4.3.6.

- (95) a. chiisa-na ie  
           small-na house  
           ‘small house’  
       b. \*Kono ie-wa           chiisa-da.  
           this house-TOP small-COP  
           (‘This House is small.’)

### 2.6.2.2 Other Analyses

Several studies discuss the Japanese clause from a cartographic perspective: Endo (2007) looks at the left periphery in Japanese, integrating Japanese into Rizzi’s account on locality and Relativized Minimality (1997). Other authors have looked at the right periphery (cf. overview in Saito 2015). Ueda (2008) develops a hierarchy of modals expressed through sentence-final particles, Saito (2013) discusses various sentence types and movement to the right periphery, and Saito and Haraguchi (2012) develop a hierarchy of complementizers and sentence-final particles which is refined in Saito (2015). What is interesting about all these studies is that they do find a hierarchy and derive generalizations that are empirically and theoretically valid, but for the clause, again highlighting the more controversial and recalcitrant nature of the DP.

### 2.6.2.3 Narrog (2009)

Narrog (2009) is a very detailed study on the ordering of modal elements in Japanese and, to my knowledge, the only one that combines classical approaches from Japanese linguistics and cartography. The author identifies twelve affixes, 20 periphrastic constructions, six verbal inflections and an infinite number of modal adverbs in Japanese (Narrog, 2009, 71–76) and establishes ordering and scope principles based on a corpus study. He compares his findings to those of Cinque’s adverbial hierarchy (1999) and eventually rejects the cartographic model. Besides several mismatches,<sup>33</sup> the biggest problem, Narrog claims, is the interaction of modal categories with those of tense and aspect in his model which differs greatly from Cinque’s version (Narrog, 2009, 233–234). In the end, Cinque’s model is deemed too abstract and “fictional” for Narrog. He claims such models can only be maintained if they are “understood as idealizations, or rough approximations to actual language data” (Narrog, 2009, 243). Furthermore, he calls into question the “psychological reality of the existence of the same universal hierarchy of categories in every sentence of any language of the world” (ibid.), again pointing towards the problem of rigidity (Larson, 2021). Even though he ends up rejecting cartography as an explanation for the problems he is

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<sup>33</sup>For example, Cinque only has one position for evidentiality, Narrog observes three which all occupy different positions in relation to modal and epistemic markers.

concerned with, his study is still relevant as it produces a very substantial hierarchy on the basis of empirical data that could in turn be checked for other languages. This hierarchy is picked up in the discussion on adverbials in Chapter 5.3.1.

#### 2.6.2.4 Takamine (2010)

Takamine (2010) looks at the order of prepositional phrases in Japanese following the theoretical framework of Cinque (1999, 2006) and Schweikert (2005). Because Japanese is a scrambling language with a variety of possible orders, it is necessary to test the order of PPs in a neutral environment. She looks at nine different categories and employs three tests for their interaction. She arrives at the following general hierarchy (see for discussion and references Takamine 2010, 58-83).

- (96) *Hierarchy of PPs in Japanese (Takamine 2010, 94)*  
TEMPORAL > LOCATIVE > COMITATIVE > REASON > SOURCE > GOAL >  
INSTRUMENTAL/MEAN > MATERIAL > MANNER

Takamine's study is relevant for two reasons. First, because she shows that word order in Japanese is not as random as it appears at first sight. Second, this hierarchy bears some resemblance to the nominal domain. This is acknowledged in Bortolotto (2016), who compares Takamine's and Rae's hierarchies (2010) to her hierarchy of relational adjectives in Romance presented above.

## 2.7 Chapter Summary

This chapter introduced the theoretical background of this study focusing on the core ideas of the cartographic account and the difference between direct and indirect modifiers. It also picked up all relevant ideas to be implemented in the account of the Japanese DP to follow. The chapter ended with a quick, condensed overview over problematic aspects of cartography and the few studies which pursued an analysis of certain aspects of Japanese syntax under a cartographic lens, highlighting the perhaps surprising lack of studies concerned with the nominal domain.

# Chapter 3

## The Japanese DP

In this chapter, I present in detail the two theoretical contributions of this book. I start out with the structure of the DP. This structure, which I assume to be universal, is in line with the assumptions in cartography outlined in the preceding chapter, but enriched with the Japanese data. It is not only better suited for handling this data, but also makes important predictions for the structure of the DP, abandoning the strict connection between direct modification and rigid modifier order. I start with an overview over the complex split-DP structure and the three relevant areas. I then briefly go over each relevant area in a bottom-up fashion. Since both, the direct and indirect domain, the main topics of this study, will each receive detailed explanations in the following two chapters, the discussion on these areas will be kept comparatively short here, but I spend some more time on the DP-external domain, i.e. the left periphery.

The second part of this chapter is dedicated to the internal structure of modifiers, with a special focus on the monomoraic elements co-occurring with non-verbal modifiers, especially NO. I will put forth an account of these elements as syntactic heads in the spirit of Rubin (1993, 1994, 1996), which license modifiers in attributive position. I thereby introduce an additional functional head in the internal structure of modifiers, informing our knowledge of the DP as a whole. I argue that all kinds of modifiers are licensed by this head, irrespective of word class and modification status, but discuss some difficult cases of CPs with overt tense heads where MOD is covert.

### 3.1 The structure of the Japanese DP

I adopt the basic cartographic assumptions of a universal spine within the clause and the DP, in which the hierarchically final element is the lexical head, i.e. N or V. This head can be subject to phrasal movement, but only to the left, never to the right (Kayne, 1994; Cinque, 2005, 2023), although, as shown in Chapter

2.1, in Japanese, and potentially other strictly head-final languages, movement of the lexical head does not occur. Rather, certain modifiers can move to a higher position, especially the DP-external domain, individually, as also argued elsewhere (cf. Kim 2019; Laenzlinger 2005, 2011). The extended projection of both lexical heads contains a (potentially) infinite amount of functional projections, that serve as merging and landing sites for various types of modifiers.

Furthermore, I assume the DP to be split, paralleling accounts on the split-CP (Rizzi, 1997; Grohmann, 2003; Laenzlinger, 2011; ?), into two large domains: The DP-external domain, essentially the left periphery of the DP, and the DP-internal domain (Kiss, 1998, 2007; Giusti, 1996; Dimitrova-Vulchanova and Giusti, 1998; Ihsane and Puskás, 2001; Aboh, 2004, 2006; Laenzlinger, 2005; Kim, 2019). The DP-internal domain is made up of two subdomains: The indirect and the direct domain.<sup>1</sup>

The complete structure of the Japanese DP is given in the figure below with the left part of the tree representing the DP-external domain and the right part the DP-internal domain, which starts where the DP-external domain ends. The thresholds between the three domains are marked in boldface. Recall that I indicate the indefinite head *d* as the threshold between the direct and the indirect domain following Cinque (2010, 2020), but that this head does not play a major syntactic role in my system since I do not assume the double-headed structure for relative clauses (see Chapter 2.4).

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<sup>1</sup>Note that the tripartite divide I assume to hold for the DP does not coincide with the tripartite structure of other authors (Laenzlinger, 2011; Grohmann, 2003) who assume a divide into an agreement and a theta domain for the lower part.

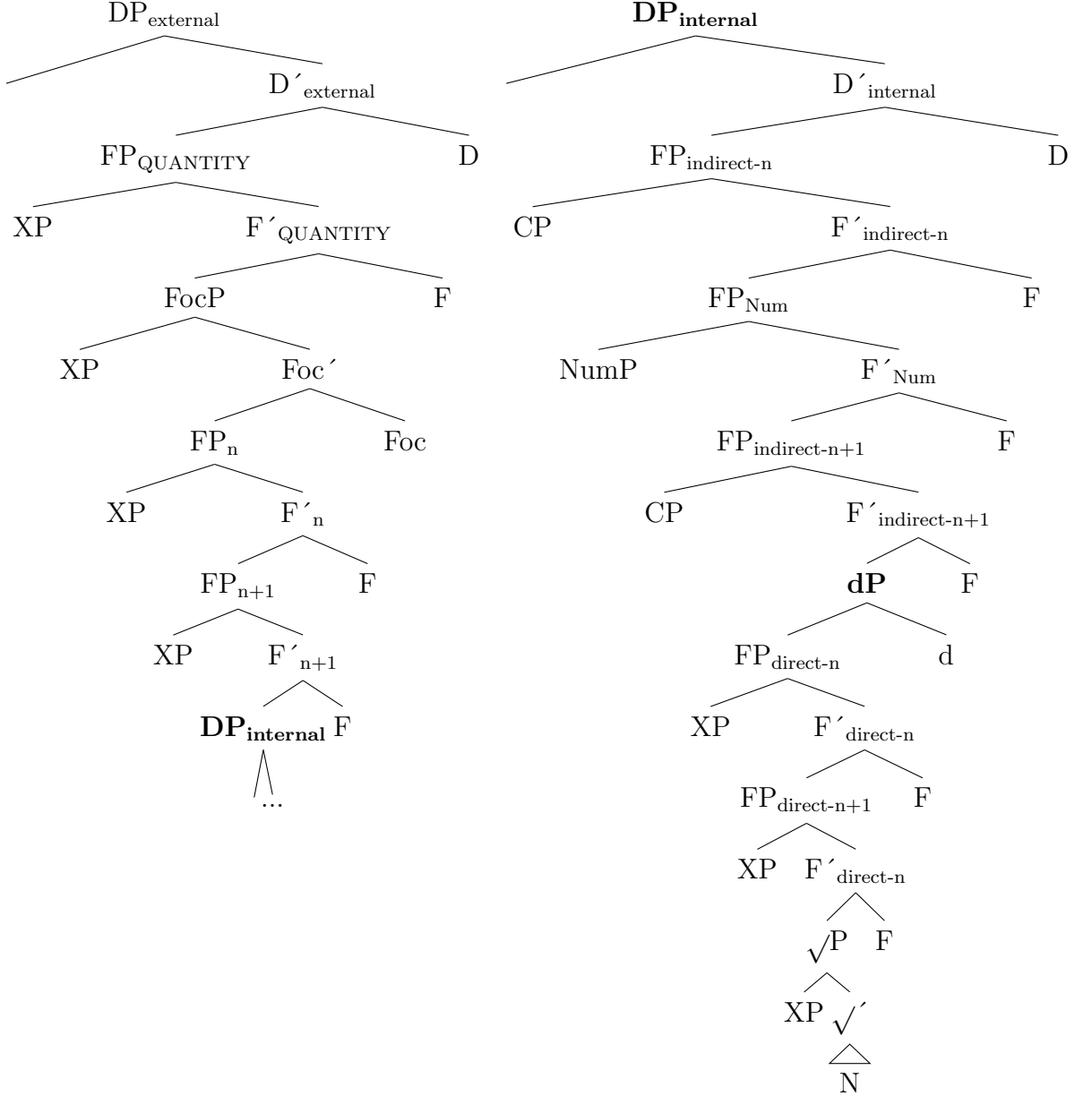


Figure 3.1: Structure of the Japanese DP

I make the following specific assumptions and predictions for the functional inventory of the three relevant areas.

- (1) a. *The DP-external domain*, i.e. the left periphery, includes functional projections that host the following elements: A projection for quantifiers, which most probably move there from the direct domain, projections for non-restrictive relative clauses, which merge in the left periphery outside the scope of D, and a projection for focused modifiers, which can optionally move there but are not required to. Furthermore, the left periphery contains an unspecified amount of available functional projections for direct as well as restrictive indirect modifiers which optionally

move there out of their respective domains. Those unspecific projections are labeled  $FP_n$ ,  $FP_{n+1}$ , in order to highlight structural hierarchy.

- b. *The indirect domain* is situated higher than the direct domain within the DP-internal domain. It contains (a potentially infinite amount of) functional projections dedicated to indirect modifiers, but those are not differentiated based on different types of indirect modifiers, i. e. full and reduced relative clauses. The indirect domain furthermore encompasses a dedicated projection for numerals and functional projections reserved for indirect modifiers precede and follow this projection.
- c. *The direct domain* is situated below the indirect domain within the DP-internal domain. It contains (a potentially infinite amount of) functional projections hosting direct modifiers. Those projections are not dedicated to modifiers of specific semantic categories. Instead, they are unspecific. Again, structural hierarchy is indicated via the labels  $FP_n$ ,  $FP_{n+1}$ , etc., but this hierarchy only becomes relevant in cases where scope restrictions play a role. The only dedicated functional projection within the direct domain is reserved for idiomatic modifiers and directly c-commands the noun.

### 3.1.1 Internal DP: Direct domain

The direct domain is situated below the indirect domain within the DP-internal domain, making it the lowest area inside the DP, directly above the noun. In the figure below, only the internal DP is displayed and the direct domain is circled.

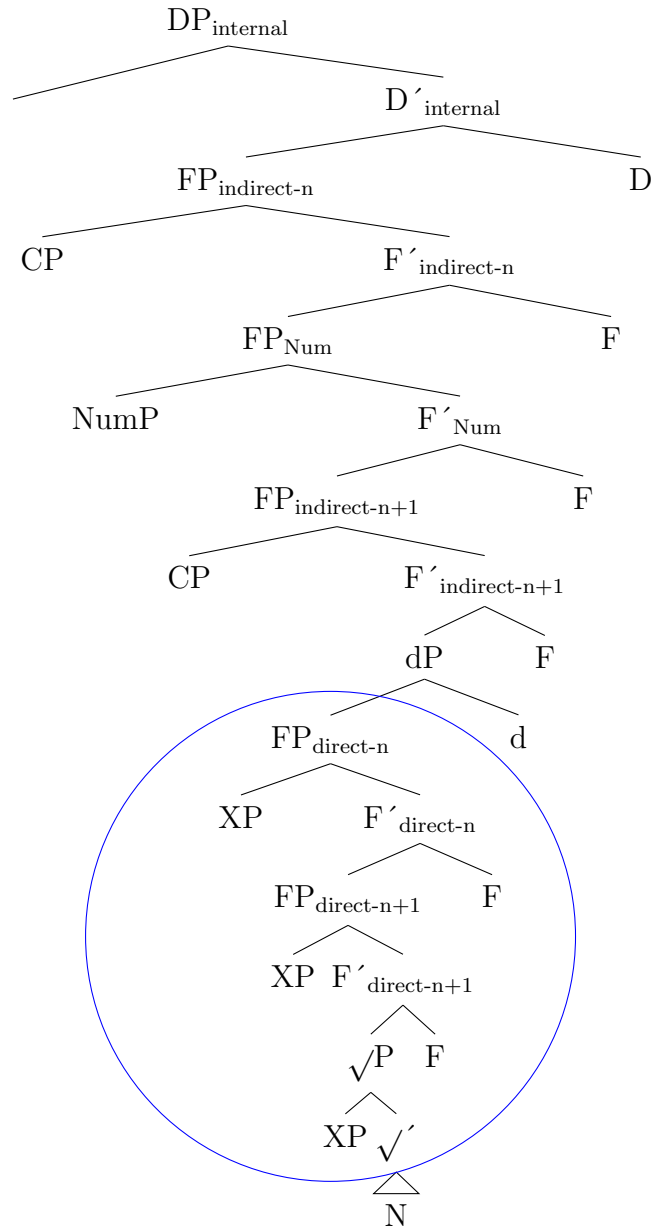


Figure 3.2: Structure of the Direct Domain in Japanese



In order to motivate a domain for direct modifiers, it needs to be proven that direct modification does exist in Japanese. As already mentioned in the preceding chapters, Japanese does possess the equivalents of non-predicative, that is direct-only, non-intersective (2) and relational modifiers (2) (Yamakido, 2005; Ayano, 2010; Nagano, 2016; Watanabe, 2017).<sup>2</sup>

- (2) a. Tarō-ga shōrai-no purosakkāsenshu-da.  
Taro-NOM future-NO pro.soccer.player-COP  
‘Taro is a future professional soccer player.’  
b. \*Ano purosakkāsenshu-ga shōrai-da.  
that pro.soccer.player-NOM future-COP  
(‘That professional soccer player is future.’)
- (3) a. Denmāku-no / chīku-no isu  
Denmark-NO / teak-NO chair  
‘Danish / teak chair’  
b. \*Ano isu-wa Denmāku-da / chīku-da.  
that chair-TOP Danish-COP / teak-COP  
(‘That chair is Danish / teak.’)

Next, building on the observation of other authors, for instance Cinque (2010) on Germanic (English) and Romance (Italian), the indirect domain is situated above the direct domain in Japanese as well. (4) exemplifies this, showing that the (non-intersective) direct-only modifier *shōrai-no* ‘future’ must be closer to the head noun than the fairly complex relative clause. The other order is significantly degraded.

- (4) a. Kare-wa [ōku-no kurabu-ga chūmoku-shi-te-i-ru<sub>RC</sub>  
he-TOP numerous-NO club-NOM pay.attention-do-GER-be-PRS  
[**shōrai-no** purosakkāsenshu]]-desu.  
future-NO pro.soccer.player-COP.POL  
‘He is a future professional soccer player whom many clubs pay attention to.’  
b. \*Kare-wa [**shōrai-no** [ōku-no kurabu-ga  
TOP future-NO numerous-NO club-NOM  
chūmoku-shi-te-i-ru<sub>RC</sub> purosakkāsenshu]]-desu.  
pay.attention-do-GER-be-PRS pro.soccer.player-COP.POL

As a crucial point of derivation from most cartographic accounts, the direct domain introduced here does not encompass functional projections dedicated to modifiers of specific semantic categories. Instead, it encompasses a possibly infinite amount of unspecific functional projections. This accounts neatly for cases of free ordering among (also in English) ambiguous qualitative modifiers, for instance of DIMENSION

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<sup>2</sup>Again, I abstract away from possible predicative use of these modifiers in contrastive and elliptical constructions or contexts in which the copula is a placeholder. See Chapter 4.2.1.

or COLOR given in (5). Crucially, it offers an explanation for the lack of ordering restrictions between direct-only modifiers, for example, the two relational modifiers *Denmāku-no* ‘Denmark, Danish’ and *chīku-no* ‘teak’ given in (6).

(5) DIMENSION vs COLOR

- a. ōki-i aka-i hon  
big-I red-I book  
‘big red book’
- b. aka-i ōki-i hon  
red-I big-I book  
‘red big book’

(6) NATIONALITY vs. MATERIAL

- a. Denmāku-no chīku-no isu  
Denmark-NO teak-NO chair  
‘Danish teak chair’
- b. chīku-no Denmāku-no isu  
teak-NO Denmark-NO chair  
‘teak Danish chair’

As I will show at length in the next chapter, Japanese modifiers also fail to exhibit ordering restrictions based on different levels of gradability and intersectivity, contrary to what is claimed in Svenonius (2008); Manzini (2025), and neither syntax-external factors such as phonological/prosodic length and word class do seem to play a role. This was already shown in (5), but is illustrated more clearly in the pair in (7): The non-gradable, intersective modifier of NATIONALITY *Chūgoku-no* ‘China, Chinese’ and the gradable subsecutive *na*-adjective of QUALITY *kirei-na* ‘beautiful’ are fine in either order. Importantly, *kirei-na* is not a direct-only modifier, but actually ambiguous and, thus, could be situated in the indirect domain. However, then it would still be expected to be merged higher, and, in neither case, to occur to the right of the direct-only NATIONALITY modifier.

(7) QUALITY vs. NATIONALITY

- a. kirei-na Chūgoku-no kabin  
beautiful-NA China-NO vase  
‘beautiful Chinese vase’
- b. Chūgoku-no kirei-na kabin  
China-NO beautiful-NA vase  
‘Chinese beautiful vase’

A significant consequence of the theory, and one point I hope to make clear in this book, is that ordering restrictions lose their status as a prerequisite for direct mod-

ification and the two issues, type of modification, direct or indirect, and ordering, become two orthogonal issues.

Crucially, Japanese differs from languages such as English which do exhibit ordering restrictions based on semantic categories within the direct domain. I argue that the reason for this lies in parametric variation, to be discussed in Chapter 4.4.2.1. Specifically, the DP-spine is universal, but there is language-specific variation as to whether functional projections based on fine-grained semantic categories are activated or not (Ramchand and Svenonius, 2014; Wiltschko, 2014). In other words, some languages possess language-specific entities to form functional projections of DIMENSION, COLOR and so on, but others do not. These language-specific entities correspond to the concept of *Units of Language* given in Wiltschko (2014, 1).

To recapitulate, ordering across two different domains is (rather) strict across languages, including Japanese, which gives evidence for a universal DP-structure incorporating a direct and a higher indirect domain (see also Ramchand and Svenonius 2014 and Wurmbrand and Lohninger 2023 for this idea, mostly with regard to the clause). However, this is not the case within one domain and this is the source of cross-linguistic differences.

In Japanese, the two possible variants of linear order between two modifiers is simply a result of merge order. The modifier merged subsequent to the noun is closer to it in linear order. On the other hand, the canonical word order of two (or more) adjectives of different semantic categories in English (and similar languages) are pre-determined in the syntax, given they are both direct.

A reviewer raises the issue on how we can determine whether a modifier is direct or indirect to begin with. The simple answer is: In most cases, we cannot. However, this is not a problem specific to the account proposed here. As illustrated in Chapter 2.2, and specifically Chapter 2.2.2, most modifiers (adjectives) in English are ambiguous as well and it is precisely their behavior in DPs with more than one modifier that can give an indication of their modification status. But this option is not operationalizable in Japanese. We can only prove that a modifier is direct in Japanese by looking at its behavior in predicative structures, alongside the other characteristics explained thoroughly in Chapter 4.2. On the contrary, clear cases of indirect modifiers in Japanese exhibit complex morphosyntactic structures, i. e. inflection for past or co-occurrence with the canonical copula.

Nominal modifiers in Japanese are therefore licensed by functional projections within the extended noun phrase, which this language shares with other languages. However, it is only direct-only modifiers that are exclusively licensed within the direct domain. And only clearly indirect modifiers are licensed within the indirect domain. For all other modifiers, there is no *a priori* indication as to their modifica-

tion status and they could be licensed by both types of functional heads. The system here is thus more versatile, compared to standard cartographic accounts. But this versatility is necessary in order to incorporate direct(-only) modifiers in the system, while still acknowledging their lack of ordering restrictions among each other.

Before closing this subsection, I will point out two cases of strict ordering observable in the direct domain in Japanese. The first concerns cases that arise when scope and other external factors play a role (see also Ramchand and Svenonius 2014 for the clause). This becomes obvious in the following minimal pair of direct-only modifiers. In a context where the speaker is identifying a vase out of a specific pool (for example at a museum), namely the single German one, *yui'itsu-no* ‘only’ must take scope over both, the following modifier and the noun, thus over the chunk *Doitsu-no kabin* ‘German vase’, and not just over *kabin* ‘vase’. In the same context, the reverse order is odd, and can only mean something akin to ‘the only vase in Germany’ or ‘the only vase produced in Germany’ where the part *German* or *in Germany* constitutes an additional piece of information (Ken Hiraiwa p.c.).<sup>3</sup>

- (9) a. [yui'itsu-no [Doitsu-no kabin]]  
           only-NO     German-NO vase  
           ‘only German vase’  
       b. ??Doitsu-no yui'itsu-no kabin  
           German-NO only-NO     vase

In cases such as (9), the modifiers occupy distinct positions and merge must have occurred in a specific order. See Chapter 4.4 for discussion.<sup>4</sup>

The only type of modifiers with a fixed position inside the direct domain are *idiomatic* modifiers, or more precisely modifiers appearing in *non-free phrases* (see Köhlich 2024). Such modifiers must appear linearly adjacent to the noun. As illustrated below, the direct-only modifier *kōki-no* ‘curious’ and the following noun

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<sup>3</sup>A factor involved here might be the nominal status of *Doitsu* ‘German(-y)’. There are similar cases reported in Shimamura (2014).

- (8) a. tsumeta-i me-no kusuri  
           cool-I     eye-NO medicine  
           ‘cool eye medicine’  
       b. \*me-no tsumeta-i kusuri  
           eye-NO cool-I     medicine (Shimamura, 2014, 163)

The noun *me* ‘eye’ modifies the noun *kusuri* first before the adjective takes scope over the whole chunk. The other order is very hard to interpret.

<sup>4</sup>This is also a good argument against scrambling as an alternative hypothesis. If scrambling was always an option, it is questionable why it is not possible when scope is involved. I am grateful to Johannes Mursell for discussion.

form the idiomatic expression *kōki-no manazashi* (alternatively *kōki-no me/shisen*) ‘curious gaze’ and no other modifier can intervene.<sup>5</sup>

- (10) a. minna-no kōki-no manazashi  
 everyone-NO curious-NO gaze  
 ‘everyone’s curious gaze’  
 b. \*kōki-no minna-no manazashi  
 curious-NO everyone-NO gaze

Following Svenonius (2008), I assume that idiomatic modifiers are embedded beyond the nP level, in a category-less root phrase  $\sqrt{P}$  (see Marantz 1997), which immediately c-commands the noun, yielding the following structure.<sup>6</sup>

- (11) a. kōki-no manazashi  
 curious-NO gaze  
 ‘curious gaze’  
 b.
- 
- ```

graph TD
  DP_internal --> B1[ ]
  DP_internal --> D_prime_internal[D'_internal]
  D_prime_internal --> FP_indirect[FP_indirect]
  D_prime_internal --> D[D]
  FP_indirect --> B2[ ]
  FP_indirect --> F_prime_indirect[F'_indirect]
  F_prime_indirect --> dP[dP]
  F_prime_indirect --> F1[F]
  dP --> FP_direct[FP_direct]
  dP --> d[d]
  FP_direct --> B3[ ]
  FP_direct --> F_prime_direct[F'_direct]
  F_prime_direct --> nP[nP]
  F_prime_direct --> F2[F]
  nP --> sqrtP[sqrtP]
  nP --> n[n]
  sqrtP --> XP_idiomatic[XP_idiomatic]
  sqrtP --> sqrt_prime[sqrt']
  XP_idiomatic --> koki_no[kōki-no]
  sqrt_prime --> NP[NP]
  sqrt_prime --> sqrt[sqrt]
  NP --> manazashi[manazashi]
  
```

In Chapter 4, I will present the criteria after which direct modifiers were identified, then outline each group of direct modifiers in Japanese, and finally, discuss the connection of universal spine and language-specific categories also in cross-linguistic dimension.

⁵This is not due to the behavior of *minna-no* ‘everyone’s’ as presented in the relevant discussion in Chapter 4.3.4.

⁶Note that I do not indicate the nP-level when giving complete structures of the Japanese DP.

3.1.2 Internal DP: Indirect domain

The indirect domain, the topic of Chapter 5, is highlighted inside the DP-internal domain in the figure below.

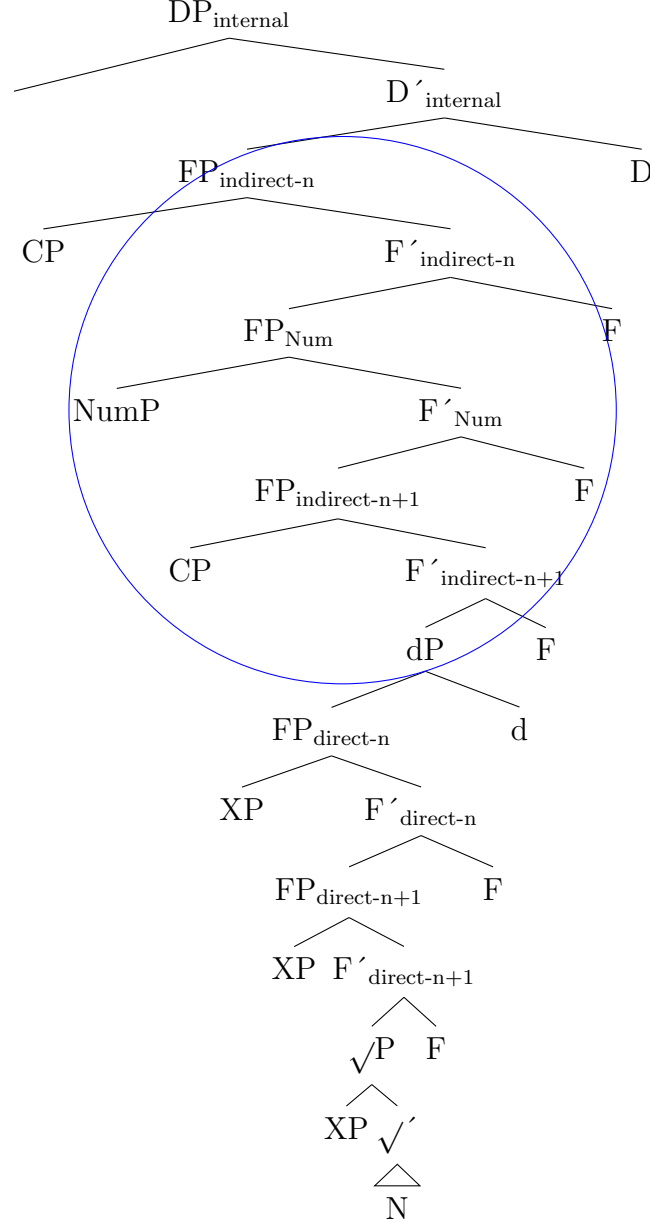


Figure 3.3: Structure of the Indirect Domain in Japanese

The indirect domain is structurally higher than the direct domain and contains a projection for numerals. Japanese provides further evidence for this as direct-only modifiers are generally disallowed from a position to the left of numerals, as also argued in Ayano (2010); Watanabe (2017), highlighting that numerals are situated within the indirect domain.

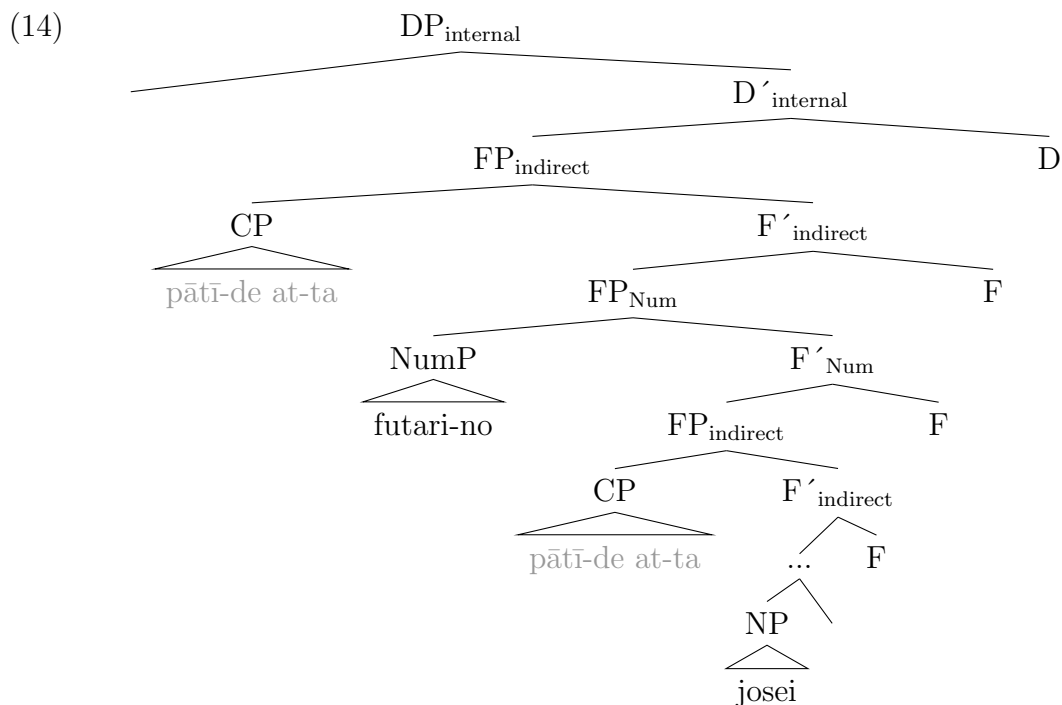
- (12) a. San-nin-no furu-i yūjin-ga ki-ta.
 three-CL-NO long.time-I friend-NOM come-PST
 ‘Three long-time friends came.’ (Ayano, 2010, 106)
- b. *Furu-i san-nin-no yūjin-ga ki-ta.
 long.time-I three-CL-NO friend-NOM come-PST

Furthermore, there is no evidence that Japanese head-external relative clauses do come in different sizes, for instance full and reduced, or CP and IP/vP (see Chapter 5.3). Instead, they are unequivocally CPs. That means that, parallel to the situation in the direct domain, there are no dedicated functional projections for indirect modifiers in the indirect domain either, in this case based on different sizes. This correctly predicts that relative clauses can freely occur to the left or to the right of numerals in linear order. As confirmed through speaker’s judgment for (13), in out-of-the-blue contexts, both orders of relative clause and numeral are equally possible.

- (13) a. [pātī-de at-ta [futari-no josei]]
 party-LOC meet-PST two-NO woman
- b. [futari-no [pātī-de at-ta josei]]
 two-NO party-LOC meet-PST woman
 ‘the two women who I met at a party’

Corpus data and speaker judgments of diverse examples reported in Chapter 5.4 confirm this observation and the assumption that available projections are situated above and below the projection hosting numerals, which do not correlate with the structural size of relative clauses.

Hence, the two orders in (13) are attributable to two different positions the relative clause can occupy, the specifier of a functional projection above or below FP_{Num} . I indicate this below with the two possible options marked in gray.

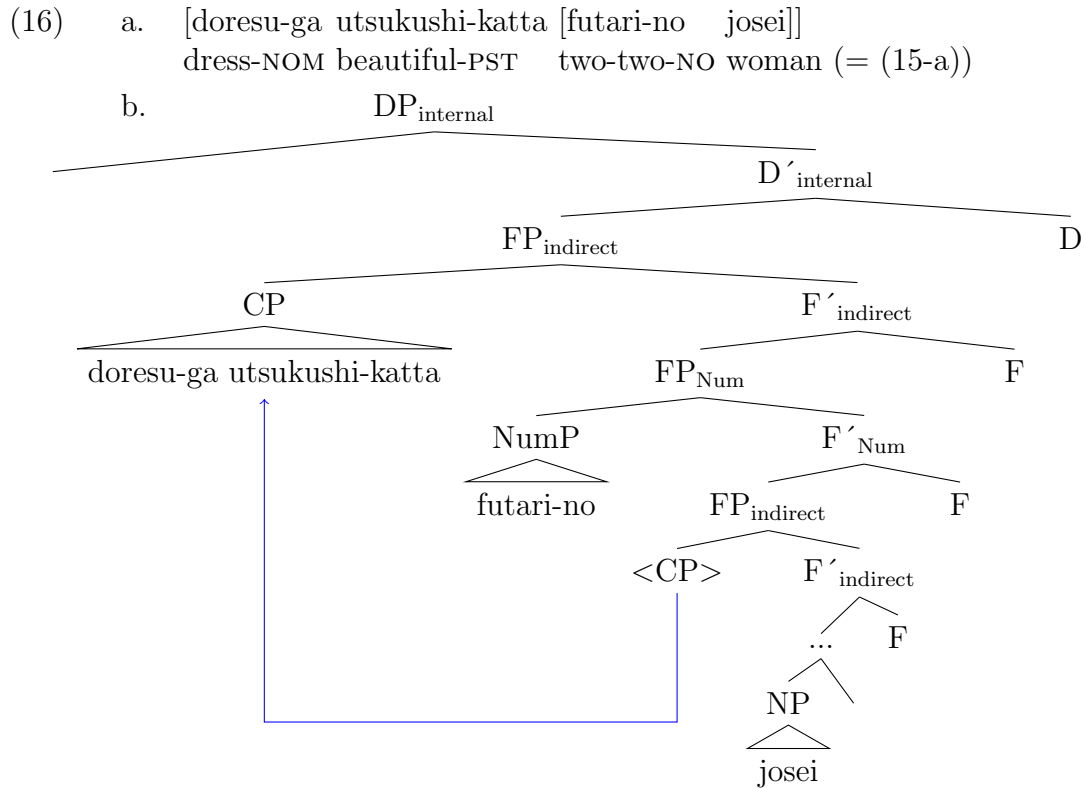


There is a syntax-external factor that influences the choice of speakers to place the relative clause in a specific position, namely to the left of the numeral, showing that their position is not entirely arbitrary. This factor is *complexity*, understood in this book both syntactically and prosodically. Syntactic complexity reflects the amount of syntactic material. Prosodic complexity, descriptively speaking, reflects length (Hawkins 2007; see Chapter 5.4). An increase in complexity can be achieved, for example, when the verbal or adjectival predicate within a relative clause co-occurs with an overt subject argument, or when it is modified by a degree adverb.

That an increase in complexity causes relative clauses to be preferred in a position to the left of numerals is illustrated in the DP in (15), which contains a numeral and a relative clause. This relative clause is adjectival in nature and exhibits an overt subject. Furthermore, the adjective is inflected for past. Most speakers I consulted clearly prefer the position of the relative clause to the left of the numeral over the reversed order. There is certain diversity in speaker judgments, although speakers either accept both orders or prefer the one with the relative clause on the left. The dispreferred order, with lower naturalness according to speakers, is indicated with the symbol ‘#’.

- (15) a. [doresu-ga utsukushi-k-at-ta [futari-no josei]]
 dress-NOM beautiful-PRED-COP-PST two-NO woman
- b. #[futari-no [doresu-ga utsukushi-k-at-ta josei]]
 two-NO dress-NOM beautiful-PRED-COP-PST woman
 ‘the two women whose dresses were beautiful’

This fact is thus suggestive of movement of the relative clause from a functional projection below FP_{Num} to a projection above in an individual fashion as depicted below.



Keeping the system simple and consistent, I assume that all (but only restrictive, see Chapter 3.1.3.3), relative clauses in Japanese merge in a position below numerals, but can potentially move up to a position above, or also to the left periphery (see below). This assumption predicts that such movement operations are open potentially to all relative clauses (Kim, 2019). The more complex a relative clause is, the more likely it is to move to a higher position. This point is illustrated with more data and further discussed in Chapter 5.4, especially in Chapter 5.4.4, and see Fn. 68 where I outline why it fares better than its alternatives.

3.1.3 The left periphery

The DP-external domain, also referred to as the left periphery, is not the main focus of this study, but will be investigated briefly here since Japanese provides evidence for this domain. Notably, this area is postulated, and investigated with regards to several languages in the works of Giusti (1996); Dimitrova-Vulchanova and Giusti (1998); Ihsane and Puskás (2001); Aboh (2004, 2006); Roca (2009); Bastos-Gee (2011); Giusti and Iovino (2016); Roehrs (2020, 2025); Kim (2019), for the integration in formal frameworks see Laenzlinger (2011) and Grohmann (2003).

I will discuss every type of modifier attested in this area in Japanese and argue that there are many insights and incentives for future research regarding this peripheral area to be gained by looking at this language.

3.1.3.1 Demonstratives

Japanese lacks articles, but has demonstratives. They can be simple, the standard ones being *kono* ‘this’ and its counterparts, which only vary in deixis - and complex, such as *koko-no* ‘the (one) here’ or *korekara-no* ‘the one from now on’. I adopt the standard assumption of demonstratives as maximal projections (Laenzlinger, 2005, 2011; Kim, 2019), but in Japanese, they are specifiers, not heads (Kuno, 1973; Kuroda, 1992). An analysis as heads would violate the strict head-finality of Japanese since demonstratives never occur postnominally.

Demonstratives first merge as specifiers of the DP projection heading the internal domain. I follow previous authors (Laenzlinger, 2011; Kim, 2019) and assume that they can optionally move to the specifier position of the DP heading the DP-external domain if modifiers have moved to a position in the left periphery. I adopt the simplistic notions DP_{internal} for the lower and DP_{external} for the higher DP projection. since there is no obvious semantic or pragmatic difference whether demonstratives occur in the lower or higher position.

3.1.3.2 Quantificational modifiers

The DP-external domain contains a projection for quantifiers, or more precisely *quantificational modifiers*, which will be labeled FP_{QUANTITY} . Most quantificational modifiers in Japanese are NO-modifiers, including *takusan-no* ‘a lot’, *kazukazu-no* ‘many’, *subete-no* ‘all’ and *amari-no* ‘few’ as well as *ōku-no* ‘many’.⁷

Cross-linguistically, there is a tendency for quantificational modifiers to appear rather high in the DP. In German, they have to precede (ambiguous) qualitative (17) and also direct adjectives (18). This behavior of German quantifiers is verified in the corpus study reported in Trost (2006a, 378), and see also Eichinger (1991).

(17) German

- a. viele strenge Präsidenten
many strict presidents
‘many strict presidents’
- b. *strenge viele Präsidenten
strict many presidents

⁷Quantification can also be expressed via Sino-Japanese prefixes such as *kaku-* ‘each, every’ or *zen-* ‘all’, which are simply juxtaposed to the head noun. *Bōdai-na* ‘giant, huge’ is a *na*-adjective. I am grateful to Ken Hiraiwa for reminding me of this.

- (18) German
- a. viele frühere Präsidenten
many former presidents
'many former presidents'
 - b. *frühere viele Präsidenten
former many presidents

This generalization partly holds in Japanese: When quantificational modifiers and direct-only modifiers co-occur, the quantifier typically occurs first, the other order being degraded.

- (19) quantificational modifier $\succ$ non-intersective modifier
- a. **ōku-no** shōrai-no purosakkāsenshu
many-NO future-NO pro.soccer.player
'many future professional soccer players'
 - b. *shōrai-no **ōku-no** purosakkāsenshu
future-NO many-NO pro.soccer.player'

- (20) quantificational $\succ$ relational modifier
- a. **takusan-no** komugi-no pan
a.lot-NO wheat-NO bread
'many wheat(-en) (pieces of) bread'
 - b. ??komugi-no **takusan-no** pan
wheat-NO a.lot-NO bread

Importantly, quantifiers in Japanese do not only readily co-occur with demonstratives, when they do, they can occur before and after them.

- (21)
- a. ano **kazukazu-no** mondai
that many-NO problem
 - b. **kazukazu-no** ano mondai
many-NO that problem
'those many problems'

- (22)
- a. ano **takusan-no** hon
that a.lot-NO book
 - b. **takusan-no** ano hon
a.lot-NO that book
'those many books'

This is a significant deviation to, for example, English where quantifiers can only co-occur with demonstratives or articles in certain contexts in the first place, for example in *those many students* or *the few students who attended the lecture*, but crucially only in this order, never in the order *quantifier* $\succ$ *demonstrative*.

- (23) *many those students

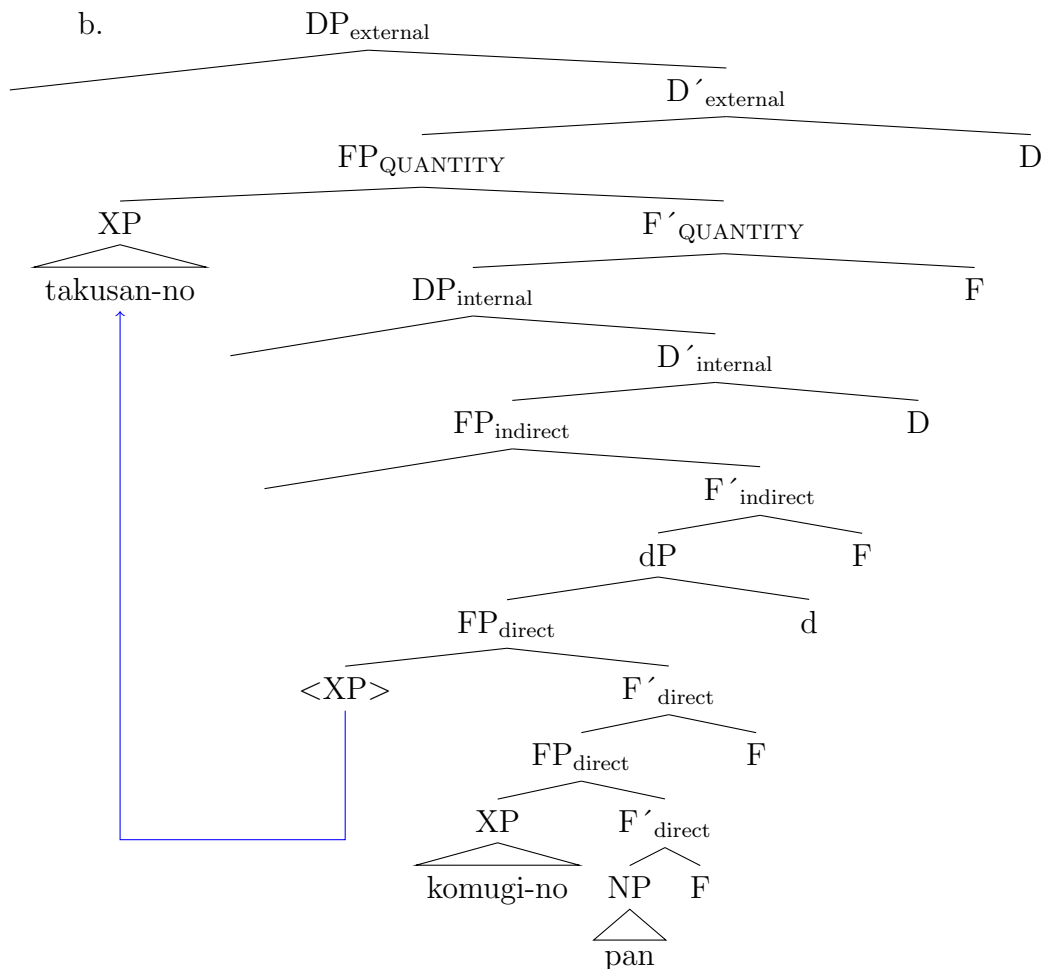
From this, I conclude that quantificational modifiers in Japanese are generally situated in the left periphery. The surface order *demonstrative* $\succ$ *quantifier* is a result of the demonstrative having moved from the internal DP to Spec, DP_{external}, similar to what Laenzlinger (2005, 2011) argues for French. That this happens regularly in Japanese, is corroborated since speaker still find this order a bit more natural.

The question remains whether quantifiers actually move to the left periphery or base-generate there (cf. Kim 2019). There is good evidence that they originate lower, probably inside the direct domain. Note that many quantificational modifiers in Japanese are exclusively used attributively, while for others this is the prototypical use. In fact, all quantificational modifiers attested in the data set are sorted in group 5, the group for direct modifiers. While some are truly exclusively direct, others, such as *takusan-no* ‘a lot’, could in principle have a predicative source, but given the ratio of 14:1 towards attributive use (4048 attributive, 172 predicative occurrences), a categorization as a prototypical attributive modifier is justified (for details see Köhlich 2023). Direct-only quantificational modifiers, those which are impossible as predicates, include *kazukazu-no* ‘many’, *shuju-no* ‘various’ and *amata-no* ‘a lot, many’ as illustrated below.

- (24) a. Kare-wa kazukazu-no shō-wo tot-ta.
 he-TOP many-NO prize-ACC take-PST
 ‘He received many prizes.’
 b. *Kare-ga tot-ta shō-wa kazukazu-desu.
 he-NOM take-PST prize-TOP many-COP.POL
 (‘The prizes he received were many.’)
- (25) a. Kare-wa shuju-no / amata-no shikaku-wo mot-te-i-ru.
 he-TOP various-NO / many-NO qualification-ACC have-GER-be-PRS
 ‘He has various / many qualifications.’
 b. *Kare-ga mot-te-i-ru shikaku-wa shuju-desu /
 he-NOM have-GER-be-PRS qualification-TOP various-COP.POL /
 amata-desu.
 many-COP.POL
 (‘The qualifications he has are various / many.’)

Based on these facts, I conclude that quantificational modifiers are base-generated in the direct domain, but have to move to the left periphery. I tentatively assume that this movement happens to check [+quantificational] features, in line with Kim (2019), and that they move to a dedicated position, call it FP_{QUANTITY}, although more concrete evidence is desirable. The derivation looks as follows.

- (26) a. takusan-no komugi-no pan
a.lot-NO wheat-NO bread
‘many wheat(-en) (pieces of) bread’ (= (20-a))



3.1.3.3 Non-restrictive and restrictive relative clauses

Another argument in favor of a split-DP structure is the fact that a certain class of relative clauses in Japanese obligatorily precedes demonstratives, namely those that are non-restrictive. First, note that all relative clauses *can* readily precede demonstratives in Japanese. This means that when no context is provided, a relative clause in this position is ambiguous and can be interpreted non-restrictively or restrictively, as depicted in (27).⁸

⁸The initial observation goes back to Kamio (1977), who pointed out a paradigmatic difference between non-restrictive pre-demonstrative relative clauses and restrictive post-demonstrative relative clauses. Ishizuka (2008), whom this section follows, has discussed in detail that pre-demonstrative relative clauses are ambiguous.

- (27) [Sakunen isha-ni nat-ta [sono musuko-ga]] kekkon-shi-ta.
 last.year doctor-LOC become-PST that son-NOM marriage-do-PST
 1. $\approx$ ‘That son who became a doctor last year got married.’ (restrictive)
 2. $\approx$ ‘That son, who happened to become a doctor last year, got married.’
 (non-restrictive)

(Ishizuka, 2008, 4, 5)

However, when the same relative clause follows the demonstrative, it can *only* be interpreted restrictively. Establishing a context that forces a non-restrictive interpretation leads to ungrammaticality as depicted in (28), see Ishizuka (2008).

- (28) *Ito-san-ni-wa musuko-ga hitori i-ru. [Sono [sakunen
 Ito-Mr./Mrs.-LOC-TOP son-NOM one be-PRS that last.year
 isha-ni nat-ta] musuko-ga] kekkon-shi-ta.
 doctor-LOC become-PST son-NOM marriage-do-PST
 (‘Mr./Mrs. Ito has one son. That son, who happened to become a doctor
 last year, got married.’)

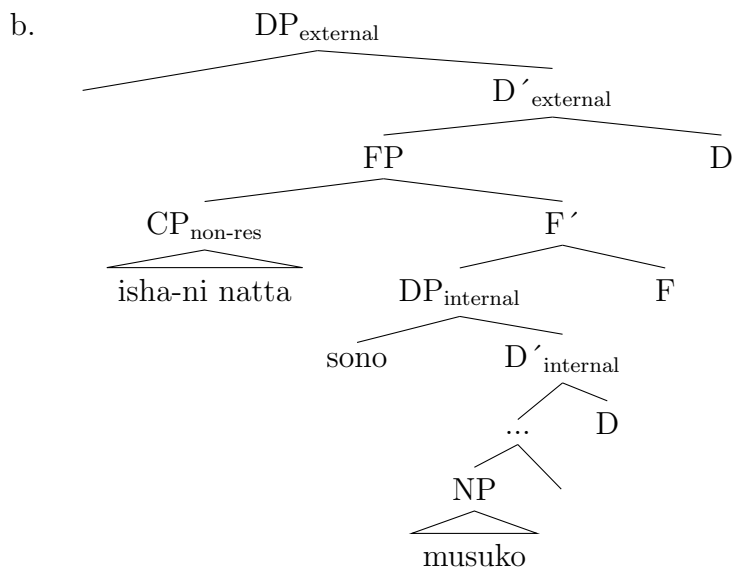
(Ishizuka, 2008, 5)

In other words, in Japanese, only restrictive relative clauses are allowed after demonstratives, non-restrictive relative clauses are confined to a position in the left periphery. Following Ishizuka (2008) and Cinque (2008b), I take this type of relative clause to be base-generated in a dedicated projection in the DP-external domain. In this position, they are outside the scope of the demonstrative, which accounts for the lack of scope effects and the obligatorily non-restrictive interpretation. This proposal is based on Kim (2019), who introduces a projection for non-integrated, *supplementary*, relative clauses (Spec, SpplP) in Korean. The difference is that, in Japanese, non-restrictive relative clauses obligatorily merge outside the internal DP, whereas in Korean a certain type of non-restrictive relative clause can also merge lower.⁹ The relevant position is depicted in the tree below.

⁹ A reviewer asks what the basis for the difference between Japanese and Korean regarding the position of non-restrictive relative clauses is. At this point, I cannot give a definitive answer. Languages seem to differ substantially as to which types of relative clauses they allow in the left periphery (if any). Whereas in Japanese, non-restrictive relative clauses have to occur in the left periphery, and in Korean a subtype of non-restrictive relative clauses, namely supplementary ones have to, German only allows restrictive relative clauses in the nominal left periphery (Köhlich, 2026). Forcing a non-restrictive interpretation, by means of the adverb *übrigens* ‘by the way’, leads to degraded results (see also Bhatt 1990; Roehrs 2020).

- (29) ??Der (übrigens) Bass singt, der Opernsänger hat für seine Vorstellung einen Preis
 the by.the.way bass sings the opera.singer has for his performance a prize
 gewonnen. (German)
 won
 ‘The opera singer, who (by the way) sings bass, has won a prize for his performance.’

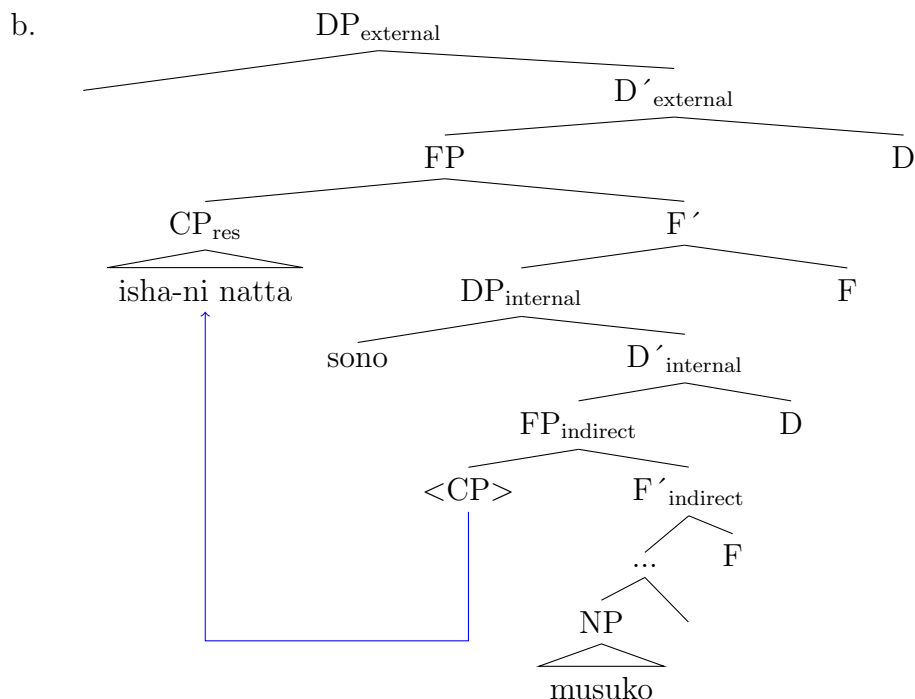
- (30) a. isha-ni nat-ta sono musuko (non-restrictive)
doctor-LOC become-PST that son



Restrictive relative clauses, on the other hand, are base-generated DP-internally, as outlined in the preceding section, presumably below NumP . Their restrictiveness is expected being under the scope of D . In addition to individual movement within the indirect domain to a position above FP_{Num} , I argue that restrictive relative clauses can even cross the internal DP border as depicted below.¹⁰

¹⁰According to Ishizuka (2008), scope effects are retained through reconstruction and the moved relative clause keeps its restrictive interpretation. It does not become entirely clear how reconstruction works in Ishizuka's system and readers are referred to her proposal. See also footnote 71 in Chapter 5.4.5.

- (31) a. isha-ni nat-ta sono musuko (restrictive)
 doctor-LOC become-PST that son



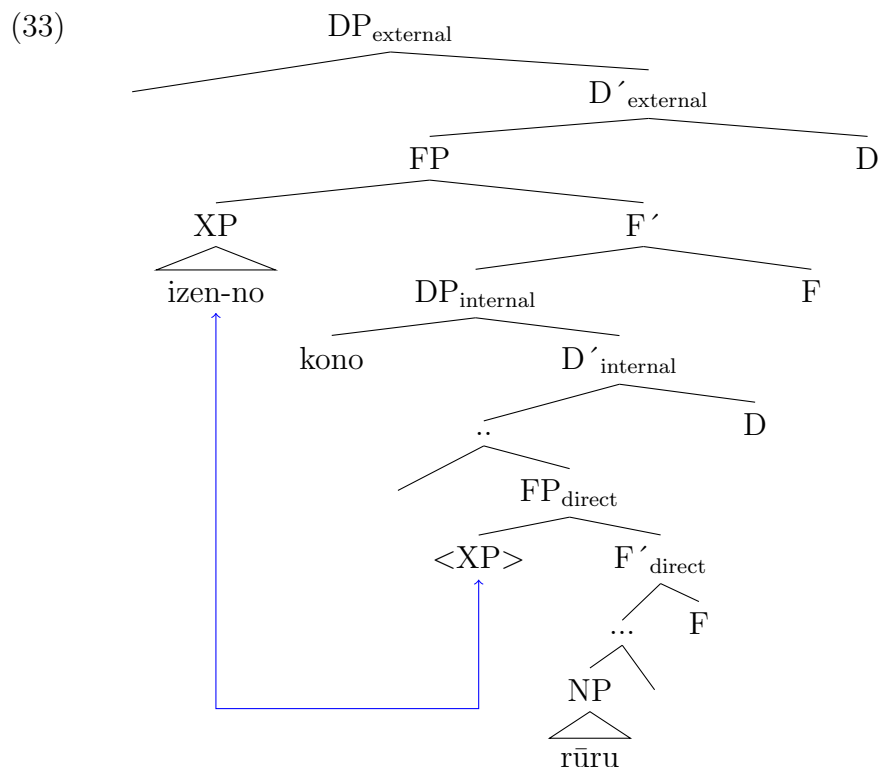
What about non-restrictive relative clauses to the right of demonstratives then? Given the fact that they, as I argued, merge in the DP-external domain, the surface order is the result of movement by the demonstrative from the DP-internal to the DP-external domain across the non-restrictive relative clause in the left periphery. This predicts, correctly, that non-restrictive relative clauses are allowed below demonstratives.

3.1.3.4 Direct modifiers

Direct modifiers can also productively appear above demonstratives, given the right circumstances, but seemingly in free alternation.

- (32) a. kono izen-no rūru
 this former-NO rule
 b. izen-no kono rūru
 former-NO this rule
 'this former rule'

From this, I conclude that direct modifiers have the option to leave the direct domain and move to the left periphery as well, in a similar fashion to quantificational modifiers. Importantly, they are not required to do so and this movement is optional.



3.1.3.5 A discussion on focused modifiers

Focused modifiers and their connection to the left periphery, both within the DP and the clause, make up a rich and controversial research area. The high occurrence of modifiers with a focus (and topic) interpretation, in some cases above D-elements, is one of the main motivations for the split-DP hypothesis (Kiss 1998, 2007; Giusti 1996; Dimitrova-Vulchanova and Giusti 1998; Ihsane and Puskás 2001; Laenzlinger 2005; Kim 2019 a. o.; cf. Alexiadou *et al.* 2007, 140–151 for discussion). The classical observation goes back to Giusti (1996); Dimitrova-Vulchanova and Giusti (1998), who point out that adjectives in Albanian, which naturally occur postnominally and are marked when they appear prenominally, can reach a position above demonstratives, analyzed as FocP, via A'-movement for focus reasons leading to DPs such as (34).

- (34) TJOTR-a grua e bukur (Albanian)
 other-the woman DET nice
 'the other nice woman'

(Giusti, 1996, 113)

Ihsane and Puskás (2001) show that a focus interpretation, with focal stress is also available to numerals and possessors in certain languages and, again, focus movement is triggered. In Hungarian, this, in turn, forces the article to subsequently move to a higher position, to check the feature [+specific], according to the authors. This

position corresponds to DP_{external} proposed here, whereas the numeral with the focus interpretation in (35) is in DP_{internal} .

- (35) a. az EGY könyv (Hungarian)
the ONE book
b. *EGY az könyv
ONE the book

(Ihsane and Puskás, 2001, 48)

Kim (2019, ch. 6.7) argues that a focus projection is universal and that exactly one FocP should be assumed in the left periphery (see also Giusti and Iovino (2016); Roehrs (2020)). This position arguably is below FP_{QUANTITY} accounting for the ordering in (36).

- (36) a. most UNSKILLFUL doctors (Kim, 2019, 137)
b. *UNSKILLFUL most doctors

Focus movement has also been invoked in order to account for non-canonical orders of adjectives such as *RED big book* (see also Svenonius 2008). Arguably, the focused adjective in such cases has move to the specifier of a focus projection, which, however, is below the DP projection in English.

- (37) [_{DP} the [_{FocP} RED [_{FP_{indirect}} ... [_{FP_{DIMENSION}} big [_{FP_{COLOR}} <red> ... [_{NP} book]]]]]]

However, in most languages, modifiers can also receive in-situ focus (Kim, 2019, 191) suggesting the absence of an EPP feature of FocP in such cases.

- (38) I need a big RED book, not a big BLUE one.

Japanese is a language (similar to, for instance, Gungbe, cf. Aboh 2004) which exhibits *focus sensitive particles*. The prototypical topic marker -WA can also acquire a focus function (39), other focus sensitive particles are -DAKE ‘only’ or -SAE ‘even’ (40) (Tomioka 2010, 2016; Vermeulen 2007, 2012, see also Chapter 5.3.3.4).

- (39) [Nihongo-wo shanai-de-**wa** tsuka-u] gaikokujin
Japanese-ACC office-LOC-FOC use-PRS foreigner
‘a foreigner who uses Japanese only in the office’

(Akaso and Haraguchi, 2011, 93)

- (40) Taro-dake/mo/sae ano mise-de nihongo-no shōsetu-wo kat-ta.
Taro-only/also/even that shop-LOC Japanese-MOD novel-ACC buy-PST
‘Only/Also/Even Taro bought Japanese novels at that shop.’
(Vermeulen, 2012, 195)

These elements mark instances of clausal focus and cannot be freely combined with DP-internal constituents in order to put focus, for instance, on numerals, shown below for DAKE ‘only’.

- (41) *Ni-satsu-dake-no hon-wo kat-ta.
two-CL-only-NO book-ACC buy-PST
(‘I bought only two books.’)

The intended meaning can be expressed if the quantifier is used as an adverbial modifier within the verbal domain, as in (42). In this case, it can optionally be left-dislocated to a clause-initial position. Clearly, it is not part of the DP, since it does not occur with NO, or any other form of the linker (see below), which we would expect if it was a nominal modifier.

- (42) Ni-satsu-dake, hon-wo kat-ta.
two-CL-only, book-ACC buy-ACC
‘I bought only two books.’ (Lit. ‘Two only, I bought books.’)

The focus sensitive particle DAKE can, however, attach to a numeral classifier and then occur DP-internally as shown in (43) where it attaches to *futari* ‘two.people’, a suppletive form of the classifier for people *-nin* and the number 2. That the focus sensitive particle is part of the DP becomes clear as NO obligatorily occurs.

- (43) futari-dake-no himitsu-ni shi-te-ok-ō to [...]
two.people-only-NO secret-DAT do-GER-prepare-EPIS COMP
‘to make it a secret shared only by two [...].’ (Lit. ‘a two-person only secret..’)¹¹

Crucially, DAKE does not add focus to the noun *himitsu* ‘secret’, but to the numeral *futari* ‘two people’, so what is focused is the amount of people. In any case, this example shows that focus is possible DP-internally which suggests the existence of FocP in Japanese (see Chapters 5.3.3.4 and 5.3.4.1).

Furthermore, in contrast to the Hungarian example presented in (35), Japanese allows additional modifiers to precede focused numeral classifiers marked by DAKE, although this might not be the default order. (44) shows that both orders of modifiers are fine.¹²

¹¹Nakanishi, Rei (2003): Yotō. Tokyo: Shinchōsha, via BCCWJ, 2023/11/09.

¹²I am grateful to Ken Hiraiwa for bringing all these facts to my attention.

- (44) a. san-nin-dake-no kura-i himitsu
 three-person-CL-only-NO dark-I secret
 b. kura-i san-nin-dake-no himitsu
 dark-I three-person-CL-only-NO secret
 ‘a dark secret shared by only three people’

These facts indicate that there seems to be no requirement for focused modifiers to occur in the left periphery, suggesting the absence of a requirement for movement. This is backed up by evidence from the position of prosodically focused lexical modifiers. Although being a pitch-accent language, Japanese can express focus via accentuation and prosody (Beckman and Pierrehumbert, 1986; Hiraiwa and Ishihara, 2012; Tomioka, 2010). Modifiers with prosodic focus can occur in a higher, DP-initial, position, but they can also remain in-situ. In (45), both answers are possible to the preceding question, emphasizing *maru-i* ‘round’.¹³

- (45) a. [Aka-i maru-i tēburu] ka, [aka-i shikaku-i tēburu] dotchi-wo
 red-I round-I table Q red-I square-I table which-ACC
 kat-ta no?
 buy-PST Q
 ‘Did you buy the red round table or the red square table?’
 b. Aka-i MARU-I tēburu-wo kat-ta yo.
 red-I round-I table-ACC buy-PST SFP
 ‘I bought the red ROUND table.’
 c. MARU-I aka-i tēburu-wo kat-ta yo.
 round-I red-I table-ACC buy-PST SFP
 ‘I bought the ROUND red table.’

Crucially, prosodic stress is in some instances not even enough to allow modifiers to move to a higher position. Recall from (4), given in Chapter 3.1.1, that direct modifiers cannot precede relative clauses. This observation still holds if the non-intersective modifier *shōrai-no* ‘future’ is focused, meaning it must remain in-situ and cannot cross the indirect domain.

¹³Japanese also bears an interesting similarity to Latin. Although this language had a comparatively free word order and modifiers on both side of the noun, adjectives were consistently placed more often in a position to the left of the noun if used contrastively as shown by Devine and Stephens (2006). However, adjectives are not obligatorily placed in this position. An even more intriguing parallel to Japanese is that “unfocused adjectives can raise to the left of the head as well as focused adjectives” (Devine and Stephens, 2006, 409), exactly paralleling the case with scrambled unfocused adjectives in the left periphery in Japanese.

- (46) a. Kare-wa [ōku-no kurabu-ga chūmoku-shi-te-i-ru
he-TOP numerous-NO club-NOM pay.attention-do-GER-be-PRS
[**SHŌRAI-no** purosakkāsenshu]]-desu.
future-NO pro.soccer.player-COP.POL
‘He is a FUTURE professional soccer player whom many clubs pay
attention to.’
- b. *Kare-wa [**SHŌRAI-no** [ōku-no kurabu-ga
TOP future-NO numerous-NO club-NOM
chūmoku-shi-te-i-ru purosakkāsenshu]]-desu.
pay.attention-do-GER-be-PRS pro.soccer.player-COP.POL

Note that movement to a non-canonical position is available in Japanese as the co-occurrence with demonstratives has shown. Such a non-canonical position is presumably also available for some direct-only modifiers in English, as shown below.

- (47) I want to meet a FORMER handsome president, not a current one.

Final evidence for the absence of an EPP-feature for FocP in Japanese comes from superlatives, another category of modifiers that has been argued to obligatorily occur within the left periphery (Cinque, 2010, 2023). In English, if an adjective in superlative co-occurs with a numeral, moving the superlatives across the numeral is necessary (cf. Cinque 2023, 11 and Guglielmo Cinque p.c.).

- (48) a. the blackest two dogs (that I’ve ever seen)
b. ??the two blackest dogs

However, this is not the case in Japanese. On the contrary, both orders are fine.

- (49) a. ichi-ban kuro-i ni-hiki-no inu
number-one black-I two-CL-NO dog
‘the blackest two dogs’
- b. ni-hiki-no ichi-ban kuro-i inu
two-CL-NO number-one black-I dog
‘the two blackest dogs’

To summarize, focused modifiers in Japanese are not required, and sometimes not allowed, to move to the left periphery, although, in most cases, they seem to be able to allow such movement operations. From this, I conclude that the left periphery contains a focus projection, which has been proven empirically adequate and necessary in cross-linguistic perspective, but that, in Japanese, this projection simply lacks an EPP feature, and its Spec position can remain empty.

3.2 The Internal structure of modifiers

In this section, I develop an account for the internal structure of Japanese modifiers, a crucial aspect that has been missing in the discussion so far. This concerns both, the monomoraic elements co-occurring with direct and ambiguous modifiers, as well as the complex internal structures available to indirect modifiers. In cartographic accounts, the internal structure of modifiers and the general DP structure are typically treated orthogonally. The account proposed in the following will yield new insights that further enrich the general understanding of the DP.

3.2.1 Introduction

In this section, I focus especially on the monomoraic elements NO, I or NA which always co-occur with non-verbal modifiers in Japanese in attributive position if those express non-past in positive contexts.

- (50)
- | | | |
|----|--|----------------------|
| a. | hisshu-*(no) yoyaku
urgent-NO reservation
'urgent reservation' | <i>no</i> -modifier |
| b. | taka-*(i) doresu
expensive-I dress
'expensive dress' | <i>i</i> -adjective |
| c. | shikku-*(na) doresu
chic-NA dress
'chic dress' | <i>na</i> -adjective |

Those elements cannot be analyzed as parts of the stem, since they disappear when the relevant lexemes, those which can, appear predicatively. Instead, the copula appears. A special behavior is exhibited by *i*-adjectives, where the vowel I also appears in predicative position.¹⁴

- (51)
- | | | |
|----|---|----------------------|
| a. | Ano haiyū-wa mumei (*no) da.
that actor-TOP unknown NO COP
'That actor is unknown.' | <i>no</i> -modifier |
| b. | Ano doresu-wa taka-i.
that dress-TOP expensive-I
'That dress is expensive.' | <i>i</i> -adjective |
| c. | Ano doresu-wa shikku (*na) da.
that dress-TOP chic NA COP
'That dress is chic.' | <i>na</i> -adjective |

¹⁴This is the reason why authors such as Nishiyama (1999) treat them as the same element (see Chapter 5.2. In this study, I propose two different analyses for I.

NO and NA are also mutually exclusive with other forms of the copula, namely the (canonical) long forms for non-past DEARU and past D(E)ATTA.

- (52)
- a. ano mumei-dearu (*no) haiyū
that unknown-COP NO actor
'that actor who is unknown' *no*-modifier
 - b. ano shikku-dearu (*na) doresu
that chic-COP NA dress
'that dress which is chic' *na*-adjective

Thus, the question arises how these elements should be treated. Especially, the element NO is of interest, since, as shown in Chapter 1.4, it is incredibly widely distributed and appears with a variety of lexical elements with different degrees of adjectival and nominal status, as well as different functional elements. Furthermore, most direct modifiers co-occur with this element, although by far not all *no*-modifiers are direct. This means an account of this recalcitrant element with enough explanatory power to either explain why it appears in so many different instances, or alternatively to reduce all these instances to one underlying function is desirable.

In this section, I will propose a unifying account of NO as a syntactic head which licenses modification, meaning I will choose such a unifying account. This account predicts neither restrictions on different word classes nor on modification types. In the next step, this analysis will be applied to the elements I and NA whose distribution is more restricted as the lexemes they co-occur with are clearly adjectives. I will then discuss why the analysis presented here fares better than alternative accounts before I outline the integration of these elements in the DP.

3.2.2 A universal syntactic head

3.2.2.1 The basic idea

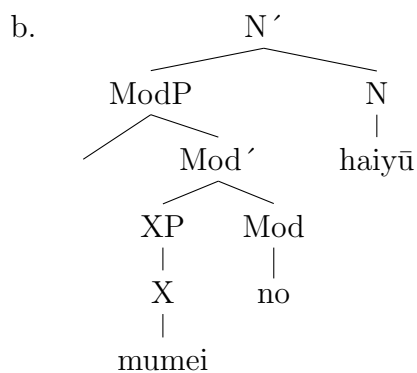
I argue that NO is the realization of a syntactic head, which I call **MOD** following Rubin (1993, 1994, 1996). This head mediates modification within the DP. The basic assumptions are as follows.

- (53)
- a. Modifiers in Japanese must be licensed by a syntactic head in attributive position.
 - b. This head can be realized overtly in Japanese. Its unspecified spell-out form is /no/.
 - c. MOD selects modifiers irrespective of modification type and word class. Whether a modifier is direct or indirect is decided in the syntax.

In this sense, MOD belongs to the category of syntactic linkers. There are several linker architectures on the market,¹⁵ but I apply the rather minimalistic account of Rubin (1994, 1996) to Japanese. Concretely, I argue that MOD is a functional head that obligatorily selects a modifier - lexical or functional - as its complement and licenses them as modifiers (Rubin, 1994, 36–40), itself being selected by the noun. In this sense, MOD is the modificational counterpart of PRED, discussed in Chapter 5.2.1.3.¹⁶

A minimal representation, as MOD is proposed in Rubin (1994), looks as follows exemplified with a *no*-modifier

- (54) a. mumei-no haiyū
unknown-MOD actor (= (50-a))



Furthermore, I argue, that this head is universal, but optionally realized, depending on the language. This sticks to the core assumptions of cartography. As languages only differ in if and how they realize lexical material mapped onto the functional structure (Rizzi, 1997; Shlonsky, 2004; Bortolotto, 2016; Cinque and Rizzi, 2010; Rizzi and Cinque, 2016), this head might be covert in most languages (see also Richards 2023). In addition to the overt realizations of this head detected by Rubin (1994, ch. 2.2 + 2.3) in Romanian, Tagalog and Mandarin, allowing a reanalysis of the element DE in Mandarin as MOD (Rubin, 1994, 97–102), I propose that it is also overtly realized inside the Japanese DP.¹⁷

¹⁵For overview over linkers in the nominal, and also clausal domain, see among others Baker and Collins 2006; Philip 2012; Franco *et al.* 2015; Richards 2023.

¹⁶Both elements are functional and universal. They are also historically related, as both Bowers and Rubin (and Nishiyama) worked at Cornell University.

¹⁷These realizations fulfill all the characteristics expected from functional categories: They are phonologically and morphologically dependent, cannot be separated from their complements and lack semantic meaning, “descriptive content” in the sense of Abney (1987, 65–66; see Rubin 1994, 34). Rubin (1994, ch. 2.2 + 2.3) also analyzes the agreement systems of Russian and German as instances of MOD. For recent discussion on German and Dutch see Zeijlstra (2020) and Alexeyenko and Zeijlstra (2021). The *ezāfe* morpheme appearing obligatorily with modifiers in Western Iranian languages including Persian and Zazaki might also be a good candidate for this analysis. See for instance Toosarvandani and van Urk (2014); Franco *et al.* (2015) and Gündoğu *et al.* (2024) for relevant analyses and the Persian data in Chapter 4.4.2.2.

3.2.2.2 Integration in The DP

The minimal structure given above in (54) of a noun phrase with MOD selecting a modifier can be directly integrated into the cartographic DP as in (55). This means what actually is embedded in the specifier position of the relevant functional projection is a modifier selected by MOD. Following *no*-modifiers, MOD is overtly realized as /no/. Recall that *mumei-no* ‘unknown’ is ambiguous, which is why I refrain from labeling the relevant functional projection as direct or indirect.

- (55) a. ano mumei-no haiyū
 that unknown-MOD actor
 ‘that unknown actor’
- b.
-
- ```

graph TD
 DP_internal[DPinternal] --- ano[ano]
 DP_internal --- D_prime_internal[D'internal]
 D_prime_internal --- FP[FP]
 D_prime_internal --- D[D]
 FP --- ModP[ModP]
 FP --- F_prime[F']
 ModP --- Mod_prime[Mod']
 Mod_prime --- XP[XP]
 Mod_prime --- Mod[Mod]
 XP --- X[X]
 X --- mumei[mumei]
 Mod --- no[no]
 F_prime --- NP[NP]
 F_prime --- F[F]
 NP --- haiyū[haiyū]

```

Crucially, this account can be applied to all kinds of modifiers, lexical or functional, independent of word class and modification type. In all cases, MOD selects the modifier and this chunk is situated in the Spec position of a functional projection within the relevant domain. Before outlining the concrete cases in Chapter 3.2.7, I will discuss the precise semantics and contribution of this element, its application to the other modifiers, some refinements and some possible alternatives.

## 3.2.3 More on the precise semantics and contribution of MOD

As mentioned in the preceding section, several linker architectures have been proposed and they mostly differ in the universality and precise contribution of the linker, as well as its semantics.<sup>18</sup> For instance, Philip (2012) in a study on a variety of typologically unrelated languages defines linkers as functional elements devoid of semantics that must intervene structurally to highlight modification. As she remarks “[...] the linker does not initiate the relationship between Head and Dependent; it

<sup>18</sup>I am grateful to an anonymous reviewer for encouraging me to include this section.

simply marks its presence.” (Philip, 2012, 65). She includes Japanese in her sample and also concludes that NO is such a linker. On the contrary, Franco *et al.* (2015) argue that linkers are not semantically vacuous, but have interpretive properties. However, the authors remain agnostic about the universality of linkers.

In the system defended here, MOD is, in fact, demanded by the syntax in order for any element to function as a nominal modifier. It thus not only marks the relationship, it establishes the relationship. In Japanese, it is, furthermore, overtly realized in all instances of non-finite (that is: tenseless) modification. I furthermore argue that it is present, but unpronounced when it selects a finite modifier, i. e. a verbal relative clause. This means that in stances such as (52) above, it is not replaced or paraphrased by the copula, and the element /no/ is not an allomorph of it. Rather, since the modifier is predicative in nature, the copula can be overtly realized, which, in turn, causes the non-pronouncement of MOD. See Chapter 3.2.7.2 for more discussion.<sup>19</sup>

However, importantly, MOD does not contribute semantics (in the strict sense) by itself and neither does it determine the type of a modifier or its contribution. In this point, I deviate from Rubin (1994), who argues that MOD necessarily contains a semantic component: Namely that of type-shifting lexemes from type  $\langle e, t \rangle$  to type  $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$ , thus from predicates to attributes (Rubin 1994, 171–177 cf. Partee 1986). However, this poses immediate problems when considering quantificational modifiers and numerals. Those obligatorily co-occur with NO in Japanese, but are not of the basic type  $\langle e, t \rangle$ , but  $\langle \langle e, t \rangle, t \rangle$  (Partee, 1986), thus would not be the right input for MOD.<sup>20</sup>

This also clearly sets MOD apart from the application of a similar syntactic head JOIN proposed in Truswell (2004, 2009) and argued for in Belk (2017). Belk proposes a very similar account to MOD, with the crucial difference that the lexical category of the input argument must be that of an adjective, which is of the basic type  $\langle e, t \rangle$  (Partee, 1986), and the output is always an attributive adjective (Belk, 2017, ch. 2.3), where *attributive* corresponds to *direct* in Cinque (2010). The upshot of this is that JOIN immediately licenses attributive modification by adjectives without having to assume set intersection. It is therefore predominantly suitable for non-intersective adjectives (Truswell, 2004, 37–38). Adjectives which are licensed via JOIN arguably come with this information in their lexicon (Belk, 2017, 157).

<sup>19</sup>Naturally, this begs the question why the forms /no/, or /na/ respectively, and /de aru/ can alternate in attributive position. I argue that the latter is the optional realization of the copula. This accounts for the fact that this option is less natural. Simply realizing the syntactic head MOD in absence of the copula is preferred.

<sup>20</sup>This also extends to several non-intersective modifiers such as *mere* or *only*, which have *per se* a quantifier-like character, and counterparts of which also only appear with NO (see Chapter 4.3.1). I am grateful to David Adger for pointing out this latter point to me.

Rather, MOD licenses modification, but the role of the modifier, whether it is direct or indirect, is determined in the syntax. This means that MOD can select direct and indirect modifiers alike. In this sense, PRED and MOD are also not distributed alongside indirect and direct modification respectively, but PRED is always present with (finite) predication, whereas MOD is always present in modification structures, those of predicative modifiers alike.

### 3.2.4 The application to *i*- and *na*-adjectives

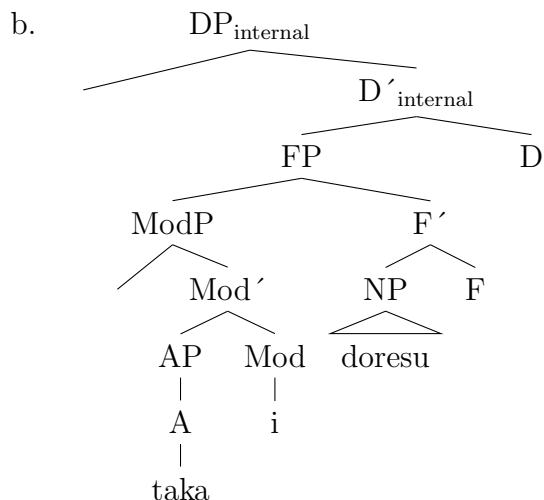
Here, I extend the analysis of MOD as a universal head to the other monomoraic elements, I and NA. All are realizations of the same element and the differences between these three elements is straightforwardly accounted for. Contrary to NO, I and NA are specified for the word classes they co-occur with, namely *i*-adjectives and *na*-adjectives respectively. This allows for the formulation of a spell-out rule for MOD which makes use of word classes.

(56) Spell-Out Rule for MOD in Japanese

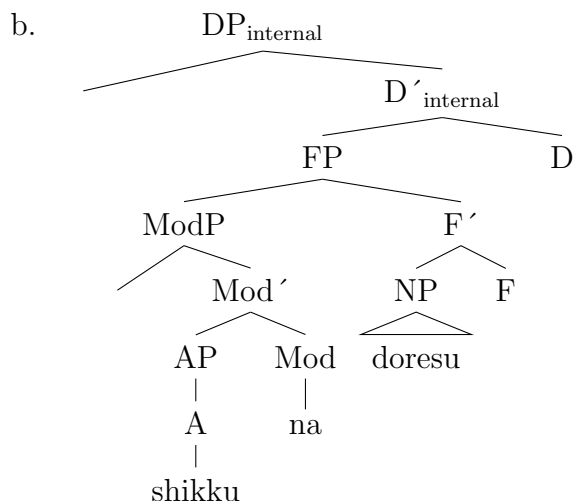
$$\begin{aligned} [Mod] \leftrightarrow & /i/ \text{ / } iA \text{ -} \\ & /na/ \text{ / } naA \text{ -} \\ & /no/ \end{aligned}$$

This rule states that MOD is realized as /i/ following *i*-adjectives, as /na/ following *na*-adjectives, and as /no/ after all other modifiers. That means that the spell-out forms /i/ and /na/ are more specified, and only occur after the word classes of *i*- and *na*-adjectives respectively, /no/ is underspecified, making it the elsewhere case. I assume that this rule is achieved via Late Insertion for functional elements (Halle and Marantz, 1993; Chomsky, 2001), ensuring that the realization of MOD can take place after it has entered syntax. Importantly, MOD is a syntactic head that is only active within the DP, therefore the rule in (56) only refers to attributive contexts. The relevant structures for lexemes belonging to the two Japanese adjective groups in attributive position thus look as follows.

- (57) a. taka-i doresu  
expensive-MOD dress  
'expensive dress' (= (50-b))



- (58) a. shikku-na doresu  
chic-MOD dress  
'chic dress' (= (50-c))



There are some potential caveats with this application worth mentioning. First, diachronic facts tell us that NO is already attested in Old Japanese where it had the function of connecting two nouns (Frellesvig 2010 among others). From there, it only later entered other domains (Sakurai 1964; Wenck 1974). On the other hand, NA is derived from the historical copula *nari* (attributive form *naru*) and developed parallel to DA, the short form of the copula confined to predicative position (Kubo 1992; Urushibara 1993; Nishiyama 1999; Miyama 2011). In the case of *i*-adjectives, their inflectional paradigm differentiated between attributive *-ki* and sentence-final *-shi*, same as other verbal categories. Arguably, the morpheme KI is the predecessor of modern *-i* as a result of sound changes and diachronic omission of the velar /k/.<sup>21</sup>

<sup>21</sup>See for an analysis Nishiyama (1999, sec. 3.3) and Chapter 5.2.3 of this book. For discussion on historical facts, see Urushibara (1993) and Frellesvig (2010).

This means that NO could also at earlier stages of the language be identified as an individual morpheme, whereas the two other elements were parts of the inflectional system of Japanese adjectives. Researchers might therefore be hesitant to apply the Mod-hypothesis to the elements I and NA.

Another potential problem is that the morpheme I does not only occur in attributive position, but also in predicative position.

- (59) a. taka-\*(i) doresu  
expensive-i dress  
'expensive dress'  
b. Ano doresu-wa taka-\*(i).  
that dress-TOP expensive-I  
'That dress is expensive.'

Since MOD only occurs in attributive position, an alternative account needs to be posited for I in predicative position. In Chapter 5.2, I will put forth an analysis of I in predicative position as an instance of the copula (Nishiyama, 1999) and discuss why this analysis is not readily applicable to attributive I.

Nevertheless, as to be discussed further below, the account of all three elements as instances of MOD makes the system adopted here consistent and avoids many of the problems of other accounts. Furthermore, it allows to reduce all three elements to the same underlying type, namely a functional head responsible for attributive modification.

### 3.2.5 Restrictions

As already visible in the examples above, the distribution of MOD is not free. It is a dependent element and needs to select an element as attributive modifier. It attaches to this modifier and is confined in its realization to the overt realization of the modifier itself. It thus behaves as expected from functional elements (cf. Rubin 1994, ch. 2.2.2). Furthermore, it needs to occur DP-internally, thus in the context of nominal modification. These two restrictions are intertwined and can be formulated in the refinement rule in (60).

- (60) Realization of MOD in Japanese, refinement  
1) MOD must select a modifier which must be overtly realized.  
2) MOD only occurs within the DP.

The requirement to select a modifier which must be overtly realized can be illustrated with instances of so-called *light nouns*. In cases where English applies *one*-substitution of nominal constituents (61), Japanese also must employ a placeholder

with a nominal character, called *light noun* in Hiraiwa (2012). Unfortunately, the most basic light noun is *no*, thus homophonous to MOD, as shown in the DP in (62).

- (61) I like red shirts, but I hate blue \*(ones).  
 (62) Aka-i shatsu-wa suki-da kedo, ao-i \*(no)-wa kirai-da.  
 red-MOD shirt-TOP like-COP but blue-MOD LN-TOP hate-COP  
 ‘I like red shirts, but I hate blue ones.’

When a noun is pronominalized with the light noun *no* ‘(the) one’, but modified by a *no*-modifier, subsequent haplology causes one of the two elements to be deleted (Hiraiwa, 2016, 13531–1352). In (63), *no* could be MOD or the light noun.

- (63) Kuruma-wa Nihon no ga suki-da.  
 car-TOP Japan MOD/LN NOM be.fond-COP  
 ‘As for cars, I like Japanese ones.’

However, the point can be made clear by using other, more specified, light nouns such as *mono* ‘thing’, which replaces the noun *kuruma* ‘car’ in the second instance in (64). It is itself modified by a nominal possessor and this possessor must be licensed by MOD, which is realized as /no/.

- (64) Kono kuruma-wa watashi-no mono desu.  
 this car-TOP I-MOD LN.thing COP.POL  
 ‘This car is mine.’

Such examples thus show that MOD must form a syntactic constituent with the modifier, hence neither of the two can be elided individually.

(60) also explains why MOD does not appear outside of modificational contexts.<sup>22</sup> This includes predicative contexts, but also accounts for clausal (floating) quantifiers mentioned in Chapter 3.1.3.5 and repeated below. Since the quantifier is part of the DP, thus is no modifier, MOD is not realized.

- (65) Ni-satsu-dake, hon-wo kat-ta.  
 two-CL-only, book-ACC buy-ACC  
 ‘I bought only two books.’ (Lit. ‘Two only, I bought books.’) (= (42))

This restriction can be further illustrated via an example of a so-called *major subject construction* (Kuroda, 1992; Heycock, 1993a; Vermeulen, 2005) given in (66) and elaborated in detail in Chapter 5.3.5.3.

<sup>22</sup>The fact that MOD is not realized when the head of I is overt, will be discussed in Chapter 3.2.7.2.

- (66) Sono josei-ga doresu-ga utsukushi-i.  
 this woman-NOM dress-NOM beautiful-I  
 ‘As for this woman, her dress (= the dress of this woman) is beautiful.’

In (66), the two nouns *doresu* ‘dress’ and *josei* ‘woman’ stand in a possessor-possessee relationship. They both appear with the nominative case marker -GA, but while the lower noun is theta marked, the higher *ga*-marked noun has a focus interpretation (see Heycock 1993a; Vermeulen 2005). Importantly, the two nouns are not syntactically embedded in a modification structure and *doresu* is devoid of an overt modifier. The underlying structure of (66) is arguably (67) in which the possessor of the dress appears overtly (Kuroda, 1992). The entity *josei* ‘woman’ appears twice making this sentence pragmatically marked. However, crucially, as it fulfills the role of a modifier of *doresu* ‘dress’, it is obligatorily followed by NO, contrary to (66).

- (67) Sono josei-ga (sono) josei-no doresu-ga utsukushi-i.  
 this woman-NOM this woman-MOD dress-NOM beautiful-i  
 ‘As for this woman, the dress of this woman is beautiful.’

The crucial part is that to arrive from (67) at (66), not only the nominal part (*sono*) *josei* ‘(this) woman’ is deleted, but also NO. There is no modifier to select, hence MOD is not present in the structure and not realized.

### 3.2.6 Discussion: Alternative accounts

In this section, I will quickly compare the idea presented here to alternative accounts and show why it fares better.

#### 3.2.6.1 Post-syntactic insertion

MOD as depicted here is very similar to the well-known *Mod-Insertion Rule* of Kitagawa and Ross (1982), and see Watanabe (2010), which became a landmark proposal for authors arguing for a single syntactic status of NO. The revised version of this rule from Watanabe (2010) is given below.

- (68) Mod Insertion Rule (Revised): [DP . . . XP(–tense) N]  
 → [DP . . . XP(–tense) Mod N], where head noun is overt and Mod = *no*  
 (Watanabe, 2010, 66)

Essentially, this rule states that the element NO is inserted post-syntactically whenever modification happens. However, it is subsequently deleted when the modifying element contains tense, which explains why it never co-occurs with relative clauses.

By including this aspect, the authors point out a crucial difference between Japanese and Mandarin, where DE does obligatorily follow relative clauses.

In a nutshell, in the account defended here, MOD is a functional category that is already present in the syntax and licenses lexical and functional elements as modifiers, whereas in the post-syntactic account of Kitagawa and Ross (1982), MOD gets inserted later after the relationship between modifier and noun has already been established. Both options are appealing, and there is no reason why the proposal of Kitagawa and Ross (1982) should be incompatible with the system defended here, which also relies on some form of Late Insertion (Halle and Marantz, 1993; Chomsky, 2001), although to ensure the correct realization of MOD.

I argue that the analysis of NO as a syntactic head is preferable from a theoretic perspective, as it fits better with the cartographic framework adopted here. Furthermore, it has the big advantage of being more language-universal, allowing to explain phenomena in a variety of different languages with regard to such syntactic heads involved in modification, contrary to the highly language-specific Mod Insertion rule.

One concrete argument against post-syntactic analyses of linkers, also provided in Philip (2012, 54–55), is that, as for instance shown in the examples (64) and following, is that MOD forms a syntactic constituent with the modifier, evidenced by the fact that neither can be elided without the other. If MOD was post-syntactically inserted, it would be odd if neither, the modifier itself or the linker, could survive elision of the other element.

### 3.2.6.2 Pre-syntactic insertion

An analysis that is worth considering, but ultimately not suitable is pre-syntactic insertion, for instance along the lines of a *word marker* hypothesis proposed prominently in Alexiadou and Müller (2008) for the inflectional endings of nouns in German, Russian and Greek, following several predecessors (Harris, 1991; Ritter, 1993b; Aronoff, 1996; Oltra Massuet, 1999; Oltra Massuet and Arregi, 2005). Word markers are pre-syntactically inserted elements, that are bestowed with a variety of binary features and these features match those of the different inflectional classes and cases of nouns in these languages (Alexiadou and Müller, 2008, 33). One could entertain the idea that NO is a word marker that is pre-syntactically added to lexical and functional elements to form a compound structure. The obvious problem is that we would never expect this word marker to be absent once the lexical element has entered syntax (Artemis Alexiadou p.c.). Recall however that NO never appears in predicative position.



- (69) Ano haiyū-wa mumei- (\*no) da.  
that actor-TOP unknown NO COP  
‘That actor is unknown.’  
(= (51))

If NO was a word marker, it should be able to appear in predicative contexts, but this is illicit. Furthermore, we would never expect such a marker to be attracted to functional elements, such as postpositions. However, NO can follow postpositions in modificational contexts, as alluded to in the introductory chapter.

- (70) ishi-de-no kōgeki  
stone-INS-NO attack  
‘an attack with stones’ (Saito *et al.*, 2008, 248)

### 3.2.6.3 Lexical compounds

One might also entertain the possibility that two adjacent elements, especially nouns, out of which the first one is followed by NO, actually do not mark cases of modification, but of lexical compound formation. A crucial fact which shows that the construction at hand is not an instance of compound formation is that the first noun can be anaphorically bound as shown in (71), which offers an ambiguous interpretation: The internal argument of the matrix verb has been dropped and either ones of the nouns in the subordinate clause, the head noun or its modifier, could serve as the antecedent, as indicated via the indices.

- (71) Sensei<sub>1</sub>-no hon<sub>2</sub>-wo yon-de, *pro*<sub>1/2</sub> suki-ni nat-ta.  
teacher-MOD book-ACC read-GER *pro* be.fond-DAT become-PST  
1.  $\approx$  ‘Having read the teacher’s<sub>1</sub> book<sub>j</sub>, *pro* became fond of it<sub>2</sub>.’  
2.  $\approx$  ‘Having read the teacher’s<sub>1</sub> book<sub>j</sub>, *pro* became fond of him/her<sub>2</sub>.’

This shows that the modifier *sensei-no* ‘teacher’s’ must have a syntactic status, and is not part of a compound. Compare this to the ungrammatical English example containing a lexical compound below. Here, the first part of the compound cannot serve as an antecedent to the pronoun *it*.

- (72) \*With my sun<sub>1</sub> glasses, I am protected from it<sub>1</sub> well.  
(Rae, 2010, 37)

### 3.2.6.4 Only predication and only attribution

Obviously, analyses that derive NO from a predicate structure are not tenable, given the many direct *no*-modifiers mentioned thus far and attested in the data set. den Dikken and Singhapreecha (2004), cf. den Dikken (2006); Moro (1997), treat NO

parallel to English DPs with the element *of* as shown below together with the underlying structure.

- (73) a. a jewel of a village (den Dikken, 2006, 162)  
 b. [RP [village][RELATOR [SIMILAR jewel]]] (den Dikken, 2006, 178)

The underlying structure contains the predicate *SIMILAR* whose head is empty and which needs to be licensed. This licensing is achieved via predicate inversion. Concretely, the predicate complement moves to the *Spec* position of the *RELATOR* which is spelled out in English as /of/ (den Dikken, 2006, ch. 5). The reason for this is to achieve a different (contrastive) information structure. In addition to the problem that the different head-parameter of Japanese leads to problems in word order, this proposal is not tenable as it predicts all *no*-modifiers to be indirect.<sup>23</sup>

On the other hand, one could be tempted to reduce *MOD* only to structures in which direct modification is at stake. I discussed this point already in Chapter 3.2.3, where I gave a reference to Belk (2017), who argues that *JOIN* (our *MOD*) selects adjectives and produces attributive adjectives, that is direct adjectives (Belk, 2017, ch. 2.3). However, this account crucially undergenerates, since *NO* also co-occurs with indirect modifiers, nominal lexemes (*Nihon-no* ‘Japan(-ese)’) and also functional elements (*doko-no* ‘where’).

### 3.2.6.5 Alternatives for I and NA

Some other noteworthy analyses for *I* and *NA* were mentioned in Chapter 1.4. The prominent analysis to be adopted for the former element, and the counterpart to *NA* in *predicative* position is that of a combination as copula and tense element (Chapter 5.2), but for the reasons mentioned above (direct modification and *MOD* being restricted to DP-internal environments), this account does not extend to the attributive occurrence of these elements. A noteworthy alternative is the case marker hypothesis defended in Yamakido (2005) and Larson and Yamakido (2008), according to which *I* and *NA* receive an analysis identical to the *ezāfe* morpheme in Western Iranian languages as case markers. The authors also propose an account that reduces the different output to the different word classes, but handle it via resorting to case. This analysis suffers from the problem that it does not become clear why *I* and *NA* are always present, even in relative clause structures, thus in larger structures where it is not clear how case is assigned.

<sup>23</sup>The same holds for Koopman and Ishizuka (2016), who propose a relative clause account for *no*-modifiers.

- (74)    kami-ga   kirei-na   hito  
          hair-NOM pretty-NA person  
          ‘a person whose hair is pretty’                      (Kaplan and Whitman, 1995, 34)

The analysis of Yamakido (2005); Larson and Yamakido (2008) does not foresee the option of the case marker to occur with CPs (or even IPs) as also criticized for instance by Samvelian (2008); Kahnemuyipour (2014) for the Persian data. It furthermore remains unclear why the respective elements should disappear, for example when modifiers inflect for past. As Watanabe (2013, Fn. 25, p. 296) notes, “This proposal is untenable for the reason that the sensitivity to tense is left unexplained.”

### 3.2.7 Application: Integration via Mod

Above, I defended the MOD-hypothesis and explained why it fares better than most alternatives. In the following subsections, I will discuss how each modifier that is selected via MOD is embedded in the DP in line with the cartographic research program. As discussed, for instance in Chapter 3.2.3, MOD does not differentiate between modification type, word class or other factors.

#### 3.2.7.1 Structure for direct modifiers

Exemplified with the non-intersective *no*-modifier *shōrai-no* ‘future’ discussed above, the structure of a direct modifier within the DP is as follows.<sup>24</sup>

- (75)    a.    ano shōrai-no    purosakkāsenshu  
              that future-MOD pro.soccer.player  
              ‘that future professional soccer player’
- b.

The relevant assumptions are simple: Since the modifier *shōrai-no* ‘future’ is direct, it is merged in the direct domain in an unspecific functional projection. It is selected

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<sup>24</sup>Recall that demonstratives such as *ano* ‘that’ are specifiers in Japanese, accounting for the correct word order.

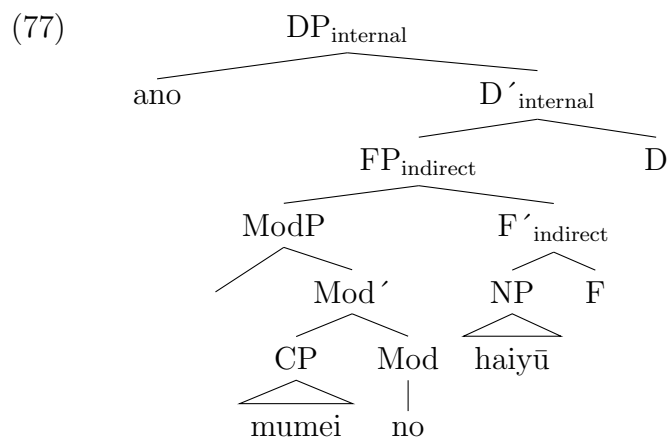
by MOD and licensed as an attributive modifier. As the word class status of *shōrai-no* is unclear, I adopt the label XP. The head of MOD is spelled out as /no/. This is the structure to be adopted for all direct modifiers.

### 3.2.7.2 Structures for indirect modifiers

More must be said about indirect modifiers. Recall that as long as a modifier can principally appear in predicative position it is ambiguous between a direct and an indirect status. This is, for example, the case for *mumei-no* ‘unknown’ repeated from (55) and given alongside the relevant predicative occurrence.

- (76) a. ano mumei-no haiyū  
           that unknown-MOD actor  
           ‘that unknown actor’  
       b. Ano haiyū-wa mumei-da.  
           that actor-TOP unknown-COP  
           ‘That actor is unknown.’

Indirect modifiers are embedded along the lines in (77) exemplified with *mumei-no* in a status as indirect modifier.



Indirect modifiers are merged in the specifier position of functional projections inside the indirect domain. Like *shōrai-no* ‘future’ in (75), indirect modifiers are licensed via MOD in attributive position. Depending on the word class of the preceding modifier, MOD is spelled out as /i/, /na/ or /no/. In this case, it licenses the *no*-modifier *mumei-no* and is realized as /no/.

A crucial difference between indirect and direct modifiers, besides the different positions inside the DP, is that indirect modifiers are embedded in larger structures. As outlined earlier in this chapter and motivated in Chapter 5.3.1, I assume indirect modifiers to be CPs.



under PredP (Bowers, 1993). First, there must at least be an IP-layer present to host the overt subjects alongside nominative case. In the examples above, this position, namely Spec, IP, is occupied by *Tarō*, *kami* ‘hair’ and *shūshoku* ‘job’ respectively.

Furthermore, there is evidence that overt tense is implied in those examples, and this tense is non-past. Including a temporal adverb with reference to the present, such as *genzai* ‘(at) present’, is unproblematic. Inserting, however, a temporal adverb denoting past, such as *kakoni* ‘in the past’, leads to ungrammaticality. This shows that overt tense, temporal reference to the present, is contained in those modifiers, even when they co-occur just with the monomoraic elements. Observe the minimal pair in (81).

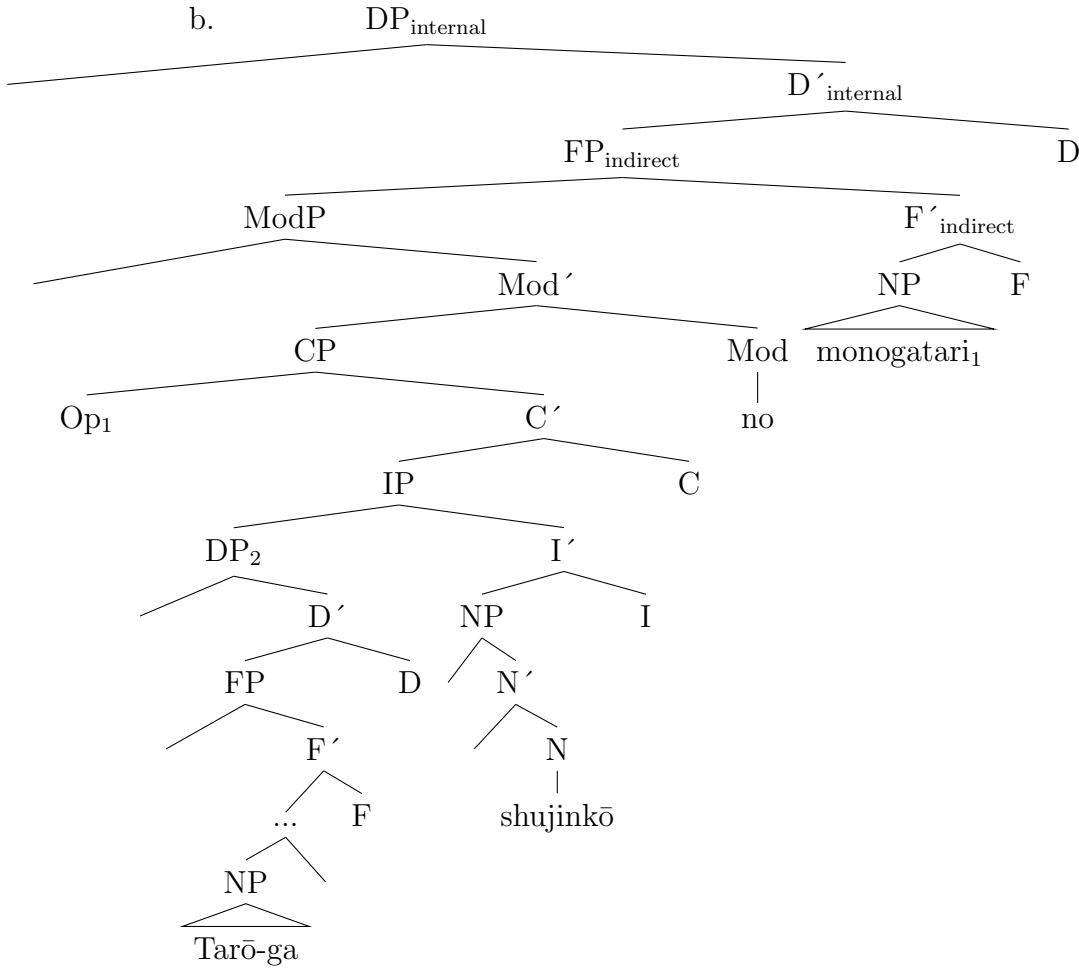
- (81) a. *Tarō-to Hanako-ga genzai shujinkō-no monogatari*  
 Taro-COM Hanako-NOM at.present main.protagonist-MOD story  
 ‘story in which Taro and Hanako are the main protagonists at present’
- b. \**Tarō-to Hanako-ga kakoni shujinkō-no*  
 Taro-COM Hanako-NOM in.the.past main.protagonist-MOD  
*monogatari*  
 story  
 (‘story in which Taro and Hanako were the main protagonists in the past’)

These arguments show that those modifiers are at least of size IP, however, no overt tense morpheme is realized, i.e. the head of IP seems to be empty. In this study, I do treat all Japanese relative clauses as involving a CP-layer (for arguments see Chapter 5.3), therefore, the relevant structure is as in (82), in which *Tarō-ga shujinkō* is a CP which is selected by MOD and licensed as attributive modifier.

- (82) [<sub>DP</sub> [<sub>CP</sub> *Tarō-ga shujinkō-no*] *monogatari*]  
 Taro-NOM main.protagonist-MOD story  
 ‘story in which Taro is the main protagonist’ (Watanabe, 2010, 66)

Adopting an analysis along the lines of operator-derivation for relative clauses (Chapter 5.3.5), the whole derivation embedded in the DP looks as follows.

- (83) a. Tarō-ga shujinkō-no monogatari  
Taro-NOM main.protagonist-MOD story (= (82))



The clausal modifier is embedded in a functional projection for indirect modifiers in the DP-internal domain. It is of size CP and selected by ModP whose head is spelled out as /no/ since the lexical head of the relative clause is a noun. The subject DP *Tarō* is situated in Spec, IP where it receives nominative case. Embedded in the relative clause is an operator *Op*, which moves to Spec, CP and is co-indexed with the noun *monogatari* ‘story’, thereby ensuring that this noun is correctly interpreted inside the relative clause, solving the Connectivity Problem (Salzmann, 2017).

The question is where the operator exactly originates. There is reason to believe that it starts in the extended projection of the modifying NP (or more precisely DP, see below) *shujinkō*. First, note that some relationship between *monogatari* ‘story’ and *shujinkō* ‘main protagonist’, one akin to possession, has to be established at some point in the derivation. This underlying relationship is given below for concreteness.<sup>25</sup>

<sup>25</sup>Note that the demonstrative *sono* is in the specifier of the DP with the head noun *shujinkō* and does not form a constituent with *monogatari*.

- (84) a. sono monogatari-no shujinkō  
 that story-MOD main.protagonist  
 ‘the main protagonist of that story’
- b.
- 

This means that the operator is a placeholder for *(sono) monogatari* ‘(this) story’, as shown in (85).

- (85) [DP [CP Tarō-ga Op<sub>1</sub> shujinkō-no] monogatari<sub>1</sub>]  
 Taro-NOM Op main.protagonist-MOD story  
 ‘story in which Taro is the main protagonist (of the story)’

This gives good evidence that the operator starts out from deep within the relative clause, namely from a position in the extended projection of the modifying noun, which then actually constitutes a DP.

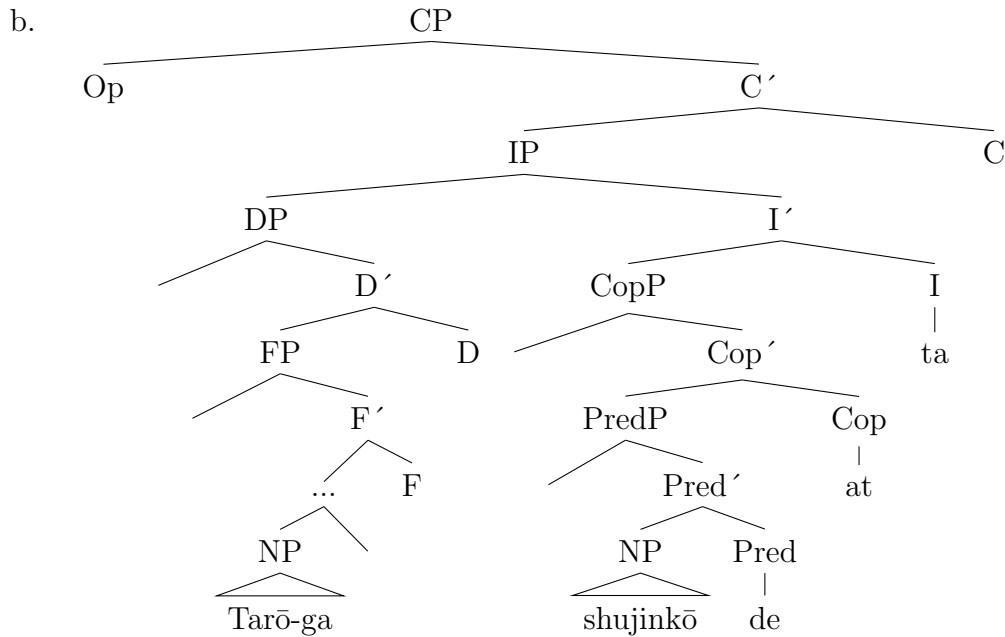
- (86)
- 

From this deep position within the modifying DP, the operator moves to Spec, CP of the relative clause and ensures correct interpretation of the head noun and the relative clause as a whole. Crucially, again, within the relative clause a second concrete spell-out of MOD is absent, as no overt modifier appears.



A remaining point left explained is why no instance of MOD appears once the head of I is filled, i.e. once the inflectional forms KATTA, DEARU or DEATTA are realized. The relevant structure of only the CP looks as follows.

- (87) a. [Tarō-ga shujinkō-de-at-ta] (\*no) monogatari  
Taro-NOM main.protagonist-PRED-COP-PST MOD story  
‘story in which Taro was the main protagonist’  
(Watanabe, 2010, 66)



Embedded in the CP are the predicative copula PRED, which is there to license predication and is realized as /de/ (Bowers, 1993; Nishiyama, 1999), alongside the dummy copula COP which offers be-support (Doron, 1983; Rapoport, 1987), realized as /ar/. Furthermore, the head of I is overtly realized via the past morpheme -TA.<sup>26</sup>

Again, overt tense, in this case past tense, is expressed. This can be evidenced via the impossibility to include a temporal adverb referring to the present, but the possibility of including a temporal adverb referring to past. Compare the minimal pair in (88) to the one in (81) above.

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<sup>26</sup>See Chapter 5.2 for justification of these heads. From now on, I will adopt the respective glosses PRED (predicative copula), COP (dummy copula) and PRS or PST respectively for the tense morpheme, when the elements DE-AR-U, D(E)-AT-TA and K-AT-TA occur.

- (88) a. \*<sub>[DP [CP Tarō-to Hanako-ga genzai  
Taro-COM Hanako-NOM at.present  
shujinkō-de-at-ta] monogatari]</sub>  
main.protagonist-PRED-COP-PST story  
(‘story in which Taro and Hanako are the main protagonists at present’)
- b. <sub>[DP [CP Tarō-to Hanako-ga kakoni  
Taro-COM Hanako-NOM in.the.past  
shujinkō-de-at-ta] monogatari]</sub>  
main.protagonist-PRED-COP-PST story  
‘story in which Taro and Hanako were the main protagonists in the past’

On a descriptive level, the overt realization of tense correlates with the absence of the overt realization of MOD. This syntactic head is not realized when tense is overtly expressed. Thus, the question arises whether MOD is still present in such circumstances, but somehow blocked from being realized, or whether CPs with an overt tense head are not licensed via MOD in the first place.

Although, at the present moment, there is no concrete evidence for either of the two options, I have already mentioned at the beginning of this subsection that I will keep the analysis of MOD as a universal syntactic head as consistent as possible. That means, I assume that it is always present structurally. However, it remains silent once the head of I is filled, i. e. once a finite indirect modifier occurs, where finiteness corresponds to the presence of tense and/or agreement (see Landau 2013, 103–108 for an overview as well as Nikolaeva 2007 and articles cited therein). I leave it to future research to determine at which point its overt output gets deleted, or if it does not get realized in the first place.<sup>27</sup>

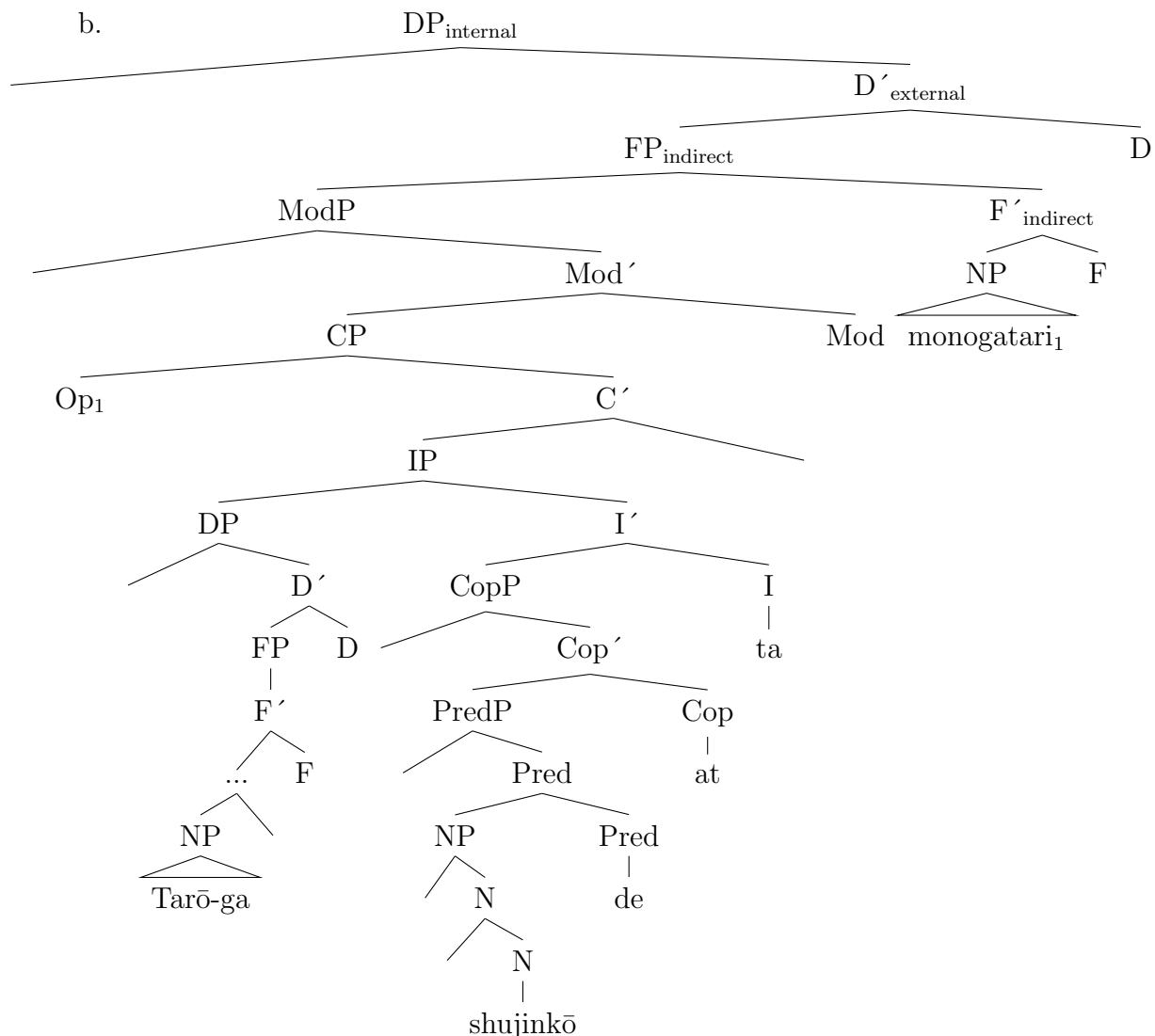
The whole DP corresponding to (87-a) looks as follows. Take note that the subject position and derivation of the operator remain the same.<sup>28</sup>

<sup>27</sup>This point extends to the realization of MOD in other languages. See the discussion at the end of this chapter.

<sup>28</sup>There are some noteworthy alternatives, which are also not entirely promising. If we assume a different CP-structure for examples such as (82), it is unclear what this different CP should look like. If we argue that MOD only selects those CPs which do not select PredP (and/or CopP), then why are verbal relative clauses not followed by NO?

Evidence from acquisition is also conflicting. Murasugi (1991, chapters 4 and 5) shows that some Japanese children in her study overgenerated NO and also used it after verbal relative clauses, a position not allowed in Standard Japanese. Some even produced it after adjectives either in addition to or instead of I and NA. This could be taken as evidence that for those children MOD is always present, but they have not learned the rules when to omit it. On the other hand, these results could also be interpreted in a way that the children did not know which CP they are producing and whether they need MOD or not. See also Yamakido (2005, ch. 3.5).

- (89) a. [Tarō-ga shujinkō-de-at-ta] monogatari  
Taro-NOM main.protagonist-PRED-COP-PST story (= (87-a))



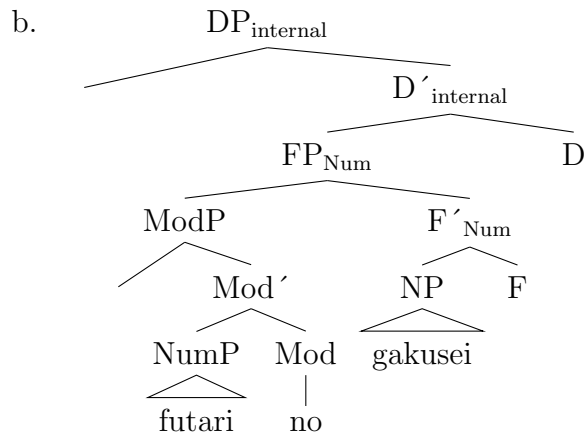
To summarize, potentially all CPs are licensed via MOD in attributive position, but MOD is only realized when the head of IP remains empty.

### 3.2.7.3 Structures for remaining modifiers

The structure for all other modifiers, including numerals and arguments, which also unequivocally appear with NO, can, again, be derived in a straightforward manner. All these modifiers are embedded in the relevant functional projection and selected by MOD, whose head is spelled out as /no/. Their syntactic role is determined in the syntax.

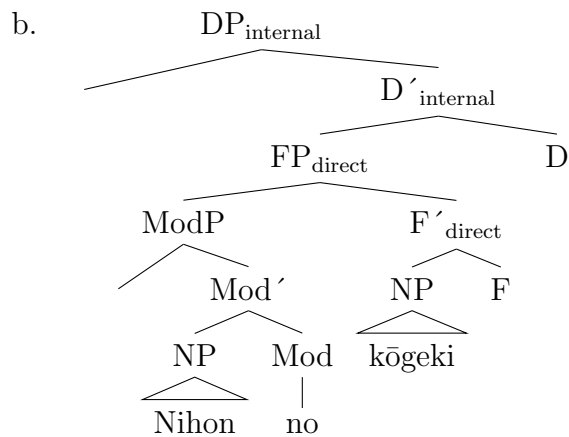
(90) A MOD-structure for numerals

- a. futari-no gakusei  
two-MOD student  
'two students'



(91) A MOD-structure for arguments

- a. Nihon-no kōgeki  
Japan-MOD attack  
'Japanese attack'



For arguments, the additional assumption will be made in Chapter 4.3.2 (and see Laenzlinger 2005, 2011 on which this is based) that they have moved from a lower position, assuming the NP of the event noun is actually more complex.

### 3.3 Chapter summary and closing remarks

This chapter presented the DP structure of Japanese and then discussed the internal structure of modifiers. The main insights are that the Japanese DP consists of a DP-external and a DP-internal domain and that the latter encompasses domains for indirect and direct modification. In the cartographic spirit, I claim that the DP structure presented in this study is universal. In other words, we need to assume a left periphery alongside a direct and an indirect domain in every language. An important deviation from other theories (Cinque, 2010, 2020; Scott, 2002; Laenzlinger, 2005, 2011; Kim, 2019) is that I have proposed that these domains do not encompass functional projections dedicated to specific semantic categories, in the direct domain, or relative clauses of different sizes, in the indirect domain. As further argued in Chapter 4.4, this is due to cross-linguistic parametric variation. Furthermore, I spent some time discussing the left periphery and essentially suggested that we should cross-linguistically assume projections for quantifiers and focused modifiers, as well as unspecified projections. An intriguing question also concerns the nature and universality of FocP as the interaction of modifiers and focus in Japanese proves rather difficult. Furthermore, this area is subject to heavy language variation.

In the second part of this chapter, I put forth an analysis of the monomoraic elements co-occurring with Japanese non-verbal modifiers as a syntactic head. This head can select all kinds of lexical and functional elements, irrespective of word class and modification type – direct, indirect, numeral, etc. – and license them as attributive modifiers. Thus what is merged in the relevant projections is not a simple modifier, but a modifier selected by MOD. I first put forth this analysis for the recalcitrant element NO and then extended it to I and NA. Problematic aspects for future research concern the fact that it is not realized once the head of IP is overt, as it needs to be shown whether other languages show a similar correlation between the realization of tense and of such a functional head, and the application to the Japanese adjective groups where we end up with two different elements spelled out as /i/.

As I have argued, the functional head MOD is universal and this assumption, a theory on the internal structure of modifiers, only enriches our understanding of the DP further. This echoes a recent contribution of Richards (2023) on the category of *linkers* (see also Richards 1997; Baker and Collins 2006; Richards 2010), but necessarily begs the question if such a head should really be assumed even for languages such as English where outside genitive-*of* no specific element appears in the DP in order to license modification. I will answer this question affirmatively, but at the same time would like to point out the necessity of future research.

## Chapter 4

# Direct Modification in Japanese

This chapter is dedicated to direct – that is, attributive – modification in Japanese. Applying valid tests for direct modification, I present empirical and theoretical evidence that this type of modification does exist in this language and discuss every group of direct modifiers in detail. In the second part of this chapter, I will focus on the lack of ordering restrictions and show how this phenomenon falls out from the theory assumed here, the existence of unspecified functional projections within the direct domain. Next, I will discuss the question, why some languages (English, German) do show strict ordering in the direct domain, some do not (Japanese) and some do only partially, namely Persian and Greek. I will offer an explanation through the *Generalization on Cross-Linguistic Ordering Restrictions* that offers a first approach to predict ordering within the direct domain. Finally, I discuss some remaining questions.

Many candidates for direct-only modifiers are scattered in various works (Yamakido 2000, 2005, 2007; Ayano 2010; Watanabe 2012; Morita 2010, 2011, 2013; Nagano and Shimada 2015; Nagano 2016, see especially Watanabe 2017; Muraki 1996, 2009, 2012), that serve as a reference for this chapter. The main insights are based on the results of my corpus study (see Chapter 1.5) and supplemented with interviews with consultants.<sup>1</sup>

### 4.1 Introduction

As was shown in the preceding chapter, the domain for direct modification is situated inside the DP-internal domain and hierarchically lower than the domain for indirect modification. Furthermore, it was argued that the functional projections inside the

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<sup>1</sup>These interviews were conducted online over Zoom in the period between May and June 2021. The procedure was as follows: I showed written examples to five native speakers and immediately asked for their judgment and feedback. Then we discussed critical points. These speakers were not trained linguists.

direct domain are not based on fine-grained semantic categories. The direct domain is circled in the following structure of the internal DP, repeated from the previous chapter.

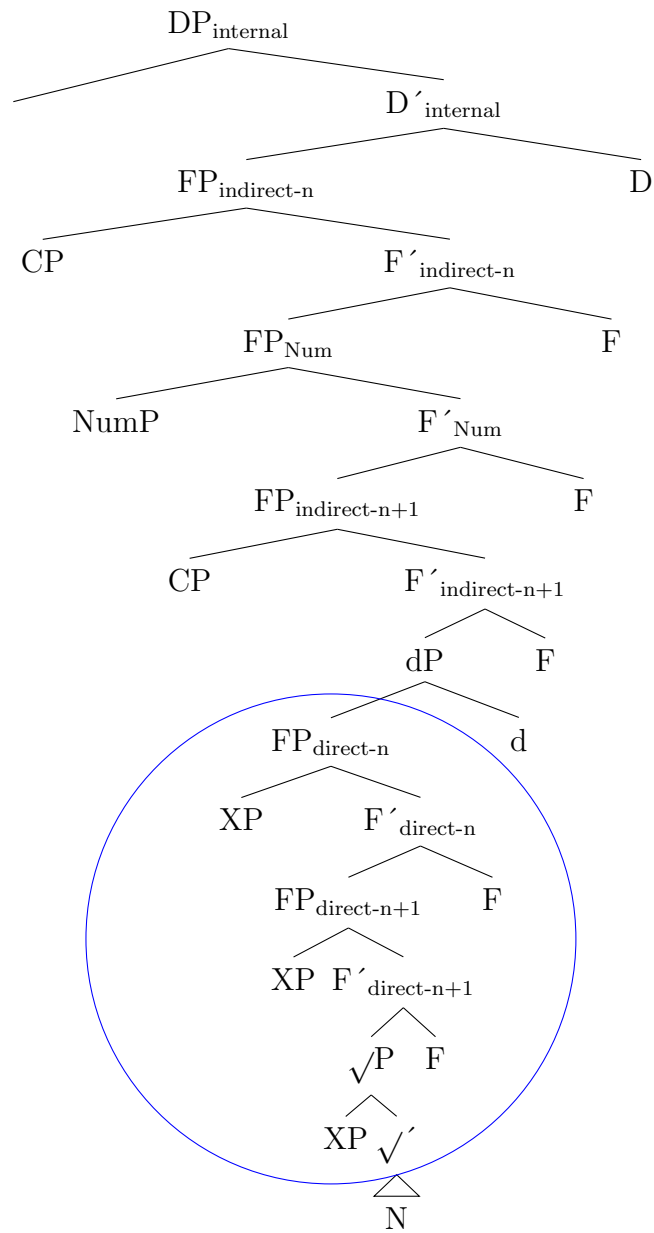


Figure 4.1: Structure of the Direct Domain in Japanese

### 4.1.1 Goals and overview

This chapter will present the following results.

- (1) a. Valid cross-linguistic as well as language-specific criteria for direct modification can be put forth in Japanese. The main criterion is *lack of predicativity*, but this is often accompanied by other criteria such as restrictions in gradability.
- b. Japanese exhibits several groups of direct modifiers: Non-intersective modifiers, relational modifiers, direct qualitative modifiers, idiomatic modifiers, quantifiers and the language-specific group of *rentaishi*.
- c. The fact that the functional projections inside the direct domain are not specified for certain semantic categories predicts the lack of ordering restrictions in linear structure.
- d. There is parametric variation as to whether a language exhibits fine-grained dedicated functional projections within the direct domain.

### 4.1.2 Arguments against direct modification

As mentioned in the introduction, many researchers have denied the existence of direct modification in Japanese (Kuno, 1973; Hinds, 1988; Whitman, 1981; Kaplan and Whitman, 1995; Sproat and Shih, 1991; Baker, 2003b; Laenzlinger, 2011). And while single instances of direct modifiers have been mentioned in the literature (see Watanabe 2017) and Japanese grammarians have long known about the category of solely attributively used lexemes, sometimes called *rentaishi* (see Chapter 4.3.6), especially in syntactic research, Japanese is characterized as a language without direct modification.

The main arguments are the missing mechanism for disambiguation between direct (attributive) and indirect (predicative) modification and the lack of ordering restrictions. Some more arguments are given in Baker (2003b). Building on Nishiyama (1999), who argues that Japanese adjectives always project a CP layer (see Chapter 5.2), Baker (2003b) explicitly claims that direct modification is lacking in Japanese and all adjectives “form relative-clause-like structures” (Baker, 2003b, 19). In his framework, direct modification is an operation limited to the adjective class and denotes the modification of a noun without an additional element (Baker, 2003a, 191). A visible output of direct modification are phi-features, which Japanese lacks in general. Furthermore, he puts forth a systematic lack of non-intersectivity in Japanese and illustrates this with the equivalent of the famous *beautiful dancer* example (Larson, 1998), which does not show the same ambiguity as in English between a non-intersective (subsecutive) and an intersective interpretation. Instead,



only the intersective interpretation is available as expected under a relative clause analysis in which the intersective reading is predicatively derived (cf. Cinque 2010 and Chapter 2.3.1). This is shown in (2) based on Baker (2003b, 3).

- (2)    utsukushi-i kashu  
          beautiful-I singer  
       1.  $\neq$  ‘beautiful as a singer’ [beautiful singer]  $\subseteq$  [singer] (subsective)  
       2.  $\approx$  ‘beautiful and a singer’ [beautiful singer] = [beautiful]  $\cap$  [singer] (intersective)

As will be shown below, not only are ordering restrictions not a prerequisite for direct modification in Japanese, there are also good counter arguments to the alleged absence of non-intersectivity, showcasing that none of the arguments given in the literature are convincing enough to deny the absence of direct modification in this language.

## 4.2 Criteria and tests for direct modification

In order to prove that direct-only modification does exist in Japanese, it is of great importance to have specific criteria as substantial guidelines to apply to potential groups of direct modifiers. When establishing such criteria, one has to take into account not only cross-linguistic, but also language-specific aspects. Typical cross-linguistic testbeds, presented throughout the preceding chapters, are *predication*, *gradability* and *tense/inflection*. These will be looked at here, sidelining the factor *ordering*, and supplemented with two language-specific factors connected to different occurrences of the copula: *coordination* and *co-occurrence with the canonical copula*.

### 4.2.1 Predication

Recall from Chapter 2.3 that the stand-out characteristic of direct modifiers, in cartography and elsewhere, is *lack of predicative use*. In English, non-verbal modifiers are licensed in predicative position via a copula, but this process is illicit with direct adjectives such as the non-intersective *former* (Siegel 1976; Larson 1998; Cinque 2010, 2014 etc.).

Recall that Japanese *na*-adjectives and *no*-modifiers appear with the copular element DA or the longer canonical copula DE-AR-U in predicative position, the latter historically derived from the former (Urushibara, 1993; Nishiyama, 1999; Narahara, 2002; Frellesvig, 2010; Miyama, 2011), as exemplified below for the *na*-adjective *nigiyaka* ‘lively’.

- (3) a. Ano machi-wa nigiyaka-da.  
that city-TOP lively-COP  
'That city is lively.'
- b. Ano machi-wa nigiyaka-de-ar-u.  
that city-TOP lively-PRED-COP-PRS  
'That city is lively.'

*i*-adjectives appear with the element *I* in predicative position which is homophonous to the output of MOD following *i*-adjectives in attributive position, as argued in the preceding chapter. I will gloss *I* in predicative position from now on as COP, matching the glossing for DA.<sup>2</sup>

- (4) Ano machi-wa ōki-i.  
that city-TOP big-COP  
'That city is big.'

Assuming indirect modification is the only kind available in Japanese, there should be no lexemes that appear as attributive modifiers, but are unable to appear in predicative position. However, several Japanese modifiers resist predicative use, for example the *na*-adjective *kanzen-na* 'complete' in (5) (Yamakido, 2005) or the *no*-modifier *Nihon-no* 'Japan(-ese)' in (6).

- (5) a. Max-ga kanzen-na baka-da.  
Max-NOM complete-MOD idiot-COP  
'Max is a complete idiot.' (Yamakido, 2005, 68)
- b. \*Ano baka-ga kanzen-da.  
that idiot-NOM complete-COP  
(('That idiot is complete.')
- (6) a. Nihon-no kuruma  
Japan-MOD car  
'Japanese car'
- b. \*Kono kuruma-wa Nihon-da.  
this car-TOP Japan-COP  
(('This car is Japanese.')

Importantly, one needs to take into account other exponents of the copula such as DESU (and the forms analyzed in Chapter 4.2.4), which is a more polite version of DA. Accordingly, the two instances of the copula behave alike. In instances where DA is impossible, DESU is as well.<sup>3</sup>

<sup>2</sup>For a precise analyses of all these elements see Chapter 5.2. See especially Chapter 5.2.3.2 on why *I* in predicative position is a copula as well.

<sup>3</sup>This element, which can never appear in attributive position, will be discussed in Chapter 5.2.7 where it will be depicted as a combination of the predicative copula, the polite form of the dummy copula and a tense marker for non-past. For now, I will apply the gloss COP.POL to highlight its function as a polite copula.

- (7) a. \*Ano baka-ga kanzen-desu.  
           that idiot-NOM complete-COP.POL  
           ('That idiot is complete. '; polite)  
       b. \*Kono kuruma-wa Nihon-desu.  
           this car-TOP Japan-COP.POL  
           ('This car is Japanese. '; polite)

Such examples have provided the main motivation for authors such as Yamakido (2005) to argue against a universal analysis of Japanese modifiers as indirect. *Predication*, especially its absence, will be used as the main criterion to check for direct modification verified via the occurrence of the copula forms DA and I.

It should be noted here that in Japanese, seemingly direct modifiers can, in specific instances, appear with the copula. First, predicative use can be facilitated easily in contrastive statements. An example is the following (and see Hoshi 2002; Shimada and Nagano 2018).

- (8) Kono kuruma-ja-na-ku-te,                      ano kuruma-ga Nihon-da.  
       this car-PRED.TOP-NEG-PRED-GER that car-NOM Japan-COP  
       'Not this car, that car is Japanese.'

Here, the direct-only modifier *Nihon* 'Japan(-ese)' seemingly appears in predicative position. See also the following example reported in Hoshi (2002) who argues against a direct status of *omoigakena-i* 'unexpected', which Yamakido (2005) argues to be direct.

- (9) Kono kyaku ja-na-ku-te,                      ano kyaku-ga omoigakena-i  
       this guest PRED.TOP-NEG-RKY-GER that guest-NOM unexpected-COP  
       no-da.  
       EMP-COP  
       'Not this visitor, that visitor is unexpected.'

(Hoshi, 2002, 11)

In the literature on relational adjectives, many such instances in which otherwise non-predicative relational adjectives appear in predicative position are reported, including class shift and the availability of a *kind*-reading. These instances are referred to as *adjective stranding* (Levi 1978; McNally and Boleda 2004; Nagano 2018; see Chapter 2.3.2). Another typical way to achieve predicative occurrence involves contrastive use, as in the Japanese examples, shown for English below.

- (10) Her infection turned out to be bacterial, not viral.                      (Levi, 1978, 254)

In (10), the relational adjective *viral* is used predicatively. In a context-neutral sentence, such a use is significantly degraded.

- (11) ??Her infection is viral.

According to Levi (1978, 255), in such instances the second noun is elided under identity at a late stage in the derivation. One piece of evidence she puts forth is that the two instances, the noun and the modifier, must be established as a set. The first instance of the noun then licenses the deletion of the second instance under phonological identity. This means (10) is derived from (12).

- (12) Her infection turned out to be (a) bacterial ~~infection~~, not (a) viral ~~infection~~.  
(Levi, 1978, 254)

Applying the data to Japanese, this means that the underlying construction of (8) is (13). Since no overt modifier is present, the syntactic head MOD is deleted alongside.

- (13) Kono kuruma-ja-na-ku-te,                      ano kuruma-wa Nihon-~~no~~  
this car-PRED.TOP-NEG-PRED-GER that car-TOP Japan-MOD  
~~kuruma~~ da.  
car COP  
'Not this car, that car is a Japanese car.'

Evidence for this claim comes from the fact that this sentence is even more natural when the noun *Nihon* 'Japan(-ese)' is followed by a so-called *bound marker*, a nominal suffix which specifies the denotation of the noun (Nagano, 2016; Shimada and Nagano, 2018). This in turn makes it easier for the two nouns to be established as a set and facilitates predicate use (see Chapter 4.3.2.8 and appendix A for more on bound markers.)

- (14) Kono kuruma-wa Nihon-sei-d-a.  
this car-TOP Japan-produce PRED-COP.PRS  
'This car is Japanese.' (Lit. 'is Japan-produce')

Furthermore, the copula can oftentimes function as a placeholder for other verbs. This is the famous *eel-construction* or *unagi-bun* coined by Okutsu (1978). In (15), the copula DA appears, but is understood as a lexical verb, for example *tabe* 'to eat', depending on the context. Semantically, it does not express an equative relation between *I* and *eel*.

- (15) Kare-wa sakana-wo tabe-ta. Boku-wa unagi-da.  
he-TOP fish-ACC eat-ACC I-TOP eel-COP  
'He ate a fish. I ate/had (the) eel.' (Lit.: 'I am eel.')

This means that, provided the appropriate context, formerly non-predicative lexemes can become acceptable in predicative position. *Unagi*-constructions are very productive and usually involve a noun directly preceding the copula (Okutsu, 1978,

23–29). Since many of the relational modifiers discussed here are nominal in nature, it is no surprise that they can appear in such constructions. In other words, it is oftentimes not clear if we are dealing with a proper predicative phrase in the relevant instances. In the discussion at the end of this chapter, I will furthermore discuss some indirect features of direct modifiers in Japanese.

### 4.2.2 Gradability

A further typical characteristic often observed with direct modifiers and generally related to their less complex nature is *non-gradability*, the inability to express a degree. Gradability can be expressed through modification by degree heads (16-a), degree adverbs (16-b) or the formation of the comparative or superlative (16-c) (cf. Baker 2003a).

- (16)    a.    a too large building  
          b.    a very large building  
          c.    a larger building than Tokyo Tower

It is a general property of the word class *adjective* to express a degree, although not all adjectives are equally freely gradable (Dixon, 2010, 70–74). A restriction in gradability is a central characteristic of both, the group of relational (17) and non-intersective adjectives (18) in most European languages (see references in the preceding chapter).

- (17)    a.    \*very / \*too / \*more nuclear energy  
          b.    \*very / \*too / \*more American attack
- (18)    a.    \*very / \*more former minister  
          b.    \*totally present editor

The semantic approach to gradability makes use of scale structures (Kennedy and McNally, 2005; Kennedy, 2007; Morzycki, 2012a). Gradable adjectives are understood as *relative* in contrast to *absolute* or *totally closed scale* adjectives. Only the former can be modified by so-called *proportional modifiers*, including the common degree adverb *very*, whereas the latter allow modification via *endpoint-oriented modifiers* (Kennedy and McNally, 2005), also called *modifiers of extension* (Fábregas, 2014) such as *completely*, denoting a maximum scale, or *slightly*, denoting a minimum scale (Kennedy and McNally, 2005, 355). The term *non-gradable modifier* will be used in this book to denote those modifiers that cannot be combined with any degree adverb and do not have access to comparative-formation.

It is important to note that lack of gradability does not automatically entail lack of predication and that gradability is not a sufficient criterion for direct modification, yet often an accompanying one. Absolute adjectives such as *pregnant* or *bent*, despite restrictions in gradability, can productively appear in predicative position in English. Cinque (2010) also provides examples of some non-intersective adjectives in English (and Italian and Bulgarian) that can be combined with degree adverbs.

- (19) the most probable winner (Cinque, 2010, 47)

Gradability in Japanese is expressed via the particle *yorī(-mo)* ‘than’ and the optional adverb *motto* ‘more’ or by various degree adverbs (see Chapter 1.4). The most prototypical one is *totemo* ‘very’, the equivalent of the proportional modifier *very* in English. Equivalents of endpoint modifiers are the maximum-standard degree adverbs *kanzenni* and *mattaku*, both ‘completely’, and the minimum-standard degree adverb *wazukani* ‘slightly’ (cf. Oshima *et al.* 2019).

It is often claimed that *no*-modifiers are non-gradable in general (Katō, 2003; Morita, 2011, 2013). And while most are incompatible with the degree adverb for relative comparison *totemo* ‘very’ (20), many are compatible with the adverb *kanari* ‘quite’, as well as the maximum- and minimum-standard degree adverbs mentioned above (21).<sup>4</sup>

- (20) ??*totemo mumei-no haiyū*  
       very    unknown-MOD actor  
       ‘very unknown actor’
- (21) a. *kanari mumei-no haiyū*  
       quite unknown-MOD actor  
       ‘quite unknown actor’
- b. *kanzenni / mattaku mumei-no haiyū*  
       completely / completely unknown-MOD actor  
       ‘completely unknown actor’
- c. *wazukani mumei-no haiyū*  
       slightly unknown-MOD actor  
       ‘slightly unknown actor’

Also note that many *no*-modifiers, for example *mumei-no* ‘unknown’, can productively appear inside a comparative construction with the comparative particle *yorī* ‘than’.

<sup>4</sup>A reviewer points out that example (21-c) is pragmatically odd, as, indeed, the standard way to express this notion would be a combination along the lines of ‘very famous’. Nevertheless, this example serves an illustrative illustration purpose.

- (22) Tanaka-san-yori (motto) mumei-no haiyū-to at-ta.  
 Tanaka-Mr.-than more unknown-MOD actor-COM meet-PST  
 ‘I met with a more unknown actor than Mr. Tanaka.’

Although this shows that gradability is restricted for *no*-modifiers, as also corroborated in the study of Oshima *et al.* (2019) in which the authors judged only 22.25% of the 200 most used *no*-modifiers on the BCCWJ to be gradable,<sup>5</sup> there are some *no*-modifiers which are compatible with relative degree adverbs of the type *totemo* ‘very’, although only a few. See (23) and Oshima *et al.* (2019), as well as their data set, for more examples.

- (23) a. totemo totsuzen-no henka  
 very sudden-MOD change  
 ‘a very sudden change’  
 b. yori ōgata-no fune  
 more large.size-MOD ship  
 ‘a ship of a larger size’ (Oshima *et al.*, 2019, 16)

Nevertheless, most non-predicative modifiers in Japanese resist gradability across the board, as exemplified for the non-intersective modifier *shōrai-no* ‘future’ in (24) and to be discussed in the following.

- (24) \*totemo shōrai-no daitōryō  
 very future-MOD prime.minister  
 (‘a very future prime minister’)

### 4.2.3 Tense and inflection

The absence of tense information is another criterion that differentiates direct from indirect modifiers. This can be easily illustrated with German where adjectives can be used attributively together with a copula appearing as a as prenominal participle, a case of a reduced relative clause (Cinque 2020, 203; Krause 2001; Douglas 2016; Harwood 2018). This is shown in (25) for the adjective *weiß* ‘white’, which can be used as a predicate. The non-intersective adjective *ehemalig* ‘former’, on the other hand, can never appear together with an inflected copula, as it can never appear with the copula in the first place (26), highlighting its direct status.

- (25) das (früher einmal) weiß gewesene Hemd (German)  
 the earlier once white be.PST.PTCP shirt  
 ‘the shirt which was (once) white’

---

<sup>5</sup>They followed up on their judgment with a questionnaire survey and identified only 8.28% of the 169 *no*-modifiers they considered as gradable on a relative scale, thus with *totemo* (Oshima *et al.*, 2019, 29).

- (26) \*der ehemalig gewesene      Präsident (German)  
       the former    be.PST.PTCP president  
       (‘the was former president’)

As shown in Chapter 3.2, Japanese modifiers can inflect for past DP-internally, but this is impossible for non-predicative modifiers such as *kanzen-na* ‘complete’ (27).<sup>6</sup>

- (27) \*ano kanzen-d-at-ta              baka  
       that complete-PRED-COP-PST idiot  
       (‘that idiot who was complete’) (Ayano, 2010, 109)

From a theoretical angle, these facts fall out naturally. The structure of direct modifiers is less complex and no IP layer is present, hence no overt tense can be expressed. In a nutshell, whereas the endings DE-AR-U, D(E)-AT-TA and K-AT-TA always carry tense information, this is not obligatorily the case for I, NA and NO. As shown in Chapter 3.2.7.2, this only applies when they are indeed part of a relative clause (see also Watanabe 2017, 793). Direct modifiers can therefore never, instead of the monomoraic endings, co-occur with the complex copula forms.

## 4.2.4 Co-occurrence with other copula forms

I will now discuss two rather language-specific tests. These correlate with the inability of predicative use as they look at co-occurrence with two different exponents of the copula, namely the canonical copula DE-AR-U and coordination forms.

### 4.2.4.1 The Canonical Copula *de-ar-u*

The complex form DE-AR-U is usually called the *standard* or *canonical* copula (Urushibara, 1993; Nishiyama, 1999; Narahara, 2002). Recall (for instance from Chapter 1.4) that this canonical copula co-occurs with *na*-adjectives and *no*-modifiers and that it is not only mutually exclusive with the copula DA, but also with the attributive exponents of MOD, namely NA and NO. This is repeated for convenience for the *na*-adjective *nigiyaka-na* ‘lively’ (28) and the *no*-modifier *shujinkō* ‘main protagonist’ (29).<sup>7</sup>

- (28) a. nigiyaka-na machi  
       lively-MOD city  
       b. nigiyaka-de-ar-u      machi  
       lively-PRED-COP-PRS city  
       ‘a city which is lively’

<sup>6</sup>That non-intersective adjectives cannot inflect for past also applies in Korean as shown by Kang (2005, 164).

<sup>7</sup>For *i*-adjectives, in order to paraphrase *i* with the corresponding *ku-ar-u*, the right context is needed. See Chapter 5.2.



- (29) a. Tarō-ga shujinkō-no monogatari  
Taro-NOM main.protagonist-MOD story
- b. Tarō-ga shujinkō-de-ar-u monogatari  
Taro-NOM main.protagonist-PRED-COP-PRS story  
'story in which Taro is the main protagonist'
- (Watanabe, 2010, 66)

That NO is not (always) an instance of the copula can be shown in the following minimal pair in which NO can yield both an equative, as well as a possessive meaning, whereas DE-AR-U, on the other hand, only causes the nominal modifier to have an equative reading, highlighting its predicative nature (see also Katō 2003 among others).

- (30) sensei-no tomodachi  
teacher-MOD friend  
1.  $\approx$  '(my) friend who is a teacher' (equative)  
2.  $\approx$  'friend of the teacher' (possessive)
- (31) sensei-de-ar-u tomodachi  
teacher-PRED-COP-PRS friend  
1.  $\approx$  '(my) friend who is a teacher' (equative)  
2.  $\neq$  'friend of the teacher' (possessive)

Cross-linguistically, there is agreement, that the copula is a licenser of non-verbal elements in predicative position, thus linking the predicate to its subject, and that it is in essence meaningless (Arche *et al.* 2019, 1–6, see Moro 1997, 2000; Bowers 2002 among others for historical overview). The appearance of the copula itself is thus strongly linked to indirect modification. On the other hand, “non-paraphrasability with a copular construction” is a sign of direct modification (Alexiadou *et al.*, 2007, 291–292).

This means that when direct-only *na*-adjectives and *no*-modifiers are used attributively, DE-AR-U, or D(E)-AT-TA, should never occur instead of the different outputs of MOD (see also Ayano 2010). This is shown below for a direct *na*-adjective (32) and a direct *no*-modifier (33). Note that this fact relates again to the lack of tense of direct modifiers and their smaller structure which does not contain projections for these complex elements.

- (32) \*ano kanzen-de-ar-u baka  
that complete-PRED-COP-PRS idiot  
'(that idiot who is complete)' (Ayano, 2010, 108)
- (33) \*Nihon-de-ar-u kuruma  
Japan-PRED-COP-PRS car  
'(car which is Japanese)'

4.2.4.2 The Coordinative Form *de*

Besides the canonical and the polite form of the copula, direct-only modifiers also cannot co-occur with the coordinative form of the copula *DE*. They are even illicit in coordinative constructions in which the first conjunct bares the copula form.

In Japanese, different word classes use different methods of coordination, meaning there is no universal coordinator at the word level similar to English *and*. Nouns are connected via the comitative/conjunction postposition *TO* (or the postposition *YA* for inexhaustive coordination).<sup>8</sup>

- (34) hon-to shinbun  
book-and newspaper  
'books and newspapers'

nominal coordination, exhaustive

Verbs (and generally inflecting word classes), on the other hand, are coordinated via the suffix *-TE*, which implies a consecution of actions and possibly additional connections between two clauses, for instance causality.<sup>9</sup> *-TE* attaches to the nominalization form (*renyōkei*) and can undergo various phonological assimilation processes. The element *koron-de* in (35-a) consists of the verb *korob* 'to fall', inflected to the nominalization form *korobi*, to which *-TE* is added and phonological assimilation takes place.<sup>10</sup> The nominalization form of vowel-stem verbs, such as *tabe* 'to eat', is syncretic with the infinitival form and the coordinative marker is always just /te/ as shown in (35-b).

- (35) a. Ken-wa koron-de nai-ta.  
Ken-TOP fall-GER cry-PST  
'Ken fell down and cried.'

verbal coordination, consonant verb (Hiraiwa, 2012, 357)

- b. Bangohan-wo tabe-te o-furo-ni hait-ta.  
dinner-ACC eat-GER HON-bath-ILL enter-PST  
'(I) ate dinner and (afterwards) took a bath.'

verbal coordination, vowel verb

<sup>8</sup>The comitative postposition *-TO* further differs from the other elements to be reviewed in that it can also appear after the last noun in a list format and even together with (case) particles. See examples and discussion in Kuno (1973) and Hiraiwa (2014) against Tatsumi and Fujiwara (2018).

<sup>9</sup>*-TE* also functions as an aspectual marker and is often glossed *PROG* or *GER(UND)*. I will adopt the label *GER*, but note that this element has nothing to do with nominalization *per se*. See Hasegawa (1996); Mihara (2013); Nakatani (2013, 2016).

<sup>10</sup>The combination /bi te/ was present still in Classical Japanese, but has become /nde/ as a result of historical sound change (*onbin*). This form can now be interpreted as analytical. In Modern Japanese, the voiced variant /de/ of the coordinative form appears after all consonant-stem verbs ending in a nasal or a voiced obstruent. A similar phonological assimilation process, gemination, happens before the past marker *-TA* (Ito and Mester, 2015).

- When *na*-adjectives and *no*-modifiers appear as the first conjunct, they exhibit the element DE, exemplified below for a *na*-adjective.<sup>11</sup>

- This element is homophonous to the allophone /de/, the voiced version of the coordination suffix TE after verbs ending in a voiced obstruent or nasal. They might look like the same element, however, they should be kept apart and DE co-occurring with *na*-adjectives and *no*-modifiers is, in fact, an instance of the copula. Several pieces of evidence for this are given in Hiraiwa (2012), and see Bloch (1970). The first piece of evidence comes from distributional patterns. As the predicative copula DA is mutually exclusive with the canonical copula DE-AR-U reviewed above, the coordinative copula DE corresponds also to a longer form: The coordinative form

(37) a. \*mazu-i                      taka-ku-te                      okashi  
                 unpleasant-MOD expensive-PRED-GER sweet  
      b. \*benri-na                      jōbu-de                      kuruma  
                 practical-MOD robust-PRED car

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of the canonical copula DE-AT-TE. Note the morphological setup, shown in (39), in which, again, the suffix -TE is added to the nominalization form.<sup>12</sup>

- (39) benri-de-at-te                      jōbu-na      kuruma  
 practical-PRED-COP-GER robust-MOD car  
 ‘a car that is practical and robust’

This means that both, DE and DE-AT-TE, as well as TO appear in contexts where nouns are coordinated. However, the former two elements are used for *clausal coordination* of a *no*-modifier with another modifier, in the sense that both exhibit predicativity. TO can never be used in clausal instances. This is illustrated below (and cf. examples in Hiraiwa 2012, 363).

- (40) a. Hanako-wa gakusei-de (-at-te) Nihon-jin da.  
 Hanako-TOP student-PRED (-COP-GER) Japan-person COP  
 ‘Hanako is a student and Japanese.’  
 b. Hanako-wa gakusei-de (-at-te) hon-wo yo-mu  
 Hanako-TOP student-PRED (-COP-GER) book-ACC read-PRS  
 koto-ga suki-da  
 COMP-NOM like-COP  
 ‘Hanako is a student and likes reading books.’  
 c. \*Hanako-wa gakusei-to hon-wo yo-mu koto-ga suki-da  
 Hanako-TOP student-and book-ACC read-PRS COMP-NOM like-COP  
 (‘Hanako is a student and likes reading books.’)

More evidence comes from the connective element *katsu*, which Hiraiwa (2012, 364) calls a *sentential conjunction marker*. *Katsu* necessarily adds an additional state of affairs, is restricted to written speech and can only follow nouns, *na*-adjectives and verbs in their nominalization form. That it must express a predication relation is illustrated in the following sentence, where a pure coordination reading, in which the person under description is the teacher of both Ken and Naomi, cannot be achieved. The only possible interpretation is ‘this person is Ken and this person is also Naomi’s teacher’.<sup>13</sup>

- (41) \*Kono hito-wa Ken katsu Naomi-no sensei-da.  
 this person-TOP Ken katsu Naomi-MOD teacher-COP  
 Under interpretation: ‘This person is Ken’s and Naomi’s teacher.’

<sup>12</sup>The historical nominalization form of verbs of the *aru*-paradigm (which *ar-* is the prime member of) was /ri/. Phonological assimilation has led to deletion of /r/ and gemination of /t/, so to a sound change from /ri te/ to /tte/. Again, the bare nominalization form DE-AR-I can be used for coordination and behaves essentially like DE-AT-TE.

<sup>13</sup>This sets *katsu* clearly apart from *shi* which has to appear with verbs inflected for tense. See also Fenger (2020).

The same holds true of -DE. This element also shows a restriction to clausal, that is predicative, contexts, giving further evidence that it is connected to indirect modification.

- (42) Kono hito-wa Ken-de Naomi-no sensei-da.  
 this person-TOP Ken-PRED Naomi-MOD teacher-COP  
 1.  $\neq$  ‘This person is Ken’s and Naomi’s teacher.’  
 2. only:  $\approx$  ‘This person is Ken, and this person is Naomi’s teacher.’

The predicative, nature of DE can be further highlighted via its interaction with *katsu*. If two entities are predicated on one noun, one of these elements has to appear, or both. The meaning is unchanged.

- (43) a. Kono hito-wa Ken-no sensei \*(katsu) Naomi-no  
 this person-TOP Ken-MOD teacher KATSU Naomi-MOD  
 sensei-da.  
 teacher-COP  
 b. Kono hito-wa Ken-no sensei-(de) Naomi-no sensei-da.  
 this person-TOP Ken-MOD teacher-PRED Naomi-MOD teacher-COP  
 c. Kono hito-wa Ken-no sensei-de (katsu) Naomi-no  
 this person-TOP Ken-MOD teacher-PRED KATSU Naomi-MOD  
 sensei-da.  
 teacher-COP  
 ‘This person is Ken’s teacher and (also) Naomi’s teacher.’  
 (Hiraiwa, 2012, 364–365)

This shows that, clearly, DE is an instance of the copula. This in turn predicts that non-predicative modifiers are banned from coordination with DE (and also DE-AT-TE), which is borne out, as illustrated below for the non-predicative *na*-adjective *kanzen-na* ‘complete’ and the *no*-modifier *Nihon-no* ‘Japan(-ese)’.

- (44) a. \*Max-ga kanzen-de baka-da.  
 Max-NOM complete-PRED idiot-COP  
 b. \*Max-ga kanzen-de-at-te baka-da.  
 Max-NOM complete-PRED-COP-GER idiot-COP  
 (‘Max is complete and Max is an idiot.’)
- (45) a. \*Nihon-de jōbu-na kuruma  
 Japan-PRED robust-na car  
 b. \*Nihon-de-at-te jōbu-na kuruma  
 Japan-PRED-COP-GER robust-na car  
 (‘a Japanese and robust car’)

An even more striking observation is that non-predicative modifiers cannot even appear in coordinative structures inside the DP as the second conjunct, regardless of whether the modifier to their left can appear in predicative position or not. As shown in (46), although *kirei-na* ‘beautiful’ can appear predicatively and can be coordinated, it cannot appear in coordination structures with a non-predicative modifier such as *Chūgoku-no* ‘Chinese’, causing the DP to be ungrammatical.

- (46) \**kirei-de*                      *Chūgoku-no* *kabin*  
       beautiful-PRED China-MOD vase  
       (‘a beautiful and Chinese vase’)

### 4.3 Direct modifiers in Japanese

The following diagnostics for direct modifiers in Japanese were established in the previous section: *Lack of predication, non-gradability, lack of tense/inflection, no co-occurrence with canonical copula, lack of coordination*. These tests will help to identify the relevant groups of direct modifiers. I will start with a discussion on those groups of direct modifiers that are cross-linguistically attested, and finally come to the language-specific group of *rentaishi*.

Recall that with the notion *direct-only*, I refer to those modifiers that truly can never appear in predicative position. For many of the modifiers discussed in the following pages, predicative use, and this extends to the other characteristics, can be achieved. For example, the relational modifier *Chūgoku* is direct when denoting an origin, ‘Chinese, from China’. Its principal meaning, denoting something related to the country ‘China’, however, is possible in predicative position, which is why modifiers of NATIONALITY are often attested in the corpus in combination with the copula. As mentioned, group 5 of the data set contains some modifiers for which predicative use is marginal, but nevertheless attested, meaning those are not direct-only. Reflecting the obvious skew, which needed to be at least 5>1, towards attributive use, I classified such modifiers as *prototypically attributive* and also sorted them to this group. Cf. Chapter 1.5.

#### 4.3.1 Non-intersective modifiers

A cross-linguistically prominent group of direct-only modifiers, outlined in Chapter 2.3.1, are non-intersective modifiers of the type *former*. Those are characterized by the missing set intersection between their denotation and that of the noun they modify and can be sub-classified into *truly non-intersective modifiers*, which can never be used predicatively, *subsective* modifiers, which can be used predicatively

but then exhibit an intersective meaning, *privative* modifiers and *adnominal degree morphemes* (Vendler, 1957; Bolinger, 1967; Larson, 1998; Morzycki, 2016).

#### 4.3.1.1 Non-Intersective Adjectives

Several non-intersective adjectives in Japanese are presented in the PhD thesis of Yamakido (2005) (cf. also Yamakido 2000, 2007), who argues explicitly against a unified relative clause analysis of Japanese adjectives, and in Ayano (2010). The authors show that ambiguous examples of the type *old friend* in English are disambiguated via two separate lexemes in Japanese. The *i*-adjective *furu-i* can be translated as ‘old’, but cannot be used to denote the age of a human entity. In other words, it only gives a subsective reading when modifying human nouns while its intersective interpretation is blocked (Yamakido, 2005, 66–67). In order to denote a person’s age, other lexemes have to be used, such as *kōrei-na* ‘old’. This is illustrated below with the relevant notation.<sup>14</sup>

- (47) *furu-i tomodachi*  
 old-MOD friend (Yamakido, 2005, 81)  
 1.  $\approx$  ‘long-time friend’ [*furu-i tomodachi*]  $\subseteq$  [*tomodachi*] (subsective)  
 2.  $\neq$  ‘aged friend’ [*furu-i tomodachi*] = [*furu-i*]  $\cap$  [*tomodachi*] (intersective)
- (48) *kōrei-na tomodachi*  
 old-MOD friend (Yamakido, 2005, 67)  
 1.  $\neq$  ‘long-time friend’ [*kōrei-na tomodachi*]  $\subseteq$  [*tomodachi*] (subsective)  
 2.  $\approx$  ‘aged friend’ [*kōrei-na tomodachi*] = [*kōrei-na*]  $\cap$  [*tomodachi*] (intersective)

Lack of intersective interpretation suggests lack of predicative use, which in turn suggests direct modification status (Larson, 1998; Cinque, 2010). As expected, this adjective cannot be used predicatively, at least not with human entities (cf. Yamakido 2005, 67).

- (49) \**Watashi-no tomodachi-wa furu-i.*  
 I-MOD friend-TOP old-COP  
 (‘My friend is old (aged).’)

Another case of ambiguity resolution presents itself in the *beautiful dancer* example. Baker (2003b, 3) is right when he says that the direct equivalent *utsukushi-i / kirei-na kashu* ‘beautiful singer’ only has an intersective reading, denoting the beauty of a person irrespective of their singing skills. However, other lexemes allow only a non-intersective, that is a subsective, interpretation of beauty pertaining to the

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<sup>14</sup>As Watanabe (2012, 511, Fn. 7) notes, the phrase *furu-kara-no* old-from-MOD ‘from a long time ago’ might be a better and more natural alternative to *furu-i*. See also Tatsuki (2009, 155–156).

dancing or singing skills of a person. Such lexemes are *takumi-na* ‘skillful’ (Ayano, 2010, 101, Fn. 2), *migoto-na* (Watanabe, 2017, 786), which can be translated as ‘beautiful’, but can only be used with reference to events, or *suteki-na* ‘wonderful’. This contrast is shown below.

- (50)    *utsukushi-i*    *kashu*  
          beautiful-MOD singer (Baker, 2003b, 3)  
          1.  $\neq$  ‘beautiful as a singer’ [beautiful singer]  $\subseteq$  [singer] (subsective)  
          2.  $\approx$  ‘beautiful and a singer’ [beautiful singer] = [beautiful]  $\cap$  [singer] (intersective)
- (51)    *Hanako-wa takumi-na / migoto-na / suteki-na kashu-da.*  
          Hanako-TOP skilfull-MOD / beautiful-MOD / wonderful-MOD singer-COP  
          1.  $\approx$  ‘beautiful as a singer’ [takumi-na kashu]  $\subseteq$  [kashu] (subsective)  
          2.  $\neq$  ‘beautiful and a singer’ [takumi-na kashu] = [takumi-na]  $\cap$  [kashu] (intersective)

This ambiguity resolution via two different lexemes points towards the fact that one is a direct, and the other an indirect modifier. Similar ambiguity resolutions have been observed in Korean (Kang, 2005, 161), and German (Maienborn, 2020, 69).

Japanese also has truly non-intersective adjectives, among those the *i*-adjective *omoigakena-i* ‘unexpected’ and the *na*-adjective *kanzen-na* ‘complete’ (Yamakido, 2005, 67–68), as well as the *i*-adjective *tayumina-i* ‘non-ceasing’ and the *na*-adjective *sakan-na* ‘vigorous’ given by Ayano (2010). *Omoigakena-i* ‘unexpected’, for example, can only express an “external adverbial reading” (Yamakido, 2005, 64). Below I illustrate the ban on predicative use, the corresponding adverbial reading and the missing intersectivity.

- (52)    a. *Omoigakena-i kyaku-ga ki-ta.*  
          unexpected-MOD guest-NOM come-PST  
          ‘An unexpected guest came.’ (Yamakido, 2005, 67)
- b. \**Ano kyaku-ga omoigakena-i.*  
          that guest-NOM unexpected-COP  
          (‘That guest is unexpected.’)
- c. *Omoigakena-ku kyaku-ga ki-ta.*  
          unexpected-ADV guest-NOM come-PST  
          ‘Unexpectedly, a guest came.’
- d. [omoigakena-i kyaku]  $\neq$  [omoigakena]  $\cap$  [kyaku]



As mentioned in the previous section, a ban on predicative use goes alongside a ban on the canonical copula DE-AR-U (53-c) (Ayano, 2010, 108-109) and the coordinative form of the copula (53-d), shown for direct-only *kanzen-na* ‘complete’.<sup>15</sup>

- (53) a. Max-ga kanzen-na baka-da.  
Max-NOM complete-MOD idiot-COP  
‘Max is a complete idiot.’ (Yamakido, 2005, 68)
- b. \*Ano baka-ga kanzen-da.  
that idiot-NOM complete-COP  
(‘That idiot is complete.’)
- c. \*ano kanzen-de-ar-u baka  
that complete-PRED-COP-PRS idiot  
(‘that idiot who is complete’)
- d. \*ano kanzen-de / de-ar-i / de-at-te kinpatsu-na  
that complete-PRED / PRED-COP-RYK / PRED-COP-GER blond-MOD  
baka  
idiot  
(‘that complete and blond idiot’)

Finally, the absence of tense is also attested, illustrated with *furu-katta* ‘was old’ (cf. also example in Ayano 2010, 109).

- (54) \*Watashi-no tomodachi-wa furu-k-at-a.  
I-MOD friend-TOP old-PRED-COP-PST  
(‘My friend was old (aged).’)

However, one needs to be careful to draw early conclusions from these results as each of those adjectival lexemes exhibits certain particularities. First, as Ayano (2010) admits, the lexemes *omoigakena-i* ‘unexpected’, *tayumina-i* ‘non-ceasing’ and *takumi-na* ‘skillful’ can in fact be inflected into past and then even be used predicatively. I report his judgments below.

- (55) a. Ano kyaku-ga omoigakena-k-at-ta.  
that guest-NOM unexpected-PRED-COP-PST  
‘That guest was unexpected.’
- b. ?Shinpo-ga tayumina-k-at-ta.  
progress-NOM non.ceasing-PRED-COP-PST  
‘Progress was non-ceasing.’
- c. ?Ano utaite-ga takumi-d-at-ta.  
that singer-NOM skillful-PRED-COP-PST  
‘That singer was skillful.’ (Ayano, 2010, 109)

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<sup>15</sup>Some corpus results contain the attributive construction *kanzen-naru* + NP. In this case, the attributive form (*rentaikei*) of the pre-modern copula *nari* occurs, thus this is still a case of attributive, not predicative use. See the discussion in Chapter 4.4.3.2.

Watanabe (2017) alludes to the very high CP status of the adverbial counterpart of *omoigakena-i* and shows that this adjective can appear above numerals. As outlined in Chapter 3.1, the functional projection hosting numerals is inside the indirect domain, therefore direct modifiers should not be able to cross this border.<sup>16</sup>

- (56) Omoigakena-i futari-no kyaku-ga soko-ni i-ta.  
 unexpected-MOD two-MOD guest-NOM there-LOC be-PST  
 ‘Two unexpected guests were there.’ (Watanabe, 2017, 790)

A further potential caveat is that many of the lexemes presented above have a rather peripheral lexematic status. *Furu-i* is banned from a combination with human entities in referring to age, possibly for pragmatic reasons.<sup>17</sup> *Migoto-na* ‘beautiful’ also cannot be used in combination with human entities, but only refers to events (Watanabe, 2017, 786). *Kanzen-na*, on the other hand, might not even be a lexical adjective but what Morzycki calls an ‘adnominal degree morpheme’ (2009; 2012a; 2012b).<sup>18</sup> *Tayumina-i* ‘non-ceasing’ and *omoigakena-i* ‘unexpected’ end in the suffix NA for negation, which inflects like an *i*-adjective.<sup>19</sup>

These modifiers therefore seem to constitute a special case, and it becomes obvious that there are only a few non-intersective modifiers among Japanese *i*- and *na*-adjectives.

#### 4.3.1.2 Non-intersective *no*-modifiers

In fact, most non-intersective lexemes – the equivalents of well-known examples such as *former*, *alleged* or *future* – are found among Japanese *no*-modifiers. Nagano and Shimada (2015, 122) give a few examples, but many more can be found in the data set. Below are given examples of truly non-intersective modifiers, and adnominal degree morphemes (Morzycki, 2014), for all of which predicative use was limited or non-existent.<sup>20</sup>

<sup>16</sup>Hoshi (2002) adds that predicative use can be ameliorated when modifiers are contrasted, although recall the examples at the beginning of this chapter for why this might be an orthogonal issue.

<sup>17</sup>The concept of honorificity might forbid this combination. As Nagano and Shimada (2015, 123) point out, *furu-i* does not correspond to *waka-i* ‘young’ but rather to *atarashi-i* ‘new’.

<sup>18</sup>In fact, Morzycki (2014) uses Japanese *kanzen-na* as one example for this category of modifiers. Also recall that the corresponding adverbial form *kanzenni* ‘completely’ is an endpoint modifier (Oshima *et al.*, 2019).

<sup>19</sup>*Tayumina-i* is derived from *tayumi*, which is a noun but has a marginal status in Modern Japanese. *Omoigakena-i* has lexicalized from the negative form of the verb *omoigakeru* ‘to expect’ (Watanabe, 2017, 803).

<sup>20</sup>Again, speakers might have the feeling that predicative use can be achieved to different extents via establishing contrastive contexts. Not only were occurrences low to non-existent in the corpus, predicative use is always tested for context-neutral utterances, as far as that is possible.

(57) *Non-intersective modifiers in Japanese with element NO*

- a. *yui'itsu-no* ‘sole, unique’
- b. *jūzen-no* ‘previous, former’
- c. *tenpu-no* ‘innate’
- d. *shōbi-no* ‘urgent’
- e. *shōrai-no* ‘future’

(58) *Adnominal degree morphemes in Japanese with element NO*

- a. *mattaku-no* ‘complete(-ly)’
- b. *hontō-no* ‘true(-ly), real(-ly)’

These modifiers cannot be used predicatively, hence fulfill the main criterion of direct modification. This is illustrated below for *shōrai-no* ‘future’ and *tenpu-no* ‘innate’.

- (59) a. *shōrai-no daitōryō*  
future-MOD prime.minister  
‘a future prime minister’
- b. \**Kono daitōryō-wa shōrai-da.*  
this prime.minister-TOP future-COP  
(‘This prime minister is future.’)

- (60) a. *tenpu-no sainō*  
innate-MOD talent  
‘innate talent’
- b. \**Kare-no sainō-wa tenpu-da.*  
he-MOD talent-TOP innate-COP  
(‘His talent is innate.’)

These modifiers are also restricted in gradability, first illustrated via combinations with the canonical degree adverb and comparative constructions, either with *motto* ‘more’ or the particle *yorī* ‘than’.

- (61) a. \**totemo shōrai-no daitōryō*  
very former-MOD prime.minister  
(‘a very future prime minister’)
- b. \**motto shōrai-no daitōryō*  
more future-MOD prime.minister  
(‘a more future prime minister’)
- c. \**Kono daitōryō-wa ano daitōryō-yori shōrai-da.*  
this prime.minister-TOP that prime.minister-than future-COP  
(‘This prime minister is more future than that prime minister.’)

Above, it was shown that the degree adverb *kanari* ‘quite, rather’ is more permissive, however the combination of *kanari* with most non-intersective modifiers seems to

be out as well, exemplified below for *shōrai-no* ‘future’, *yui’itsu-no* ‘unique’ and the nominal degree morpheme *hontō-no* ‘true’.

- (62) a. \*kanari shōrai-no daitōryō  
           quite future-MOD prime.minister  
           (‘a quite future prime minister’)  
       b. \*kanari yui’itsu-no hōhō  
           quite sole-MOD method  
           (‘a quite sole method’)  
       c. \*kanari hontō-no tomodachi  
           quite true-MOD friend  
           (‘a quite true friend’)

Encoding a tense interpretation is also ungrammatical.

- (63) \*shōrai-d-at-ta daitōryō  
       future-PRED-COP-PST prime.minister  
       (‘a prime minister who was future’)

Finally, there is one privative modifier, namely *nise-no* ‘fake’, for which predicative use is degraded but not entirely ungrammatical, as is the case also with the corresponding *fake* in English (Cinque, 2014).

- (64) ?Kono pisutoru-wa nise-da.  
       this gun-TOP fake-COP  
       ‘This gun is fake.’

This shows that non-intersective modifiers do exist in Japanese and most of them appear with the element NO. They exhibit many characteristics of their cross-linguistic counterparts: they cannot be used predicatively, are not gradable and are not complex, all features that make them candidates for direct-only modifiers.<sup>21</sup>

### 4.3.1.3 Hierarchies and Ordering

Recall from Chapter 2.5.1 that Svenonius (2008) argues that it is only modifiers of different gradability and intersectivity that exhibit ordering preferences and assigns two different projections, SortP for subsective, but gradable modifiers, and nP for non-gradable intersective modifiers (see also Dékány 2011; Truswell 2009; Kim 2019). He predicts that the latter type are closer to the noun (in linear order, shown through the symbol ‘>’). The examples below illustrate this point.

<sup>21</sup>Other instances of non-intersective modifiers can be found among the group of *rentaishi* discussed in Chapter 4.3.6.

- (65) a. tiny red hats (gradable/subsective  $\succ$  non-gradable/intersective)  
 b. \*?red tiny hats (non-gradable/intersective  $\succ$  gradable/subsective)  
 (Svenonius, 2008, 36)

But what about truly non-intersective modifiers? In German, *früher* ‘former’, *heutig* ‘today’s’ or *gestrig* ‘yesterday’s’ among others, systematically have to precede intersective and subsective adjectives (see corpus study in Trost 2006a, as well as Eichinger 1991).<sup>22</sup>

- (66) a. der gestrige schöne Tag (German)  
 the yesterday beautiful day  
 ‘yesterday’s beautiful day’  
 b. \*der schöne gestrige Tag  
 the beautiful yesterday day

Structural precedence of non-intersective adjectives has been tested for Greek and English by Panayidou (2013) based on scope effects. Assuming both adjectives are direct, she detects differences in scope between the ordinary intersective adjective *bright* and the truly non-intersective adjective *former* in English depending on their order. When both adjectives modify the noun *student* in the order *former*  $\succ$  *bright*, two readings are possible: A reading where *former* takes scope over the combination of *bright* and the noun *student* at the same time, denoting a student who is no longer bright, and a reading where both modifiers take individual scope over the noun and *former* skips the intervening adjective, denoting a person who was once a student and is or was bright, irrespective of their affiliation (Panayidou, 2013, 106).

- (67) the former bright student (ambiguous)  
 1.  $\approx$  ‘a student who is no longer bright’  
 2.  $\approx$  ‘a person who is no longer a student and is/was bright’

If the order is reversed to *bright*  $\succ$  *former*, *bright* cannot modify solely the noun without skipping over the intervening adjective and only the first reading is available (Panayidou, 2013, 106).

- (68) the bright former student (only reading 1)  
 1.  $\approx$  ‘a bright person who is no longer a student’  
 2.  $\neq$  ‘a person who was bright as a student and is no longer one’

This shows that we are dealing with a marked order. Since a common reason for marked orders – besides different kinds of modification (which is refuted in Panayi-

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<sup>22</sup>In German linguistics, such modifiers are also called ‘*referentiell*’, meaning ‘referential’ (Trost, 2006a), or ‘*Situativa*’ (Eichinger, 1979, 1991), because they situate a referent in space and time.

dou 2013, Chapter 3.5.2) – is movement out of canonical positions, the author concludes that *former* > *bright*, and therefore *non-intersective* > *intersective* is the underlying structural hierarchy (Panayidou, 2013, 107–111). This yields the following preliminary structural hierarchy based on intersectivity.

- (69) Hierarchy based on Intersectivity (Trost, 2006a; Svenonius, 2008; Panayidou, 2013)  
 Non-intersective > Subjective > Intersective > N

On the contrary, Ayano (2010) argues that non-intersective adjectives in Japanese have to appear lower than other adjectives (70) as well as lower than numerals (71) as in the following examples (*furu* will be glossed as ‘long.time’ now to avoid confusion).

- (70) a. monoshizuka-na furu-i yūjin  
 quiet-MOD long.time-MOD friend  
 ‘quiet long-time friend’ (Ayano, 2010, 104)  
 b. \*furu-i monoshizuka-na yūjin  
 long.time-MOD quiet-MOD friend  
 c. monomidaka-i kanzen-na baka  
 burningly.curious-MOD complete-MOD idiot  
 ‘burningly curious complete idiot’ (Ayano, 2010, 105)  
 d. \*kanzen-na monomidaka-i baka  
 complete-MOD burningly.curious-MOD idiot
- (71) a. San-nin-no furu-i yūjin-ga ki-ta.  
 three-CL-MOD long.time-MOD friend-NOM come-PST  
 ‘Three long-time friends came.’ (Ayano, 2010, 106)  
 b. \*Furu-i san-nin-no yūjin-ga ki-ta.  
 long.time-MOD three-CL-MOD friend-NOM come-PST  
 c. futari-no kanzen-na baka  
 two-MOD complete-MOD idiot  
 ‘two complete idiots’ (Ayano, 2010, 107)  
 d. \*kanzen-na futari-no baka  
 complete-MOD two-MOD idiot

From this, he concludes that non-intersective modifiers in Japanese are systematically in a lower domain than all other direct modifiers and presents his own hierarchy.

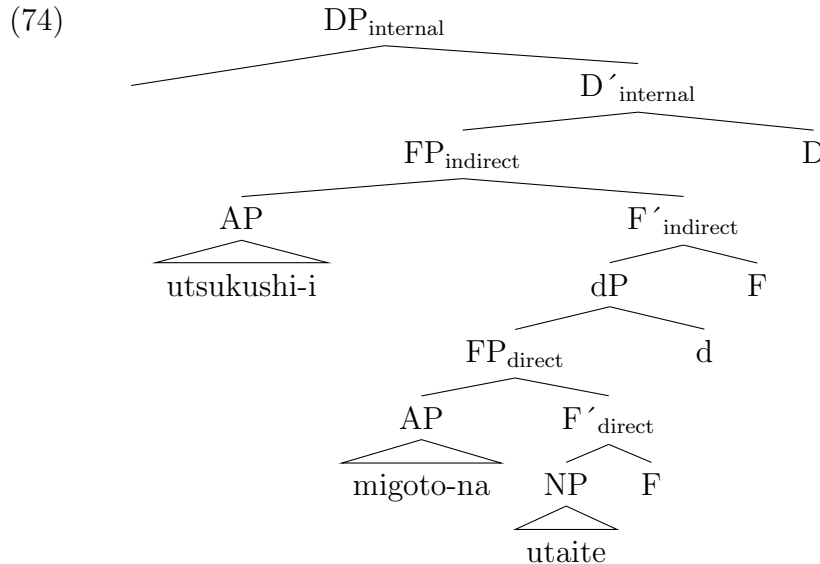
- (72) Hierarchy for Japanese based on Intersectivity (Ayano, 2010)  
 Intersective > Non-Intersective > N

Watanabe (2017) adds the following piece of data regarding the order of *utsukushi-i* (intersective) and *migoto-na* (subjective), those modifiers that divide the meaning of

‘beautiful’ as described above. He judges the subsecutive modifier best closest to the noun but mentions that this contrast “may not be as sharp as we want” (Watanabe, 2017, 790).

- (73) a. *utsukushi-i migoto-na utaite*  
 beautiful-MOD wonderful-MOD singer  
 ‘beautiful person who sings beautifully/wonderfully’  
 b. ??*migoto-na utsukushi-i utaite*  
 wonderful-MOD beautiful-MOD singer (Watanabe, 2017, 790)

The observations in Ayano (2010) and Watanabe (2017) can, however, be explained by adhering to the direct/indirect dichotomy, rather than a structural hierarchy inside the direct domain. First, that non-intersective modifiers appear lower than numerals is expected since numerals are merged in the indirect domain, explaining (71). The fact that the non-intersective modifier *kanzen-na* ‘complete’ is best closer to the noun in (70) follows naturally if we assume that *monoshizuka-na* ‘quiet’ and *monomidaka-i* ‘burningly curious’ are indirect.<sup>23</sup> Concerning the example in Watanabe (2017), the order *utsukushi-i*  $\succ$  *migoto-na* is also expected under the assumption that these modifiers are indirect and direct respectively, in line with Cinque (2010, 22–24). Similar to the *beautiful beautiful dancer* or *visible visible stars* examples (Chapter 2.2), the modifier closer to the noun is direct. Their respective positions in the DP are illustrated below (omitting ModP for readability).



<sup>23</sup>There might even be a completely different reasons at stake here, namely phonological length or prosody. *Monoshizuka-na* ‘quiet’ is twice as long as *furu-i* ‘long time’ (6 vs. 3 morae), and it is not clear why the author did not use the more common *shizuka-na* ‘quiet’ with only 4 morae. However, in my survey, I could not find significant results for the effect of mora size on ordering as reported in Chapter 4.4.3.1.

The same structural positions are at stake in the following minimal pair. The non-intersective modifier *shōrai-no* ‘future’ must be closer to the noun than the *na*-adjective *majime-na* ‘hardworking’. Again, an indirect, or at least ambiguous, modifier must precede the direct-only modifier showing its higher position within the DP.

- (75) a. *majime-na shōrai-no purosakkāsenshu*  
hardworking-MOD future-MOD football.player  
‘a hardworking person who will be a professional football player in the future’  
b. ??*shōrai-no majime-na purosakkāsenshu*  
future-MOD hardworking-MOD pro.football.player

Crucially, a hierarchy based on intersectivity can hardly be established in Japanese and rather all these examples seem rather to involve a direct/indirect dichotomy. Recall from Chapter 3.1 that truly non-intersective modifiers also do not engage in ordering hierarchies with other types of direct modifiers, for example the equivalents of relational adjectives, to be reviewed below. This is exemplified in (76) with the non-intersective modifier *tada-no* ‘mere’. (77) shows that this modifier cannot occur predicatively.<sup>24</sup>

- (76) non-intersective vs. relational modifier  
a. *tada-no Doitsu-no gakusei*  
mere-MOD German-MOD student  
b. *Doitsu-no tada-no gakusei*  
German-MOD mere-MOD student  
‘a mere German student’  
(77) \**Kono gakusei-wa tada-da.*  
this student-TOP mere-COP  
(‘This student is mere.’)

This discussion has shown that non-intersective modifiers do exist in Japanese and that they predominantly can be found in the group of *no*-modifiers. Furthermore, it was shown that the well-known ambiguous cases in English are resolved in Japanese, which is a substantial problem for Baker (2003b) who correlates the apparent lack of non-intersectivity with a lack of direct modification. Finally, since non-intersective modifiers are direct, they are merged lower than indirect modifiers and appear closer to the noun, explaining the mismatches between the assumptions of Svenonius (2008) and Ayano (2010).

<sup>24</sup>Only with the meaning ‘mere’ is it never possible in predicative position. *Tada-no* can also have the meaning ‘for free’, in which case it can be used predicatively. Therefore it is sorted in group 1 in the data set.



### 4.3.2 Relational modifiers

The second cross-linguistic group of direct-only modifiers are *relational adjectives*, although in Japanese it might be better to speak of *relational modifiers*. Those modifiers were discussed in Chapter 2.3.2 with regard to European languages (cf. Fábregas 2007; Rainer 2013; Bortolotto 2016).

Japanese, besides Arabic, was one of the two Non-European languages looked at to verify this category cross-linguistically in Bisetto (2010), which explains the surprising amount of literature on this category with regard to Japanese (Nagano and Shimada, 2015; Nagano, 2016, 2018; Shimada and Nagano, 2018; Morita, 2010, 2011, 2013). Bisetto (2010), highlighting derivational factors, analyzes the suffix -TEKI 的 ‘-ic’ as the derivational suffix of relational adjectives in Japanese.<sup>25</sup> However, as shown in Nagano (2016, 48–50), these derivatives are not true equivalents of relational adjectives, since they in fact can be used predicatively and are gradable. Rather, the equivalents of relational adjectives appear in Japanese unequivocally with the element NO and are not subject to morphological processes. Those equivalents seem to be nominal not only on the deep, but also on the surface level.

In the following, I will first show that the equivalents of *thematic* and *classificatory* relational adjectives exist in Japanese and then give an overview of the categories named in Nagano (2016), discussing some of them in sufficient detail in order to show which are candidates for direct modifiers, and where deviations might occur.

#### 4.3.2.1 Thematic and classificatory relational modifiers

In Chapter 2.3.2, it was outlined that relational adjectives in European languages come in two types: *classificatory* and *thematic*. Only the equivalents of classificatory relational adjectives are discussed in the literature with regard to Japanese. An example is given below. Observe that, contrary to the English equivalent *environmental crisis*, no derivational processes are observable on the morphological level.

- (78)    kankyō-no            kiki  
          environment-MOD crisis  
          ‘environmental crisis’

As observable above, the equivalents of the classificatory adjective *environmental* is a *no*-modifier which expresses a certain relationship to the noun it modifies. Crucially however, *kankyō* is a noun itself. Besides the absence of morphological derivation,

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<sup>25</sup>Examples include *keizai-teki* ‘economical’ derived from *keizai* ‘economy’ or *shukan-teki* ‘subjective’ derived from *shukan* ‘subjectivity’, cf. also Bisetto (2010, 71–73).

this can also be exemplified on the syntactic level via productive use with case particles.

- (79) Kankyō-ga daiji-da.  
 environment-NOM important-COP  
 ‘The environment is important.’

As far as competing constructions go, *of*-phrases (*problems of the economy*) or PPs are not available, yet Japanese makes frequent use of lexical compounds, often Sino-Japanese four character lexical compounds, with the same meaning.

- (80) kankyō-kiki (環境危機)  
 environment-crisis  
 ‘environmental crisis’

Thematic relational adjectives have not been identified as a separate type of relational modifiers with regard to Japanese, probably because such modifiers are nominals in the first place. Recall that thematic relational adjectives are, in languages such as English, restricted to the modification of event or action nouns (Grimshaw, 1990). A direct equivalent of such examples would be (81) in which *machi-no* ‘urban’ could be interpreted as a *thematic relational modifier* since it expresses a subject role.

- (81) machi-no hatten  
 city-MOD development  
 ‘urban development / development of the city’

Event nouns in Japanese usually belong to the category of so-called *Verbal Nouns* (Teramura, 1982; Oshima *et al.*, 2019). Such nouns can either appear as true nouns, as in (81), or as predicates of a clause together with the light verb *su* ‘to do’. (82) is a paraphrase of (81), in which *hatten* serves as the predicate of the matrix clause and *machi* now appears as a noun with the nominative case particle.

- (82) Machi-ga hatten-shi-ta.  
 city-NOM development-do-PST  
 ‘The city developed.’

Since such modifiers appear very nominal, no special discussion on their character seems necessary at first. Nevertheless, in Japanese, neither the equivalents of classificatory (83-a), nor of thematic relational adjectives (83-b) can serve as the predicate of a sentence together with the copula.

- (83) a. \*Kiki-ga kankyō-da.  
           crisis-NOM environment-COP  
           ('The crisis is environmental.')
- b. \*Hatten-ga machi-da.  
       development-NOM city-COP  
       ('The development is urban.')

#### 4.3.2.2 Different categories

A further obstacle for the precise analysis of relational modifiers in Japanese is the abundance of possible categories scattered in the literature, for which it is not clear if all really fulfill the relevant syntactic criteria. A very detailed list of categories is presented in Nagano (2016, 53–54), who understands relational adjectives in Japanese as non-predicative modifiers with a nominal character. In the table below, I give her categories and compare them to the categories presented in hierarchies of relational adjectives written in the cartographic research field (Rae, 2010; Bortolotto, 2016), to older approaches to relational adjectives in English (Downing 1977; Levi 1978; Warren 1984), to standard hierarchies of qualitative adjectives, and to the PP hierarchy of Takamine (2010).<sup>26</sup>

| Nagano (2016) | In other hierarchies                                                                                                                                                                                                                    | Example                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Material      | (Scott, 2002; Laenzlinger, 2005, 2011): <i>Material</i> (QA)<br>Rae (2010); Bortolotto (2016): <i>Material</i> (RA)<br>Takamine (2010): <i>Material</i> (PPs)                                                                           | <i>komugi-no pan</i><br>‘wheat(-en) bread’  |
| Nationality   | Sproat and Shih (1991): <i>Provenance</i> (QA)<br><br>Scott (2002); Cinque (2010); Laenzlinger (2005, 2011): <i>Nationality/Origin</i> (QA)<br>Alexiadou and Stavrou (2011); Arsenijević <i>et al.</i> (2014): <i>Ethnic Adjectives</i> | <i>Chūgoku-no kabin</i><br>‘Chinese vase’   |
| Shape         | Sproat and Shih (1991): <i>Shape</i> (QA)<br>Scott (2002); Cinque (2010); Laenzlinger (2005, 2011): <i>Shape</i> (QA)                                                                                                                   | <i>sankaku-no heya</i><br>‘triangular room’ |
| Style/Pattern | no direct equivalent                                                                                                                                                                                                                    | <i>shima-no uwagi</i><br>‘striped jacket’   |

*Continued on next page*

<sup>26</sup>This table is most of all meant to serve as a reference. See also Bortolotto (2016, 93–94) as well as Rainer (2013, 16) for similar comparisons.

Table 4.1 – *Continued from previous page*

| Nagano (2016)     | In other hierarchies                                                                                                                                                                                         | Example                                                                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Type              | potentially Downing (1977): <i>comparison</i> (RA)<br>potentially Warren (1984): <i>comparant-compared</i> (RA)                                                                                              | <i>Yōroppa-no keizai</i><br>‘European economy’ (= European type of economy) |
| Tendency towards  | no direct equivalent                                                                                                                                                                                         | <i>ame-no kisetsu</i><br>‘rainy season’                                     |
| Line/Family/Group | close to <i>group adjectives</i> (Grimshaw, 1990; Cetnarowska, 2015)                                                                                                                                         | <i>Surabu-no gengo</i><br>‘Slavic languages’                                |
| Location          | Downing (1977): <i>place</i> (RA)<br>Levi (1978): <i>in</i> (RA)<br>Warren (1984): <i>place-object</i> (RA)<br>Rae (2010); Bortolotto (2016): <i>Location</i> (RA)<br>Takamine (2010): <i>Location</i> (PPs) | <i>umi-no seibutsu</i><br>‘marine life’                                     |
| Time              | Downing (1977): <i>time</i> (RA)<br>Levi (1978): <i>in</i> (RA)<br>Warren (1984): <i>time-object</i> (RA)<br>Rae (2010); Bortolotto (2016): <i>Time</i> (RA)<br>Takamine (2010): <i>Time</i> (PPs)           | <i>yoru-no hōmon</i><br>‘nocturnal visit’                                   |
| Topic             | Levi (1978): <i>about</i> (RA)<br><br>Rae (2010); Bortolotto (2016): <i>Matter</i> (RA)                                                                                                                      | <i>keitai-no mondai</i><br>‘morphological problem(s)’                       |
| Status            | Downing (1977): <i>occupation</i> (RA)<br><br>Warren (1984): <i>affected object-actor</i> (RA)<br>included in <i>group adjectives</i> (Grimshaw 1990; Cetnarowska 2015)                                      | <i>fuku-daitōryō-no ninki</i><br>‘vice-presidential term’                   |
| Possession        | potentially <i>part-whole</i> in Warren (1984)                                                                                                                                                               | <i>yotsu-ashi-no dōbutsu</i><br>‘four-legged animal’                        |

*Continued on next page*

Table 4.1 – *Continued from previous page*

| Nagano (2016) | In other hierarchies                                                                                                                       | Example |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------|
|               | Nikolaeva and Spencer (2012, 2020); Spencer and Nikolaeva (2017): non-canonical modification carried out by nouns or relational adjectives |         |

Table 4.1: Categories of Japanese Relational Modifiers compared to other studies

It would take us too far afield to discuss each of these categories above. Instead, I will concentrate on a sample of relevant categories, most importantly NATIONALITY, MATERIAL and SHAPE. The reason for this is that these categories also appear in well-known hierarchies for *qualitative* adjectives. It will be shown that modifiers of NATIONALITY and MATERIAL have a strong direct character in Japanese. Modifiers of SHAPE, on the other hand, do not. In the second part of this section, I will put forth data concerning combination with the different forms of the copula and discuss ordering and possible hierarchies.

#### 4.3.2.3 NATIONALITY and ethnic modifiers

Modifiers of NATIONALITY are quite intriguing in Japanese. This category is quite low in ordering hierarchies of qualitative adjectives in the literature (cf. Scott 2002). Recall, on the other hand, the group of *ethnic adjectives*, which has been put forth for various languages as toponymic relational adjectives denoting an origin (Bartning 1984; Bosque and Picallo 1996; Gunkel and Zifonun 2008; Alexiadou and Stavrou 2011 a. o.). It is unclear whether the term *ethnic adjectives* is applicable only with regard to thematic, or also to classificatory adjectives denoting an origin. Recall that the latter type can be used predicatively in English (84-a), while NATIONALITY adjectives with a thematic reading, those modifying event nouns, cannot (84-b) (cf. Alexiadou and Stavrou 2011).

- (84) a. These bed sheets are Italian.  
b. \*This attack was Italian.

In the literature on Japanese, only equivalents of NATIONALITY modifiers with object nouns are given (Morita, 2011; Nagano, 2016), for example (85).<sup>27</sup>

<sup>27</sup>Note that in this example, the modifier *Nihon-no* does not necessarily denote an origin, or nationality, but might denote several other relations to the head noun, including LOCATION on a reading ‘in Japan’.

- (85) Nihon-no kuruma  
 Japan-MOD car  
 ‘Japanese car’

Similar to thematic relational modifiers, the equivalents of thematic ethnic adjectives have probably not been discussed due to the fact that they just seem to highlight a case of modification by a noun. Arguably, an example could be the following, given in passing by Nagano (2016, 48).

- (86) Nihon-no kōgeki  
 Japan-MOD attack  
 ‘Japanese attack’

It is a crucial observation that NATIONALITY modifiers in Japanese, also those with a classificatory meaning, are only used attributively, never predicatively (87-b) and are non-gradable (87-c), (87-d). Importantly, gradability is not even possible with endpoint modifiers as shown in (87-e). This crucially differentiates them from English classificatory adjectives.

- (87) a. Nihon-no kuruma  
 Japan-MOD car  
 ‘Japanese car’  
 b. \*Kono kuruma-wa Nihon-da.  
 this car-TOP Japan-COP  
 (‘This car is Japanese.’)  
 c. \*totemo Nihon-no kuruma  
 very Japan-MOD car  
 (‘very Japanese car’)  
 d. \*Watashi-no kuruma-wa sensei-no kuruma-yori Nihon (no  
 I-MOD car-TOP teacher-MOD car-than Japan MOD  
 kuruma) da.  
 car COP  
 (‘My car is more Japanese than my teacher’s (car).’)  
 e. \*kanzenni Nihon-no kuruma  
 completely Japan-MOD car  
 (‘completely Japanese car’)

What holds for classificatory, also holds for the equivalents of thematic ethnic modifiers as shown below.

- (88) a. Nihon-no kōgeki  
Japan-MOD attack  
'Japanese attack'
- b. \*Kono kōgeki-wa Nihon-da.  
this attack-TOP Japan-COP  
(‘This attack is Japanese.’)
- c. \*totemo / kanzenni Nihon-no kōgeki  
very / completely Japan-MOD attack  
(‘very / completely Japanese attack’)

While in English, THEME readings of ethnic adjectives have been argued to be impossible (Kayne, 1981; Giorgi and Longobardi, 1991),<sup>28</sup> no such restrictions are applicable to the Japanese equivalents, presumably for the fact that they are simply nouns.

Furthermore, object and subject roles can both be expressed inside the same sentence as in (91) (cf. Murasugi 1991). In such cases, the first modifier is usually interpreted as the subject and the second, the one adjacent to the noun, as the object. As expected, Japanese exhibits a rather free behavior of word order here as well and the reverse interpretations are not impossible.<sup>29</sup>

- (91) Amerika-no Itaria-no hakai  
America-MOD Italy-MOD destruction  
'destruction of Italy by America'

*Amerika*=subject, *Itaria*=object

Finally, modifiers of NATIONALITY can modify human nouns as the following example given in Morita (2011, 91) shows. Again, predicative use is impossible, and so is gradability.

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<sup>28</sup>Nevertheless such readings have been marginally reported in English (Arsenijević *et al.*, 2014) and seem productive in Polish (Cetnarowska, 2015).

- (89) the French defeat (Arsenijević *et al.*, 2014, 23)

- (90) chiń-s-ka porażka (Polish)  
China-RA-NOM.SG.F destruction.NOM.SG.F  
'destruction of China' (Cetnarowska, 2015, 136)

Rainer (2013, 20) mentions that online he found “dozens of examples of the type *the Russian invasion by Napoleon, the Afghan invasion by Russia, the Polish invasion by Germany, the Chinese invasion by Japan*, and the like, many of which apparently occur in authoritative documents”.

<sup>29</sup>Many other thematic roles can be expressed within the DP, but for most, an additional post-position needs to appear before MOD.

- (92) a. Nihon-no haiyū  
Japan-MOD actor  
'Japanese actor'
- b. \*Kono haiyū-wa Nihon-da.  
this actor-TOP Japan-COP  
(‘This actor is Japanese.’)
- c. \*totemo Nihon-no haiyū  
very Japan-MOD actor  
(‘very Japanese actor’)
- d. \*Kono haiyū-wa sono haiyū-yori Nihon-no haiyū da.  
this actor-TOP that actor-than Japan-MOD actor COP  
(‘This actor is more Japanese than that actor.’)
- e. \*kanzenni Nihon-no haiyū  
completely Japan-MOD actor  
(‘completely Japanese actor’)

This might be different for other languages. Arsenijević *et al.* (2014) have argued that all ethnic modifiers have an underlying origin-reading which arguably permits predicative use, as in the following example in English.

- (93) Guillem is French. (Arsenijević *et al.*, 2014, 23)

In German, as in Japanese, such examples are clearly marked as also mentioned in Kotowski (2016).

- (94) ??Guillem ist französisch. (German)  
Guillem is French  
'Guillem is French'

In order to express the nationality of a human entity in predicative position in Japanese, the suffix *-jin* 人, lit. ‘human’ needs to be added. Again, the same holds true in German (96).

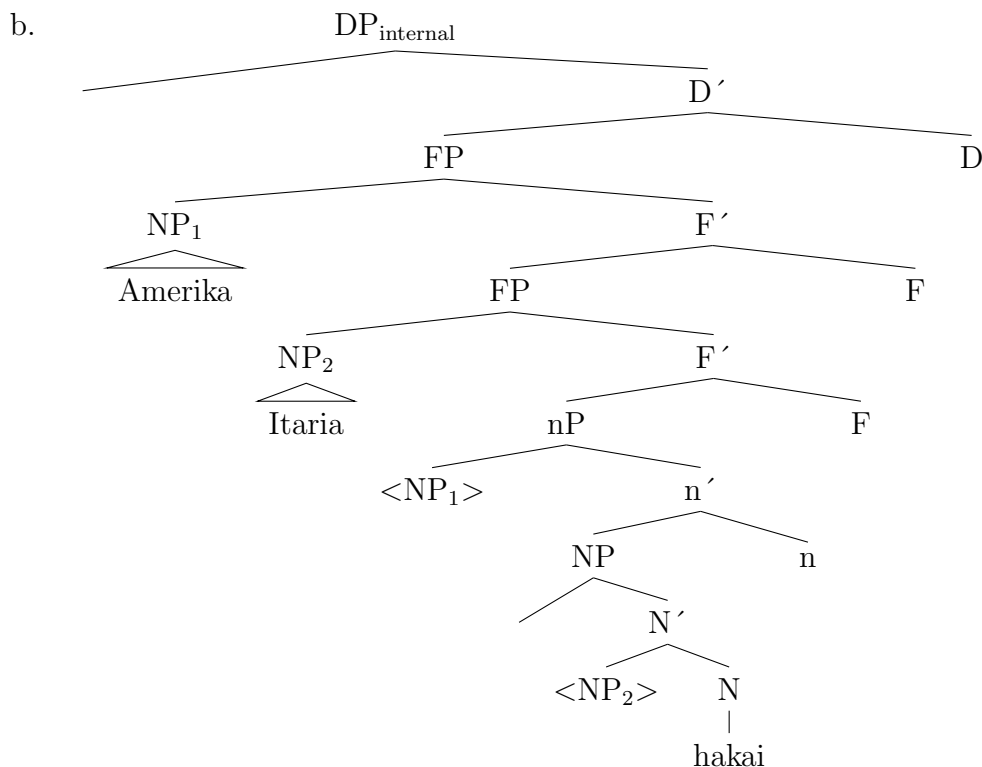
- (95) Kono gekai-wa Nihon-\*(jin)-da. (Japanese)  
this surgeon-TOP Japan-person-COP  
'This surgeon is Japanese.' (= a Japanese person)
- (96) Guillem ist Franzose. (German)  
Guillem is Frenchman  
'Guillem is French' (= a French person)

A remaining question is how the thematic relation, the theta roles within DPs such as (91) are being established. In the cartographic research literature, dedicated projections are given for modifiers that express a THEME or AGENT reading (Cinque, 1993; Laenzlinger, 2005, 2011; Bortolotto, 2016). As for their derivation, Laen-



zlinger (2011) predicts that modifiers in the specifier positions of these projections have moved there from lower positions. Taking the parallel to the verbal domain seriously, nouns with a THEME reading start out as complements of N, those with an AGENT reading as specifiers of an additional nP projection (see also Adger 2003). One derivation could look as follows. Recall that the first modifier is typically interpreted as the agent, but that the reading can also be reversed, which underlines the unspecific nature of the functional projections within the direct domain.

- (97) a. Amerika-no<sub>Agent</sub> Itaria-no<sub>Theme</sub> hakai  
 America-MOD Italy-MOD destruction (= (91))



To summarize, modifiers of NATIONALITY are a productive source of direct modification in Japanese and the relevant lexemes clearly have a nominal status.

#### 4.3.2.4 Modifiers of MATERIAL

MATERIAL is in Japanese also exclusively expressed via *no*-modifiers. These cannot be used predicatively (98-b) and are not gradable (98-c), as judged unequivocally so by all speakers I consulted in relevant constructions. One example is *komugi-no* ‘wheat(-en)’, and recall also the examples with *chiku-no* ‘teak’ from Chapter 3.1.

- (98) a. komugi-no pan  
wheat(-en)-MOD bread  
'wheaten bread'
- b. \*Kono pan-wa komugi-da.  
this bread-TOP wheat(-en)-COP  
(‘This bread is wheat(-en).’)
- c. \*totemo komugi-no pan  
very wheat(-en)-MOD bread  
(‘very wheaten bread’)

In contrast to modifiers of NATIONALITY, modifiers of MATERIAL can be modified via endpoint modifiers (Kennedy and McNally, 2005; Fábregas, 2014; Oshima *et al.*, 2019) Cf. Chapters 2.3.2 and 4.2.2.

- (99) kanzenni komugi-no pan  
completely wheat(-en)-MOD bread  
'completely wheaten bread' (= bread consisting completely of wheat)

Finally, note that also modifiers of MATERIAL are nominals and occur productively with case particles as shown below.

- (100) Ano komugi-wa oishi-i.  
that wheat-TOP tasty-COP  
'That wheat is tasty.'

Restrictions in predication and gradability are the reason why modifiers of MATERIAL are closer to relational than to qualitative modifiers (Nagano, 2016; Morita, 2011). Their syntactic restrictions qualify them as direct modifiers.

#### 4.3.2.5 Modifiers of SHAPE are not direct-only

The category SHAPE is not a part of any study on relational adjectives in European languages, but is rather a central category of qualitative adjectives in Sproat and Shih (1991); Cinque (1993, 2010), Scott (2002) and Laenzlinger (2005, 2011). Why is it treated then as a relational modifier in Nagano (2016) and also Morita (2010, 2011)? First, note that the category SHAPE is expressed in Japanese via *i*- and *na*-adjectives, but also via *no*-modifiers and that there are lexemes, such as *maru* ‘round’ and *shikaku* ‘square’, which can appear with either I or NO in attributive position.<sup>30</sup>

<sup>30</sup>The BCCWJ yields 405 results with -I and 74 with -NO for *shikaku*, and even 33 with -NA. It is a matter of debate whether we should predict to different spell-out forms of MOD or simply two different lexemes which differ in word class status and thus lead to a different form of the syntactic head. The latter option seems more attractive in the system presented here, but, at the moment, nothing hinges on this.

- (101) a. maru-i / maru-no tēburu  
           round-MOD / round-MOD table  
           ‘round table’  
       b. shikaku-i / shikaku-no heya  
           square-MOD / square-MOD room  
           ‘square room’

The authors predict that both gradability and predicativity are marked for the *no*-variant of these lexemes, making them essentially equivalents of true relational adjectives. However, not only are there examples of predicative use in the corpus,<sup>31</sup> predicative use is also accepted by many speakers, including the ones I asked, as shown below.

- (102) a. Kono heya-wa shikaku-da.  
           this room-TOP square-COP  
           ‘This room is square.’  
       b. Kono tēburu-wa maru-da.  
           this table-TOP round-COP  
           ‘This table is round.’

Regarding gradability, the *no*-modifier-variant is marked in combination with the degree adverb *totemo* ‘very’, compared to the *i*-variant, showing that this is not for semantic reasons. None of my consultants accepted the combination of the degree adverb with *maru-no* (103-a), and only two out of five judged *totemo shikaku-no* ‘very square’ to be marginally acceptable (103-b).

- (103) a. \*totemo maru-no tēburu  
           very round-MOD table  
           ‘very round table’  
       b. ??totemo shikaku-no heya  
           very square-MOD room  
           ‘very square room’

However, the following combinations involving gradability are perfectly fine.

- (104) a. motto shikaku-no heya  
           more square-MOD room  
           ‘more square room’  
       b. omot-ta-yori maru-no tēburu  
           think-PST-than round-MOD table  
           ‘a table (which is) rounder than X thought’

---

<sup>31</sup>12 occurrences for *shikaku-da*, which is still very little making it a prototypical direct modifier, but 88 for *maru-da*.

Indeed, there are some prototypical direct *no*-modifiers in the data set that arguably express shape, such as *hishigata-no* ‘diamond-shaped, rhombic’ (37 attr., 4 pred.; cf. also Morita 2011, 98) and *yūkei-no* ‘tangible’ (29 attr., 0 pred.), suggesting that direct SHAPE modifiers do exist. However, importantly this is not the case for all modifiers of this category, unlike NATIONALITY or MATERIAL.

This discussion extends to modifiers of the category COLOR: The four basic color terms *aka* ‘red’, *ao* ‘blue’, *shiro* ‘white’ and *kuro* ‘black’ can also appear with -I and -NO, that is as adjectives or *no*-modifiers.<sup>32</sup> Again, both versions are possible predicatively, contrary to what is argued in Morita (2011), but the *no*-variant is only marginally gradable (105-c).<sup>33</sup>

- (105) a. *shiro-i* / *shiro-no* *kōto*  
           white-MOD / white-MOD coat  
           ‘white coat’ (Koopman and Ishizuka, 2016)
- b. *Kono kōto-wa shiro-da.*  
      this coat-TOP white-COP  
      ‘This coat is white.’
- c. \**totemo shiro-no kōto*  
      very white-MOD coat  
      (‘very white coat’)

#### 4.3.2.6 Other copula forms

As outlined above, the absence of predicative use correlates with the impossible co-occurrence with other copula forms, including DE used for coordination. This seems to be borne out for non-predicative relational modifiers in Japanese as exemplified below. Crucially, as noted in Chapter 4.2.4.2, these modifiers can neither appear as the second conjunct in coordinative structures (107).<sup>34</sup>

- (106) \**Chūgoku-de kirei-na kabin*  
          China-PRED beautiful-MOD vase  
          (‘a Chinese and beautiful vase’)
- (107) \**kirei-de Chūgoku-no kabin*  
          beautiful-PRED China-MOD vase  
          (‘a beautiful and Chinese vase’)

<sup>32</sup>A complementary distribution pointed out in the literature (Sawada, 1992; Kishita, 2005) is that the *no*-variant, the color noun (Okami, 2012, 2016), serves the purpose of identification and must receive a “contrastive interpretation” (Koopman and Ishizuka, 2016) thus is infelicitous in contexts where the referent does not need to be identified.

<sup>33</sup>Note that direct-only modifiers of COLOR do appear in the data set to be discussed in Chapter 4.3.3.

<sup>34</sup>One apparent counterexample was the material modifier *komugi* ‘wheat(-en)’ where DE could be understood as a postposition, denoting the material of origin so *komugi-de kata-i pan* could have the interpretation ‘hard bread made from wheat’.

These data points highlight that DE has to appear in phrasal contexts and always involves a predication structure (Bowers 1993, see Chapter 5.2). Even if only one modifier involved in the coordination structure does is banned from predication, coordination becomes ungrammatical.

Furthermore, as exemplified in (108), in no instance was the canonical copula DE-AR-U allowed in combination with non-predicative relational modifiers. This again speaks against analyses that treat NO as some predicative element.

- (108) \*komugi-de-ar-u pan  
wheat-PRED-COP-PRS bread  
(‘bread which is wheat(-en)’)

#### 4.3.2.7 Hierarchy and ordering of relational modifiers

Bortolotto (2016) establishes an ordering hierarchy for relational modifiers based on Romance, given here in terms of structural selection.

- (109) *Hierarchy of Relational Adjectives (based on Bortolotto 2016, 180)*  
TIME > LOCATION > AGENT > INSTRUMENT > MATTER (>) THEME > N

Also recall that it is a characteristic of relational adjectives to be strictly adjacent to their head noun and not to allow permutation with qualitative adjectives, as shown below for English.

- (110) a. big wooden table  
b. \*wooden big table (Bisetto, 2010, 66)

In the DP structure presented in this book, no specific functional projections dedicated to the different categories of relational modifiers are included inside the direct domain. This predicts (correctly) that relational modifiers in Japanese are not ordered according to a specific hierarchy. However, Watanabe (2012) argues in favor of such a hierarchy. First, he mentions that modifiers of NATIONALITY and MATERIAL have to be closer to their head noun than modifiers of other categories.<sup>35</sup>

- (111) DIMENSION  $\succ$  MATERIAL  
a. chiisa-na ki-no hashi  
small-MOD wood-MOD bridge  
b. ??ki-no chiisa-na hashi  
wood-MOD small-MOD bridge  
‘small wooden bridge’ (Watanabe, 2012, 507)

---

<sup>35</sup>I give his judgments below, but it is not clear whether this is an issue of grammaticality or preference. Note also that he uses as contrast the non-predicative modifier *chiisa-na* ‘small’ (see Chapter 4.3.6).

- (112) DIMENSION  $\succ$  NATIONALITY
- a. *chiisa-na Chūgoku-no kabin*  
small-MOD China-MOD vase
  - b. ??*Chūgoku-no chiisa-na kabin*  
China-MOD small-MOD vase  
'small Chinese vase' (Watanabe, 2012, 507)

Next, the author argues that the categories NATIONALITY and MATERIAL reflect the structural hierarchy NATIONALITY  $>$  MATERIAL in linear order when occurring together.

- (113) NATIONALITY  $\succ$  MATERIAL
- a. *hokuō-no ki-no isu*  
North.Europe-MOD wood-MOD chair
  - b. \**ki-no hokuō-no isu*  
wood-MOD North.Europe-MOD chair  
'North European wooden chair' (Watanabe, 2012, 508)

- (114) NATIONALITY  $\succ$  MATERIAL
- a. *Chiri-no kin-no kubikazari*  
Chile-MOD gold-MOD necklace
  - b. \**kin-no Chiri-no kubikazari*  
gold-MOD Chile-MOD necklace  
'Chilean gold necklace' (Watanabe, 2012, 508)

However, first this assessment is not shared by all authors. Nagano (2016, Fn. 14) emphasizes that she does share the judgment for (111) and also for (113), but not for (112) and (114), meaning she does not have a preferred order. She suggests that the ordering restrictions in the other cases might be phonologically conditioned (*ibid.*). More precisely, speakers including Nagano might prefer a shorter modifier to the right of a longer one. This is the case for *chiisa* 'small' (3 morae) and *hokuō* 'North European' (4 morae) preceding *ki* 'wooden' (1 mora). It is not however for *chiisa* 'small' vs. *Chūgoku* 'China, Chinese' (3 vs. 4 morae) and *Chiri* 'Chile, Chilean' vs. *kin* 'gold(-en)' (both 2 morae).

Furthermore, *ki* 'wooden' is the only monomoraic modifier in the data set and might be the only one that triggers such ordering restrictions (cf. Chapter 4.4.3.1). Concerning (114), Ken Hiraiwa (p.c.) mentioned to me that some speakers might think of *kin-no kubikazari* 'gold(-en) necklace' as a set phrase which is why separating the two entities is judged as not good.

I have tested for several ordering preferences among relational modifiers and in combinations with qualitative modifiers and could not detect any ordering restrictions. Importantly, in order to exclude the factors *phonology* (the number of morae)

and *word class status* (*i*-, *na*-adjective, *no*-modifier), I tried to verify ordering restrictions for different combinations and scenarios (see Chapter 4.4.3.1), however, in no case did I trigger an increase or decrease in preference for any speaker in any pair. In contrast, note the awkward English translation for almost all b) examples in the examples given below.

(115) NATIONALITY vs. MATERIAL

- a. Denmāku-no chīku-no isu  
Denmark-MOD teak-MOD chair  
'Danish teak chair'
- b. chīku-no Denmāku-no isu  
teak-MOD Denmark-MOD chair  
'teak Danish chair'

(116) QUALITY vs. NATIONALITY

- a. kirei-na Chūgoku-no kabin  
beautiful-MOD China-MOD vase  
'beautiful Chinese vase'
- b. Chūgoku-no kirei-na kabin  
China-MOD beautiful-MOD vase  
'Chinese beautiful vase'

(117) MATERIAL vs. QUALITY (SHAPE)

- a. komugi-no kata-i pan  
wheat-MOD hard-MOD bread  
'hard wheat(-en) bread'
- b. kata-i komugi-no pan  
hard-MOD wheat-MOD bread  
'wheat(-en) hard bread'

(118) COLOR vs. NATIONALITY

- a. ao-i Chūgoku-no kabin  
blue-MOD China-MOD vase  
'blue Chinese vase'
- b. Chūgoku-no ao-i kabin  
China-MOD blue-MOD vase  
'Chinese blue vase'

Note also the following contrast given in Nagano (2016), which is identical to Watanabe's example (112) with the difference that the *i*-adjective *chiisa-i* 'small' is used instead of the *rentaishi chiisa-na* 'small'.

(119) DIMENSION vs. NATIONALITY

- a. chiisa-i Chūgoku-no kabin  
small-MOD China-MOD vase  
'small Chinese vase'
- b. Chūgoku-no chiisa-i kabin  
China-MOD small-MOD vase  
'Chinese small vase'

(Nagano, 2016, 57)

These facts emphasize that neither *ordering restrictions* nor *adjacency to the head noun* are valid criteria for the equivalents of relational modifiers in Japanese, which is expectable considering the broad direct domain of Japanese.

#### 4.3.2.8 Further issues: Nominal status and predicative use

There are two remaining issues concerning relational modifiers. First, since relational modifiers are rather nominal, they make up a prototype of direct nominals in Japanese as also mentioned in Watanabe (2012) for modifiers of NATIONALITY and MATERIAL. This is shown below via productive combination with case particles (120) and the inability to be nominalized (121).

- (120) Chūgoku-ga kawat-ta.  
China-NOM change-PST  
'China has changed.'

- (121) \*Chūgoku-sa  
China-ness

A further piece of evidence for nominal status is the fact that these modifiers can be modified themselves leading to bracketing ambiguities. As shown in (122), the *i*-adjective *kata-i* 'hard' can either modify *pan* 'bread', skipping over *komugi* 'wheat(-en)' which modifies *pan* 'bread'. On an alternative reading, *kata-i* modifies *komugi* 'wheat(-en)' yielding the DP *kata-i komugi* 'hard wheat' which in turn modifies *pan*.<sup>36</sup>

- (122) kata-i komugi-no pan  
hard-MOD wheat-MOD bread  
1.  $\approx$  'hard wheat(-en) bread'  
2.  $\approx$  'bread out of hard wheat'

Another cross-linguistic phenomenon concerns predicative use. Recall that relational adjectives in various languages can be used predicatively, either as a result of a

<sup>36</sup>Such bracketing ambiguities can be compared to cases of what Nikolaeva and Spencer (2020) call *syntagmatic mixing*. In languages such as Selkup and other Samoyedic languages, relational adjectives, albeit being subject to derivational operations, can still be modified. In the Japanese case however, this seems closer to what Spencer (2005) calls *indeterminacy*.



class shift to a qualitative adjective, or when they have a *kind-reading* (McNally and Boleda, 2004). Another mechanism was noun ellipsis, facilitated in contrastive statements. In Japanese, as briefly mentioned in Chapter 4.2.1, relational modifiers also become possible in predicative position when they are connected to a certain kind of suffixes, referred to as *bound markers* or *relational nouns* in Nagano (2016); Shimada and Nagano (2018). When these markers, which roughly correspond in their meaning to relational modifiers, are added to a relational modifier, a new entity called *expanded modifiers* in Nagano (2016, 54) is formed. These expanded modifiers are not only more specified in their meaning, they are also productive in predicative position. This is illustrated below where the pair *chīzu* ‘cheese’ and *iri* ‘put inside’ form *chīzu-iri* ‘(with) cheese put inside’. As Levi (1978) has argued for similar cases in English, establishing modifier and modified noun as a set facilitates predicative use. Interested readers are referred to Appendix A for an overview.

- (123) a. *chīzu-iri-no*            *pan*  
           cheese-put.in-MOD bread  
           ‘bread with cheese put inside’  
       b. *kono pan-wa*    *chīzu-\*(iri)-da*  
           this bread-TOP cheese-put.in-COP  
           ‘This bread is with cheese inside.’

Nouns to which the adjective-forming suffix *-TEKI* is added also become possible in predicative position and gradable. As mentioned at the beginning of this chapter, this suffix was argued originally by Bisetto (2010) to form relational adjectives in Japanese, but the examples below, illustrated with NATIONALITY adjectives with a THEME reading, clearly show that a class shift to a qualitative adjective occurred (see also Nagano 2016, 48).

- (124) a. *Ano kōgeki-wa*    *America-\*(teki)-d-at-ta.*  
           this attack-TOP America-ADJ-COP-PRED-PST  
           ‘This attack was (typical) American.’  
       b. *totemo America-\*(teki)-na kōgeki*  
           very    America-ADJ-na    attack  
           ‘very American (= typical for Americans) attack’

#### 4.3.2.9 Summary: Characteristics of relational modifiers in Japanese

Taking all these facts into account, Japanese does exhibit a category that is equivalent to the cross-linguistic category of relational adjectives. Due to their unclear lexical status, they are referred to as *relational modifiers*. They show the following characteristics:

- (125)    a.    They generally cannot be used predicatively.  
          b.    They are not gradable.  
          c.    They cannot be coordinated.  
          d.    They can be used as real nouns.  
          e.    However, they do not show the typical ordering restrictions and strict adjacency to the head noun associated with relational adjectives in other languages.

### 4.3.3 Direct qualitative modifiers

Those authors who denied the existence of direct modification in Japanese (Sproat and Shih, 1991; Laenzlinger, 2011) only investigated modifiers of categories such as DIMENSION, SHAPE and QUALITY, those categories which are related to *qualitative* modification in European languages, where *qualitative* is used contra *relational*. In a sense, those categories form the most basic adjectival categories. The fact that no direct modifiers seem to be available for these categories in Japanese was thus directly related to the alleged absence of direct modification. Importantly, as also mentioned in the introduction, most modifiers of these categories are not direct-only in other languages either, and the status of a certain qualitative adjective in English does only become obvious in contexts of ordering (Cinque, 2010; Laenzlinger, 2011).

However, Japanese exhibits several lexemes that one could classify as direct-only qualitative modifiers and these cases are worth discussing.<sup>37</sup> Some potentially direct modifiers denoting SHAPE were discussed above (*hishigata-no* ‘diamond-shaped, rhombic’; *yūkei-no* ‘tangible’). Examples of direct modifiers of the category DIMENSION are the lexemes *ōki* ‘big’ and *chiisa* ‘small’. Those can appear either with I or with NA, without a substantial change in meaning. However, they never appear with the copula DA (126-d), only the *i*-variant can be used predicatively.

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<sup>37</sup>I have annotated (some) modifiers for their semantic category, but, of course, this is sometimes quite subjective. In any case, since the relevant functional projections are not active in Japanese, the semantic category of a modifier is not crucial.

- (126) a.  $\bar{o}ki-i$  /  $chiisa-i$  ie  
big-MOD / small-MOD house  
'big / small house'
- b. Kono ie-wa  $\bar{o}ki-i$  /  $chiisa-i$ .  
this house-TOP big-COP / small-COP  
'This house is big / small.'
- c.  $\bar{o}ki-na$  /  $chiisa-na$  ie  
big-MOD / small-MOD house  
'big / small house'
- d. \*Kono ie-wa  $\bar{o}ki-da$  /  $chiisa-da$ .  
this house-TOP big-COP / small-COP  
(‘This house is big / small.’)

There is also no use as adverbials or inchoatives attested for the *na*-version of these lexemes, contrary to the corresponding *i*-adjective, where I inflects to KU.

- (127)  $\bar{O}ki-k\mathbf{u}$  / \* $\bar{o}ki-n\mathbf{i}$  nat-ta.  
big-ADV / big-ADV become-PST  
'Became big.'

Therefore, the *na*-variants of these lexemes are also classified as *rentaishi*, the topic of Chapter 4.3.6. An additional factor to discuss is that they are in fact gradable as illustrated below. Remember, however, that non-gradability is an accompanying factor of non-predicative modifiers, not a prerequisite (Cinque, 2010, 47).

- (128) Ichi-ban  $\bar{o}ki-na$  no-wo eran-da.  
number-one big-MOD LN-ACC choose-PST  
'X chose the biggest one.' (Sekine, 1937, 1: 28)

Potential other direct modifiers of DIMENSION are documented in the data set, for instance *ōgata-no* 'large-scale' (709 attr., 73 pred.) and *atsude-no* 'thick' (189 attr., 2 pred.). However, both are, in principle, possible in predicative position, albeit only marginally attested. This holds especially for *atsude-no* whose possibility in predicative position was corroborated by some speakers despite its very few occurrences.

Clearer cases of direct modifiers are found among the category QUALITY, which is *per se* one of the more, perhaps the most, versatile category. For instance, *gokujō-no* 'of highest quality' (142 attr. 5 pred.) or *seirai-no* 'innate' (92 attr., 0 pred.) yield few to none predicative occurrences.

Direct modifiers of the category COLOR are *shikkoku-no* 'pitch black' (165 attr., 2 pred.) and *konpeki-no* 'deep blue' (46 attr., 0 pred.).

It is interesting to note that direct modifiers of some less canonical categories given in the very fine-grained hierarchy of Scott (2002, 106, 114), can be found as well. Examples include *gokkan-no* 'freezing cold' (44 attr., 5 pred.) representing

the category TEMPERATURE or *kyūsoku-na* ‘fast, rapid’ (599 attr., 4 pred.) for the category SPEED, illustrated below for *gokkan-no*.

- (129) a. *gokkan-no*      *kisetsu*  
           freezing.cold-MOD season  
           ‘freezing cold season’  
       b. ??*Ima-no*    *kisetsu-wa* *gokkan-da*  
           now-MOD season-TOP freezing.cold-COP  
           ‘The season at the moment is freezing cold.’

Finally, there are some direct-only modifiers that do not fit any semantic category present in established hierarchies, such as *kidai-no* ‘rare’ (71 attr., 0 pred.) and *kaishin-no* ‘spot-on; critical’ (58 attr., 0 pred.).

- (130) a. *kaishin-no*    *ronbun*  
           spot.on-MOD essay  
           ‘a spot-on paper; a paper written to one’s full satisfaction’  
       b. \**Kono ronbun-wa kaishin-da.*  
           this    paper-TOP spot.on-COP  
           (‘This paper is spot-on.’)  
       (131) a. *kidai-no*    *sainō*  
               rare-MOD talent  
               ‘rare talent’  
           b. \**Kare-no sainō-wa*    *kidai-da.*  
               he-MOD talent-TOP rare-COP  
               (‘His talent is rare.’)

Since the functional projections inside the direct domain are not governed by semantic categories, the problem in which functional projections to sort these modifiers disappears. They are merged in the direct domain, but in an unspecific functional projection.

- (132) a. *kidai-no*    *sainō*  
           rare-MOD talent  
           ‘rare talent’ (= (131-a))  
       b.
- ```

graph TD
    DP_internal --> Branch1[ ]
    DP_internal --> D_prime_internal[D'_internal]
    D_prime_internal --> FP_direct[FP_direct]
    D_prime_internal --> D[D]
    FP_direct --> ModP[ModP]
    FP_direct --> F_prime_direct[F'_direct]
    ModP --> Mod_prime[Mod']
    Mod_prime --> XP[XP]
    XP --> kidai[kidai]
    Mod_prime --> Mod[Mod]
    Mod --> no[no]
    F_prime_direct --> NP[NP]
    NP --> saino[sainō]
    F_prime_direct --> F[F]
    
```

4.3.4 Idiomatic expressions

Another source for direct modification are modifiers involved in idiomatic expressions, although it might be better to speak of *modifiers in non-free phrases*.³⁸ Recall that in English, only prenominal adjectives can elicit an idiomatic reading with their corresponding head noun, as for example *white* in *white lie* (Cinque, 2010, 87–88). The only idiomatic expression with a single modifier given in the literature is as follows, given in Nagano and Shimada (2015, 122). As expected, predicative use is impossible with the idiomatic meaning.

- (133) a. aka-no tan'in
red-MOD stranger
'a total stranger' (Lit. 'a red stranger')
- b. *Kono tan'in-wa aka-da
this stranger-TOP red-COP
only meaning: 'This stranger is red.'

Several more idiomatic modifiers in Japanese can be identified, where “idiomatic” refers to modifiers that only co-occur with a certain noun, or a certain group of nouns with similar meaning. Thus, it is no prerequisite that a new meaning is formed, as in the example *white lie*. One example of an idiomatic modifier, already mentioned in Chapter 3.1, is *kōki-no* ‘curious’. This lexeme yields 69 results in attributive position, against zero with the copula. 61 out of those 69 occurrences are with nouns with the meaning ‘gaze, look’, among them *me* (also ‘eye’), *manazashi* ‘gaze’ and *shisen* ‘gaze, look’. The restriction in predicativity, and also gradability, is shown below.³⁹

³⁸See Mel'čuk (2012); Köhlich (2024) for the difference between idiomatic expressions and collocations.

³⁹Compared to *me*, *manazashi* and *shisen* can be described as Sino-Japanese nouns with the same meaning from a higher register. Ken Hiraiwa has told me that to his native speaker ears the modification of *manazashi* ‘gaze’ via *kōki-no* ‘curious’ seems to be truly idiomatic.

- (134) a. Shikashi kare-ga kōki-no me-de mi-rare, ōku-no
 but he-NOM curious-MOD eye-INS see-PASS many-MOD
 hito-ga kare-no hanashi-wo kiki-ta-gat-ta koto[...]
 people-NOM he-MOD story-ACC hear-VOL-seem-PST fact
 ‘But the fact that he was viewed with curious eyes (a curious gaze)
 and many people wanted to hear his story [...]’⁴⁰
- b. *Kare-ga mi-rare-ta me-wa kōki-dat-ta.
 he-NOM see-PASS-PST eye-TOP curious-COP-PST
 (‘The eye/gaze he was viewed with was curious.’)
- c. *totemo kōki-no me
 very curious-MOD eye
 (‘very curious gaze’)

Other examples include the modifiers *higō-no* ‘unnatural’ (47 attr., 0 pred.) which occurs only with *shi* ‘death’ and *saigo* ‘end’ (also in the sense of ‘death’) (135) and *anmoku-no* ‘tacit’ (350 attr., 0 pred.), mostly with *ryōkai* ‘agreement’.⁴¹

- (135) a. Nanninka-no Kirisuto-kyōtō-wa higō-no
 several.people-MOD Christianity-believer-TOP unnatural-MOD
 shi-wo toge-ta.
 death-ACC meet-PST
 ‘Several Christians met an unnatural death.’⁴²
- b. *Nanninka-no Kirisuto-kyōtō-ga toge-ta shi-wa
 several.people-MOD Christianity-believer-NOM meet-PST death-TOP
 higō-d-at-ta.
 unnatural-PRED-COP-PST
 (‘The death several Christians met was unnatural.’)

These data points suggest three things: First, idiomatic expressions do exist in Japanese. Second, they seem to offer another source for direct modification. Third, they are mostly found in the domain of *no*-modifiers.

As shown in Chapter 3.1, Svenonius (2008) suggests that idiomatic modifiers are merged as specifiers of category-less root phrases $\sqrt{P}$, embedded under nP (Marantz, 1997). They, thus, modify the very core at the noun (Svenonius, 2008, 36–37).

⁴⁰Ogawa, Ryō (2002): *Doreishōnin Soniē - 18-seiki Furansu no doreikōki to Afurika-shakai*, Tokyo: Yamakawa Shuppansha, via BCCWJ, 2023/11/09.

⁴¹Two things are noteworthy here: First, the combination of *higō* ‘unnatural’ and the two nouns are in 32 out of 47 cases part of the larger expression *higō-no shi/saigo-wo toge-ru* ‘to meet an unnatural death’. Second, the modifier *anmoku-no* ‘tacit’ does not yield a single instance of predicative use in the BCCWJ-corpus, but some instances can be found on Google. Nevertheless, it is also given in the literature (Muraki, 2012; Lehmann and Nishina, 2015) as a non-predicative modifier.

⁴²Ōno, Kazumichi (2001): Translation of Michelet, Jules: *Bible de l’humanité*. Tokyo: Fujiwara shoten, original: 1864, via BCCWJ, 2023/11/09. Note that potentially, sticking to the religious dimension of the modifier, a translation along the line of ‘undignified’ might be more appropriate.

Hence, their position is the lowest inside the direct domain, and therefore within the whole DP.

Concrete evidence for this claim comes from the fact that nothing can intervene between the idiomatic modifier and the head noun, i.e. the idiomatic modifier cannot be higher in the structure. This is shown for English in Svenonius (2008, 36–37), and also holds in Japanese (136).

- (136) a. minna-no kōki-no manazashi
 everyone-MOD curious-MOD gaze
 ‘everyone’s curious gaze’
 b. *kōki-no minna-no manazashi
 curious-MOD everyone-MOD gaze

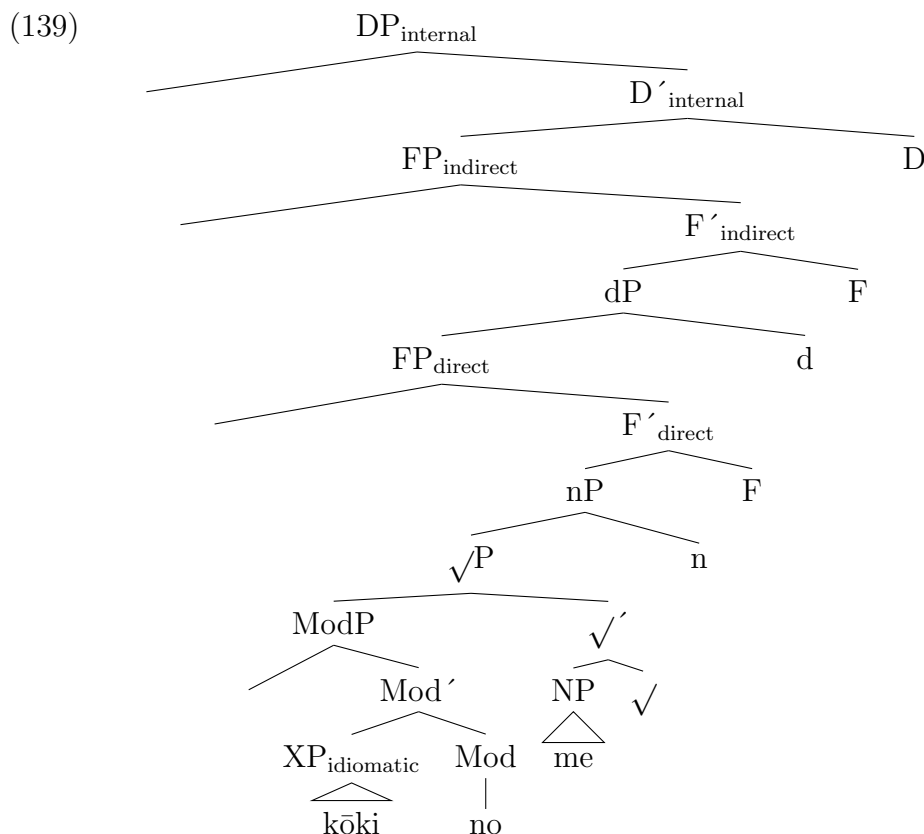
Crucially, this behavior is not due to *minna-no* ‘everyone’s’. As shown below, this modifier can intervene between non-idiomatic modifiers and the head noun.

- (137) a. minna-no aka-i doresu
 everyone-MOD red-MOD dress
 b. aka-i minna-no doresu
 red-MOD everyone-MOD dress
 ‘everyone’s red dresses’

The adjacency facts apply to other idiomatic modifiers as well, for example the combination *aka-no tan’in* ‘red stranger’, tested with an intervening relative clause.

- (138) a. [dare-mo shira-na-i] aka-no tan’in
 everyone-EMP know-NEG-MOD red-MOD stranger
 ‘the total stranger that no one knows’
 b. *aka-no [dare-mo shira-na-i] tan’in
 red-MOD everyone-EMP know-NEG-MOD stranger

This thus yields the following position for idiomatic modifiers repeated from Chapter 3.1, now including the internal structure of the idiomatic modifier under ModP.



Two more things are noteworthy. First, this root phrase *only* hosts idiomatic modifiers in Japanese. Kim (2019, 130), who adopts the category $\sqrt{P}$ from Svenonius (2008), argues that this phrase hosts idiomatic modifiers, but also modifiers which receive an internal theta role from the noun, thus compound-like constructions, in Korean. As evidence, she shows that no other modifier can intervene between such modifiers and the head noun in Korean either, parallel to idiomatic constructions. In the following example, *kyengcey* ‘economy’ is the internal argument of *kayhyek* ‘renovation’ and must be immediately adjacent to the noun (Kim, 2019, 130), reflecting essentially the category of relational modifiers discussed above.

- (140) a. sin kyengcey kayhyek (Korean)
 new economy renovation
 ‘a new renovation of the economy’
 b. *kyengcey sin kayhyek
 economy new renovation (Kim, 2019, 130)

However, this is not the case in Japanese, as the translation of the examples shows. Here, permutation is easily possible.⁴³

⁴³This also extends to some examples given by Laenzlinger (2011), such as (141). Here, a thematic modifier with a THEME role (*machi-no* ‘city’) and another modifier (*saikin-no* ‘recent’), according to Laenzlinger (2011) have to appear in this order, thus showing the opposite pattern to Kim’s.

- (142) a. arata-na keizai-no kaikaku
new-MOD economy-MOD renovation
b. keizai-no arata-na kaikaku
economy-MOD new-MOD renovation
'new renovation of the economy'

Second, Okami (2012) has argued that those COLOR modifiers which can appear with -I and -NO can only receive an idiomatic reading in the *i*-variant. She gives the following examples.

- (143) a. Tarō-to Hanako-wa aka-i / *aka-no ito-de
Taro-and Hanako-TOP red-MOD / *red-MOD string-INS
musub-are-te-i-ru.
bind-PASS-GER-be-PRS
'Taro and Hanako are destined to be together.' (Lit. 'bound by a red string')
b. Kare-ni-wa kuro-i / *kuro-no uwasa-ga ar-u.
he-DAT-TOP black-MOD / *black-MOD rumor-NOM be-PRS
'There are dark rumors about him.' (Okami, 2012, 96)

Interestingly, such color items do allow permutation when occurring with another modifier, as opposed to the examples above.

- (144) a. Kare-ni-wa hen-na kuro-i uwasa-ga ar-u.
he-DAT-TOP strange-MOD black-MOD rumor-NOM COP-PRS
b. Kare-ni-wa kuro-i hen-na uwasa-ga ar-u.
he-DAT-TOP black-MOD strange-MOD rumor-NOM COP-PRS
'There are strange dark rumors about him.'

This might suggest that these are in fact not idiomatic modifiers, or at least they are not merged in the same position, the root phrase.

4.3.5 Quantificational modifiers

As already shown in Chapter 3.1.3.2, another source for direct modifiers are quantifiers or more precisely quantificational modifiers. Many quantificational modifiers are direct-only, while some are only prototypically attributive. Examples of quantificational modifiers which cannot be used predicatively include *kazukazu-no* 'many', *amata-no* 'a lot, many' and *shuju-no* 'various'.

-
- (141) machi-no saikin-no hakai
city-MOD recent-MOD destruction
'the recent destruction of the city' (Laenzlinger, 2011, 227–228)

However, this DP is probably not very natural to begin with. Furthermore, without context, permutation is easily possible as well.

- (145) a. Kare-wa kazukazu-no shō-wo tot-ta.
 he-TOP many-MOD prize-ACC take-PST
 ‘He received many prizes.’
 b. *Kare-ga tot-ta shō-wa kazukazu-desu.
 he-NOM take-PST prize-TOP many-COP.POL
 (‘The prizes he received were many.’)
- (146) a. Kare-wa shuju-no / amata-no shikaku-wo
 he-TOP various-MOD / many-MOD qualification-ACC
 mot-te-i-ru.
 have-GER-be-PRS
 ‘He has various / many qualifications.’
 b. *Kare-ga mot-te-i-ru shikaku-wa shuju-desu /
 he-NOM have-GER-be-PRS qualification-TOP various-COP.POL /
 amata-desu.
 many-COP.POL
 (‘The qualifications he has are various / many.’)

In Chapter 3.1.3.2, it was argued that quantificational modifiers are merged first inside the direct domain, since they exhibit direct behavior, but can subsequently move to the left periphery, as they can precede not only demonstratives (147-a) but also other direct-only modifiers (147-b). Probably they do this to check off a [quantificational] feature.

- (147) a. kazukazu-no ano mondai
 many-MOD that problem
 ‘those many problems’
 b. takusan-no komugi-no pan
 a.lot-MOD wheat-MOD bread
 ‘many wheat(-en) (pieces of) bread’

4.3.6 Rentaishi

Finally, I will discuss the controversial word class of *rentaishi* 連体詞, also called *fukutaishi* 副体詞, arguably a language-specific source for direct-only modifiers. This word class is comprised of several lexemes with different morphology, derived from other word classes and now fossilized so that they are only used as attributive modifiers. This is the reason why *rentaishi* are typically named by many Japanese linguists working in more traditional frameworks as non-predicative lexemes whose only function is to modify nouns making them the only group of direct modifiers acknowledged in the literature for a long time (see the newer works mentioned throughout this chapter).

The word class *rentaishi* makes up a controversial case, first, because it was created more upon historical and ideological than on syntactic grounds at the beginning

of the 20th century, and second, since it is not accepted as an individual word class by all linguists, both in- and outside of Japan. This is also the reason why many linguists would probably not classify the lexemes presented in the following pages as direct-only modifiers, since they strongly resemble fixed expressions. Nevertheless, I will show that they are quite close to idiomatic, and sometimes also non-intersective, modifiers; and since the relevant lexemes only serve as attributive modifiers, they deserve a discussion here. I will give a brief introduction, and then discuss different subgroups. For more information the reader is referred to Köhlich (2020, to appear).

4.3.6.1 Introduction: Terminology, diachrony, criteria, subgroups

The term *rentaishi* literally translates to *adnominal-noun-word* and is in English usually translated as ‘adnoun’ or ‘prenoun’. German equivalents are *Adnomen*, sometimes *Adnominalie* (Rickmeyer, 1995). *Rentaishi* were incorporated into the Japanese word class system at the beginning of the Meiji-Period (1868-1912). The important linguists of the time, especially MATSUSHITA Daizaburō (Matsushita 1976 (1928) and also Yamada 1908; Mitsuya 1932; Sekine 1937; Kieda 1938; Tokieda 1941; Yuzawa 1979 (1940)), realized that there are several lexemes in Japanese that are restricted to attributive use, do not inflect and whose sole function is to modify nouns. They also realized that such lexemes bear striking similarities to the *defective* adjectives in Western (mostly Dutch and English) languages, which only inflect dependent on the noun they modify (Sugimoto, 1967, 1991; Kim, 2006). The new word class *rentaishi* was created to incorporate all those defective lexemes, becoming an equivalent of the Western adjective.

Another important criterion of this word class is the morphological diversity of its members. Being fossilized derivatives from other word classes and frozen in their morphological form and usage domain, explains not only the restriction to attributive position, it also predicts the absence of other inflectional forms. This is a major point of deviation from all major Japanese word classes which since the language reform of the Meiji-Period have been predominantly categorized on the basis of their shared morphological paradigms (Nitta *et al.*, 2000).

Starting with an example, there are several lexemes which look verbal on a morphological level – because they are indeed derived from verbs – but only appear as prenominal modifiers. In fact, all those verbal *rentaishi* are derived from verbs fossilized in their attributive inflectional form (*rentaikei*). They include verbal derivatives ending in the simple non-past form such as *aru* ‘(a) certain’ (148),⁴⁴ derivatives end-

⁴⁴This *aru* is homophonous to the existential/copular verb *ar-* which is why some authors (Lehmann and Nishina, 2015) treat these two elements as the same. However, the copula verb *ar-* is generally incompatible with animate nouns, therefore it cannot be the same element as *aru* in (148).

ing in the morpheme -IU, which signaled passive, potential or spontaneity, and itself inflected for attributive form, such as *iwayuru* ‘so-called’ (149), and those ending in the past morpheme TA, such as *taishita* ‘considerable’ (150). None of these modifiers can be used predicatively or is gradable, highlighting the role of *rentaishi* for the present study.

- (148) a. aru gakusei
 certain student
 ‘a certain student’
 b. *Gakusei-wa aru.
 student-TOP certain
 (‘This student is a certain.’)
 c. *motto / totemo aru gakusei
 more / very certain student
 (‘a more / very certain student’)
- (149) a. iwayuru gakusei
 so-called student
 ‘a so-called student’
 b. *Kono gakusei-wa iwayuru.
 this student-TOP so-called
 (‘This student is so-called.’)
 c. *motto / totemo iwayuru gakusei
 more / very so-called student
 (‘a more / very so-called student’)
- (150) a. taishita sa
 considerable difference
 ‘a considerable difference’
 b. *Sa-wa taishita.
 difference-TOP considerable
 (‘The difference is considerable.’)
 c. *motto / totemo taishita sa
 more / very considerable difference
 ‘more / very considerable difference’

Other potential members of *rentaishi* are derived from adjectives and appear in the attributive form of the premodern copula *nari* or *tari*.⁴⁵ Again, the relevant lexemes in Modern Japanese are restricted to attributive use and are non-gradable.

⁴⁵*tari* can be seen as an allomorph which was restricted to Sino-Japanese lexemes.

- (151) a. tannaru machigai
 mere misunderstanding
 ‘a mere misunderstanding’
 b. *Machigai-wa tannaru
 misunderstanding-TOP mere
 (‘The misunderstanding is mere.’)
 c. *motto / totemo tannaru machigai
 more / very mere misunderstanding
 (‘more / very mere misunderstanding’)
- (152) a. shutaru gen’in
 main reason
 ‘the main reason’
 b. *Gen’in-wa shutaru.
 reason-TOP main
 (‘The reason is main.’)
 c. *motto / totemo shutaru gen’in
 more / very main reason
 (‘more / very main reason’)

Recall the lexemes *ōki* ‘big’ and *chiisa* ‘small’ which can either co-occur with I or with NA, but never with the copula DA. The *na*-variant of these lexemes has traditionally also been called *rentaishi*.⁴⁶

- (153) *Kono ie-wa ōki-da.
 this house-TOP big-COP
 (‘This house is big.’) (= (126-d))

Finally, there are some lexemes that are counted as *rentaishi* which end in GA, namely *waga* ‘my/our’ and *onoga* ‘my’, and some that end in NO such as *honno* ‘only, mere’.⁴⁷

In this sense, it is debatable, whether one should analyze *honno* as a fossilized lexeme, with /no/ being part of the stem, or whether one should maintain the analysis presented here of /no/ as a syntactic head. An argument for the former is that the first result for *honno* on the Corpus of Historical Japanese (CHJ) dates back to the year 900. Maintaining an analysis of *rentaishi* as a closed class of historical lexemes, I tentatively opt for this analysis, although nothing immediately hinges on this. Note however, that this also means that not every direct-only *no*-modifier

⁴⁶Another such case is *okashi* ‘strange’. The question remains whether there is one stem with two possible attributive morphemes, or if the respective lexemes are derived from an *i*-adjective-counterpart, have become fossilized and are now lexicalized as uninflectable *rentaishi*. Both analyses are available (see also Oshima and Hayashi 2021a, Fn. 15), but I will assume that *ōki-i* and its counterparts are *i*-adjectives, whereas *ōki-na* belong to a different word class. This conditions the spell-out of MOD. See also Chapter 4.4.3.2.

⁴⁷Now the nominative case particle, GA is the archaic genitive marker, accounting for the archaic flavor of DPs in which the relevant *rentaishi* appear.

(sorted in group 5) in the data set is a *rentaishi*, and *rentaishi* is here not treated as a feature, but as a word class of historical derivatives, despite recent efforts of several authors to argue for exactly that (Muraki, 1996, 2009, 2012, 2015; Oshima and Hayashi, 2021a,b,a; Abe *et al.*, 2022). See also Köhlich (to appear) for discussion.

The following list gives an overview of the morphological subgroups of *rentaishi*.⁴⁸

Group	Examples	Translation
verbal <i>rentaishi</i>	verbs inflected for attributive form, non-past	<i>aru</i> ‘(a) certain’; <i>saru</i> ‘previous’; <i>akuru</i> ‘following’; <i>kitaru</i> ‘ahead’
	verbs combined with morpheme -IU, inflected for attributive form	<i>iwayuru</i> ‘so-called’; <i>arayuru</i> ‘all kinds of’
	verbs ending in -TA, past	<i>taishita</i> ‘considerable’; <i>tonda</i> ‘remarkable’
adjectival <i>rentaishi</i>	three lexemes in combination with element -NA	<i>ōkina</i> ‘big’; <i>chiisana</i> ‘small’; <i>okashina</i> ‘strange’
	lexemes ending in attributive forms of premodern copula -NARU/TARU	<i>tannaru</i> ‘mere’; <i>shutaru</i> ‘main’
<i>rentaishi</i> with <i>ga</i> and <i>no</i>	lexemes ending in -GA, often in certain archaic sayings	<i>waga</i> ‘my/our’; <i>onoga</i> ‘my’
	lexemes ending in -NO	<i>rei-no</i> ‘ordinary, aforementioned’; <i>hon-no</i> ‘only, mere’

Table 4.2: List of morphological groups among the group of *rentaishi*

⁴⁸This list is compiled from Mitsuya (1932); Kieda (1938); Suzuki (1972); Komatsu (1973); Tokieda (1941); Mio (1958); Kai (1980); Kim (2006); Matsubara (2009), see Köhlich (2020, 181–182). Traditionally, also the demonstratives *kono* ‘this’ and *kon-na* ‘such a’ have been analyzed as *rentaishi* in the relevant studies. I will not include them in the present study where they are treated as demonstratives.

4.3.6.2 Summary

The morphologically diverse group of *rentaishi* “created” in the beginning of the past century constitutes another group of direct-only modifiers in Japanese, and a language-specific one as such. From a syntactic point of view, it is striking that these relics from Classical Japanese serve an important role in modern syntactic theory. They also fit an important characteristics argued by Cinque (2010) for direct-only modifiers: they are *functional*, meaning they make up a closed group. This is expectable given the fact that they are fossilized. Furthermore, from a perspective of word class status, these lexemes resist a unifying analysis based on morphological criteria and rather the absence thereof lends itself as a criterion.

Also, note that there are several *rentaishi* with similar meanings to non-intersective modifiers (Kamp and Partee, 1995; Panayidou, 2013) discussed at the beginning of this section. These include *aru* ‘a certain’ *tannaru* ‘mere’ and *shutaru* ‘main’. It is remarkable that an awareness of the solely attributive character of these lexemes existed long before the concept of non-intersectivity had been established.

In a nutshell, this group of historically fossilized attributive modifiers is highly relevant for the discussion on direct modification and brings in a language-specific dimension, adding to the equivalents of cross-linguistic groups of direct modifiers looked at so far.

4.3.7 Summary

The preceding discussion has shown that Japanese does exhibit direct modifiers, contrary to what has often been claimed in the literature. When taking stock and applying a large scale empirical validation, the number of direct modifiers is much larger than it is suggested according to the few studies concerned with them. Furthermore, it was again highlighted that the vast majority of direct modifiers can be found among the group of *no*-modifiers.

The following groups of direct modifiers could be established.

- (154)
- a. Non-intersective modifiers
 - b. Relational modifiers
 - c. Qualitative modifiers of standard categories (QUALITY etc.), potentially of different categories
 - d. Idiomatic modifiers
 - e. Quantificational modifiers
 - f. *Rentaishi*
-

4.4 Implications and discussion

Having established the groups of direct modifiers present in Japanese, I will now discuss the structure of the direct domain more closely, with special focus on the consequences and implications Japanese offers for cartography.

First, the following characteristics of the direct domain were identified:

- (155)
- a. The direct domain does not consist of functional projections dedicated to specific semantic categories, but of unspecific functional projections.
 - b. This has the consequence that no hierarchical order according to these specific functional projections exists, hence no ordering restrictions of modifiers are expected in linear order either.
 - c. Idiomatic modifiers occupy a specific position, directly adjacent to the noun.
 - d. Quantificational modifiers are direct, but typically move out of the direct domain to the left periphery.

These points inevitably raise critical questions on a cross-linguistic dimension and for any cartographic account. An immediate question is, why is Japanese so lenient when it comes to ordering, but other languages are not, i.e. why do, for instance, English or German dictate strict ordering on their modifiers implicating that structural hierarchies are at stake in such languages? But then what is the reason that such hierarchies are active in English, but not in Japanese? Since it is a core principle of cartography that structures are uniform, such deviations should not exist, and thus pose a serious challenge. This also extends to the concept of UG in general. From a UG perspective, the question arises whether this distribution is simply random and, furthermore, whether we only find these two extremes cross-linguistically. Or could it be possible that there exist languages that exhibit a mixed behavior, that are situated between English and Japanese? If so, how would children learn to determine whether they need to acquire ordering restrictions of direct modifiers based on specific categories or not?

These issues will be addressed in this section. I will argue that although the general spine, the structure of the direct domain, is uniform, its inventory is subject to parametric differences. Furthermore, I will show that there are in-between cases and demonstrate this with two languages: Persian and Modern Greek. These observations will then lend themselves to cross-linguistic generalizations. In the next subsection, I will discuss ordering in Japanese in more detail.

4.4.1 More on ordering in Japanese

As presented in Chapter 2.1, a core assumption of cartography is that direct qualitative modifiers (generally adjectives) show ordering restrictions when two or more occur within the same DP (see Scott 2002; Cinque 2010; Laenzlinger 2011). This means that whether modifiers adhere to the expected ordering principles or not directly allows to draw generalizations regarding their modification type. The non-canonical order of a DIMENSION and a COLOR adjective in languages such English, for instance, can straightforwardly be analyzed as the result of *red* being merged in the indirect domain, thus higher. Otherwise, the order *big* > *red* should be impossible given an out-of-the-blue sentence with no specific context. This is shown below.

(156) [red_{indirect} [big_{direct} [book]]]

On the other hand, it was sufficiently demonstrated that ordering and modification type are orthogonal issues in Japanese and that (most) modifiers, whether ambiguous or direct-only, are not expected to show ordering restrictions. Thus, the situation given above does not hold automatically.

The general freedom in ordering between such qualitative modifiers was quantitatively measured in an earlier study (Köhlich, 2018). In this study, which was carried out on the BCCWJ, I tracked combinations of modifiers belonging to the categories QUALITY, DIMENSION, SHAPE and COLOR inside and across the two canonical adjective groups within the same DP in order to establish whether their semantic category, as well as their word class, has an effect on ordering.

The results are displayed in the following table. The leftmost columns depict the relevant categories of the first adjective in a given DP, the second adjective corresponds to the labels written horizontally in the top row.

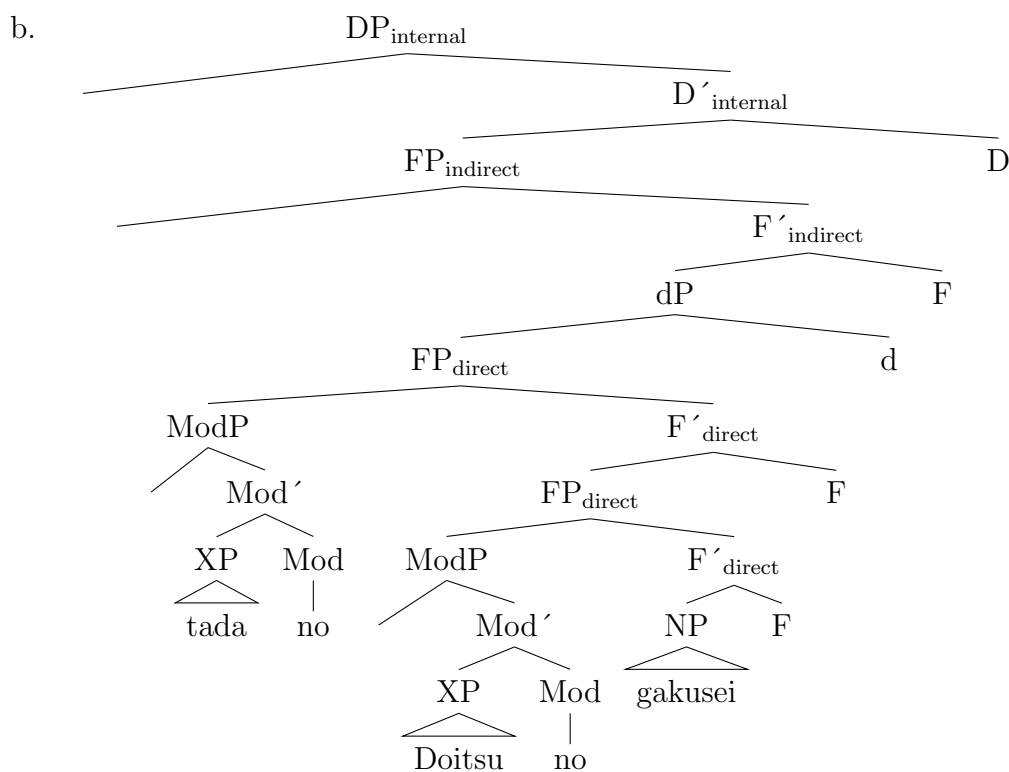
Order	QUALITY	DIMENSION	SHAPE	COLOR
QUALITY	290	40	16	33
DIMENSION	45	57	16	18
SHAPE	8	4	5	6
COLOR	15	19	10	5

Table 4.3: Ordering Among Japanese Adjectives on the BCCWJ

Although admittedly there are not many co-occurrences in the first place, which has to do with the design and size of the corpus (see Köhlich 2023), the results confirm a lack of ordering restrictions and confirm the overall theme of the present study.⁴⁹

The present study adds to this observation that ordering restrictions are also absent amongst and across direct-only modifiers. This holds for pairs of relational modifiers, combinations of relational and qualitative modifiers, and relational with non-intersective modifiers respectively. The different positions modifiers exhibit in linear order on the surface are simply the results of the different positions they occupy inside the DP and the order of merge. For instance, in example (76) above, the non-intersective modifier *tada-no* ‘mere’ precedes the relational modifier *Doitsu-no* ‘German’ in linear order if it occupies a higher position, i.e. if it is merged later.

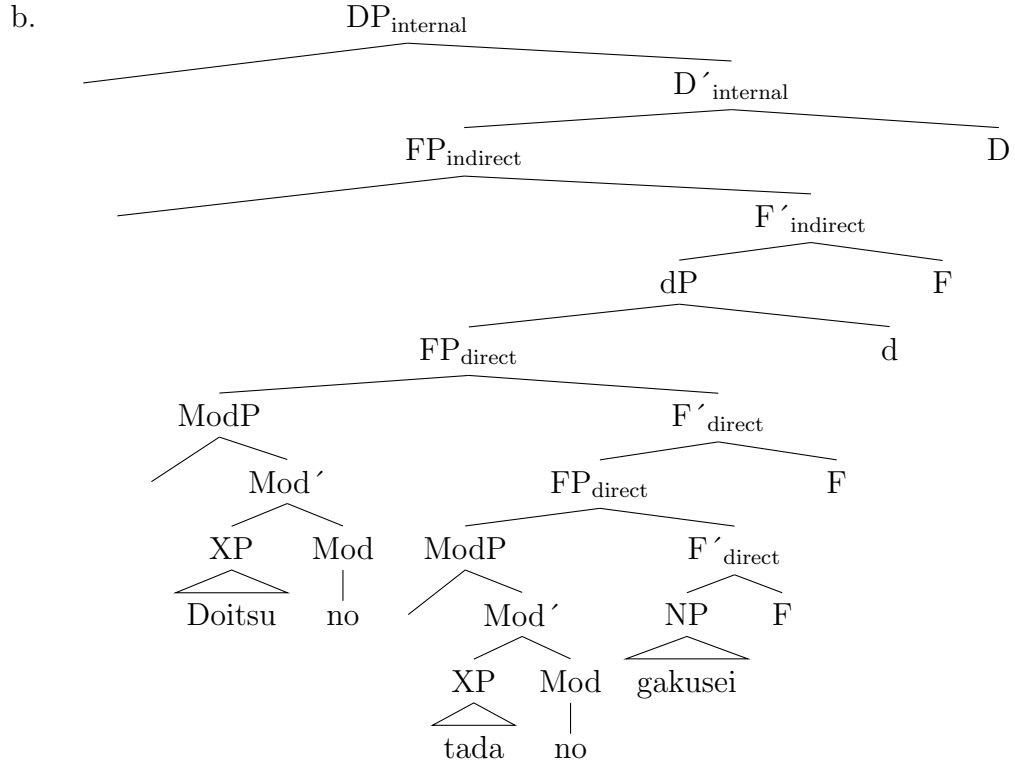
- (157) a. *tada-no Doitsu-no gakusei*
 mere-MOD German-MOD student
 ‘a mere German student’ (= (76-a))



On the other hand, the reverse order *Doitsu-no tada-no gakusei*, lit. ‘German mere student’, ungrammatical in English, is a result of *tada-no* occupying a lower position.

⁴⁹Seemingly relevant is the skew towards QUALITY > COLOR over the reverse order, but then again, both categories do not show restrictions with regard to DIMENSION. On the other hand, QUALITY seems to precede SHAPE. In general, the numbers presented here are too little to draw any substantial conclusions from.

- (158) a. Doitsu-no tada-no gakusei
 German-MOD mere-MOD student
 ‘a mere German student’ (= (76-b))



Although both modifiers are situated in the direct domain, which has been verified viz. their status as direct modifiers, their exact position is not predetermined and this scenario is at stake for all direct modifiers with interchangeable ordering.

4.4.2 Language variation in ordering

Assuming a broad direct domain explains the absence of ordering restrictions in Japanese. But it yet fails to explain stricter ordering restrictions in languages such as English or German, which over the years have justified the inventory of functional projections based on specific semantic categories (Cinque, 2010, 2025; Laenzlinger, 2011). The question is: what makes Japanese so different from English, i.e. what determines whether a language has a broad domain with unspecific, or a more articulated domain with specific functional projections?

4.4.2.1 Universal Spine Hypothesis and Units of Language

To answer these questions, I draw ideas from Wiltschko’s *Universal Spine Hypothesis* (USH, Wiltschko 2014, 24–26)j, which offers a radically different perspective from cartography. I argue that this hypothesis enriches this research program in various

ways and offers a solution to the questions presented above.⁵⁰ I adopt the following main ingredients (cf. Wiltschko 2014, ch. 1, 2).

- (159)
- a. There is a universal syntactic spine which holds across all languages and is part of UG.
 - b. Across this spine, we find a universal inventory of hierarchically ordered categories.
 - c. Whether these categories surface in a specific language, however, is subject to language-specific factors.
 - d. Each language has its own language-specific entities, called *Units of Language* in Wiltschko (2014, 1), which derive language-specific categories from universal categories.

The basic idea is as that of a universal DP-spine with three different domains and functional heads within these domains licensing modifiers and giving landing sites. This general DP-spine and its three domains are uniform cross-linguistically: the left periphery, the indirect domain, and the direct domain. The nature of the functional heads are shared with standard cartographic assumptions as well. Functional heads within the direct and indirect domain license direct and indirect modifiers respectively and the left periphery serves mainly as a landing site for modifiers which move there for various reasons.

Furthermore, while the general spine is fixed, and thus variation limited across domains cross-linguistically, variation in the architecture, and thus in the surface order, comes into play within the same domain and this variation is language-specific (for a similar idea in the clause see Ramchand and Svenonius 2014). Specifically, depending on its language-specific inventory, or its Units of Language (UoLs; Wiltschko 2014), each language realizes the categories in each domain in a different way, albeit those differences are systematic.

Languages such as English possess the Units of Language to realize specific categories of direct modifiers such as QUALITY and COLOR. They then combine those with the cross-linguistic spine and inventory of functional projections for direct modifiers to form the language-specific categories F_{QUALITY} and so on. These functional heads license the respective modifiers in the structure and are ordered hierarchically. In linear order, this predicts the fixed order of these direct modifiers.

Japanese, on the other hand, lacks the specific Units of Language for QUALITY and similar categories, therefore does not realize specific functional projections which license modifiers of QUALITY and which are ordered hierarchically. However, the universal DP is shared and this includes the direct domain. The difference between

⁵⁰For the specific spine and its projections proposed by Wiltschko as an alternative to the ones proposed here, see Wiltschko (2014, ch. 2).

the functional heads and projections this domain contains and those of the English type is that the former are unspecific, and not specified for modifiers of specific semantic meanings, word classes, or levels of gradability, and other factors. This means, the functional heads within the Japanese direct domain are also not ordered in a specific hierarchy, and this correctly predicts the absence of ordering restrictions on the surface.

From a cross-linguistic perspective, and a perspective of UG, this means that children must acquire the relevant Units of Language required by the language they are learning in order to see which categories need activating and are embedded into a hierarchy.

Nevertheless, it is evident that the general hierarchy of functional projections in a given domain, if a given language realizes them, is uniform as well, considering the fact that the cross-linguistically attested ordering restrictions exemplified by modifiers of different categories are very similar across languages and variation can be explained by the application of different movement operations (Cinque, 2023, 2025). Therefore, the process of acquiring the relevant hierarchy is not random, but conditioned on acquiring the relevant functional categories.

As argued in Wiltschko (2014), the Universal Spine Hypothesis fares better than its alternatives in a lot of scenarios. The Universal Base Hypothesis (UBH) of early generativism (Chomsky, 1965) assumes uniformity, universality and identical structural make-up across all languages, the very core idea of present cartography (Cinque and Rizzi, 2010, 68). However, it faces two concrete problems: The first are overly specific categories in a certain language that should be present in all languages, yet are not attested. Equally problematic are cases of missing universal categories in a specific language (cf. Wiltschko 2014, 10–19 and ch. 2.), the exact problem observable in Japanese, when it comes to ordering, and essentially the problem of plenitude (Larson, 2021).

The No Base Hypothesis (NBH), on the other hand, attributes a special status to all languages in the sense that every category is language-specific. There are several counter arguments to this hypothesis, but only considering the data points here, it would not explain why the category of direct modifiers is attested in almost every language and why we observe such similar patterns in ordering restrictions across languages. Such a striking commonality does not come about by chance, but instead by cross-linguistic grammatical factors. Without a grammatical baseline, Wiltschko (2014, 19–23) asks, how come learners of so many different languages all acquire the same patterns?

One might object that the Universal Base Hypothesis is, nevertheless, active in Japanese, but that this language makes more use of movement than other languages. There is a logical problem with this objection. If there was an underlining order

of modifiers, then we should be able to identify one order as the base order, or, alternatively, identify factors that ameliorate or deteriorate one order over the other. It should not be the case that both orders are equally perfectly acceptable. I.e., if we ask speakers, and especially if we look at corpus data, we should see that one order is still prevailing. That is because movement is always an additional option that the speaker has to perform and the listener has to parse. Therefore, I argue, the option of two different underlying orders is much more attractive than assuming one underlying order plus movement.

Having established that Japanese lacks the language-specific inventories to realize fine-grained functional projections for direct-only modifiers, while languages such as English do realize said projections, leading to a strict marking of ordering in the latter language, the question is how far language variation is allowed. Can languages do whatever they want? Or are there some principles they have to adhere to? Taking the strict English-type and the loose Japanese-type as two extremes, one would expect that there is some “middle-type” or even a continuum across languages. This is essentially what is predicted by the claims made above. There should be languages that have the Units of Language for some categories but not for others.

If we find such a language, the next question is which intermediate levels of ordering emerge. That means, if some categories are ordered strictly, but some are not, which are ones that show strict ordering? Are they always the same or can they be different? Such an in-between language would offer the necessary data for a generalization. In the following, I will show that direct-only, non-intersective and relational modifiers, cross-linguistically are more likely to demand strict ordering, than qualitative modifiers, which are ambiguous in the first place in a lot of languages. In order to argue for this, I will focus on two languages in which qualitative modifiers are ordered quite freely, but once direct-only modifiers are involved, ordering restrictions apply. These languages will be Persian and Modern Greek. The reason for these choice are the existence of previous accounts on ordering in the DP in both languages and the comparatively easy testability.

4.4.2.2 Persian

I start the discussion with Persian.⁵¹ The relevant predictions are easy to test because modifiers with a few exceptions (demonstratives, superlatives, etc.) occur only postnominally. They are in this position always connected to the noun via the

⁵¹Specifically Fārsi, the Persian variant spoken in Iran. I wish to thank Nasimeh Bahmanian very much for providing many of the examples and long discussions. For previous discussion and similar examples see also Kahnemuyipour (2014). However, some of his examples were judged as not so clear, which is why I elicited additional examples.

so called *ezāfe*-morpheme (which in fact could be an instance of MOD as it shows a similar distribution to NO; see Toosarvandani and van Urk 2014; Gündoğu *et al.* 2024).

Like Japanese, but different from English, Persian does not show ordering restrictions among qualitative modifiers, illustrated with the familiar example of a DIMENSION and a COLOR modifier. Modifiers of these categories, which most likely have an adjectival status, are acceptable in either order in Persian and no order seems to be more grammatical or preferred over the other.

- (160) DIMENSION vs. COLOR
- a. miz-e qermez-e bozorg
table-EZ red-EZ big
 - b. miz-e bozorg-e qermez
table-EZ big-EZ red
'big red table'

Similar facts hold for modifiers of MATERIAL and DIMENSION, as well as NATIONALITY and QUALITY respectively.

- (161) DIMENSION vs. MATERIAL
- a. sandali-ye felezzi-ye bozorg
chair-EZ metal-EZ big
 - b. sandali-ye bozorg-e felezzi
chair-EZ big-EZ metal
'big metal chair'

- (162) QUALITY vs. NATIONALITY
- a. goldān-e qašang-e žāponi
vase-EZ beautiful-EZ Japanese
 - b. goldān-e žāponi-ye qašang
vase-EZ Japanese-EZ beautiful
'beautiful Japanese vase'

Note that modifiers of MATERIAL and NATIONALITY can be used predicatively in Persian, in contrast to Japanese. Therefore, I analyze them as qualitative adjectives, analogous to the hierarchy of Scott (2002), albeit not as direct-only modifiers.

- (163) Sandali felezzi ast / žāponi ast.
chair metal COP / Japanese COP
'The chair is metal(-ic) / Japanese.'

However, direct-only modifiers do exist and they demand ordering restrictions. This will be shown first for relational adjectives, labeled as such since they are in fact

adjectival derivatives from nouns, formed by the suffix *-i*. One example is *haste'i* ‘nuclear’ derived from the noun *haste* ‘nucleus, core’. This modifier, same as relational adjectives in other languages, cannot be used predicatively.

- (164) **Fizikdān haste-i ast.*
 physicist nuclear-RA COP
 (‘The physicist is nuclear.’)

In (165), it is shown, with an example from Kahnemuyipour (2014), who draws the same conclusion, that the relational adjective has to occur closer to the noun than the qualitative adjective *javān* ‘young’, which is not a direct-only adjective. The reversed order is illicit. Assuming the postnominal position is a result of NP-movement in a pied-piping fashion (Ross, 1967; Cinque, 2010), the order of the postnominal modifiers reflects the mirror image of the underlying hierarchy.

- (165) a. *fizikdān-e haste-i-ye javān*
 physicist-EZ nuclear-RA-EZ young
 ‘young nuclear physicist’
 b. **fizikdān-e javān-e haste-i*
 physicist-EZ young-EZ nuclear-RA (Kahnemuyipour, 2014, 18)

In the following, I illustrate two more ordering pairs of relational and qualitative adjectives. In each case, the relational adjective must be adjacent to the head noun. Additionally, I show the illicit predicative use of the relational adjective.

- (166) a. *doktor-e omun-i-ye javān*
 doctor-EZ general-RA-EZ young
 ‘young general practitioner’
 b. **doktor-e javān-e omun-i*
 doctor-EZ young-EZ general-RA
 c. **Doktor omun-i ast.*
 doctor general-RA COP
 (‘The doctor is general.’)

- (167) a. *kāx-e pādšāh-i-ye qašāng*
 palace-EZ royal-RA-EZ beautiful
 ‘beautiful royal palace’
 b. ??*kāx-e qašāng-e pādšāh-i*
 palace-EZ beautiful-EZ royal-RA
 c. **Kāx pādšāh-i ast.*
 palace royal-RA COP
 (‘The palace is royal.’)

Finally, when a relational adjective co-occurs with a non-intersective modifier such as *qabli* ‘former’, also direct-only in Persian as shown in (169), the surface order is *relational* $\succ$ *non-intersective*.

- (168) a. dānešmand-e haste-i-ye qabli
 scientist-EZ nuclear-RA-EZ former
 ‘former nuclear scientist’
 b. ??dānešmand-e qabli-ye haste-i
 scientist-EZ former-EZ nuclear-RA

- (169) *Dānešmand qabli ast.
 scientist former COP
 (‘The scientist is former.’)

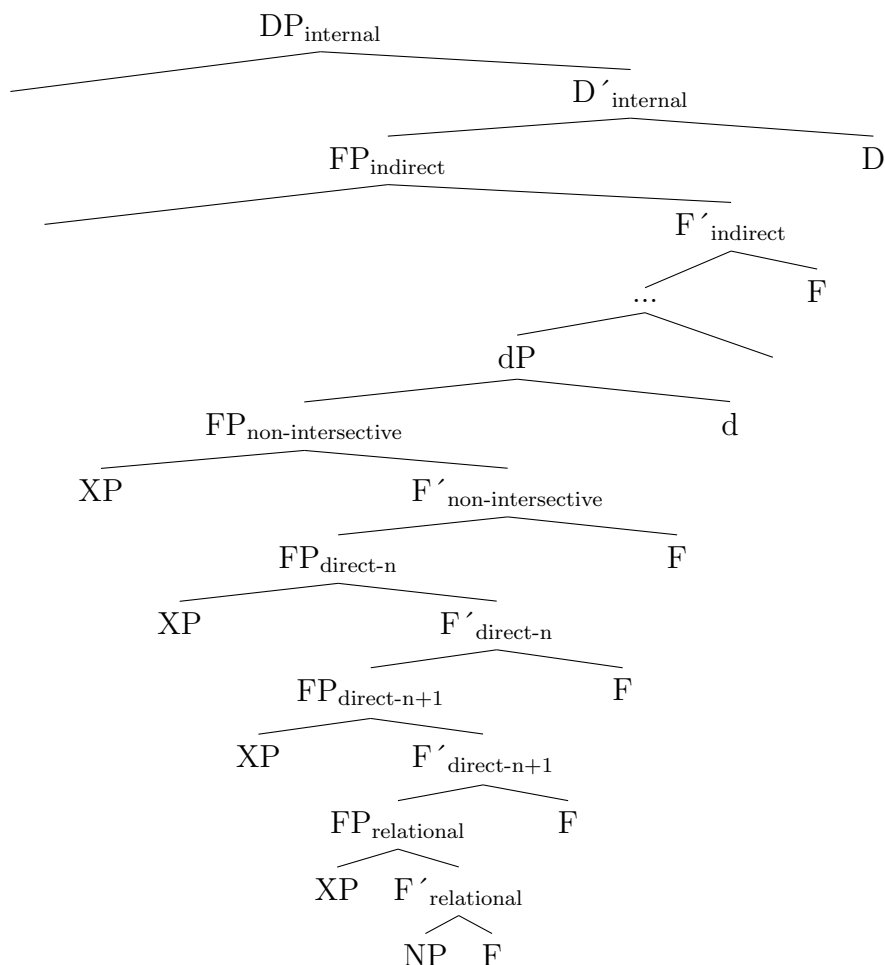
These facts are in line with the literature and findings in other languages. Recall that, truly non-intersective modifiers precede intersective direct modifiers in German and English (Trost 2006a; Panayidou 2013; see Chapter 4.3.1.3), suggesting a higher position for non-intersective modifiers. On the other hand, relational modifiers are hierarchically very low inside the direct domain (Bortolotto 2016; see Chapter 2.3.2.3).

In general, these findings suggest that Persian has the language-specific inventory to form specific projections for relational, and potentially non-intersective, modifiers, but that it lacks the inventory to realize specific functional projections reserved for qualitative modifiers of DIMENSION, QUALITY and so on. In other words, it possesses only the Units of Language responsible for direct-only modifiers (cf. also Kahnemuyipour 2014, 18).

I therefore suggest that the direct domain in Persian looks as follows. The general DP structure is the same as in Japanese. Furthermore, the direct domain also contains several functional projections which are not based on specific categories, another characteristic shared with Japanese. The important point of deviation is that Persian does have certain functional projections which are based on specific categories, namely the projections that host relational adjectives. These are placed in a fixed position closer to the head noun. Tentatively, I also assume projections for non-intersective modifiers, which must be higher than those for relational modifiers, judging from (168). Note that the surface order is the result of roll-up movement of the whose-pictures type after which all modifiers appear in the mirror-image order (Cinque, 2017, 2023; Kahnemuyipour, 2014). I do not display the *ezāfe*-morpheme as a separate element, although it is also a very good candidate for a syntactic

head selecting modifiers as also argued in Philip (2012); Toosarvandani and van Urk (2014); Franco *et al.* (2015) and Gündoğu *et al.* (2024).⁵²

(171) DP-Structure for Persian



4.4.2.3 Greek

Modern Greek behaves similarly to Persian, although predictions are a bit more difficult to test.⁵³ This is so because modifiers can, in principle, appear on both sides of the noun, and because they can appear with an additional determiner, a construction dubbed *determiner spreading* (Androutsopoulou, 1995). This option

⁵²Intriguingly, some combinations of relational with qualitative adjectives were judged better in the unexpected order, potentially for unrelated reasons such as phonology and/or prosody. According to Nasimeh Bahmanian, the following example can be uttered in both orders, but interestingly, the qualitative adjective *šadid* ‘severe’ seems to be a bit better closer to the head noun than the relational adjective *zistmohiti* ‘environmental’.

- (170) a. bohrān-e šadid-e zistmohit-i
 crisis-EZ severe-EZ environmental-RA
 b. bohrān-e zistmohit-i-ye šadid
 crisis-EZ environmental-RA-EZ severe
 ‘severe environmental crisis’

⁵³I am very grateful to Jim Kotopoulos for providing the examples in this section and taking the time to discuss them with me. Unless indicated otherwise, the examples are his.

becomes necessary with postnominal modifiers (Alexiadou and Wilder, 1998; Alexiadou, 2003; Campos and Stavrou, 2004; Panagiotidis and Marinis, 2011; Lekakou and Szendrői, 2012; Panayidou, 2013). An illustration is the following example where the two postnominal modifiers must occur with their own determiner. This is optional when they appear in prenominal position (adapted from Alexiadou and Wilder 1998, 303).

- (172) a. to vivlio *(to) kokino *(to) meghalo
 DET book DET red DET big
 b. to meghalo (to) kokino (to) vivlio
 DET big DET red DET book
 ‘the big red book’

Determiner spreading has often been argued to instantiate a case of predicative modification (see especially Cinque 2010, A3 and references therein), but matters are not so clear (see Lekakou and Szendrői 2012 for examples of adjectives that are intersective but resist predicative use). Hence, I will limit myself to the relevant facts for the discussion. While free order of modifiers has often been pointed out in relevant structures, a welcoming fact for an analysis as indirect modifiers (Alexiadou and Wilder, 1998; Alexiadou *et al.*, 2007; Cinque, 2010), it seems to be the case that the order among standard categories of qualitative modifiers is also significantly less strict in structures not involving determiner spreading, compared to English or German. This is pointed out in Laenzlinger (2011, 174) and was confirmed by Jim Kotopoulos (p.c.). In the following example, as in Persian and Japanese, the modifiers of DIMENSION and COLOR, leaving factors such as focus, topicalization and emphasis aside, can appear in any order without deteriorating acceptability. Note that we only need to look at the prenominal position since in postnominal position, determiner spreading would be obligatory.

- (173) DIMENSION vs. COLOR
 a. to meghalo kokino vivlio
 DET big red book
 b. to kokino meghalo vivlio
 DET red big book
 ‘the big red book’

As in Persian, when direct-only modifiers are involved, a strict ordering can be observed. The surface order between the qualitative adjective *neari* ‘young’ and the relational adjective *piriniki* ‘nuclear’, derived via the suffix -I, must be *qualitative adjective* $\succ$ *relational adjective* $\succ$ *noun*, the reverse order (174-b) is ungrammatical. Crucially, *piriniki* is impossible in constructions involving determiner spreading

(174-c), therefore can never appear postnominally (174-d). It is also not readily available in predicative position (174-e). These facts are interconnected and underline its direct status.⁵⁴

(174) relational vs. qualitative modifier

- a. i neari piriniki fisikos
DET young nuclear physicist
'the young nuclear physicist'
- b. *i piriniki neari fisikos
DET nuclear young physicist
- c. *i piriniki i fisikos
DET nuclear DET physicist
'(the nuclear physicist)'
- d. *i fisikos i piriniki
DET physicist DET nuclear
'(the nuclear physicist)'
- e. *I fisikos ine piriniki.
DET physicist COP nuclear
'(The physicist is nuclear.)'

An intriguing case is made by the modifier *theatriki* 'theatrical', for example modifying the noun *parastasi* 'play'. *Theatriki* can (at least for some speakers including Jim Kotopoulos) occur in determiner spreading-structures, but only in prenominal position, never postnominally (175-b). Furthermore it can appear in predicative position (175-c).

- (175)
- a. i theatriki (i) parastasi
DET theatrical DET play
'the theatrical play'
 - b. *i parastasi i theatriki
DET play DET theatrical
 - c. I parastasi ine theatriki.
DET play COP theatrical
'The play is theatrical.'

Nevertheless, even when determiner spreading is invoked, there are still ordering restrictions observed when it co-occurs with a qualitative adjective such as *orea* 'nice'. Then, *parastasi* has to occur closer to the noun. I show this below for a DP without (176-b), and a DP with determiner spreading (176-c).

⁵⁴According to Jim Kotopoulos, *piriniki* can, in some instances, occur predicatively in which case it has a strong flavor of an elliptical construction. One reason for this might be that, contrary to English, Greek does not need an article when a noun appears predicatively, so only the noun needs to be deleted. A similar phenomenon in Japanese was shown at the beginning of this chapter.

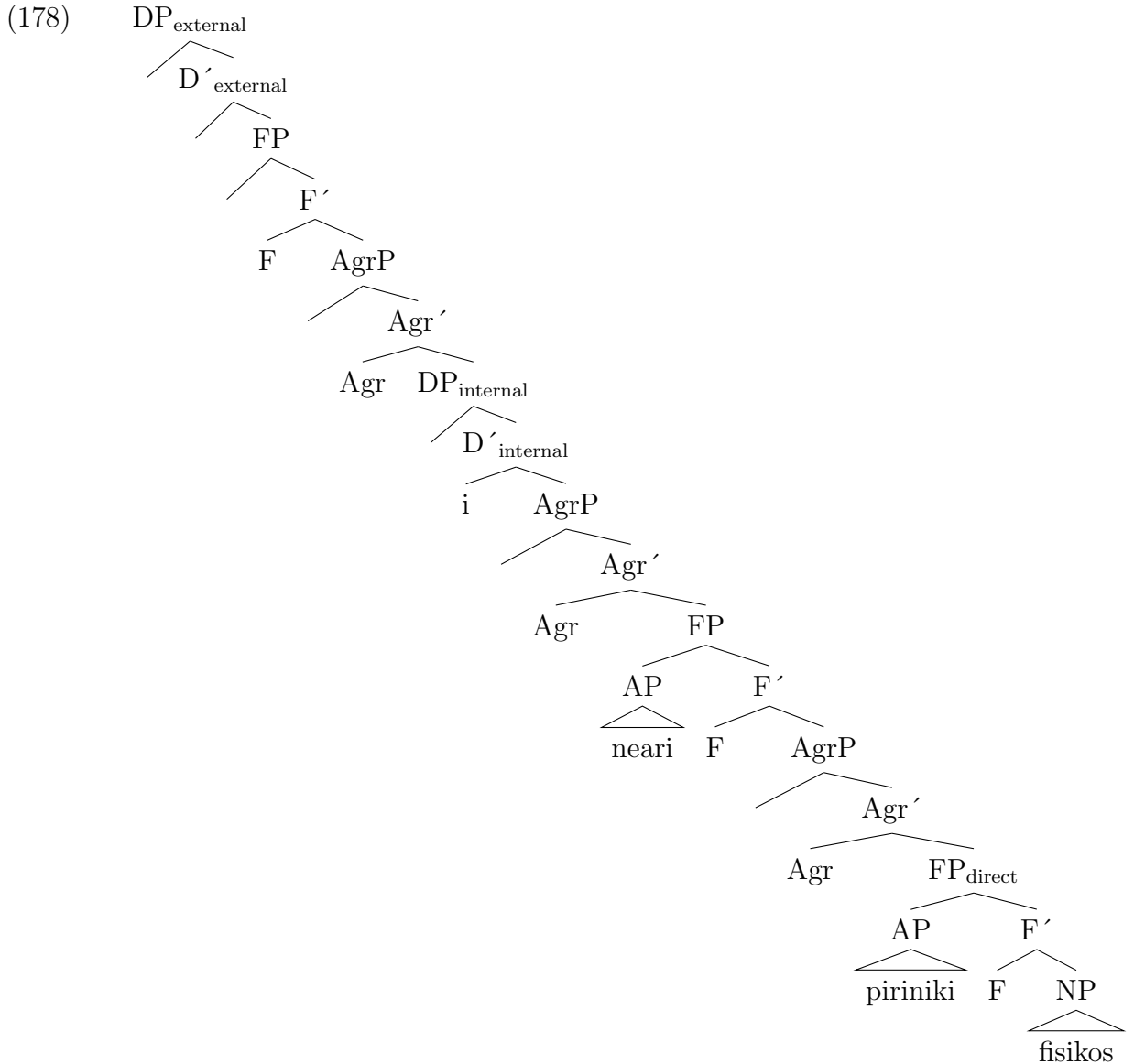
- (176) a. *i orea (i) theatriki (i) parastasi*
DET nice DET theatrical DET play
'the nice theatrical play'
- b. **i theatriki orea parastasi*
DET theatrical nice play
- c. **i theatriki i orea (i) parastasi*
DET theatrical DET nice DET play

These facts are highly challenging for existing theories. They suggest that the same, or at least a similar, inventory of Units of Language is at stake for direct modification in Greek as in Persian. Both languages, then, do not have in their inventory functional projections dedicated to categories of qualitative modifiers such as DIMENSION. However, they do realize functional projections dedicated to specific direct modifiers, namely relational modifiers. This explains the observed ordering principles. The DP structure for these two languages should thus look quite similar.

A remaining question is how cases in Greek can be explained where one, presumably direct, modifier occurs prenominally and another one occurs postnominally with determiner spreading.

- (177) *i piriniki fisikos i neari*
DET nuclear physicist DET young
'the young nuclear physicist'

This receives a nice explanation under the structure presented here and the assumption of two DPs (as also argued in Laenzlinger 2011), a lower position of the relational modifier and a combination of two movement operations. The underlying structure is presumably the following. Assuming *piriniki* is a direct modifier, it is merged in the direct domain, thus lower in the DP. The exact status of *neari* cannot be determined, but it seems to be higher in any case.



The derivation likely proceeds as follows: First the NP moves in a pictures-of-whom type movement operation to the specifier of a higher functional head (AgrP in Cinque 2017, 2023), dragging along the relational modifier *piriniki*, but crucially not affecting the order. Then this chunk moves across the qualitative modifier *neari* to the left periphery across the determiner. Crucially, this modifier remains in-situ and we get the order *piriniki* $\succ$ *fisikos* $\succ$ *i* $\succ$ *neari*. Next, the determiner moves to the DP-external domain and leaves a copy in the internal DP, as, for instance, argued in Laenzlinger (2011) and see Grohmann (2003); Grohmann and Panagiotidis (2015).

4.4.2.4 Summary

The preceding discussion has shown that the DP structure presented in this study is applicable to other languages, emphasizing its universal status, but that parametric variation is permitted. Importantly, this variation is not random. There are lan-

languages which do not show ordering restrictions of categories such as QUALITY, but do show ordering restrictions when it comes to direct-only modifiers. On the other hand, the reverse case is (at least so far) missing from the paradigm. Languages such as English or German show ordering restrictions in all scenarios.

On the basis of these facts, I propose the following generalization.

(179) *Generalization on Cross-Linguistic Ordering Restrictions*

If a language shows ordering restrictions among qualitative modifiers such as QUALITY and DIMENSION, it will also show ordering restrictions among direct-only modifiers, for example relational and non-intersective modifiers, among each other and with other categories. In other words, if a language has the language-specific inventory to realize functional projections for categories of QUALITY, DIMENSION, etc., it will also realize functional projections for direct-only modifiers.

This means that there are indeed languages between the English type, in which all categories adhere to specific, preferred orders in context-neutral utterances, and the Japanese type, in which there is the same universal spine observable, but not the language-specific entities for functional projections of fine-grained semantic categories. It is for future researchers to prove or refute this claim and to see whether more points can be added to the continuum and whether an even more fine-grained generalization can be put forth.

4.4.2.5 Two more languages

Before closing this subsection, I would like to draw attention to two languages that prove the above made claim in different ways. First, Brown (2024) shows that in Naasioi, a Papuan language, direct adjectives do not adhere to ordering restrictions. Adjectives in this language can appear pre- or postnominally, seemingly without restrictions. Furthermore, direct can be differentiated from indirect adjectives via presence of an additional article in the latter case, making this construction appear very similar to determiner spreading in Modern Greek. First, direct modifiers must appear closer to the noun than indirect modifiers, illustrated in N-A-A sequences below.

- (180) a. Te mosi pangkaing te mutaanu aatu-maang. (Naasioi)
 DET dog big DET black sleep-PRS.PROG
 ‘The big dog which is black is sleeping.’
 b. *Te mosi te mutaanu pangkaing aatu-maang.
 DET dog DET black big sleep-PRS.PROG

(Brown, 2024, 432)

Assuming all modifiers that appear without the additional article *te* are direct, they should adhere to ordering restrictions in contexts with two or more direct modifiers. Alas, they do not, as Brown (2024) argues based on fieldwork data. This generalization seems to hold, at least, for modifiers of DIMENSION, COLOR and QUALITY at least and is exemplified below with a string of three adjectives, which, according to Brown (2024) are fine in either order.

- (181) Te orara pangkaing arasi kiring dua-'ar-ing.
 DET bad big brown breadfruit fall-PL-IMM.PST
 'The bad big brown breadfruits fell.'

In other words, this language poses a problem for standard cartography as well. In fact, since it seems to pattern with the languages depicted above (although no immediate conclusion can be drawn about relational and nonintersective modifiers), the same facts seem to be at stake, namely a direct domain with unspecific projections, in which various movement operations can take place.

Containment relations akin to the one presented for Persian and Greek are given for Ojibwe, an Algonquian language, in Rosen (2016). This language seems to have a small class of direct modifiers, which are preminally bound to the noun. The interesting part is that pairs of direct adjectives of COLOR and NATIONALITY as well as COLOR and DIMENSION do not adhere to ordering restrictions, but pairs of DIMENSION and NATIONALITY must do so (NI = inanimate noun).

- (182) COLOR vs. NATIONALITY: No ordering restrictions
- a. misko-Anishinaabe-mashkimod
 red-Ojibwe-bag.NI
 - b. Anishinaabe-misko-mashkimod
 Ojibwe-red-bag.NI
 'a red Ojibwe bag' (Rosen, 2016, 161)
- (183) DIMENSION vs. COLOR: No ordering restrictions
- a. gichi-misko-waakaa'igan
 big-red-house.NI
 - b. misko-gichi-waakaa'igan
 red-big-house.NI
 'a big red house' (Rosen, 2016, 161)
- (184) DIMENSION vs. NATIONALITY: Ordering restrictions
- a. gichi-Anishinaabe-mashkimod
 big-Ojibwe-bag.NI
 'a big Ojibwe bag'
 - b. *Anishinaabe-gichi-mashkimod
 Ojibwe-big-bag.NI (Rosen, 2016, 162)

Furthermore, for some reason, NATIONALITY adjectives can appear to the left of DIMENSION adjectives if an adjective of COLOR intervenes (Rosen, 2016, 162).

- (185) Anishinaabe-misko-gichi-adoopowin
 Ojibwe-red-big-table.NI
 ‘a big red Ojibwe table’ (Rosen, 2016, 162)

This order string NATIONALITY $\succ$ COLOR $\succ$ DIMENSION is the mirror-image order of modifiers compared to their assumed merge position. Naturally, it is unforeseen by any hierarchy. It should be impossible with modifiers in prenominal position according to Greenberg’s U20 and cannot be explained if we adopt the assumption that only the lexical head, the noun, can move. In other words, this surface order can only be explained if either, individual movement of modifiers, or, unspecific functional projections and free Merge positions are assumed. However, the latter assumption would have the big disadvantage that the strict order DIMENSION $\succ$ NATIONALITY is unaccounted for.

Needless to say more work is necessary here, but one, very tentative, conclusion could be that, all things being equal, this language also does not realize functional heads based on specific semantic categories explaining the, in most cases, free order. However, there is a ban on the merge order of a modifier of DIMENSION followed by a modifier of NATIONALITY explaining why a string of two adjacent modifiers of those categories in that order is never available. I thus interpret these facts similar to the Japanese case, but with the twist that modifiers of DIMENSION might never be merged directly after modifiers of NATIONALITY.

4.4.3 Remaining questions

This section discusses two remaining question, namely whether there is complete freedom of ordering in Japanese, and whether complex direct modifiers do exist.

4.4.3.1 #1: Is there complete optionality in ordering?

If Japanese is a language, potentially one of many, with a broad direct domain lacking specific functional projections to dictate ordering, does that mean that no ordering principles exist at all? In other words: Is ordering of modifiers completely optional in this language? It was established that the semantic category of a modifier cannot be a factor, but in the following, I discuss potential other factors.

Word Class/Morphology

I took care in balancing out the word classes that I tested for ordering preferences. Not only do the two Japanese adjective groups not adhere to ordering restrictions within and across one another (Köhlich, 2018), the present study showed that direct *no*-modifiers, not only among one another, but also together with *i*- and *na*-adjectives, also do not lead to any ordering preferences in speakers. This clearly suggests the absence of the factors *word class* or *morphology*, which readers can verify in the examples above.

Phonology

It has been argued in Nagano (2016) that phonology might play a role for ordering of modifiers, which is exactly why I have chosen to test modifiers with different amounts of morae. As it turns out, phonology does seem to be a factor, but only to a limited extent. For example, in none of the pairs *Denmāku-no* ‘Denmark, Danish’ vs. *chīku-no* ‘teak’ (5 vs. 3 morae) or *ao-i* ‘blue’ vs. *Chūgoku-no* ‘China, Chinese’ (2 vs. 4 mora) did ordering restrictions emerge. Note that I counted mora without the monomoraic elements following.

The only strict ordering seems to be dictated by the single monomoraic modifier of MATERIAL *ki-no* ‘wood(-en)’. Recall the example given in Watanabe (2012), who argues that the combination with a trimoraic modifier leads to ungrammaticality if it is not placed closest to the noun. The relevant example is repeated from (113) below.

- (186) a. hokuō-no ki-no isu
 North.Europe-MOD wood-MOD chair
 ‘North European wooden chair’
 b. *ki-no hokuō-no isu
 wood-MOD North.Europe-MOD chair (Watanabe, 2012, 508)

Now, this extends to the combination of *ki* ‘wooden’ with other modifiers. The speakers I consulted prefer *ki-no* closer to the head noun, but do not find the opposite example to be completely ungrammatical, just very marked. This is shown below.

- (187) a. Denmāku-no ki-no isu
 Denmark-MOD wood-MOD chair
 b. #ki-no Denmāku-no isu
 wood-MOD Denmark-MOD chair
 ‘Danish wood chair’

Note that it really is phonology that is the factor at stake here, not semantic category, as the equivalent *mokusei-no* ‘wooden’ is possible in either position and not marked.

- (188) a. Denmāku-no mokusei-no isu
Denmark-MOD wooden-MOD chair
b. mokusei-no Denmāku-no isu
wooden-MOD Denmark-MOD chair
‘Danish wood chair’

While this means that phonology does have an effect on ordering to a certain extent, it is questionable how big its effect actually is. As noted, *ki-no* is the only monomoraic modifier in the data set. This means that we cannot test similar modifiers to establish strict paradigms. On the other hand, with bimoraic modifiers such as *ao-i* ‘blue’, I did not detect such an effect.⁵⁵

Pragmatic Factors, Scope

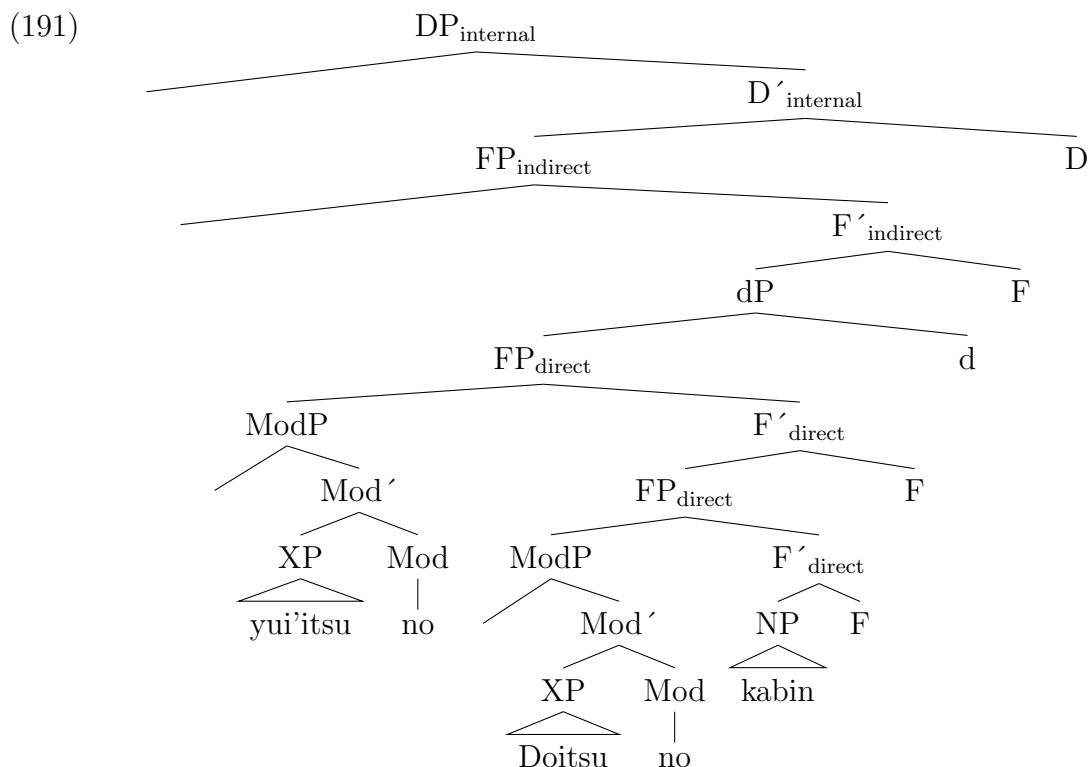
As already mentioned in Chapter 3.1, the only cases in which fixed order emerge involve different scope readings. In (190), the speaker is restricting the domain of German vases, not the domain of vases *per se* to a concrete *German vase* of which only one is available in the discourse. The reverse order, given the same context, is significantly degraded.

- (190) a. yui’itsu-no Doitsu-no kabin
only-MOD German-MOD vase
‘the only German vase’
b. ??Doitsu-no yui’itsu-no kabin
German-MOD only-MOD vase

This is thus a case in which the two modifiers occupy distinct merge positions (at least in relation to one another, the precise projection being not important), although both inside the direct domain, as illustrated below.

⁵⁵ A comment made by an anonymous reviewer, alongside a provided example, fits well with this observation. According to their judgment the following minimal pair, with a bimoraic modifier of NATIONALITY does, all things being equal, also not offer a contrast.

- (189) a. Tai-no mokusei-no isu
Thailand-MOD wooden-MOD chair
b. mokusei-no Tai-no isu
wooden-MOD Thailand-MOD chair
‘Thai wood chair’



Connected to this is the strong nominal character of modifiers of NATIONALITY. Some, but not all, speakers reported that the second order to them suggests a reading like ‘the only vase IN Germany’. This also becomes obvious when combining such modifiers with possessive modifiers as in the following. Here, speakers again find a specific order more natural when they want to identify the particular referent: That in which the NATIONALITY modifier is closer to the noun.

- (192) a. *watashi-no Doitsu-no kabin*
 I-MOD German-MOD vase
 ‘my German vase(-s)’
 b. ??*Doitsu-no watashi-no kabin*
 German-MOD I-MOD vase

Seemingly, scope is at play again. The possessive modifier *watashi-no* ‘my’ takes scope over *Doitsu-no kabin*, not just *kabin* alone, restricting the domain to a specific vase, namely one belonging to the speaker. Again, the reverse order does not involve scope and means something like ‘my vase(-s), which happen(s) to be in German / form/in Germany’.

These facts highlight that ordering in Japanese is not completely unrestricted, but that there are influencing factors, among them, to a minor extent, phonology, and, to a larger extent, scope restrictions. Importantly, these factors mostly seem to be *syntax-external*, as also reported for similar cases in the clause in Ramchand and Svenonius (2014). Another such factor, presented in Chapter 5.4 for indirect modifiers, is *complexity*. This factor might not be really testable for direct modifiers,

one of whose defining characteristics is to be, generally, less complex than their indirect counterparts. However, a related phenomenon will be discussed in the next section.

4.4.3.2 #2: Complex direct modifiers and the situation in Classical Japanese

Another remaining issue to address concerns the possible existence of complex, that is clausal, direct modifiers. It is one of the characteristics of direct modifiers that they exhibit a less complex structure than indirect modifiers, which are arguably always embedded in a CP (Cinque 2010, 2020, see Chapter 3.2.7.2). However, there is reason to believe that direct modifiers can also be complex. First, direct modifiers are not only functional, but can also be *phrasal*. Evidence for this comes from the fact that direct adjectives can, in some languages, appear with overt adjectival arguments or adjuncts. This is shown below with an example from Bulgarian involving the direct (non-intersective) adjective ‘main’ and a prepositional complement (cf. Cinque 2010, ch. 4.1.2 for more data).

- (193) glavnata po značenie pričina (Bulgarian)
 main.the in significance reason
 ‘the main reason for importance’ (Cinque, 2010, 47)

In Japanese, some direct modifiers are clearly *clausal*. This concerns especially structures in which such modifiers co-occur with an overt subject argument. Some relevant data points are given in Hoshi (2002), who shows that some of the alleged examples of non-intersective modifiers given in Yamakido (2000) do not only exhibit a predicative character, but also become intersective, highlighting this predicative character, when a subject appears (see Yamakido 2005 for discussion). Observe (194).

- (194) Pītā-wa [tsukiai-ga furu-i] tomodachi-da.
 Peter-TOP association-NOM long.time-MOD friend-COP
 ‘Peter has been a friend for a long time.’ (Lit.: ‘Peter is a friend whose association is old.’)
 (Hoshi, 2002, 11)

Here, the alleged direct modifier *furu-i* ‘long.time’, which in combination with human entities only yields a non-intersective reading (Chapter 4.3.1), appears in a clausal structure with an overt subject. Furthermore, it now has acquired an intersective interpretation *and* is possible in predicative position, evidenced in (195).

- (195) Pītā-wa [tsukiai-ga furu-i], soshite Pītā-wa
 Peter-TOP association-NOM long.time-COP and Peter-TOP
 tomodachi-da.
 friend-COP
 ‘The association with Peter has been long and Peter is a friend.’
 (Hoshi, 2002, 11)

Now admittedly, as has been argued, *furu-i* can appear in predicative position as long as it is predicated on inanimate entities, which means it is not a direct-only modifier in the first place. However, a clausal construction is also possible for relational modifiers, as shown below for *Chūgoku-no* ‘China, Chinese’.

- (196) seizōkoku-ga Chūgoku-no kuruma
 country.of.production-NOM China-MOD car
 ‘a car whose country of production is China’

In this case, the modifier also allows paraphrasing NO with DE-AR-U and inflection for past.

- (197) a. seizōkoku-ga Chūgoku-de-ar-u kuruma
 country.of.production-NOM China-PRED-COP-PRS car
 ‘a car whose country of production is China’
 b. seizōkoku-ga Chūgoku-d-at-ta kuruma
 country.of.production-NOM China-PRED-COP-PST car
 ‘a car whose country of production was China’

Two things are of importance here: Again, note that the modifiers in the preceding examples, different from the examples in Cinque (2010), are not only phrasal but clausal. Second, what *Chūgoku* here actually refers to is the country *China*, but not the notion *from China* as in examples such as *Chūgoku-no kabin* ‘Chinese vase’ presented in Chapter 4.3.2. *Chūgoku-no* is *per se* not a direct-only modifier either.

More tricky is the following example with the *rentaishi ōki-na* ‘big’, which, different to *ōki-i*, cannot appear in predicative position. However, it can take a subject argument in attributive position.

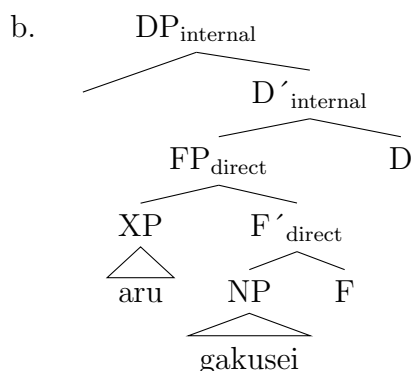
- (198) yane-ga ōki-na ie
 roof-NOM big-MOD house
 ‘house whose roof is big’

This means, the structural make-up, at least on the surface, for such constructions is similar to the following example, repeated from Chapter 3.

Furthermore, (198) could find another explanation which relates to the nature of *ōki-na* as a *rentaishi*. Recall that all verbally derived *rentaishi* are fossilized in their attributive form (*rentaikei*), and this also extends to adjectival *rentaishi* co-occurring with the premodern copula *nari/tari*, inflected for *-naru/taru* respectively, and potentially *na*. As also argued in Chapter 5.3.4, the attributive form of verbs in Classical Japanese clearly projected a C layer. Evidence for this comes from the occurrence in embedded wh-questions where it served the role of the complementizer (Kinsui, 1995; Kaplan and Whitman, 1995; Hiraiwa, 2005). This suggests that such modifiers might indeed still project a CP-layer explaining the clausal behavior.⁵⁶

This discussion also extends to other *rentaishi* in Modern Japanese such as *aru* ‘a certain’. There are (at least) two analyses possible for such lexemes: Either, they are analyzed as *bona fide* (but fossilized) direct modifiers which do not project a higher functional layer, which are, somehow, not licensed via MOD. Under this analysis, the ending *ru* is taken to be part of the stem.

- (204) a. *aru gakusei*
 certain student
 ‘a certain student’ (= (148))

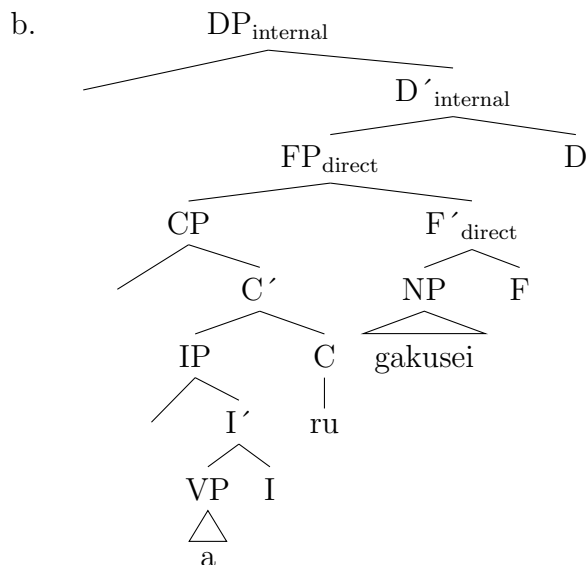


Alternatively, staying closer to the structure such modifiers presumably had in Classical Japanese, they are direct modifiers, but project higher IP and CP layers. The ending *ru* is the complementizer head. Note that I abstract away from the possible presence of MOD.

⁵⁶In this respect, note also the following example with a verbally derived *rentaishi* and a nominal subject, which prompted discussions already in the middle of the 20th century. Several authors have argued that the transition from verb to *rentaishi* had not been fully achieved accounting for the possibility of overt subjects. See Sekine (1937); Matsushita (1976 (1928)) and Kieda (1938).

- (203) *kimi-ga iwayuru shingi*
 you-NOM so-called loyalty
 ‘your so-called loyalty’ (Lit. ‘what you call loyalty’) (Sekine, 1937, 1:28)

- (205) a. a-ru gakusei
 certain-COMP student
 ‘a certain student’ (= (148))



This discussion also concerns Classical Japanese, where all verbally and adjectivally derived lexemes analyzed as *rentaishi* in Modern Japanese were still inflecting parts of speech in Classical Japanese. This means, in attributive position, they arguably could perform a role as direct modifiers casting doubt on the analysis of the attributive form (*rentaikei*) in Classical Japanese – although probably projecting a C layer (Kinsui 1995; Kaplan and Whitman 1995; Hiraiwa 2005; cf. Chapter 5.3.4) – as being necessarily tied to indirect modification. Under this account, all modifiers had access to both kinds of modification at this stage of the language as well. For example, *i*-adjectives, whose attributive form was represented via the suffix -KI, were presumably structurally ambiguous between an direct and an indirect status, similar to the situation in Modern Japanese.

- (206) taka-ki mono
 high-RTK thing
 ‘a high thing’

Classical Japanese

To summarize, the existence of clausal direct modifiers provides a challenge to the theory presented here, but could be explained via either, assuming the existence of complex direct modifiers, or, the special status of certain lexemes that are only direct in certain usage domains. A bigger challenge is posed by direct modifiers historically derived from verbs, and this extends to the situation in Classical Japanese. This is a fruitful area for future research.

4.4.4 Summary

In this section, I argued that Japanese lacks the language-specific categories to realize specific functional projections. This explains the lack of strict ordering and shows that direct modification and ordering are orthogonal issues. I then discussed the cross-linguistic implications of this claim and put forth a generalization on ordering principles. Finally, I showed that ordering in Japanese is in fact not entirely ungoverned either and discussed the possibility of complex direct modifiers.

4.5 Chapter summary and closing remarks

This chapter started with a discussion on why direct modification has so far been neglected, or entirely dismissed, in Japanese. Then, criteria for direct modification were introduced and it was shown that the lack of predicativity is the leading criterion, which is often accompanied by other criteria such as non-gradability. The following observations emerged in the course of the discussion:

- (207)
- a. Japanese exhibits a variety of different direct-only modifiers: The equivalents of cross-linguistically well-known categories of non-intersective and relational modifiers, as well as cases of direct qualitative modifiers, idiomatic modifiers, direct quantificational modifiers and the language-specific group of *rentaishi*.
 - b. Almost all cases of direct-only modifiers are found among *no*-modifiers.
 - c. There is a lack of ordering restrictions which is due to a lack of projections dedicated to specific categories inside the direct domain.
 - d. The reason for this is that Japanese does not have the language-specific entities to realize such dedicated functional projections.

I believe that the challenges these findings pose for cartography are also chances to expand this research field and adapt it in a way so that the very elaborate predictions it makes become even more powerful. Postulating a broader, more fluid set of functional projections inside the direct domain preempts the common criticism regarding superfluous projections as well as alleged violations of ordering restrictions. The concept of cross-linguistic variation enriches our understanding of the DP and generates new insights for other languages.

It is definitely a desideratum for future research to establish more cases of clear-cut ordering restrictions in Japanese, besides scopal relations or restrictions among direct and indirect modifiers. This would not only help expanding the hierarchy of direct modifiers proposed in this book in a language-specific and cross-linguistic

perspective, but also could expand the structure of the direct domain I have proposed here and add a more fine-grained dimension to it.

Chapter 5

Indirect Modification in Japanese

This chapter discusses the indirect domain and focus will be switched to indirect, that is predicative, modification. Although this kind of modification is often assumed to be the only one available in Japanese (Sproat and Shih, 1991; Baker, 2003b; Laenzlinger, 2011), no systematic analysis has been presented for indirect modifiers in this language, either. This concerns their fine-grained structure and the integration of this structure into the DP, their size and their position. This chapter thus aims at closing those research gaps and at broadening our understanding of this part of the DP, and indirect modifiers in Japanese and in general. After pointing out the characteristics of indirect modifiers in Japanese, I will discuss the issues mentioned above in turn.

5.1 Introduction

Recall that the indirect domain is situated inside the DP-internal domain, higher than the direct domain. It exhibits several functional projections for indirect modifiers as well as a projection that hosts numerals. As in the direct domain, these projections are not dedicated to a specific type of indirect modifiers, in this case to different sizes. The structure of the indirect domain inside the DP-internal domain is circled in the structure below.

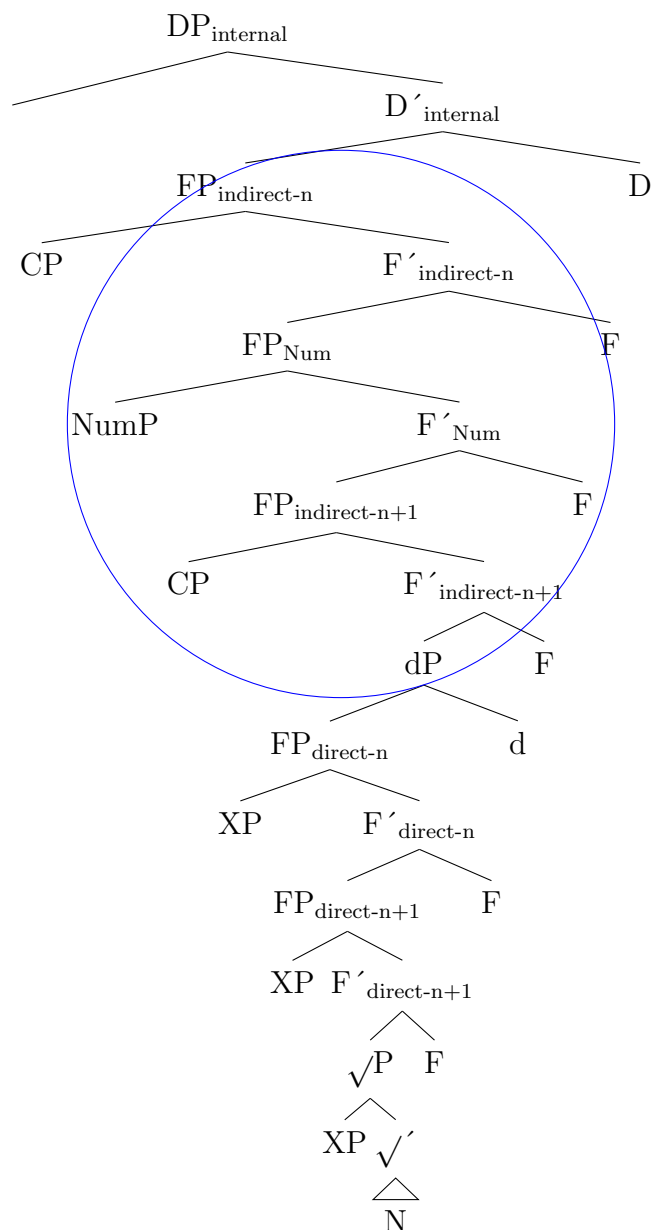


Figure 5.1: Structure of the indirect domain in Japanese

5.1.1 Goals and overview

This chapter deals with the following questions relevant for indirect modifiers in Japanese:

- (1)
 - a. What are the complex morphological elements co-occurring with indirect modifiers, especially adjectives such as *taka-katta* ‘was expensive’, in Japanese?
 - b. What is the size and merging position of indirect modifiers?
 - c. What influences the position of indirect modifiers?

These questions will be addressed in turn and yield the following answers:

- (2)
- a. The complex morphological elements such as K-AT-TA are made-up of three elements: two copulas, and a tense morpheme. This analysis can be adapted to seemingly ‘simpler’ elements such as DA.
 - b. There is no difference in Japanese between full and reduced relative clauses; meaning indirect modifiers are only of one size: All relative clauses in Japanese are CPs and derived via operator movement (Miyamoto, 2017).¹
 - c. The only factor which has an effect on the position of modifiers inside the indirect domain is syntactic and prosodic complexity. More complex modifiers move to a position above numerals.

5.1.2 Characteristics of Japanese indirect modifiers

Recall that relative clauses are the prime example of indirect modifiers (Cinque 2010, 2020; among others). In Chapter 2.4, it was mentioned that languages such as English overtly display two types of relative clauses differing in size, structure and position, namely *full* and *reduced* relative clauses, but that this difference is generally hard to determine till non-existent in prenominal-RC languages. Such languages usually do not show two different types of relative clauses overtly, but, instead, exhibit other types of noun-modifying clauses that do not have direct counterparts as relative clauses in English and other well-known European languages (Keenan and Comrie, 1977; Lehmann, 1984; de Vries, 2002; Wu, 2011).²

Japanese fits the description of other prenominal-RC languages. This language only exhibits one morphosyntactic type of relative clause, but it offers several constructions that do not correspond to relative clauses in European languages (Comrie, 1998; Matsumoto, 2017). Furthermore, it exhibits the following characteristics which further complicate the analysis.

¹In fact, nothing hinges on whether they have to be CPs or IPs (although CP-status grants a landing position for the operator). The important point is that relative clauses in Japanese only come in one size.

²For a coherent and contemporary typological overview of prenominal relative clauses see Wu (2011), who also points out that prenominal relative clauses “have essentially only been studied in a few well-known and often-quoted languages such as Basque, Japanese, Mandarin, and Turkish” (Wu, 2011, 569).

- (3)
- a. Lexemes exhibit (with the exceptions of nouns and *na*-adjectives in non-past) no overt morphological difference between their role as modifier and in predicative position.
 - b. Modifiers can exhibit a fairly complex morphological make-up in attributive position, again mirroring their predicative forms.
 - c. Japanese exhibits a rather free word order within the DP, and this extends to indirect modifiers.
 - d. Relative clauses can appear above demonstratives and can in principle be restrictive and non-restrictive in this position.

The observation that, with a few exceptions, the morphology of lexemes in attributive position is identical to the one displayed in predicative position further differentiates Japanese from other East Asian prenominal-RC languages. For example, looking at Mandarin (DE), as well as other Sino-Tibetan languages, or Korean ((NU-)N, relative clauses exhibit certain elements which connect them to the noun and which are absent in non-embedded root clauses. Japanese lacks such element. the surface form of an indirect modifier matches that of its predicative counterpart. This is, once more, exemplified below for a verbal relative clause.³

³It might be tempting to deduce the size of Japanese relative clauses from the fact alone that modifiers appear in a *morphologically finite* form. Already Keenan (1985, 156) mentioned “We may note that the prenominal RCs in Japanese do not put Vrel in a non-finite or specifically relative form, but the Japanese case appears to be the exception among prenominal RCs here”. However, the notion of finiteness is not very well understood even in European languages and lacks a uniform definition. See the discussion in Hudson (1973); Stanton (2011); Belikova (2008); Belk (2017) with regard to reduced relative clauses in English, Dutch and Russian among others and Nikolaeva (2007) for an overview; see, for instance, Wurmbrand (2005, 2014) and Landau (2004, 2013) for various phenomena in English.

The biggest obstacle is that finiteness can be approached from different perspectives. Overt tense is interpreted as morphological finiteness, especially in the Latin-Greek tradition. The classical syntactic understanding of finiteness is the assignment of nominative case (Chomsky, 1981), but also the morpho-syntactic marking of tense, person, politeness and other categories (see Nikolaeva 2007 and articles therein). Its association with the C-domain has been put into question in recent literature as well, see Adger (2007); Bisang (2007) and especially Wurmbrand *et al.* (2020), who argue that it can be encoded at different levels of the clause, ranging from C to T even to v. From a semantic point of view, finiteness is related to the distribution of overt tense and the indication of the “existence of a predication at the time of the utterance” (Platzack, 1995, 201). As shown for example in Ogiwara (1994) and Miyagawa (2012), Japanese relative clauses can, but do not need to, be dependent on the higher clause.

To summarize, finiteness is not helpful in determining the status and syntactic size of a relative clauses and will be sidelined here as a criterion. Nevertheless, note that I used the notion of finiteness, in the sense that tense is projected, to argue for the possible non-occurrence of realizations of MOD.

- (4) a. Kare-ga hon-wo kat-ta.
 he-NOM book-ACC buy-PST
 ‘He bought a book.’
 b. kare-ga kat-ta hon
 he-NOM buy-PST book
 ‘the book (which) he bought’

This extends to all non-verbal modifiers inflected for past. *na*-adjectives and *no*-modifiers can co-occur with the canonical copula forms DE-AR-U and D(E)-AT-TA both in attributive and predicative position, and *i*-adjectives can co-occur with K-AT-TA in order to show that they are inflected for past, as shown below.

- (5) a. Ano doresu-wa taka-k-at-ta.
 that dress-TOP expensive-PRED-COP-PST
 ‘That dress was expensive.’
 b. ano taka-k-at-ta doresu
 that expensive-PRED-COP-PST dress
 ‘that dress which was expensive’ *i*-adjective, past

I will start with a discussion of these elements before discussing the size and position of indirect modifiers.

5.2 The Elements PRED, COP and T

In this section, I present a precise analysis of the complex morphological elements Japanese modifiers can co-occur with. Following Nishiyama (Nishiyama 1999, and see also Miyama 2011; Watanabe 2013; Yamada 2023), I will step by step outline that they are composed of a predicative copula (PRED) in the sense of Bowers (1993), a dummy copula and a tense marker, starting with the final element, and translate the data into the cartographic structure adopted here. This analysis will then be applied to the shorter version of the copula, I and DA, and I will explain why an adaption to the monomoraic elements occurring in attributive position is not justified.

5.2.1 Nishiyama’s Analysis

5.2.1.1 Tense marker

The final element TA contained in K-AT-TA and D(E)-AT-TA and also occurring with verbs straightforwardly receives an analysis as a tense marker for past.⁴ TA is subject to different phonological processes, namely gemination, and also devoicing yielding the allomorph /da/. The first generalization to be drawn is thus as follows.

⁴TA is likely derived from the premodern auxiliary verb *-tari*, which was added to the nominalization form of inflecting categories and was itself inflecting.

- (6) TA = I (past)

The analogous element for non-past is -U.⁵

- (7) U = I (non-past)

5.2.1.2 Dummy Copula

When added to adjectives and *no*-modifiers, TA is obligatorily preceded by the element AR, which Nishiyama analyzes as the dummy copula. In this sense, the copula is *semantically vacuous*, a “dummy element whose function is merely to support tense” (Nishiyama, 1999, 187). It thus performs a function of *be*-support, similar to English *do*-support.

This view goes, among others, back to Bach (1967) and Doron (1983) and was formalized in Rapoport (1987). In English, the copula *be*, although it can serve the semantically important function of identification (*That woman is Tali.*; cf. Rapoport 1987, 150), is typically devoid of meaning, yet necessary in predicative and equative sentences involving non-verbal elements (examples based on Rapoport 1987, 156).

- (8) a. Xeli *(is) a nut.
b. Tali *(is) that woman over there.

As a contrast, it is not necessary in small clause predication environments involving a lexical verb.

- (9) I consider Xeli a nut. (Rapoport, 1987, 152)

Rapoport (1987) first showed that there is no semantic requirement for the copula, that it is semantically empty, hence must be a syntactic element that satisfies a syntactic requirement. This requirement is that every sentence has to contain a verb, since every sentence needs to express tense (Rapoport, 1987, 157). This can be achieved by a lexical verb (9), or, alternatively, by the copula (8). In absence of a verb, the copula supports the tense (and agreement) features of I (*ibid.*).⁶

In many languages, the semantically vacuous copula can be omitted in the default tense non-past/present, but is needed in other contexts, for example to express past

⁵When /u/ is added to the stem of vowel-stem verbs such as *tabe* ‘to eat’, an epenthetic alveolar /r/ is inserted. Note that there are also consonant-stem verbs whose stem ends in /r/ and which share their paradigm with the dummy copula (see below). In this case, the alveolar consonant is part of the stem. Before the past ending and the suffix -TE, /r/ is deleted and /t/ is geminated. These groups of verbs should thus not be mixed up. They can be further differentiated based on their inflectional paradigms. For discussion see Ito and Mester (2015, 364–365). In any case, the morpheme signaling non-past remains the same.

⁶It does not become clear whether *consider* indeed is a lexical verb in this analysis. See Bowers (1993) for an analysis of *consider* and similar verbs in English.

where it bears tense inflection. This underlines its supporting character for the I node and lack of semantic contribution. Observe the following scenario in Russian (cf. Babby 2010; cf. Doron 1983; Rapoport 1987 for Hebrew).

- (10) a. Pagoda xorošaja. (Russian)
 weather.NOM.SG.F good.NOM.SG.F
 ‘The weather is good.’ present
- b. Pagoda *(byla) xorošaja.
 weather.NOM.SG.F was.SG.F good.NOM.SG.F
 ‘The weather was good.’ past

The semantically vacuous copula in Japanese is identified in Nishiyama (1999) and Urushibara (1993) as -AR- and given the syntactic label *V*. To avoid confusion, I will use the label COP.⁷

- (11) AR = COP

AR can also be used as an independent verb, which expresses existence, identical to English *be*.

- (12) Kuruma-wa niwa-ni ar-u.
 car-TOP garden-LOC COP-PRS
 ‘The car is in the garden.’

It also appears in other scenarios *be* appears in, including equation (13) and identification (14). In both case, another element, namely DE (the predicative copula, see below) is required.

- (13) Hanako-wa sensei-de-ar-u.
 Hanako-TOP teacher-PRED-COP-PRS
 ‘Hanako is a teacher.’

- (14) Ano ko-wa Hanako-de-ar-u.
 that girl-TOP Hanako-PRED-COP-PRS
 ‘That girl over there is Hanako.’

It needs to be made clear that the sole, but very strong, piece of evidence that AR is a (dummy) copula is its requirement to support tense, in other words to host the tense node. Non-verbal elements can never express overt tense, with an overt non-past or past morpheme, without the presence of AR, as given below.

⁷Watanabe (2013) assigns the dummy copula the light verb status *v*. Recall that before the past tense morpheme TA, obligatory gemination happens.

- (15) *Ano doresu-wa shikku-(de)-u / (de)-ta.
 that dress-TOP chic-PRED-PRS / PRED-PST
 ('That dress is/was chic.')

This means, the dummy copula is a different element from DE, which is not sufficient to support tense and will be reviewed below.

5.2.1.3 Predicative Copula

However, a tense marker and the dummy copula are not enough to license an adjective or a *no*-modifier in predicative position either. Additionally, the element DE, in the case of *na*-adjectives and *no*-modifiers, or KU, in the case of *i*-adjectives, are required. Following Nishiyama (1999), I will analyze this element as another, a more meaningful kind of, copula, namely as the syntactic head for predication PRED (Bowers 1993, there PR; cf. also Svenonius 1996; Bowers 2002; den Dikken 2006; Heycock 2013). This element is the real bearer of predication.⁸

Evidence for this analysis and for the differences between the two copulas are visible in the following data points. Precisely, whenever a predication relation between a non-verbal *na*-adjective or *no*-modifier and a verb is established, the predicative copula obligatorily appears, but the dummy copula is absent. This is visible in (16), where *no*-modifiers are used as depictives of a verb. See especially Koizumi (1995), to whom most of these data points and observations can be attributed.

- (16) a. Jon-ga sakana-wo hadaka-*(de) tabe-ta.
 John-NOM fish-ACC naked-PRED eat-PST
 'John ate the fish naked.' (Nishiyama, 1999, 188)
- b. Jon-ga katsuo-wo nama-*(de) tabe-ta.
 John-NOM bonito-ACC raw-PRED eat-PST
 'John ate the bonito raw.' (Koizumi, 1995, 27)

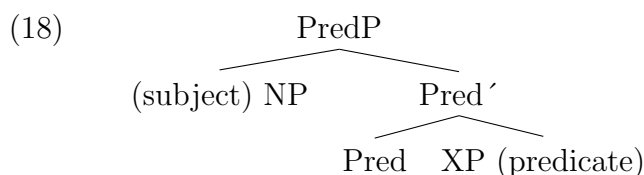
In these examples, the predication relation is established via PRED, which is spelled-out as /de/. While predication needs to be established, the tense node already has a host, namely, in both sentences the verb *tabe* 'to eat'. Therefore, the dummy copula is not needed.

This means the two copulas differ in the following ways:

⁸A similar view can also be found in the work of one of the first Western linguists specialized in Japanese: Bernard Bloch. Bloch assessed in the 1940's that the Japanese copula is an inflectional element and that every sentence needs such an inflectional element as its nucleus, making the copula an essential ingredient of the sentence (Bloch 1970, 35, see Nishiyama 1999, 187–188).

- (17) a. The dummy copula is semantically meaningless, it is simply there to support tense. The predicative copula is the bearer of predication.
 b. The dummy copula cannot occur independently and is not required when a verb is present. The predicative copula can independently co-occur with modifiers in tenseless environments, in which a verb establishes tense.

Following Bowers (1993) and its application to Japanese in Nishiyama (1999), I define PRED as a functional category with a syntactic as well as a semantic dimension. Semantically, it fulfills the function *Pred(ication)* in the sense of Chierchia and Turner (1988). Syntactically, it is a functional head which projects the phrase PredP and selects the maximal projection of any lexical category (AP, NP, PP, etc., given as XP below) as its external argument in its specifier position. From there, a predication relation with the predicate argument in its complement position is established (Bowers, 1993, 595–596). This means PRED is the predicative counterpart of MOD (Rubin 1994; Chapter 3.2, see especially Chapter 3.2.3). The general structure is given below.



There are many syntactic points of justification and empirical arguments why the category PRED is necessary, and why, besides its semantic dimension, it should be kept apart from VP. These include the following (see especially Bowers 1993, 598–599):⁹

⁹As with regard to the syntactic head MOD, the adoption of a universal head PRED is not without controversy. For discussion see Bowers (1993, 2002); Svenonius (1996); den Dikken (2006); Heycock (2013) as well as Matushansky (2019) and Bruening and Al Khalaf (2020), who have spoken decisively against this hypothesis. Besides theory-internal problems, the authors argue that the relevant head is redundant, as many problems it meant to solve in the original theory can now be solved via other mechanisms, and this is indeed the case. For instance, unlike categories can also be coordinated via a Boolean phrase.

Furthermore, Matushansky (2019) tries to show that the alleged manifestations of PRED in several languages do not hold, because each language shows exceptions, where a copular element does not appear even though one would expect it to. In Edo, the alleged PRED elements do not appear in resultatives, in Welsh they do not appear in PPs, in the Berber language East Riffian they are restricted to adjectives of Arabic origin (Matushansky, 2019, sec. 4).

While this is problematic for the universality of the Pred-hypothesis, Matushansky's analysis does not prove that no PRED-like elements exist. What holds for MOD also holds for PRED: The relevant element is attested in Japanese and sticking with the idea of language-uniformity, I acknowledge its value and argue that it is present in Japanese and by extension cross-linguistically.

- (19) Having a functional head PRED allows for the coordination of unlike lexical categories.

Bowers (1993) explained cases such as *I regard him* $[[_{DP} \text{a fool}] \text{ and } [_{AP} \text{unable to do his job.}]]$ without having to resort to non-binary structures. Both coordinates are being mediated by the predication head.

- (20) It enables movement of lexical categories, with the possibility that the subject can remain in Spec, PredP.

In a sentence such as *How stupid do* $[_{PREDP} \text{I } [_{PRED} \text{consider him?}]]$, the subject remains in-situ, while phrasal material moves above it. This was a significant contribution at the time (cf. Matushansky 2019, 68–69 for discussion).

- (21) PredP provides a syntactic position for the subject in expletive-construction and various other small clauses, fulfilling the EPP even in absence of thematic relations.

Exemplifying with tenseless small clauses such as *I consider* $[_{him} \text{stupid.}]$, no obvious subject position is available. This subject position is, however, offered by PRED in addition to licensing predication.

- (22) Subject and object position are kept apart, the subject being in Spec, PredP, the object in Spec, VP.

This relates to the previous point in the sense that also non-thematic subjects, for example those in *tough* constructions, receive a subject position, although this position is not related to thematic role assignment, i.e. that of the agent.

- (23) Many languages have overt realizations of PRED.

Probably the strongest piece of evidence for PRED is offered by its lexical and syntactic manifestations in various languages. This includes *as* and *for*, which appear in small clauses with matrix verbs such as *regard* in absence of a copula in English (24-a) (Bowers, 1993, 596). Similar analyses have been proposed for the cross-linguistic counterparts of these elements, for example Russian *kak* ‘as’ (24-b) (or *za* ‘as’ or *v* ‘in’; cf. Bailyn (2002)).¹⁰

- (24) a. I regard $[_{SC} \text{John as crazy}]$.
 SUB V OBJ PRED A (Bowers, 1993, 596)

¹⁰Further possible manifestations of PRED include instrumental case in Russian (Bailyn, 2001), and copular elements in a variety of African languages including Edo and Chichewa (Baker, 2003a), as well as Welsh (Bowers, 2002) and Greek (Campos and Stavrou, 2004).

- b. On vygladit [kak durak]. (Russian)
 he look PRED fool
 ‘He looks like a fool.’ (Bailyn, 2002, 39)

This also includes the overt manifestations of PRED in Japanese. Besides depictives, presented in (16), another instance where the predicative copula is manifested in absence of the dummy copula are coordination structures. As argued in Chapter 4.2.4.2, DE must either autonomously or together with the sentential conjunction marker KATSU establish coordination of two nominals. Such examples were analyzed as clausal coordination in which two instances of the copula must appear, one at the end of the sentence as the main clause predicate, one after the first conjunct. An example is repeated below (cf. Hiraiwa 2012, 364, from which examples are adapted).

- (25) Kono hito-wa Ken-no sensei *(katsu) / *(de) Naomi-no
 this person-TOP Ken-MOD teacher and / PRED Naomi-MOD
 sensei-da.
 teacher-COP
 ‘This person is Ken’s teacher and Naomi’s teacher.’

Recall also that direct modifiers, such as *Chūgoku-no* ‘Chinese’, on the other hand, can never appear in such coordination structures, not even as the second modifier in the DP. As they can never co-occur with the copula in predicative position, it is expected that they can therefore never co-occur with DE either, if this element has copular status as well.

- (26) a. *Chūgoku-de kirei-na kabin
 China-PRED pretty-na vase
 b. *kirei-de Chūgoku-no kabin
 beautiful-PRED China-MOD vase
 (‘a Chinese and beautiful vase’)

Finally, there are some historical pieces of evidence for the copular status of DE. This form in fact goes back to an amalgamation of the premodern copula NI – the short form of the premodern copula inflected for nominalization form *nari* – and the coordinative suffix TE. These two elements began in Classical Japanese to phonologically assimilate and have completed merging into one element, DE, in Modern Japanese (Frellesvig, 2010; Miyama, 2011).

This means that /de/ in DE-AR-U and in DE-AT-TA is the predicative copula. The representation of all three elements thus looks as follows.

- (27) a. Yoru-ga shizuka-de-ar-u.
 night-NOM quiet-PRED-COP-PRS
 ‘The night is quiet.’
 b. Yoru-ga shizuka-de-at-ta.
 night-NOM quiet-PRED-COP-PST
 ‘The night was quiet.’ (Nishiyama, 1999, 189, 194)

What about /d/ in D-AT-TA, in which the vowel /e/ is not present? The vowel /e/ is deleted here (Nishiyama, 1999, 203), probably to avoid the clashing of two vowels, a phenomenon not uncommon in Japanese, both diachronically and synchronically.

Coming now to *i*-adjectives, those exhibit, in certain contexts, the same complex tripartite of morphemes parallel to the complex form DE-AR-U. This complex form is KU-AR-U. It is straightforward, and the analysis adopted in Nishiyama (1999), that KU corresponds to DE. All things being equal, these two complex forms only differ in one aspect, namely the realization of the predicative copula. Japanese thus provides evidence for a second possible realization of PRED.

One thing to consider is that KU-AR-U is restricted in its distribution and rather marked in canonical root clauses.

- (28) ??Ano doresu-wa taka-ku-ar-u.
 that dress-TOP expensive-PRED-COP-PRS
 ‘That dress is expensive.’

A prominent context in which this complex form is allowed, however, are emphatic expressions containing the focus-sensitive particles MO ‘even’, SAE ‘even’ or BAKARI ‘only’. Interestingly, those intervene between KU and AR-U.¹¹

- (29) Ano doresu-wa taka-ku mo ar-u.
 that dress-TOP expensive-PRED EMP COP-PRS
 ‘That dress is even expensive.’

There are further instances where *i*-adjectives must co-occur with these morphological complex forms: Before the modals *beki* ‘should’ and *mai* ‘will not be’ and in subordinate clauses with the complementizer *koto* (Nishiyama 1999, 199 and Miyama 2011, 51). In these cases, these elements form a direct string.

- (30) a. Ano doresu-wa taka-ku-ar-u beki-da / mai.
 that dress-TOP expensive-PRED-COP-PRS should-COP / will.not.be
 ‘That dress should be / will not be expensive.’
 b. Ano doresu-wa taka-ku-ar-u koto-wo wasure-ta.
 that dress-TOP expensive-PRED-COP-PRS COMP-ACC forget-PST
 ‘I forgot that that dress is expensive.’

¹¹This was noted as early as Miyata (1948), see Nishiyama (1999, 185) and Kubo (1992). It is unclear where exactly those focus markers emerge and are situated.

In the same constructions, *na*-adjectives (and *no*-modifiers) must co-occur with the corresponding form of the canonical copula DE-AR-U (or past DE-AT-TA), in fact DA is not possible (Nishiyama, 1999, 186). As shown in (32), their use in emphatic contexts further highlights the parallelism between K(U) and DE.

- (31) Ano doresu-wa shikku-de-ar-u beki-da / *shikku-da beki-da.
 that dress-TOP chic-PRED-COP-PRS should-COP / chic-COP should-COP
 ‘That dress should be chic.’
- (32) Ano doresu-wa shikku-de mo ar-u.
 that dress-TOP chic-PRED EMP COP-PRS
 ‘That dress is even chic.’

K(U) also appears in other contexts in which DE appears, for instance in coordinative structures involving *i*-adjectives. Recall from Chapter 4.2.4.2 that it is optionally followed by -TE, yielding the more canonical option. Now, we can safely identify KU in these contexts as the predicative copula as well.

- (33) aka-ku-(te) utsukushi-i doresu
 red-PRED-GER beautiful-MOD dress
 ‘a red and beautiful dress’

The predicative copula is used in tenseless constructions as well, for example in resultative, inchoative and causative constructions. As expected, it is not followed by the dummy copula in these contexts, because a verb is present to establish tense. This can be put forth as a further piece of evidence that Japanese has a requirement for PRED (different to, for example, English, visible in the translations); for similar examples and discussion also see Yamada (2023, 8–10). Finally, the adverbial inflectional form of *i*-adjectives is also /ku/ and receives the same analysis as PRED (Nishiyama, 2005), but there is some controversy (Namai, 2002).

- (34) a. Jon-ga kabe-wo aka-ku nut-ta.
 John-NOM wall-ACC red-PRED paint-PST
 ‘John painted the wall red.’
 resultative (Nishiyama, 2005, 153)
- b. utsukushi-ku nar-u
 beautiful-PRED become-PRS
 ‘become beautiful’ inchoative
- c. utsukushi-ku sur-u
 beautiful-PRS do-PRS
 ‘make beautiful’ causative
- d. haya-ku arui-ta
 fast-PRED walk-PST
 ‘walked fast’ adverbial

Next we come to past contexts. Consider first the following example in which an *i*-adjective is inflected for past and the morphological complex form is KU-AT-TA. A focus sensitive particle occurs between the predicative and the dummy copula, as in the examples above.

- (35) Hanako-wa kanashi-ku sae at-ta.
 Hanako-TOP sad-PRED EMP COP-PST
 ‘Hanako was even sad.’ (Kubo, 1992, 109)

In past contexts, when the three elements occur uninterrupted, *i*-adjectives co-occur with the complex form K-AT-TA, not KU-AT-TA, as given below.

- (36) Ano doresu-wa taka-k-at-ta.
 that dress-TOP expensive-PRED-COP-PST
 ‘That dress was expensive.’

We can conclude that KU-AT-TA, or K-AT-TA, contain the same elements as DE-AT-TA, or D-AT-TA. However, there is no free alternation and K-AT-TA has a default character. Instead of formalizing a parallel rule of contraction and vowel reduction, Nishiyama (1999, 192–193) treats [u] as an epenthetic vowel and simply analyzes the consonant part /k/ as PRED. This analysis receives support from phonological observations. As Japanese favors open syllables, an epenthetic [u] is often inserted in word-final position as well (Kubozono, 2005; Ito and Mester, 2015). The analysis also becomes relevant for the treatment of the simple element -I (see Chapter 5.2.3) and will be adopted in the spell-out rule below.

Before closing this subsection, I point out two more scenarios in which the predicative copulas DE and K(U) appear. One concerns negation. Given what we know about predication and its syntactic realization in Japanese, it is expected that the predicative copula has to appear in contexts of negation in order to license predication.

Negation is expressed via the negative element NA, which can probably be regarded as the negative form of the dummy copula.¹² In the case of *na*-adjectives, additionally the topic marker WA intervenes between this element and the predicative copula (39-a).¹³ In colloquial speech, DE and WA are contracted to JA (39-b).¹⁴

¹²Since it inflects exactly like an *i*-adjective, it is also often classified as such. Note that -I, as it appears in predicative position, will be analyzed as an amalgamation of copulas and tense markers following *i*-adjectives below, and is therefore not glossed for now.

¹³Note that WA could very well also be analyzed as a focus marker here, as done in Yamada (2023, 6), but nothing hinges on this.

¹⁴The predicative copula is also obligatory for the polite form of negation, in which the existential verb, i.e. the dummy copula AR, is followed by the polite auxiliary element -MAS-.

- (37) Ano doresu-wa shikku-de-wa-ari-mase-n.
 that dress-TOP chic-PRED-TOP-COP-AUX.POL-NEG

- (38) Ano doresu-wa utsukushi-ku-nai.
that dress-TOP beautiful-PRED-NEG
‘That dress is not beautiful.’ negation, *i*-adjective
- (39) a. Ano doresu-wa shikku-de-wa-nai.
that dress-TOP chic-PRED-TOP-NEG
b. Ano doresu-wa shikku-ja-nai.
that dress-TOP chic-PRED.TOP-NEG
‘That dress is not chic.’ negation, *na*-adjective

On a final note, PRED has another realization, namely /ni/, as argued in Nishiyama (1999). This element co-occurs with *na*-adjectives in tenseless environments, such as resultative, inchoative and causative constructions, but not in depictives. Under such an analysis, it can be given the same status as K(U), meaning that *na*-adjectives show an alternation in the realization of PRED that *i*-adjectives do not show (Nishiyama 1999, 205–208).¹⁵

- (40) a. Jon-ga kabe-wo makka-ni nut-ta.
John-NOM wall-ACC crimson.red-PRED paint-PST
‘John painted the wall crimson red.’
resultative (Nishiyama, 1999, 207)
- b. shizuka-ni nar-u
quiet-PRED become-PRS
‘become quiet’ inchoative
- c. shizuka-ni sur-u
quiet-PRED do-PRS
‘make quiet’ causative
- d. shizuka-ni arui-ta
quiet-PRED walk-PST
‘walked quietly’ adverbial (Nishiyama, 2005, 150)

To summarize, Japanese provides lexical evidence for the head PRED, which is conditioned in its realization by word class, syntactic environment and phonology. I propose the following spell-out rules adopted from Nishiyama (1999, 207), where [+V] represents resultative, inchoative and causative use of *na*-adjectives respectively. These rules mean that PRED is always realized as /k/ when preceded by an *i*-adjective, as /ni/ when preceded by a *na*-adjective in a tensed environment and otherwise as /de/.

‘That dress is not chic.’

¹⁵Again, I present Nishiyama’s analysis of adverbially used *na*-adjectives as co-occurring with PRED at face value without comments, knowing there is yet to be found empirical proof for these claims. An alternative could be to treat NI as an instance of the homophonous dative marker as suggested to me by Jim Kotopoulos.

- (41) Spell-out rule for PRED in Japanese

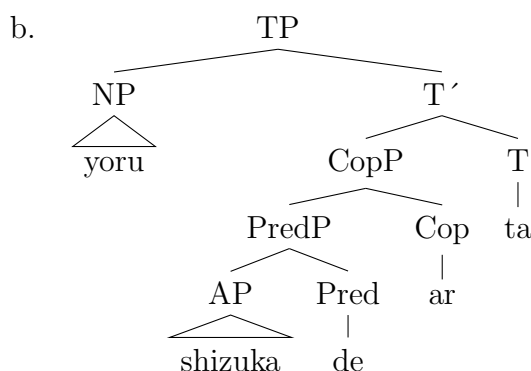
$$\begin{aligned}
 [Pred] &\leftrightarrow /k/ \ / \text{ iA } - \\
 [Pred; +V] &\leftrightarrow /ni/ \ / \text{ naA } - \\
 &\quad \quad \quad /de/
 \end{aligned}$$

5.2.1.4 Structural representation of predicative use

Having established the ingredients of the morphologically complex forms DE-AT-TA/D-AT-TA, DE-AR-U and K-AT-TA, I will now derive the relevant structure, which I adopt from Nishiyama (1999, 189–194). I will start with the predicative structure for past.

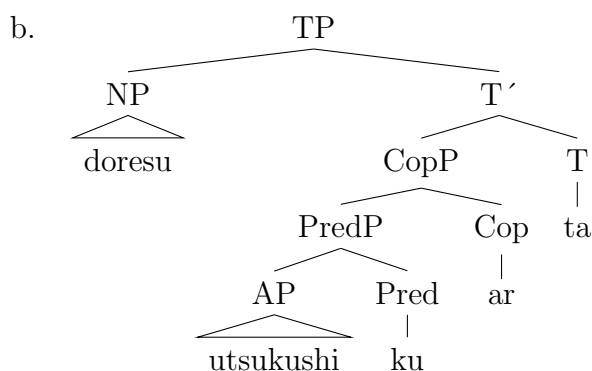
- (42) Predicative
- na*
- adjectives in past (Nishiyama, 1999, 189, 194)

- a. Yoru-ga shizuka-d(e)-at-ta.
 night-NOM quiet-PRED-COP-PST
 ‘The night was quiet.’



- (43) Predicative
- i*
- adjectives in past (Nishiyama, 1999, 190, 192)

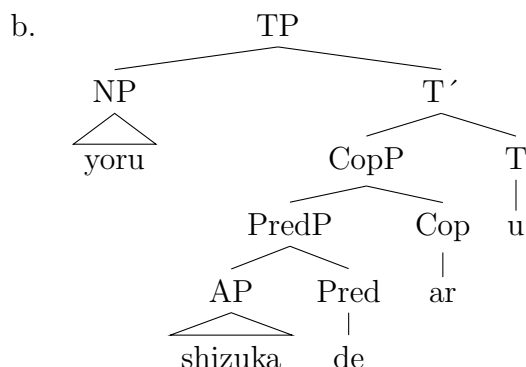
- a. Doresu-ga utsukushi-k-at-ta.
 dress-NOM beautiful-PRED-COP-PST
 ‘The dress was beautiful.’



These structures are exactly parallel, the only difference being the different spell-out of PRED. The structure for non-past DE-AR-U is also exactly analogous, mind the different tense marker for non-past.

- (44) Predicative *na*-adjectives in non-past (Nishiyama, 1999, 189, 194)

- a. Yoru-ga shizuka-de-ar-u.
 night-NOM quiet-PRED-COP-PRS
 ‘The night is quiet.’

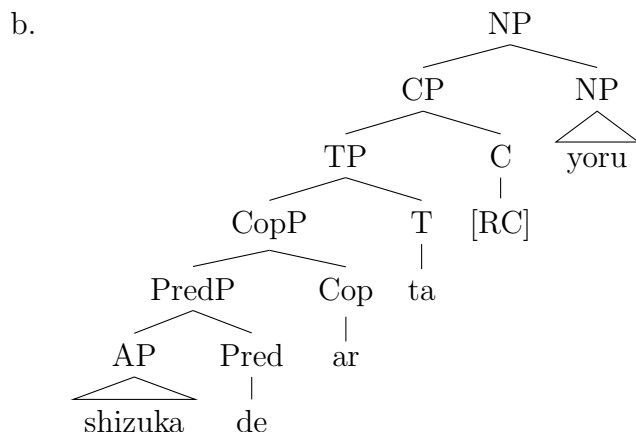


5.2.2 Application to attributive use

As the past forms K-AT-TA and D(E)-AT-TA as well as the non-past form DE-AR-U can be used attributively without morphological difference to their predicative counterparts, the predicative structure can simply be translated into an attributive structure. According to Nishiyama, attributively used adjectives project an additional CP layer whose head is empty and contains a relative clause feature (Nishiyama 1999, 197; cf. Kaplan and Whitman 1995). In the tree below, I depict the structure for a DP containing a *na*-adjective inflected for past. The structures for K-AT-TA and DE-AR-U are analogous apart from the different realization forms of PRED and I.

- (45) Attributive *na*-adjectives in past (Nishiyama, 1999, 197)

- a. shizuka-d(e)-at-ta yoru
 quiet-PRED-COP-PRS night
 ‘the night which was quiet’



5.2.3 Application to predicative *da* and *i*

Next, the analysis can be extended to the predicative elements DA and I.

5.2.3.1 Predicative *da*

It is a standard assumption that DA is a variant of the canonical copula DE-AR-U (Urushibara, 1993) and these elements have been treated similarly in this book. There are several pieces of evidence for this claim (cf. Miyama 2011). First, those elements are almost equally distributed, and they are mutually exclusive. However, while DE-AR-U can appear in almost all environments (except before sentence-final particles where it is marked, Miyama 2011, 45–46), DA cannot. Instances where it is not possible include attributive structures, and, as mentioned above, before certain grammatical elements such as *beki* ‘should’, as well as in emphatic constructions. Second, that DA is likely historically derived from DE-AR-U is suggested by the in-between stages DE-AR and DEA (Narahara, 2002).

From a morphosyntactic perspective, a corresponding analysis of DA to DE-AR-U seems to be pretty straightforward as well – at least to a certain point. Morphologically, the initial consonant /d/ can receive an analysis as PRED, exactly parallel to /d/ in D-AT-TA. The vowel /a/ is a realization of the dummy copula COP.

It is crucial to note that /da/ cannot be an allomorph of /de/, since the two forms are not paraphrasable, not even in emphatic constructions. Compare (46) to (32).

- (46) *shizuka-da mo ar-u
 quiet-DA EMP COP-PRS
 (‘it is even quiet’) (Nishiyama, 1999, 186)

What is striking is the alleged absence of a tense marker. Based on the fact that DA and DE-AR-U are, in most environments, interchangeable, Nishiyama simply treats the two as allomorphs and assumes that all three elements – predicative copula, dummy copula and tense marker for non-past – are simply combined in /da/ as shown below (Nishiyama, 1999, 196–197).

- (47) Yoru-ga shizuka-da.
 night-NOM quiet-PRED.COP.PRS
 ‘The night is quiet.’ (Nishiyama’s analysis)

Nothing immediately hinges on this, but I will assume here that the predicative copula is the consonant /d/ and that the dummy copula and the tense marker for present are merged in /a/ as shown below.

- (48) D-A = PRED-COP.PRS

This means that DA is not derived from DE-AR-U, but simply a shorter form, in which all relevant elements, including the tense marker, are still present.

5.2.3.2 Predicative *i*

A similar structure can be derived for *i*-adjectives, although predicative -I, at first glance, does not seem to bear any resemblance to KU-AR-U. First, note that non-past is the only environment in which the velar consonant /k/ is absent in the inflectional paradigm of *i*-adjectives. In Classical Japanese, however, the attributive form of the inflectional paradigm of *i*-adjectives was -KI, thus included this consonant (cf. Nishiyama 1999, 189–190). Nishiyama (1999) accounts for these facts as follows: Modern I is a direct successor of KI derived via a historical devoicing process (cf. Frellesvig 2010). The consonant /k/ was an instance of PRED and continued this role in Modern Japanese, although it is not pronounced. He formalizes this through a *Velar-filter* (Nishiyama, 1999, 191).

In other words, the head of PredP is structurally present, but phonologically reduced.¹⁶ *i*-adjectives thus co-occur with the underlying elements κ -I, which constitute a combination of the predicative copula and a tense marker and are realized as /i/ (cf. Nishiyama 1999, 199–201; cf. also Baker 2003b, 2).

(49) Nishiyama’s analysis for *i*-adjectives in non-past

- a. Ano doresu-ga taka-~~k~~-i.
that dress-NOM expensive-PRED-PRS
‘That dress is expensive.’

But, under this analysis no dummy copula is present. Recall that this is also the case in other languages in default non-past contexts for instance Hebrew (Doron, 1983; Rapoport, 1987) and Russian (Babby, 2010), as in the example in (10) repeated below. This can thus be applied to Japanese (Nishiyama 1999, 191–192, and see Urushibara 1993).

- (50) Pagoda xorošaja.
weather.NOM.SG.F good.NOM.SG.F
‘The weather is good.’ Russian

An obvious challenge to this account is the question how tense is supported, if no dummy copula is present. As explained above, the dummy copula is needed to support the I node and establishes some kind of *be*-support. And while no overt

¹⁶Such phonological processes are not uncommon diachronically, as the consonant /k/ has been systematically reduced in front of the vowel /i/ (Frellesvig, 2010, 340), but also attested synchronically with certain verb forms. For example, the past form of *kak* ‘to write’ is *kai-ta*, not **kaki-ta* (Nishiyama, 1999, 191).

tense is present in the Russian example, Nishiyama argues that non-past information is included in I. Furthermore, how can the parametric difference between *i*- and *na*-adjectives, that necessarily arises, be accounted for? For lack of better alternative, I assume for predicative I almost the same structure as for DA: The predicative copula is represented via the filtered consonant /k/, and the dummy copula and the tense morpheme are merged in /i/.

(51) (K)-I = PRED-COP.PRS

5.2.4 Why the application to attributive *i* and *na* fails

This section refutes Nishiyama's analysis of the attributive forms for non-past, I and NA, as copular elements. It is not surprising that Nishiyama (1999) puts forth the same analysis for I in attributive contexts as for I in predicative contexts, thus treats those as the same elements. The only difference is that, the adjective projects an additional CP layer. He suggests the following (Nishiyama, 1999, 199–202).

(52) Attributive *i*-adjectives in non-past according to Nishiyama (not correct)

- a. taka-(k-)i doresu
 expensive-PRED-PRS dress
 'dress which is expensive'
- b.
-
- ```

graph TD
 NP1[NP] --- CP[CP]
 NP1 --- NP2[NP]
 CP --- TP[TP]
 CP --- C[C]
 TP --- PredP[PredP]
 TP --- T[T]
 PredP --- AP[AP]
 PredP --- Pred[Pred]
 AP --- taka[taka]
 Pred --- k[k]
 T --- i[i]
 C --- RC["[RC]"]
 NP2 --- doresu[doresu]

```

For *na*-adjectives, Nishiyama (1999, 196–199) analyzes NA as the very same element as DA, thus as a combination of predicative copula, dummy copula and tense marker for non-past. The only difference in consonant realization is conditioned by the presence of a CP. This means that the labor between predicative and attributive use is strictly divided in *na*-adjectives via DA and NA, which are allomorphs and not interchangeable (Nishiyama, 1999, 197). He gives the following phonological transformation rule in the spirit of Distributed Morphology (Halle and Marantz, 1993, 1994).

- (53) [pred.cop, dum.cop, -past, rel.cl]  $\leftrightarrow$  /na/ / NA -  
(Nishiyama, 1999, 197)

Nishiyama (1999) treats all attributive adjectives in Japanese as embedded in a CP layer and therefore as indirect modifiers.<sup>17</sup> Baker (2003b) and Hoshi (2002) adopt his analysis and explicitly deny the possibility of direct modification in Japanese.

Nishiyama not only encourages the long-existing analysis of NA as the attributive form of the copula, (Teramura, 1982; Miyagawa, 1987; Murasugi, 1991; Kubo, 1992; Urushibara, 1993), but also manages to unify *i*- and *na*-adjectives by analyzing I in a parallel fashion. This is a significant contribution. However, this analysis over-generalizes as presented in abundance throughout this book. Although sometimes up for discussion, the last chapter mentioned many direct *i*- and *na*-adjectives such as the following, and see especially Yamakido (2005) or Ayano (2010) from where these examples are repeated.

- (54) a. Omoigakena-i kyaku-ga ki-ta.  
unexpected-MOD guest-NOM come-PST  
'An unexpected guest came.'  
(Yamakido, 2005, 67)
- b. \*Ano kyaku-ga omoigakena-i.  
that guest-NOM unexpected-PRED.COP.PRS  
(‘That guest is unexpected.’)
- (55) a. Max-ga kanzen-na baka-d-a.  
Max-NOM complete-MOD idiot-PRED-COP.PRS  
'Max is a complete idiot.'  
(Yamakido, 2005, 68)
- b. \*Ano baka-ga kanzen-d-a.  
that idiot-NOM complete-PRED-COP.PRS  
(‘That idiot is complete.’)

This means I and NA in attributive position cannot be universally analyzed as copular elements which is why I deny this analysis in favor of MOD (Chapter 3.2). In turn, this means there are two different instances of I: A copular element in predicative position and the syntactic linker MOD in attributive position.

### 5.2.5 Application to *no*-modifiers

The analysis presented here can be straightforwardly extended to *no*-modifiers, which share their entire inflectional paradigm with *na*-adjectives with the exception of NO. All the elements receive the same analysis, illustrated first with DA and

<sup>17</sup>Nishiyama himself does consider the possibility of direct modification. He states “[...] we know that natural language allows adjectives to act as direct modifiers (as in English). Thus, unless there is evidence to the contrary, the null hypothesis is that Japanese adjectives also act as direct modifiers [...]” (Nishiyama, 1999, 201) and mentions MOD presented in Chapter 3.2 as an alternative.



DE-AR-U in combination with a clear noun (56), and a more adjectival *no*-modifier (*no*-adjective, (57)). See also Nishiyama (1999, 195).

- (56) a. Hanako-wa sensei-d-a.  
Hanako-TOP teacher-PRED-COP.PRS  
b. Hanako-wa sensei-de-ar-u.  
Hanako-TOP teacher-PRED-COP-PRS  
'Hanako is a teacher.' noun
- (57) a. Haishizun-ni-wa hayame-no yoyaku-ga  
high.season-LOC-TOP early-MOD reservation-NOM  
hisshu-d-a.  
necessary-PRED-COP.PRS  
b. Haishizun-ni-wa hayame-no yoyaku-ga  
high.season-LOC-TOP early-MOD reservation-NOM  
hisshu-de-ar-u.  
necessary-PRED-COP-PRS  
'During high season, early reservation is necessary.'<sup>18</sup>  
*no*-adjective

The analysis also carries over to DE-AT-TA, which can-occur with a *no*-modifier as well.

- (58) Hanako-wa sensei-de-at-ta.  
Hanako-TOP teacher-PRED-COP-PST  
'Hanako was a teacher.' noun, past

Naturally, this extends to the attributive use of DE-AR-U and DE-AT-TA. Same as DA and DE-AR-U are mutually exclusive in predicative position, NO and DE-AR-U are mutually exclusive in attributive position.

- (59) sensei-de-ar-u Hanako  
teacher-PRED-COP-PRS Hanako  
'Hanako, who is a teacher'

Although Nishiyama does not focus on NO-modifiers, he proposes a spell-out rule for *no* as well, which looks as follows.

- (60) [pred.cop, dum.cop, -past, rel.cl]  $\leftrightarrow$  /no/ \_  
(Nishiyama's analysis of NO; Nishiyama 1999, 197)

This spell-out rule says that NO is a manifestation of the predicative copula, dummy copula and non-past tense marker and appears in relative clause contexts. It is realized in the elsewhere-case, meaning when the modifier is not a clear *na*-adjective.

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<sup>18</sup>ISHIMORI Hiromi: *Tabi, chotto senchimentaru*. Tankobon, 2002. Retrieved via BCCWJ.

As in the case of the other modifiers, this cannot be the right analysis given the abundance of direct modifiers among the group of *no*-modifiers. Instead, I have put forth an analysis in Chapter 3.2 of NO as an instance of MOD which is blocked when DE-AR-U or other elements with an overt I-head appear.

However, it is worth emphasizing that the existence of such cases, those in which a *no*-modifier has a clear predicative function, show that not all *no*-modifiers are direct. This, in turn, refutes a claim made in Nagano and Shimada (2015), who argue, “Japanese adjectives differ from English adjectives in formally disambiguating direct and indirect modification” (Nagano and Shimada, 2015, 124). According to the authors, it is only *no*-modifiers which are responsible for direct modification, all other modifiers in Japanese being indirect. However, this is simply not true as shown in many examples throughout this book and in the data set.

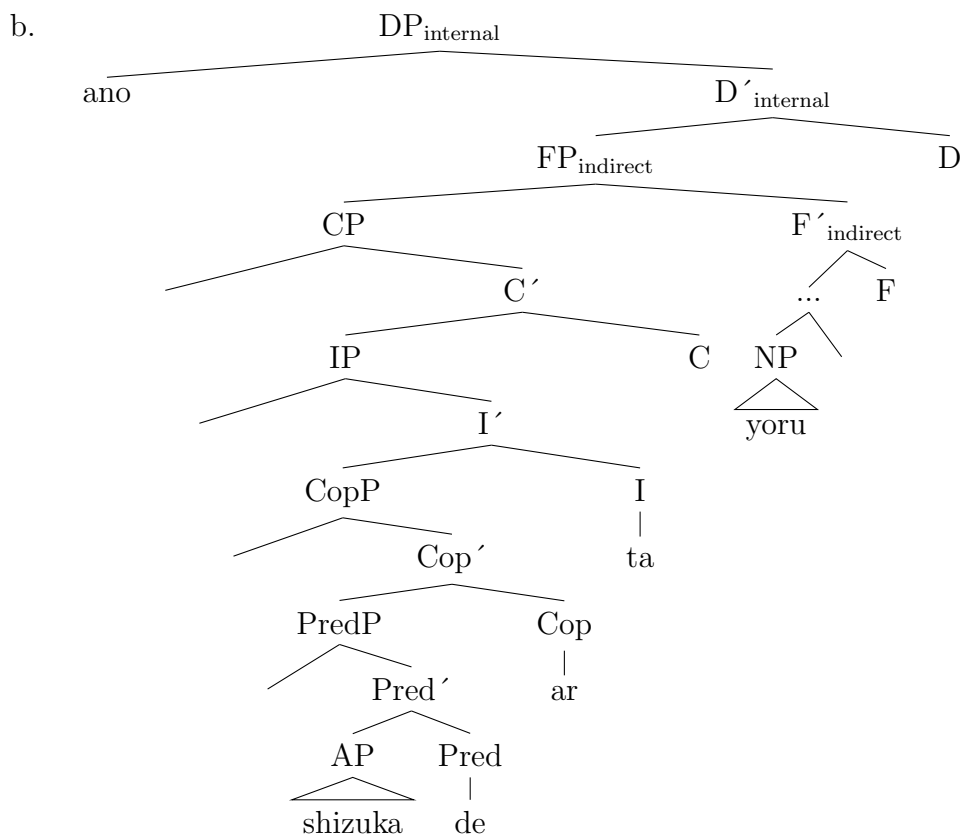
### 5.2.6 The integration of indirect modifiers into the DP

The complex morphological elements deduced in this section can be neatly integrated along the modifier into a cartographic structure of the DP. This is exemplified below with a *na*-adjective inflected for past, but the derivation is exactly parallel for the other elements for past (D-AT-TA for *no*-modifiers, K-AT-TA for *i*-adjectives) as well as DE-AR-U for non-past. Indirect modifiers are merged in a functional projection in the indirect domain and are embedded in a complex set of functional layers whose heads spell-out the predicative copula, the canonical copula and the tense morpheme respectively.<sup>19</sup>

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<sup>19</sup>In this chapter, I will not include the universal syntactic head MOD in the visualizations of relative clauses in cases where it is not realized. The CP status of relative clauses will be defended in the following.

- (61) a. ano shizuka-d(e)-at-ta yoru  
 that quiet-PRED-COP-PST night  
 ‘that night which was quiet’



### 5.2.7 A note on the polite elements

At this point, the question remains why the polite copula DESU, glossed as COP.POL up to now, and its counterpart for past DESHITA are mutually exclusive with the copular forms DE-AR-U and DE-AT-TA, but follow I and also K-AT-TA.<sup>20</sup> Observe some relevant examples.

- (62) a. Ano doresu-wa shikku (\*d-a) desu.  
 that dress-TOP chic PRED-COP.PRS DESU  
 ‘That dress is chic.’ polite non-past, *na*-adjective
- b. Ano doresu-wa shikku (\*d-at-ta) deshi-ta.  
 that dress-TOP chic PRED-COP-PST DESU-PST  
 ‘That dress was chic.’ polite past, *na*-adjective

<sup>20</sup> Assuming there are three stages of politeness in Japanese (Minami, 1977), DESU expresses the medial level and is used when talking to strangers and superiors. In the verbal domain, the corresponding politeness is achieved via the polite auxiliary -MAS-, in past MASHI-TA. Also note that some other grammatical elements, for example the epistemic suffix *mitai* ‘seem’, occur after I but replace DA (Kubo, 1992; Urushibara, 1993; Nishiyama, 1999).

- (63) a. Ano doresu-wa taka-\*(i) desu.  
 that dress-TOP expensive-PRED.COP.PRS DESU  
 ‘That dress is expensive.’ polite non-past, *i*-adjective
- b. Ano doresu-wa taka-k-at-ta desu.  
 that dress-TOP expensive-PRED-COP-PST DESU  
 ‘That dress was expensive.’ polite past, *i*-adjective

(63-b) is the only way to express politeness and past tense on an *i*-adjective at the same time. The polite past form DESHITA never co-occurs with *i*-adjectives.

- (64) \*Ano doresu-wa taka-i deshi-ta.  
 that dress-TOP expensive-PRED.COP.PRS DESU-PST  
 (‘That dress was expensive.’) (polite)

In a recent analysis, Yamada (2023), and see also Miyagawa (2022), argues that Modern Japanese has two separate instances of DESU, which correspond to two different grammars. This explains the different distribution. In a nutshell, the element DESU following *na*-adjectives (and nouns/*no*-modifiers) consist of the familiar tripartite of elements. The element DESU following *i*-adjectives, on the other hand, is just a single marker of politeness and crucially follows the copula (Yamada, 2023, 2–4).

In the first case, DESU/DESHITA consist of the identical elements as DE-AR-U/DE-AT-TA, but significantly, politeness is marked. Translated into the analysis proposed here, I treat -U and -TA as tense morphemes, DE as the predicative copula, and the medial elements -S- and -SHI- as the polite version of the dummy copula.<sup>21</sup> This means that the dummy copula is not simply semantically vacuous, but has a semantic and pragmatic dimension. Nevertheless it needs to support tense. Therefore, I maintain the gloss COP.POL for the medial part, highlighting that it is a polite version of the dummy copula, as shown below.

- (65) a. Yoru-ga shizuka-de-s-u.  
 night-NOM quiet-PRED-COP.POL-PRS  
 ‘The night is quiet.’ polite non-past, *na*-adjective
- b. Yoru-ga shizuka-de-shi-ta.  
 night-NOM quiet-PRED-COP.POL-PST  
 ‘The night was quiet.’ polite past, *na*-adjective

<sup>21</sup>Note that the difference in consonant quality between the consonants /s/ and /sh/ is simply phonologically governed (see Ito and Mester 2015). The epenthetic vowel [i] has to be inserted after /s/, preserving this consonant as well as the CV mora structure favored in Japanese. [i] becomes the coda of the new mora [ɛi] (/shi/). The consonant shift of the preceding consonant from [s] to [ɕ] is systematic in the sense that the fricatives [s] and [z] always become alveolo-palatal before [i]. The mora [si] simply is never realized in Japanese (Lewin, 1989; Vance, 2008).

- There are some pieces of evidence for this claim. First, DESU in (66) is syntactically not necessary, in contrast to its combination with *na*-adjectives. This means it has a syntactic dummy status, albeit a more semantic contribution. Furthermore, DESU in such instances is never paraphrasable with DE-AR-U as shown below (and see Yamada 2023, 10–11).

- Furthermore, use of the grammaticalized version of DESU has in recent times increased. It is also attested after D-AT-TA, where the copular elements are already present as well (Yamada, 2023).

- Yamada (2023) also gives examples where DESU appears multiple times inside one sentence, each time followed by a sentence-final particle (69).

- These examples show that this instance of DESU has been grammaticalized into an element that simply marks politeness and co-occurs with the copula. To summarize:<sup>23</sup>

<sup>23</sup>Note that the question is left open why DESU can never appear in attributive position.

- (72) a. DE-S-U = PRED-COP.POL-PRS (*na*-adjectives and nouns)  
 b. D(E)-SHI-TA = PRED-COP.POL-PST (*na*-adjectives and nouns)  
 c. DESU = POL (*i*-adjectives and elsewhere)

To summarize, the element DE-S-U is another form of the copula, and, as shown in the previous chapter, equally incompatible with direct modifiers. It contrasts with another element DESU used to mark politeness.<sup>24</sup>

### 5.2.8 Summary

This section has adopted the landmark analysis of Nishiyama for the complex elements Japanese (non-verbal) modifiers co-occur with showing they are combinations of a tense marker and two different copulas. This analysis can readily be incorporated into the cartographic framework assumed here. Next, I will discuss the size of indirect modifiers in Japanese.

## 5.3 The size of indirect modifiers in Japanese

As showed in the introduction to this chapter, the size of Japanese relative clauses, whether they are IPs or CPs, is difficult to determine. Interest in this matter has

- (70) \*ano shikku desu doresu  
 that chic DESU dress  
 ('that chic dress') (polite)

Tentatively, I assume this really is a pragmatic ban, restricting politeness to the matrix clause. Some authors (Horie, 2017) have shown that verbs in a polite speech style are in some instances possible in attributive position, especially then when the root verb is in the very polite speech style. Apparently, such constructions are common in the service domain.

- (71) [O-mōshikomi-ni nari-mashi-ta] kādo-wo  
 HON-apply-DAT become-AUX.POL-PST card-ACC  
 hakkō-sase-te-itadaki-mas-u.  
 issue-cause.POL-GER-humbly.receive-AUX.POL-PRS  
 'We shall respectfully issue a card [for which you had sent your application form].'  
 (Horie, 2017, 51)

It is not necessary however to use the polite form of the verb in the embedded clause, and potentially this is a new trend to be as polite as possible that just has not reached the adjectival domain yet (Ken Hiraiwa p.c.).

<sup>24</sup>The analysis of the element *desu* as monostructural performed here is, in principle, consistent with the analysis given in Miyagawa (2022, 64–67). He analyzes this element as *modal* and the head of a Mod(al)P, which, as is evident from the structural precedence, occurs higher than TP.

Note that the analysis given here can be carried over to the epistemic element D(E)ARŌ, which is, like DESU, mutually exclusive with the copula D-A, but has to follow I and any tense morpheme when co-occurring with *i*-adjectives and verbs respectively. Tentatively, I propose the following segmentation.

- (73) a. D(E)-AR-Ō = PRED-COP-EPIS (*na*-adjectives and nouns)  
 b. D(E)ARŌ = EPIS (*i*-adjectives and elsewhere)

increased since the end of the 1980's. At that time, the null hypothesis was that cross-linguistically all relative clauses are CPs. In her doctoral thesis and subsequent work, Murasugi (1991, 2000) explicitly advocated the IP-hypothesis for Japanese. This was picked up by Kayne (1994, and see Kayne 2005), who argued that relative clauses in prenominal-RC languages are derived via movement of the remnant IP to the prenominal position. On the other hand, the CP-hypothesis is explicitly defended for example in Kaplan and Whitman (1995) and Miyamoto (2017).<sup>25</sup> Furthermore, some authors (Krause, 2001; Miyagawa, 2011, 2012, 2017) have argued for a non-unified status, some (Sells, 2007) proposed to abandon the difference in structural size in Japanese altogether, while others (Matsumoto, 2017) take the position that Japanese relative clauses should be analyzed entirely differently from relative clauses in European languages.

In this section, I show that relative clauses in Japanese are only of one size, and propose a CP status. I first demonstrate they must be at least IPs, then dismiss accounts which argue that they cannot be bigger as well as those which argue they come in different sizes, before eventually showing why they are CPs. The most important point to take away, however, is that Japanese relative clauses come only in one size, meaning there is no dichotomy CP-IP or full-reduced. This has serious consequences for cartography. Instead of distinct merging positions for reduced and full relative clauses, a unifying analysis of relative clauses predicts that they are unordered, crucially also with regard to numerals. This aspect will then be discussed in the following section.

### 5.3.1 Japanese relative clauses are at least IPs

#### 5.3.1.1 High adverbials

A popular, and reliable, test used to determine the size of (especially reduced) relative clauses in several European languages is their interaction with different adverbs. It is well-known that adverbs attach at different levels of the clause, the CP, IP or VP-domain (cf. for instance Ramchand and Svenonius 2014) and take scope over different elements. Below, I give a reconstruction of the hierarchy of adverbs proposed by Cinque (1999, 109) where everything above *Asp<sub>perfect</sub>*, marked in boldface, is the CP-domain.<sup>26</sup>

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<sup>25</sup>See the overview and discussion in Hoshi (2004) and Miyamoto (2017).

<sup>26</sup>For other hierarchies see Laenzlinger (2011); Morzycki (2016).

- (74) [Mood<sub>speech act</sub> *frankly* [Mood<sub>evaluative</sub> *fortunately* [Mood<sub>evidential</sub> *allegedly* [Mod<sub>epistemic</sub> *probably* [T<sub>past</sub> *once* [T<sub>future</sub> *then* [Mod<sub>irrealis</sub> *perhaps* [Mod<sub>necessity</sub> *necessarily* [Asp<sub>habitual</sub> *usually* [Asp<sub>repetitive</sub> (I) *again* [Asp<sub>frequentative</sub> (I) *often* [Mod<sub>volitional</sub> *intentionally* [Asp<sub>celerative</sub> (I) *quickly* [T<sub>anterior</sub> *already* [Asp<sub>terminative</sub> *no longer* [Asp<sub>continuative</sub> *still* [**Asp<sub>perfect</sub>** *always* [Asp<sub>retrospective</sub> *just* [Asp<sub>proximative</sub> *soon* [Asp<sub>durative</sub> *briefly* [Asp<sub>generic/progressive</sub> *characteristically* [Asp<sub>prospective</sub> *almost* [Asp<sub>sg.completive</sub> (I) *completely* [Asp<sub>pl.completive</sub> *all* [Voice *well* [Asp<sub>celerative</sub> (II) *fast/early* [Asp<sub>repetitive</sub> (II) *again* [Asp<sub>frequentative</sub> (II) *often* [Asp<sub>sg.completive</sub> (II) *completely* ]

Douglas (2016) and Harwood (2018) independently predict that reduced relative clauses in English and French respectively should be compatible with all adverbs at least below Adv<sub>generic/progressive</sub> and possibly even as high as Asp<sub>perfect</sub>, if they indeed have a status as big as IP. That is because these projections are associated with tense and also host auxiliary verbs, elements of the IP-domain. However, this should also predict that “any adverb higher than [...] Asp<sub>perfect</sub> will be impossible in RRCs” (Douglas 2016, 223) given that this is the CP domain. As it turns out, reduced relative clauses in the languages they test are not only compatible with the predicted kind of adverbs. In fact, they are even compatible with very high speaker-oriented, evaluative and evidential, thus clausal, adverbs. This is shown by Douglas (2016, 226–230; 239–245) for English and French, and the same findings are observable in Italian (cf. Cinque 2020, 214).

- (75) The cakes [unfortunately<sub>evaluative</sub> / probably<sub>evidential</sub> (being) eaten by the guests] will cause stomach upsets. (English) (Douglas, 2016, 226)
- (76) L'étudiant [malheureusement<sub>evaluative</sub> / probablement<sub>evidential</sub> arrêté  
DET.student unfortunately / probably arrested  
par la police] est un étranger. (French)  
by DET police be.3.SG a foreigner  
'The student [probably/unfortunately arrested by the police] is a foreigner.'  
(Douglas, 2016, 241)
- (77) I film [sfortunatamente<sub>evaluative</sub> / presumibilmente<sub>evidential</sub> persi  
DET films unfortunately / presumably lost  
durante l'ultima guerra] sono davvero molti. (Italian)  
during DET.last war are really many  
'The films [unfortunately/presumably lost during the last] war are really many.'  
(Cinque, 2020, 214)



That means that despite being able to prove IP (over for example vP) status, it cannot be determined whether reduced relative clauses in those languages are just IPs or potentially bigger, that is CPs.<sup>27</sup>

This test can be applied in Japanese. First note that all types of adverbs do exist in Japanese: VP (*hayaku* ‘fast’), IP/TP (*kinō* ‘yesterday’) and CP adverbs (*zannennagara* ‘unfortunately’). Important differences are first that the canonical position for CP adverbs is very high in the clause, although they can appear in different positions. More importantly, these adverbs can never be the target of negation, which is very low, as shown in Koizumi (2017). Compare (78), in which the TP adverb *kinō* is in the scope (c-command domain) of negation, to (79), in which the CP adverb *osoraku* ‘possibly’ is not in the scope of negation.

- (78) [Kinō [[hashit-ta] wake-de-wa-na-i]].  
 yesterday run-PST case-PRED-TOP-NEG-PRED.COP.PRS  
 ‘It is not the case that (I) ran yesterday.’
- (79) Osoraku [[[hashit-ta] wake-de-wa-na-i] darō].  
 possibly run-PST PRED-TOP-NEG-PRED.COP.PRS EPIS  
 ‘Probably, it is not the case that (he or she) ran.’  
 ≠ ‘It is not the case that (he or she) probably ran.’  
 (Koizumi, 2017, 556)

Since different types of adverbs exist in Japanese, it should be possible to deduce a hierarchy as well. This is attempted in Narrog (2009), who argues for an infinite number of modal adverbs in Japanese and, based on ordering and scope principles, which he investigates in a corpus study (see Narrog 2009, 71–76), gives the following hierarchy, which also contains the categories tense, aspect and negation. As already mentioned in (Chapter 2.6.2.3), this hierarchy turns out not to be a perfect match with the hierarchy proposed in Cinque (1999) and Narrog identifies several problems.

- (80) *Narrog’s hierarchy of Japanese modal categories (Narrog 2009, 235)*  
 MoodP speech act I (imperative) > MoodP speech act II (hortative) >  
 ModP epistemic I (speculative) > TP past I > ModP evidential I (reportive)  
 > TP past II > ModP epistemic II (epistemic possibility) > TP past III  
 > NegP IntNeg I > ModP evidential II inferential evidentiality > TP past  
 IV > NegP IntNeg II > ModP epistemic III epistemic necessity > TP past  
 V > NegP IntNeg III > ModP deontic I weak deontic necessity > TP past  
 VI > NegP IntNeg IV > AspP result/compl I > ModP deontic II strong  
 deontic necessity > TP past VII > NegP IntNeg V > AspP result/compl III

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<sup>27</sup>Furthermore, different languages might allow reduced relative clauses of different sizes and therefore combinations with different kinds of adverbs, which is also hinted at in Krause (2001, 76–77), but not further elaborated.

> ModP volition > TP past VIII > NegP IntNeg VI > AspP result/compl IV > ModP evidential III future-oriented > NegP InstNeg VII > AspP result/progress/compl II > ModP root possibility I NegP IntNeg VIII > AspP result/progress/compl V

Nevertheless, high CP adverbs are attested in Japanese and can be tested in relative clauses in a similar vein. This yields the same picture as in English or French. Relative clauses combine rather freely with high subject- and speaker-oriented adverbs. This is shown below first for evaluative adverbs (81), in a direct translation of the example in Cinque (2020, 214), and then for evidential adverbs (82). As the brackets indicate, the adverbs are understood as part of the relative clause.<sup>28</sup>

- (81) [**Zannennagara/Oshikumo**<sub>evaluative</sub> sensō-de ushinaw-are-ta] eiga-wa  
unfortunately/regrettably war-LOC lose-PASS-PST movie-TOP  
ō-i.  
many-PRED.COP.PRS  
‘The films [unfortunately/regrettably lost during the war] are many.’
- (82) [**Osoraku/Dōyara/Tabun**<sub>evidential</sub> sensō-de ushinaw-are-ta] eiga-wa  
possibly/apparently/probably war-LOC lose-PASS-PST movie-TOP  
ō-i.  
many-PRED.COP.PRS  
‘The films [possibly/apparently/probably lost during the war] are many.’

Such constructions can also be constructed for relative clauses with an adjectival nucleus as shown in (83) with the evaluative adverb *saiwaini-mo* ‘fortunately’.

- (83) [Shinpai-shi-ta kedo, **saiwaini-mo**<sub>evaluative</sub> genki-d-at-ta]  
worry-do-PST but fortunately-EMP healthy-PRED-COP-PST  
josei-wa Hanako d-at-ta.  
woman-TOP Hanako PRED-COP-PST  
‘The woman [I was worried about but who fortunately was fine] was Hanako.’

These data points in comparison with the ones illustrated above for English, French and Italian lead to the same conclusion: Relative clauses in Japanese are also at least IPs, but, on the other hand, it is not clear whether they can be bigger or not.

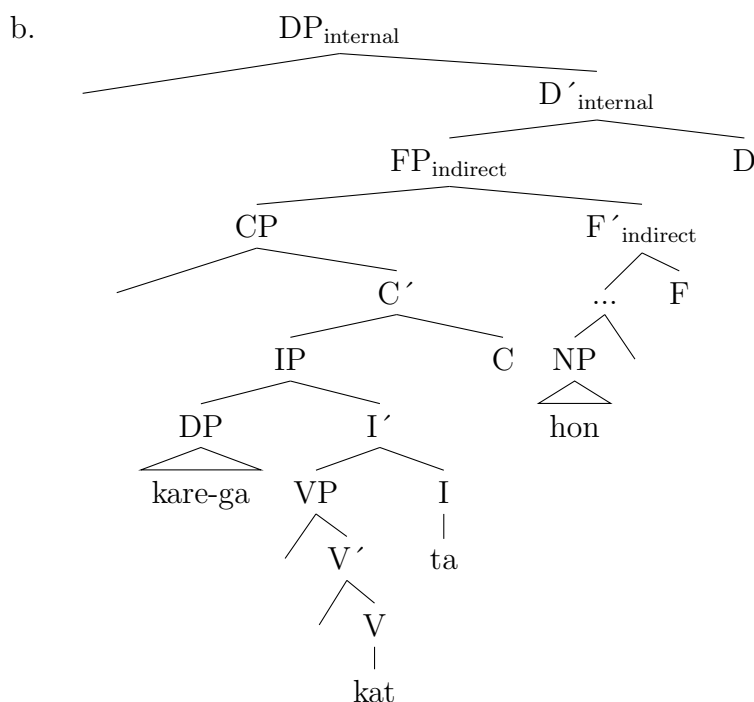
### 5.3.1.2 Exponence of an overt subject

Another indication of IP status is the availability of a subject position, which is standardly assumed to be Spec, IP (Chomsky 1986; Koopman and Sportiche 1991,

<sup>28</sup>Finding evidential adverbs can prove difficult, since Japanese speakers would rely on modal suffixes such as *rashi-i* ‘seem’ to express the speaker’s opinion. I thank C. Takeuchi for providing valuable judgments and examples of relevant adverbs.

see McCloskey 1997). It was already mentioned that prenominal-RC languages generally exhibit an overt subject inside relative clauses and many relevant examples were shown in Japanese in Chapter 3.2. This is thus another indication for IP-status, although, in principle, nothing is said about a potential higher CP layer (although see Fukui 1986 and subsequent work for an alternative analysis for Japanese). I give a verbal relative clause with filled Spec, IP for illustration, to be refined later.

- (84) a. kare-ga kat-ta hon  
 he-NOM buy-PST book  
 ‘the book (which) he bought’



### 5.3.2 Why Japanese relative clauses are not only IPs

The question is whether there is a way to prove that relative clauses in Japanese are *only* IPs. Below, I evaluate arguments from the literature and eventually falsify them. These include the analysis in Cinque (2020) for an unequivocal IP status of relative clauses in Mandarin Chinese and arguments from Murasugi (1991).

#### 5.3.2.1 Individual-level vs. stage-level

Cinque (2020, ch. 3.6) argues that relative clauses in Mandarin Chinese, which also only exhibit one surface type, are IPs. He bases his analysis on the alleged adherence to ordering restrictions between stage-level (transitory) and individual-level (non-transitory) relative clauses presented in Larson and Takahashi (2007) (and del Gobbo 2005) for various prenominal RC-languages, among them Mandarin and Japanese. The authors argue that when these types of relative clauses co-occur,

the individual-level relative clause has to occur closer to the head noun than the stage-level relative clause (85). However, when two relative clauses with the same interpretation co-occur, (86) and (87), the order is free. First, some of the relevant examples in Mandarin are given (Larson and Takahashi 2007, 104, taken from del Gobbo 2005, 298).<sup>29</sup>

(85) *stage-level*  $\succ$  *individual-level*

- a. [Wo zuotian kanjian de]<sub>s-level</sub> [xihuan qu yinyuehui de]<sub>i-level</sub> ren  
I yesterday meet DE like go concert DE person  
shi Lisi. (Mandarin)  
COP Lisi  
'The person who I met yesterday who likes to go to concerts is Lisi.'
- b. \*[Xihuan qu yinyuehui de]<sub>i-level</sub> [wo zuotian kanjian]<sub>s-level</sub> de ren  
like go concert DE I yesterday meet DE person  
shi Lisi.  
COP Lisi

(86) *individual-level* + *individual-level*

- a. [Hui shuo Yidaliyu de]<sub>i-level</sub> [xihuan qu yinyuehui de]<sub>i-level</sub> ren  
can speak Italian DE like go concert DE person  
shi Lisi. (Mandarin)  
COP Lisi  
'The person who can speak Italian who likes to go to concerts is Lisi.'
- b. [Xihuan qu yinyuehui de]<sub>i-level</sub> [Hui shuo Yidaliyu de]<sub>i-level</sub> ren  
like go concert DE can speak Italian DE person  
shi Lisi.  
COP Lisi

(87) *stage-level* + *stage-level*

- a. [Cong Yidali huilai de]<sub>s-level</sub> [wo zuotian kanjian de]<sub>s-level</sub> ren  
from Italy return DE I yesterday meet DE person  
shi Lisi. (Mandarin)  
COP Lisi  
'The person who returned from Italy who I met yesterday is Lisi.'
- b. [Wo zuotian kanjian de]<sub>s-level</sub> [cong Yidali huilai de]<sub>s-level</sub> ren  
I yesterday meet DE from Italy return DE person  
shi Lisi.  
COP Lisi

Crucially, Mandarin relative clauses share this restriction with full, but not reduced, relative clauses in English. First, observe the examples of postnominal relative

<sup>29</sup>Recall that verbs and nouns in this language are always connected to the head noun via the element DE, as presented in Chapter 2.2 as an indication of indirect modification. See also Fn. 10 in that chapter for various analyses, and Chapter 3.2 for why it could be an instance of MOD.

clauses of different kinds of transitoriness in English below. As visible, no ordering restrictions apply (see also Baker 2003b).

- (88)    a.    the person [who I met]<sub>s-level</sub> [who smokes]<sub>i-level</sub>  
           b.    the person [who smokes]<sub>i-level</sub> [who I met]<sub>s-level</sub>
- (Cinque, 2020, 239)

Reduced relative clauses, on the other hand, do uphold such ordering preferences, as visible in the following example.

- (89)    the Wednesday Thursday lecture
1. stage-level  $\succ$  individual-level
  2. \*individual-level  $\succ$  stage-level            (Larson and Takahashi, 2007, 112)

In (89), the first modifier must have a deictic, stage-level, interpretation and refer to a lecture on a particular Wednesday, whereas the second modifier must have a generic, individual-level, interpretation and denote the usual lecture on Thursdays (Cinque, 2020, 239). From these facts, Cinque (2020) deduces that relative clauses in Mandarin cannot be of the same type as full, but as reduced relative clauses in English, thus presumably IPs.

Apart from the fact that it is not clear whether these observations can be extended to all relative clauses in Mandarin, things are more complicated. Direct counterarguments come from Kim (2019), who acknowledges the empirical adequacy of the examples in Larson and Takahashi (2007) for Korean, but puts forth examples in which the order *individual-level*  $\succ$  *stage-level* is actually preferred over the order *stage-level*  $\succ$  *individual-level*, (90), and examples in which ordering restrictions are observed among the same type of relative clauses (91).<sup>30</sup>

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<sup>30</sup>These observations are in Kim (2019) explained via a constraint-based account, which predicts obligatory movement of relative clauses with more familiar topics to the discourse participants (the speaker singing a song or remembering something) to a sentence-initial position, presumably at PF. The surface order is arguably the result of an output filter (Prince and Smolensky, 1993) which ranks the best order based on various semantic and pragmatic constraints (Kim, 2019, ch. 5), those being *episodic event description*, *earlier event time* and *higher familiarity to interlocutors* (Kim, 2019, 202). She establishes the hierarchy *supplementary*  $\succ$  *non-restrictive*  $\succ$  *restrictive* (Kim, 2019, 120). Importantly, such constraints bear no relation to the interpretation as stage or individual-level relative clause (Kim, 2019, 239–240) and also to any difference full-reduced or CP-IP. See also Kotowski (2016), who did not find a significant tendency in a questionnaire-study on the order between stage- and individual-level adjectives in German.

- (90) *individual-level*  $\succ$  *stage-level* as preferred order in Korean (Kim, 2019, 85)
- a. [Mina-ka phyengso culkye-pwuru-n-un,]<sub>i-level</sub> [chilsipnyentay-ey  
Mina-NOM usually enjoy-singing-IPFV-REL 1970's-LOC  
ywuhaynghay-ss-te-n]<sub>s-level</sub> noray-nun leylipti-i-Ø-ta.  
be.popular-ANT-RTRO-REL song-TOP let.it.be-COP-PRS-DECL  
'The song that Mina usually enjoys singing that was popular in the  
1970's is "Let it be".'
- b. ?[Chilsipnyentay-ey ywuhaynghay-ss-te-n,]<sub>s-level</sub> [Mina-ka phyengso  
70's-LOC be.popular-ANT-RTRO-REL Mina-NOM usually  
culkye-pwuru-n-un]<sub>i-level</sub> noray-nun leylipti-i-Ø-ta.  
enjoy-singing-IPFV-REL song-TOP let.it.be-COP-PRS-DECL
- (91) *Two stage-level relative clauses, no free ordering in Korean* (Kim, 2019, 86)
- a. [Mina-ka kacang cal kiekha-n-un,]<sub>s-level</sub>  
Mina-NOM most well remember-IPFV-REL  
[kwusipnyentay-chopan-ey iss-ess-te-n]<sub>s-level</sub> il-un  
1990's-early-LOC exist-ANT-RTRO-REL event-TOP  
emma-hako-uy yehayang-i-Ø-ta.  
Mom-with-GEN trip-COP-PRS-DECL  
'The event that Mina remembers the best which happened in the early  
1990's is the trip with her Mom.'
- b. \*[Kwusipnyentay-chopan-ey iss-ess-te-n,]<sub>s-level</sub> [Mina-ka kacang  
1990's-early-LOC exist-ANT-RTRO-REL Mina-NOM most  
cal kiekha-n-un]<sub>s-level</sub> il-un emma-hako-uy  
well remember-IPFV-REL event-TOP Mom-with-GEN  
yehayang-i-Ø-ta.  
trip-COP-PRS-DECL

These data points are problematic for Larson and Takahashi (2007) and by extension also for Cinque (2020) but, as Kim (2019) admits, it might be far-fetched to draw any immediate fundamental conclusions from them.

Coming now to Japanese, Larson and Takahashi (2007, 102) report identical judgments regarding the order of stage- and individual-level relative clauses as given below.

(92) *stage-level*  $\succ$  *individual-level* in Japanese after Larson and Takahashi (2007, 102)

- a. [Watashi-ga kinō at-ta]<sub>s-level</sub> [tabako-wo su-u]<sub>i-level</sub> hito-wa  
I-NOM yesterday meet-PST tobacco-ACC inhale-PRS person-TOP  
Tanaka-san-de-s-u.  
Tanaka-Mr.-PRED-COP.POL-PRS  
‘The person who I met yesterday who smokes is Mr./Mrs. Tanaka.’
- b. \*[Tabako-wo su-u]<sub>i-level</sub> [watashi-ga kinō at-ta]<sub>s-level</sub> hito-wa  
tobacco-ACC inhale-PRS I-NOM yesterday meet-PST person-TOP  
Tanaka-san-de-s-u.  
Tanaka-Mr.-PRED-COP.POL-PRS

(93) *Two individual-level relative clauses* in Japanese according to Larson and Takahashi (2007, 102)

- a. [Tabako-wo su-u]<sub>i-level</sub> [sake-wo no-mu]<sub>i-level</sub> hito-wa  
tobacco-ACC inhale-PRS alcohol-ACC drink-PRS person-TOP  
Tanaka-san-de-s-u.  
Tanaka-Mr.-PRED-COP.POL-PRS  
‘The person who smokes who drinks alcohol is Mr./Mrs. Tanaka.’
- b. [Sake-wo no-mu]<sub>i-level</sub> [tabako-wo su-u]<sub>i-level</sub> hito-wa  
alcohol-ACC drink-PRS tobacco-ACC inhale-PRS person-TOP  
Tanaka-san-de-s-u.  
Tanaka-Mr.-PRED-COP.POL-PRS

(94) *Two stage-level relative clauses* in Japanese according to Larson and Takahashi (2007, 102)

- a. [Watashi-ga kinō at-ta]<sub>s-level</sub> [sake-wo non-de-i-ta]<sub>s-level</sub>  
I-NOM yesterday meet-PST alcohol-ACC drink-GER-be-PST  
hito-wa Tanaka-san-de-s-u.  
person-TOP Tanaka-Mr.-PRED-COP.POL-PRS  
‘The person who I met yesterday who was drinking alcohol is Mr./Mrs. Tanaka.’
- b. [Sake-wo non-de-i-ta]<sub>s-level</sub> [watashi-ga kinō at-ta]<sub>s-level</sub>  
alcohol-ACC drink-GER-be-PST I-NOM yesterday meet-PST  
hito-wa Tanaka-san-de-s-u.  
person-TOP Tanaka-Mr.-PRED-COP.POL-PRS

Speakers I asked judged the example in (92) as more natural if the stage-level relative clause preceded the individual-level relative clause. However, the reversed order *smokes*  $\succ$  *likes to go to concerts* seems not to be ungrammatical, contrary to what is argued in Larson and Takahashi (2007). Regarding the counterexamples reported in Kim (2019), I have attempted as best as possible to give Japanese translations of her examples and discuss them with consultants as out-of-the-blue examples in order to see if her observations also hold in this language. As it turns out, my consultants

did not detect any ordering restrictions and judged both orders as equally natural, exemplified below.

(95) *Free ordering between stage-level and individual-level relative clause in Japanese translated from Korean from Kim (2019, 85)*

- a. [Mīna-ga itsumo uta-u,]<sub>i-level</sub> [1970-nen-ni  
Mina-NOM always sing-PRS 1970's-year-LOC  
ninki-dat-t-ta]<sub>s-level</sub> kyoku-wa Let-it-be-d-a.  
be.popular-PRED-COP-PST song-TOP let.it.be-PRED-COP.PRS  
'The song that Mina always sings that was popular in the 1970's is "Let  
it be".'
- b. [1970-nen-ni ninki-dat-t-ta,]<sub>s-level</sub> [Mīna-ga itsumo  
1970's-year-LOC be.popular-PRED-COP-PST Mina-NOM always  
uta-u]<sub>i-level</sub> kyoku-wa Let-it-be d-a.  
sing-PRS song-TOP let.it.be PRED-COP.PRS

This discussion has shown first of all that a deduction of IP status on the grounds of ordering restrictions among stage- and individual-level relative clauses is hard to establish. It also emphasizes free order of modifiers of the same type in Japanese and corroborates the findings from previous chapters. A final disclaimer: I am aware that a deeper analysis of the order of relative clauses in Japanese needs to be conducted than can be granted here. Speakers generally had a hard time determining which order among the two relative clauses I presented was best. There might be factors ameliorating one order over the other that can be brought about by context.<sup>31</sup> Aside from syntactic and phonological/prosodic complexity, which will be addressed in the next section, I could not find determining factors to ameliorate one order over the other, so this is an endeavor I leave for future research.

### 5.3.2.2 Reason and manner clauses

An advocator for the IP status of relative clauses in Japanese is Murasugi (1991). One argument she puts forth is the lack of long distance dependencies in *reason* and *manner* relative clauses. See the following examples.

- (96) [Marī-ga [Jon-ga kaet-ta to] omot-te-i-ru] riyū  
Mary-NOM John-NOM return-PST COMP think-GER-be-PRS reason  
1. ≈ 'the reason Mary thinks that John returned'  
2. ≠ 'the reason why John returned according to what Mary thinks'  
(Murasugi, 1991, 185)

<sup>31</sup>This was also pointed out to me by Min-Joo Kim to whom I am grateful for discussing the relevant examples.



- (97) [Jon-ga [Marī-ga mondai-wo toi-ta to] it-ta] hōhō  
 John-NOM Mary-NOM problem-ACC solve-PST COMP say-PST way  
 1.  $\approx$  ‘the way Mary said that John solved the problem’  
 2.  $\neq$  ‘the way John solved the problem according to what Mary said’  
 (Murasugi, 1991, 133)

In (96), the speaker can only refer to the reason why Mary thinks that John returned, not to the actual reason of his return, according to what Mary thinks. Similarly, in (97), the speaker can only refer to the way Mary said that John solved the problem, not to the way he actually solved it. This contrasts with relative clauses in which locative and temporal nouns are relativized. These relative clauses in fact allow two readings, a higher reading and a lower reading. This shows that they do allow long distance relativization (Murasugi, 1991, 133–134).

- (98) [Marī-ga [Jon-ga sentaku-shi-ta to] it-ta] hi  
 Mary-NOM John-NOM washing-do-PST COMP say-PST day  
 1.  $\approx$  ‘the day on which Mary said that John did laundry’  
 2.  $\approx$  ‘the day on which John did the laundry according to what Mary said’  
 (Murasugi, 1991, 134)

- (99) [Jon-ga [Marī-ga mondai-wo toi-ta to] it-ta] tokoro  
 John-NOM Mary-NOM problem-ACC solve-PST COMP say-PST place  
 1.  $\approx$  ‘the place where Mary said that John solved the problem’  
 2.  $\approx$  ‘the place where John solved the problem according to what Mary said’  
 (Murasugi, 1991, 134)

Following the framework of Lasnik and Saito (1992), Murasugi (1991, ch. 3.4) assumes a *pro*-account for relative clauses and that relativization is only possible for arguments, never adjuncts, to which *reason*- and *manner*-nouns belong, according to her, and which must be clause bound. This contrasts them with *time*- and *place*-nouns which are arguments. In favor of *pro*, she dismisses operator-movement based accounts. As she assumes Japanese relative clauses to be IPs, no projection (Spec, CP) is available to host an operator in the first place.

Abstracting away from the fact that *pro*-based accounts have been dismissed on several grounds (Salzmann, 2017, 36–37), rather than having to assume an IP status for such relative clauses, the problem seems to be the peculiar behavior of reason and manner-relative clauses, as also argued in Miyamoto (2017). In this light, note another contrast between reason/manner relative clauses, on the one hand, and locative/temporal relative clauses on the other. The latter can appear as major subjects in a major subject construction (Kuroda 1992 and already Kuno 1973; cf. Chapter 5.3.5.3).

- (100) Sono hi-ga [Hanako-ga [Tarō-ga hashit-ta to] it-ta].  
 that day-NOM Hanako-NOM Tarō-NOM run-PST COMP say-PST  
 ‘The day is such that Hanako said that Taro ran.’  
 (Miyamoto, 2017, 628)

As in (98), this sentence can refer to the time that Hanako said that Tarō ran, or the time that he ran according to Hanako. The corresponding (major subject) construction with the noun *riyū* ‘reason’ is not only unavailable under the desired reading (‘the reason Tarō ran according to Hanako’), it is entirely ungrammatical.

- (101) \*Sono ryū-ga [Hanako-ga [Tarō-ga hashit-ta to] it-ta].  
 that reason-NOM Hanako-NOM Tarō-NOM run-PST COMP say-PST  
 (‘The reason is such that Hanako said that Taro ran.’)  
 (Miyamoto, 2017, 628)

Miyamoto (2017) draws the conclusion that relative clauses as in (98) are derived via operator-movement out of major subject constructions. Since the underlying structure is not available for manner nouns in the first place, the relative clause in (96) does not yield the desired long-distance reading. The conclusion to draw from these facts is that the unavailability of long-distance dependencies in reason- and manner-relative clauses in Japanese is not related to the size of the relative clause, but to the nature of the relativized noun and the possible derivation of Japanese relative clauses in Japanese out of a major subject position (see Chapter 5.3.5.3).

### 5.3.3 Relative clauses are not of different sizes

In this subsection, accounts will be reviewed according to which relative clauses in Japanese come in different sizes. These are mostly based on the phenomenon *Nominative Genitive Conversion* (NGC, also *ga-no-conversion*). I will review some of the arguments from the literature and then report on how my consultants judged the data as well as some additional examples. It will be argued that not enough evidence can be put forward to justify an analysis along such lines.

#### 5.3.3.1 Nominative-Genitive-Conversion

Japanese can mark subjects within a relative clause with the nominative case particle, but alternatively also with NO, the genitive case particle.<sup>32</sup> As exemplified in (102), both options – nominative and genitive – are available and this choice does not affect the meaning. Note that I adopt the gloss GEN for this instance of NO here

<sup>32</sup>Genitive case marking of the subject in a relative clause is a phenomenon not unique to Japanese, but also exhibited in other languages, for example Turkish, Mongolian and Dagur (Krause, 2001).

as it cannot be an instance of MOD in accordance with the refinement rule given in Chapter 3.2, i. e. since it does not follow a nominal modifier and is involved in nominal modification.

- (102)    kare-ga / kare-no kat-ta    hon  
          he-NOM / he-GEN buy-PST book  
          ‘the book (which) he bought’

In the 1970’s, Harada (1971) argued that the different markers are purely a result of idiolectal differences. He sparked a discussion on the linguistic dimension of this phenomenon, which was picked up in the generative literature from the 90’s onwards. In fact, although the two respective markers – -GA for nominative and -NO for genitive – seem interchangeable in most scenarios, various restrictions are observable with genitive-marked subject relative clauses. The following have been pointed out in the literature (Watanabe, 1996; Hiraiwa, 2001b; Miyagawa, 2011, 2012, 2017; Ochi, 2004, 2017, 2021; Akaso and Haraguchi, 2011; Nambu and Nakatani, 2014, 2023).

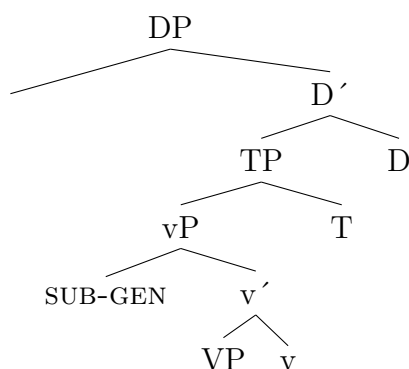
- (103)    a.    A restriction on CP adverbs  
          b.    Intervention effects: A restriction on intervening adjuncts  
          c.    Focus is impossible with genitive marked subjects.  
          d.    Genitive marked subjects give rise to potential Principle B violations.  
          e.    Genitive marked subjects can give rise to scope ambiguities.  
          f.    Genitive marked subjects are not compatible with accusative marked objects. (Transitivity restriction)

There are two main approaches to this phenomenon, which both relate the licensing of case to the size of the relative clause: The ‘C-licensing approach’, which assumes a uniform analysis and that a CP is always present, which is where case is licensed (Watanabe 1996 and Hiraiwa 2001a,b), and the ‘D-licensing approach’ (Miyagawa, 2011, 2012, 2017), which postulates different analyses of nominative and genitive marked subject relative clauses, namely as CPs and TPs. I will concentrate mainly on the latter account, since it correlates with the two different sizes of relative clauses. After outlining the main idea, I will show – focusing on the first three points mentioned above – that 1. these arguments are not strong enough to convincingly argue for a different status, and 2. they can also be handled by not resorting to different analyses. For a complete overview of all the patterns, the reader can consult the relevant literature and especially Ochi (2017).

## 5.3.3.2 The idea

Concretely, Miyagawa (2011, 2012, 2017) treats genitive marking of subjects inside a relative clause as a version of Exceptional Case Marking (ECM, see Chomsky 1986, 2001). The reason why the subject is marked by genitive and not by nominative is because T is defective (Miyagawa, 2011, 1271). It is defective because it does not have a higher node, namely C, from which it would receive case. Therefore, it cannot be a case-assignor, and since the subject cannot receive case from T, it must look for another case-assigning category in a higher node. It finds this category in D, which is name-bearing for the approach (*ibid.*). The fact that the relative clause is TP is thus a prerequisite for genitive case-marking. This is illustrated below.

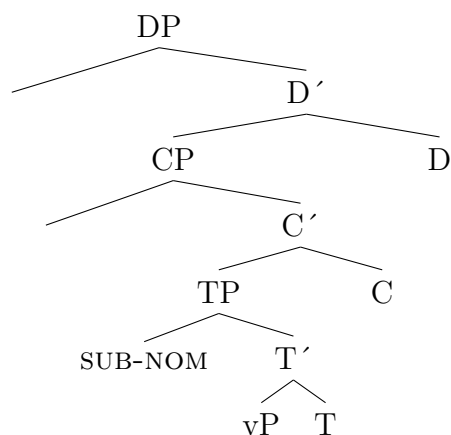
(104) Genitive (Miyagawa, 2017, 147)



Nominative subjects on the other hand do receive their case from T. As mentioned above, T needs to receive case from somewhere in the first place to later assign it to the subject. This means, it must be selected by a higher CP. This T, which has received case from C, is not defective, but properly assigns nominative case to the subject (Miyagawa, 2011, 1271).

Miyagawa (2011, 1271–1272) assumes that C is a phase and, intervening between T and D, stops D from assigning case to the subject, which is why the nominative subject cannot get its case from D in this scenario. This means that in Miyagawa's model, a CP layer has to be present when a subject appears in nominative case. This alternative option is shown below.

- (105) Nominative (Miyagawa, 2017, 147)



### 5.3.3.3 CP-elements and adjacency

A central argument for Miyagawa's claim that genitive marked relative clauses lack a CP layer comes from the alleged incompatibility between genitive-marked relative clauses and CP-elements. Recall from Chapter 5.3.1 that high speaker- and subject-oriented CP adverbs are freely compatible with Japanese relative clauses. The relevant example is repeated for convenience below (= (81)).

- (106) [Zannennagara/Oshikumo<sub>evaluative</sub> sensō-de ushinaw-are-ta]  
 unfortunately/regrettably war-LOC lose-PASS-PST  
 eiga-wa ō-i.  
 movie-TOP many-PRED.COP.PRS  
 'The films unfortunately/regrettably lost during the war are many.'

Miyagawa (2011, 1273) argues that in subject relative clauses, modal TP adverbs such as *kitto* 'certainly' are possible with both types, those in which the subject is marked by nominative and genitive (107). However, on the contrary, CP adverbs are only possible with nominative-marked, never with genitive-marked, subject relative clauses, which is because no CP layer is present in the latter case. He shows this using the evaluative adverb *saiwaini* 'fortunately' (108).

- (107) [kitto Tarō-ga / no yon-da] hon  
 certainly Tarō-NOM / GEN read-PST book  
 'the book [that Tarō certainly read]' (Miyagawa, 2011, 1273)
- (108) [saiwaini Tarō-ga / \*no yon-da] hon  
 fortunately Tarō-NOM / GEN read-PST book  
 'the book [that Tarō fortunately read]' (Miyagawa, 2011, 1273)

Recall that I have argued that these tests merely show that relative clauses are IP/TP, but that they cannot readily be used as indication whether they are IPs or CPs. A further problem with the example (108) is that the adverb *saiwaini* 'fortu-

nately’ was not accepted by any of my consultants in the first place and arguably should be replaced by the more natural *saiwaini-mo* as given below.

- (109) [saiwaini-mo Tarō-ga / no yon-da] hon  
 fortunately-EMP Tarō-NOM / GEN read-PST book  
 ‘the book [that Tarō fortunately read]’

Comparing (109) with (107), the speakers I consulted either judged both DPs as acceptable, regardless of whether the nominative or genitive case marker occurred, or they judged only the nominative-marked relative clauses acceptable, but crucially for *both* examples. This means they did not provide varying judgements depending on the two adverbs and, if they showed preference it was a strong preference for nominative in the first place.

A general problem with such examples is that neither context nor matrix clause are provided. Attempting to embed this relative clause in a matrix clause however runs the risk that the adverb is understood as referring to the matrix verb, which follows from its status as a CP adverb. In the following sentence in (110), the reference of the adverb is semantically and pragmatically controlled for. A reading where *saiwaini-mo* refers to the matrix clause – and also to the second subordinate clause – is not syntactically excluded, but pragmatically marked, since it would be strange to associate ‘fortunately’ with losing something or being in trouble, compared to a fortunate finding of ones wallet. Hence, this adverb should attach at the CP-layer of the embedded relative clause. This is indicated by the brackets. The important part is that, according to the speakers I consulted, both versions, with GA and with NO, are equally acceptable.<sup>33</sup>

- (110) [[Saiwaini-mo Tarō-ga / no mitsuke-te-kure-ta] saifu-wo  
 fortunately-EMP Tarō-NOM / GEN find-GER-do.favor-PST wallet-ACC  
 mata nakushi-te-shimat-ta node,] komat-te-i-ru.  
 again lose-GER-suffer.from-PST because be.in.trouble-GER-be-PRS  
 ‘[Because I have lost the wallet again [that Tarō fortunately has found for me]] I am in trouble.’

The fact that both case markers are possible seems surprising and is problematic for Miyagawa’s analysis. This argument relates directly to another piece of data already mentioned in Harada (1971), namely the degradation of genitive marked subject relative clauses with intervening material between the verb and the subject as in the following example.

<sup>33</sup>I am deeply grateful to Ken Hiraiwa for providing this sentence and for him as well as C. Takeuchi for discussion of these patterns.

- (111) [kodomo-tachi-ga / \*no minnade ikioi-yoku kake-nobot-ta] kaidan  
 kid-PL-NOM / GEN everyone vigorous-ADV run-climb-PST stairs  
 ‘the stairs [which the kids vigorously ran up all together]’  
 (Harada, 1971, 30)

The judgments of my consultants with regard to (111) again suggest that genitive is rather marked when additional elements intervene, or in other words, when the subject is not adjacent to the noun.<sup>34</sup> This observation was backed up experimentally in a rating and a self-paced reading study in Nambu and Nakatani (2014, 2023), who show a significant degradation of genitive-marking with intervening TP or CP (compared to VP) adjuncts, whereas acceptance remained stable for the nominative-marked subject condition. Interestingly, the authors also note in their investigation of reading times a significant degradation of genitive with intervening adjuncts in general. Besides CP and TP adjuncts, they also detected this for locative (more than temporal or manner) PPs, which are arguably in the VP domain (Nambu and Nakatani, 2023, 15–23). It seems as if there is a general intervention effect happening with genitive case but it is questionable whether this is due to a smaller status of relative clauses containing a genitive-marked subject.<sup>35</sup>

An alternative piece of data is offered by Ochi (2021) and illustrated in (112). Here, a genitive-marked relative clause with three adjuncts appears. This is odd, but not unacceptable, and for some reason less odd than if the subject appeared after *kinō* ‘yesterday’ according to the speakers the author has consulted.<sup>36</sup>

- (112) (?) [kodomo-tachi-no kinō ōzei kyōshitsu-de sawai-da] koto  
 children-PL-GEN yesterday many classroom-LOC clamor-PST fact  
 ‘the fact [that many children clamored in the classroom yesterday]’  
 (Ochi, 2021, 326)

Generally speaking, there seems to be a tendency for speakers of Modern Japanese to prefer nominative over genitive in relative clauses, as already mentioned in Harada (1971). The study in Nambu and Nakatani (2014, 2023) can be interpreted along these lines as well, as they do show this tendency independent of the condition

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<sup>34</sup>The intervention patterns are analyzed in Miyagawa (2011, sec. 4.2.) along an economy of movement analysis. The genitive subject (that has received case from D) remains in Spec, vP and does not raise to Spec, TP since T is defective and thus has no EPP feature. Therefore, the genitive marked subject cannot appear high up in the clause because there is no motivation for it to move there. Low (VP) adverbs, however, can be easily accommodated. It remains unclear what a notion such as *economy* can refer to in a language such as Japanese in which scrambling is so productive.

<sup>35</sup>The authors suggest a processing-based account for locative PPs (Nambu and Nakatani, 2023, 25).

<sup>36</sup>In his account, all Japanese relative clauses are IPs. The different factor is simply the subject, which can optionally move up to D where it receives genitive case, which it does in (112).

*adjacent/non-adjacent*, as also mentioned by the authors. Therefore, all these facts could find a different explanation.

### 5.3.3.4 Focus and genitive

The interaction between *focus* and relative clauses in Japanese gives further important insights. Assuming a projection related to focus is part of the (extended) CP domain, the left periphery of the clause (Rizzi, 1997), predicts that focus is impossible inside genitive case-marked subject relative clauses, if they are not CPs as such a layer would not be accessible. And this is exactly the analysis argued for in Miyagawa (2017).

Recall from Chapter 3.1.3 that focus is expressed in Japanese either prosodically, or via focus sensitive particles, the topic particle -WA, which can express a focus function (113), and emphatic particles such as -DAKE ‘only’ or -SAE ‘even’ (114) (Tomioka, 2010, 2016; Vermeulen, 2007).

- (113) [Nihongo-wo shanai-de-wa tsuka-u] gaikokujin  
 Japanese-ACC office-LOC-FOC use-PRS foreigner  
 ‘a foreigner [who uses Japanese only in the office]’  
 (Akaso and Haraguchi, 2011, 93)

- (114) [Tarō-dake-ga non-da] kusuri  
 Tarō-only-NOM drink-PST medicine  
 ‘the medicine [(that) only Tarō took]’ (Akaso and Haraguchi, 2011, 96)

Postponing a precise analysis of focus marking inside relative clauses to Chapter 5.3.4.1, the crucial point is that those focus particles are generally banned from genitive-marked subject relative clauses, as argued in Akaso and Haraguchi (2011) and Miyagawa (2017). Compare (114) with (115).

- (115) \*[Tarō-dake-no non-da] kusuri  
 Tarō-only-GEN drink-PST medicine  
 (‘the medicine [(that) only Tarō took]’) (Akaso and Haraguchi, 2011, 96)

Unlike many of the contrasts mentioned above, this contrast seems to be more systematic and my consultants had a clearer opinion and shared the judgments of the literature. In other words, the arguments presented in Miyagawa (2017) seem to hold.

Nevertheless, there are some challenges to this observation. First, note that the genitive particle NO and focus sensitive particles such as DAKE ‘only’ are not totally incompatible. The following examples show that they can co-occur. Note that the focus particle, in this case, follows an adverb (116) and a PP (117).



- (116) [kinō-dake Tarō-no non-da] kusuri  
 yesterday-only Tarō-GEN drink-PST medicine  
 ‘the medicine [that Tarō drank only yesterday]’ (Ochi, 2017, 692)

- (117) [omote-dake-ni nori-no tsui-te-i-ru] shīto  
 surface-only-LOC glue-GEN stick-GER-be-PRS sheet  
 ‘a sheet [where glue is only stuck to the surface]’

Furthermore, as shown in Ochi (2017), in some Kyūshū dialects, nominative-genitive conversion has even been retained in main clauses. In these dialects, -DAKE is infelicitous with genitive-marked subjects in general, thus also in main clauses, which are CPs.

- (118) ??Hanako-dake-no furansugo-no hana-se-ru to yo. (Kyūshū J.)  
 Hanako-only-GEN French-GEN speak-POT-PRS COMP SFP  
 ‘(Listen), only Hanako can speak French!’  
 (Ochi, 2017, 692)

These data points show that focus and genitive-marking are not mutually exclusive. Nevertheless, the intuition seems to be strong that genitive-marked subject relative clauses resist focus marking. The question is if this is really due to a different size of the relative clause. In the following section, focus will be used as one indication why Japanese relative clauses must be CPs in general.

### 5.3.3.5 Summary

This subsection discussed arguments, mainly from Miyagawa (2011, 2012, 2017) for relative clauses of different sizes based on the phenomenon of Nominative-Genitive-Conversion. This discussion has shown that the data points of Miyagawa are either problematic or can be handled under a different explanation rather than resorting to the systematic distinction between CP and IP status. In other words, the assumption that relative clauses come in different sizes in Japanese does not withstand closer scrutiny.

## 5.3.4 Japanese relative clauses are CPs

Essentially, this leaves us with one option, namely the null hypothesis that all relative clauses in Japanese are CPs. I will provide some arguments here. One very convincing piece of evidence comes from the behavior of focused elements just viewed above. Other arguments come from historical evidence on the presence of complementizers. Finally, I will present the derivation of relative clauses along an operator movement account, which gives further evidence for CP status.

## 5.3.4.1 Focus and relative clauses in Japanese

As outlined in the examples above, the possible occurrence of focused and topicalized elements inside relative clauses is a rather convincing piece of evidence for CP status. Adopting a left-periphery of the clause, a high area within the split CP (Rizzi 1997; and see Grohmann 2003; Laenzlinger 2011), which contains the projections FocP and (several) TopP among others (for instance Kiss 1998, 2007; Aboh 2014) offers a position for elements in (high) non-canonical positions. This is shown below where such a focused element in Italian appears in a non-canonical position. Focused elements are displayed in capital letters.<sup>37</sup>

- (119) Il TUO LIBRO ho letto (,non il suo). (Italian)  
 the your book have read not the his  
 ‘YOUR BOOK, I have read (, not his).’ (Rizzi, 1997, 286)

As Cinque (2020) points out, reduced relative clauses in Italian are incompatible with focus marking, which suggests that they lack a CP alongside a higher FocP layer (Cinque, 2020, 218–219).

- (120) \*Il libro A MARIA dato (non a Carlo) era molto costoso. (Italian)  
 DET book to Maria given not to Carlo was very expensive  
 (‘The book given TO MARIA (not to Carlo) was very expensive.’)  
 (Cinque, 2020, 219)

Above, I already presented examples in which relative clauses in Japanese contain focus sensitive particles. Those can be attached to the subject, but they can also be used to put emphasis on adverbs (116) or PPs (117). Below, to complement the picture, I give an example of an adjectival relative clause in which the subject, which is the possessee of the relativized head noun, is focused (121). In this DP, it is *only* the dress of the woman which is highlighted as beautiful, contrasting with other items, or possibly the woman itself.

- (121) doresu-dake-ga utsukushi-k-at-ta josei  
 dress-only-NOM beautiful-PRED-COP-PST woman  
 ‘the woman such that only her dress was beautiful’ (Lit. ‘only the dress was beautiful woman’)

Such a contrastive focus, involving a *set of alternatives* (Zimmermann and Onea, 2011), can also be expressed by the particle *WA*, which is typically used as the topic marker but can express focus (Vermeulen, 2007, 2012). In addition to (113), where

<sup>37</sup>See Chapter 3.1.3 for the left periphery in the DP; for focus within the DP, see Chapter 3.1.3.5.

this marker was added to a postposition, it can also attach to the subject of a relative clause and replace the nominative (or genitive) case marker (122).

- (122) [Tarō-wa tabe-na-k-at-ta] ryōri-wa dore de-s-u  
 Tarō-TOP eat-NEG-PRED-COP-PST dish-TOP which PRED-COP.POL-PRS  
 ka?  
 Q  
 ‘Which one is the dish [that TARŌ did not eat]?’

Moreover, WA inside a relative clause does not necessarily entail contrastive focus, but can also yield presentational focus (123).<sup>38</sup>

- (123) Kore-wa [mō waka-i hito-wa yoma-na-i]  
 this-TOP any.longer young-MOD person-TOP read-NEG-MOD  
 hon-de-s-u.  
 book-PRED-COP.POL-PRS  
 ‘This is a book [such that it is young people who do not read it any longer].’

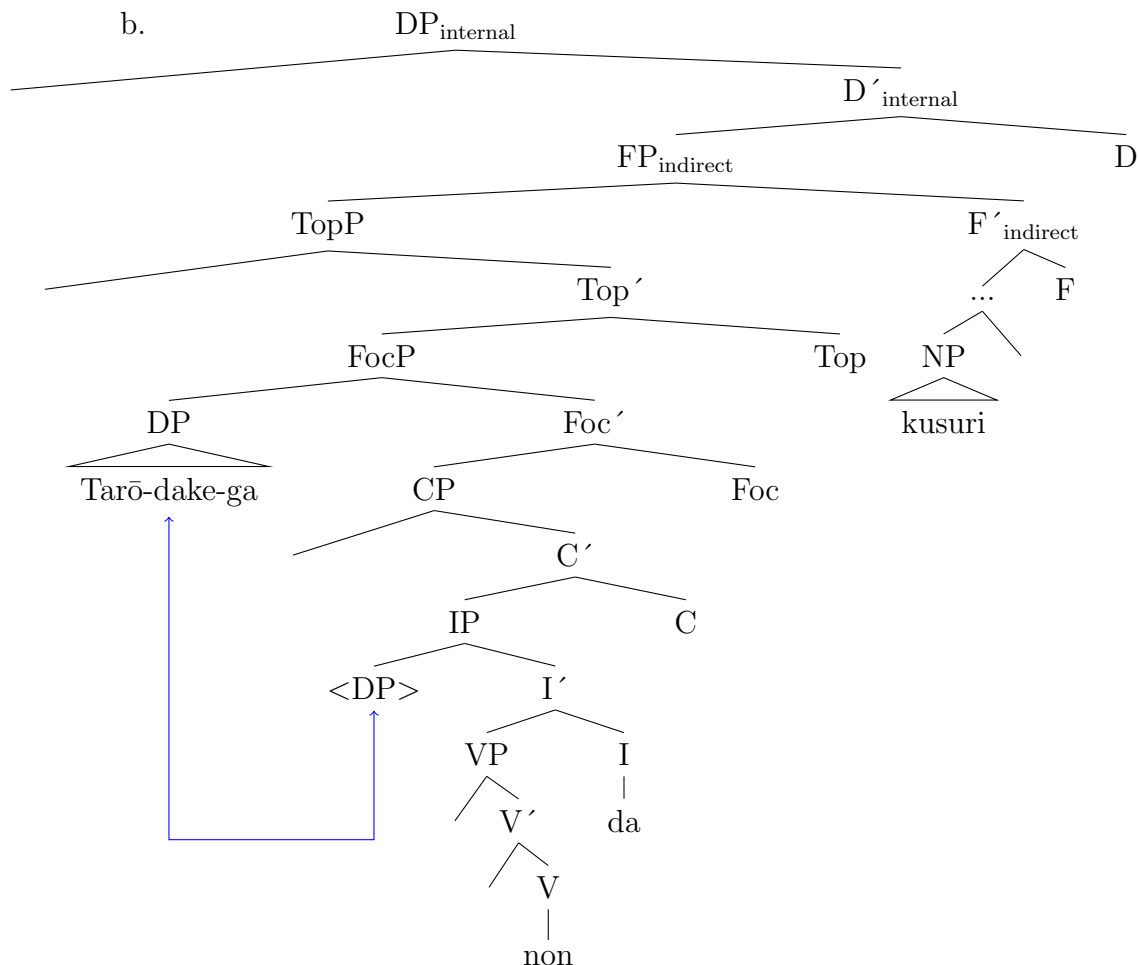
The possibility of focus (and topic) marking within relative clauses suggests the presence of a Focus layer (and a Topic layer). This, then, is good evidence of the CP status of relative clauses in Japanese. As with DP-related focus, we do not know whether focused constituents obligatorily move to a higher FocP layer or whether they remain in-situ. Recall that movement to a focus position was by no means a requirement for modifiers within the DP (Chapter 3.1.3.5). Assuming a movement operation would look as follows where the subject DP *Tarō-ga*, having received nominative case in Spec, IP, moves to Spec, FocP. But of course, the focused subject DP could also receive in-situ focus through the higher FocP layer.<sup>39</sup>

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<sup>38</sup>This type of focus is less defined by an existing set of alternatives and its contrastive nature, but more by the intention of the speaker to “express the most important part of the utterance, or what is new in the utterance” (Krifka, 2007, 23).

<sup>39</sup>I have not indicated an additional CP layer above FocP. In the analysis of (Cinque, 2020, 57–65), it is argued for some languages that this is necessary and why.

- (124) a. [Tarō-dake-ga non-da] kusuri  
 Tarō-only-NOM drink-PST medicine  
 ‘the medicine [(that) only Tarō took]’ (= (114))



### 5.3.4.2 Complementizers

Kaplan and Whitman (1995) provide morphological evidence for a CP status of relative clauses in Classical Japanese and Korean which they apply to Modern Japanese. Essentially, their argument is based on the fact that in Classical Japanese and Korean, verbs in attributive structures necessarily co-occur with overt complementizers, which project a CP layer. Their example and analysis of Korean is depicted in the example below.

- (125) [ecey pro manna-ass-te-n] salam (Korean)  
 yesterday pro meet-PST-RTRO-COMP person  
 ‘the person [*pro* met yesterday]’ (Kaplan and Whitman, 1995, 30)

It is the element  $-(NU)N$  that the authors interpret as an affixal complementizer. The argument is that it appears after all tensed elements as would be expected from complementizers, with C being higher than T (Kaplan and Whitman, 1995, 30).

- Another convincing piece of evidence that the attributive form fulfilled the role of a complementizer in Classical Japanese is the phenomenon of *focus concord* (Japanese: *Kakari-Musubi*, Lit. ‘dependent-concluding’), as presented in Kinsui (1995) and advocated for in Hiraiwa (2005, ch. 3.5).<sup>42</sup> *Kakari-Musubi* refers to a phenomenon in which certain elements in the clause affect the inflectional form of the matrix verb.

<sup>40</sup>Note that the adnominal form in some of the nine attested inflection classes of Classical Japanese is expressed by an additional, distinct, morpheme, which can be straightforwardly analyzed as a complementizer, but in some classes, sentence-final form and adnominal form exhibit different inflectional morphemes. This is the case for elements following the *ri*-inflectional class, to which also -KERI belongs, and also for the historical inflectional paradigm of *i*-adjectives which alternated between -SHI (sentence-final form) and -KI (adnominal form) as shown below.

- (126)      a.    taka-shi  
                high-SHK  
                ‘is high’*sentence-final*  
       b.    taka-ki    mono  
                high-RTK thing  
                ‘a high thing’*attributive*

See the discussion in Chapter 4.4.3.2 on why -KI, although it can be analyzed as a complementizer, does not necessarily only entail indirect modification.

<sup>41</sup>I gloss -RU as COMP here, but it can be glossed as RTK (*rentaikei*) with more emphasis on the inflection form.

<sup>42</sup>For descriptions on focus concord in Classical Japanese, see among others Morishige (1971); Funagi (1987). See Narrog (2019) for a recent contribution.

For example, clause-internal *wh*-elements, and focus elements trigger inflection of the verb in sentence-final position to attributive, not sentence-final, form. This is shown in example (129), in which the *wh*-element *izure-no* ‘when’ triggers an embedded question highlighted via the question particle *-KA* and the matrix verb occurs in attributive form (Hiraiwa 2005, 125–127 and see Nomura 1995).

- (129) Miyuki fu-ru koshi-no Ohoyama yuki-sugi-te izure-no  
 snow fall-PRS.RTK mount-MOD Ōyama go-pass-GER when-MOD  
 hi-ni-**ka** wa-ga sato-wo mi-**mu**.  
 day-LOC-Q I-GEN home-ACC see-COMP  
 ‘Crossing the snowy Mt. Ōyama, on which day can I see my home?’  
 Classical Japanese (Hiraiwa 2005, 125 from Manyōshū 3153)

When verbs appear as complements of matrix verbs, they are also inflected for adnominal form (Kinsui, 1995, 10–11). In other words, the adnominal form performs the complementizing function in absence of a complementizer.

- (130) [Tomo-no empō-yori ki-tar-**u**]-wo yoroko-bu.  
 friend-MOD away-ABL come-PST-COMP-ACC delighted.at-SHK  
 ‘*pro* is delighted at (the fact) that a friend came all the long way.’  
 Classical Japanese (Kinsui, 1995, 10)

These pieces of data give strong evidence that in Classical Japanese the adnominal form served the purpose of a complementizer.

Kaplan and Whitman (1995) argue that the complementizer layer has been preserved in Modern Japanese, although it is not filled overtly, but via a null complementizer (Kaplan and Whitman, 1995, 33–35). Indeed, it does seem convenient to not have to assume a categorial change of the category of relative clauses in the course of the language from CP to IP. The authors fledge out their analysis with the element *-NA*, the non-past attributive form of *na*-adjectives, and argue that *NA* is a copula that originates in *V* and raises through *I* to *C* “where it supports the null complementizer” (Kaplan and Whitman, 1995, 34). They give the following structure for a relative clause with an adjective where a nominal subject is present that relates to possession, similar to the examples mentioned in Chapter 3.2.7.2.

- (131) [NP [CP [IP kami-ga kirei t<sub>V</sub> t<sub>I</sub>] [C [I [V na]]] hito]  
 hair-NOM pretty COP person  
 ‘a person whose hair is pretty’  
 Modern Japanese (Kaplan and Whitman, 1995, 34)

While I agree that these arguments in favor of a CP layer, in Classical and in Modern Japanese for relevant instances, are rather convincing, the analysis of *NA* always

projecting such a layer cannot be maintained for the same reasons why Nishiyama's analysis cannot be universally maintained, as explained in Chapter 5.2.4. Besides the fact that it is not clear what triggers this movement to C other than the necessity to support the null complementizer, this analysis denies the possibility of direct modification same as Nishiyama's. We saw examples in Chapter 4.4.3.2, where an additional CP layer, at least the possibility, was not immediately excluding direct modification, but this does not mean all modifiers always project a CP.

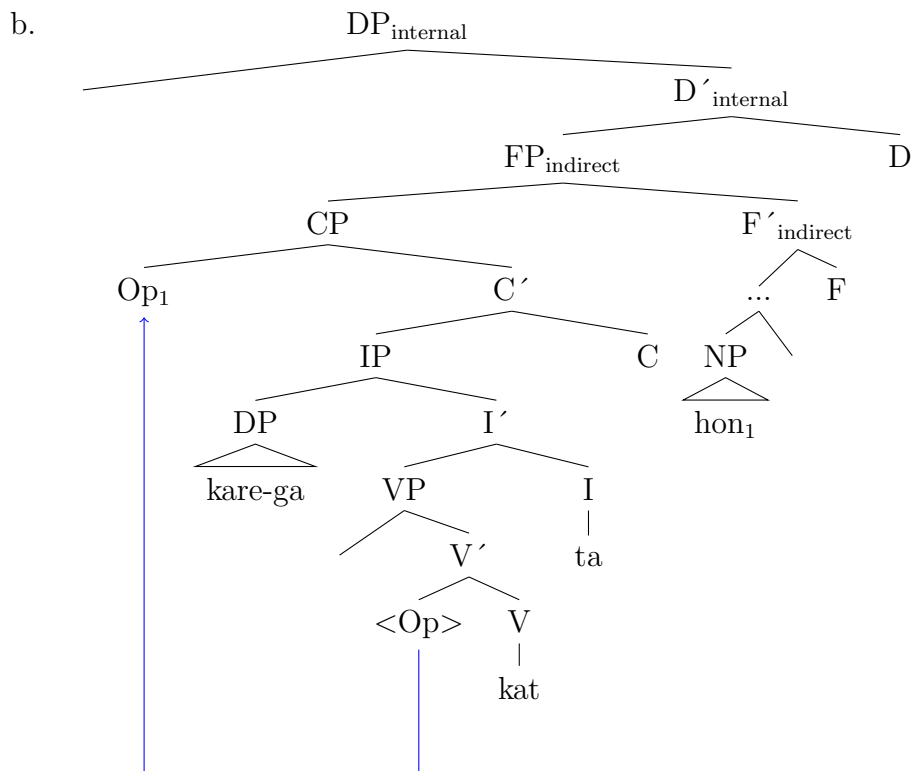
### 5.3.5 Derivation of indirect modifiers

Further evidence for CP status comes from the derivation of Japanese relative clauses adopted here. Recall that the intriguing thing about the category of relative clauses in general is that the head noun needs to be active, somehow represented, both in the matrix clause as well as inside the relative clause (the *Connectivity Problem*, Salzmänn 2017). Recall furthermore that there are many technical implementations to tend to this problem. In Japanese, the standard analysis is to assume a (silent) relative operator which ensures correct interpretability of the head noun inside the relative clause (see Miyamoto 2017 a.o.), and this will be presented below (for general accounts on Operator movement within relative clauses see, among others, Chomsky 1977; Jackendoff 1977 and Salzmänn 2017, ch. 2 for an overview).

#### 5.3.5.1 Operator movement

Schematically, I assume that a head-external relative clause with a verbal nucleus is derived along the following lines.

- (132) a. kare-ga  $Op_1$  kat-ta hon<sub>1</sub>  
 he-NOM  $Op_1$  buy-PST book<sub>1</sub>  
 ‘the book (which) he bought’



Besides the assumptions that relative clauses are clear indirect modifiers and are underlining single-headed (for reasons of simplicity, but see Cinque 2020; Chapter 2.4), I adopt an operator-account (Ishii, 1991; Miyamoto, 2017), according to which it is a relative clause operator which ensures the correct interpretability of the noun within the relative clause and mediates the relationship between noun and relative clause, solving the *Connectivity-Problem* (Salzmann, 2017). This operator  $Op$  starts from within the relative clause, in this case, from the internal argument position of the verb since it is an object relative clause. It then raises to Spec, CP, which means such a position must be available as a landing site. It is exactly this operation, the raising of the operator, for which we can find evidence to be reviewed below.

At first glance, similar results can be achieved by assuming a *raising* analysis within a double-headed framework with the difference that the noun itself is the internal head of the relative clause, in addition to an identical counterpart (the external head in the matrix clause) and then moves to Spec, CP. The external head is then deleted following this operation. This derivation likewise predicts movement effects in an identical fashion (see Cinque 2020, 22–36 for arguments and Cinque 2020, 65–71 for a discussion on prenominal RC-languages).

Complications arise when trying to apply this derivation to other kinds of relative clauses in Japanese, for instance head-internal relative clauses (Hiraiwa, 2012; Hiraiwa *et al.*, 2017). Furthermore, some sort of remnant movement of the relative



The operator-derivation can be motivated by identifying various reconstruction effects, which, in turn, provide clear evidence for movement (Schachter, 1973; de Vries, 2002), most of which have been reported for Japanese in Ishii (1991) and are discussed in Miyamoto (2017) and Whitman (2013). A prominent reconstruction effect involves pronominal anaphor binding with the heavy pronouns *karejishin/kanojishin* ‘himself/herself’ or *jibunjishin* ‘oneself’ (Miyamoto 2017, 615 quoting Ishii 1991, 29), similar to English *the picture of himself<sub>1</sub> [which John<sub>1</sub> likes e]]*, (Salzmann, 2017). Observe (133).

- In order for the reflexive *karejishin* ‘himself’ to find its antecedent *Jon*, the two elements, or some instances of them (the operator), must have been in a c-command relation at some point in the derivation. The underlining matrix clause in (134) confirms that movement must have happened (Miyamoto, 2017, 615–616).<sup>43</sup>

- Furthermore, even though Japanese has been argued, for example in Kuno (1973), to lack island effects in relative clauses in general, it does show them clearly with object relative clauses (Miyamoto 2017, 623–626, see also Whitman 2013, 369–373).<sup>44</sup>

- <sup>43</sup>Allegedly the same holds with raised quantifiers, according to Whitman (2013) then even with the canonical reflexive pronoun *jibun* ‘self’.

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The corresponding root clauses of (135) are repeated below for convenience.

- (136) Gakusha<sub>1</sub>-ga hon<sub>2</sub>-wo kai-ta. Bill-ga sono hon<sub>2</sub>-wo  
 scholar<sub>1</sub>-NOM book<sub>2</sub>-ACC write-PST Bill-NOM that book<sub>2</sub>-ACC  
 yakushi-te-i-ru.  
 translate-GER-be-PRS  
 ‘A scholar<sub>1</sub> wrote a book<sub>2</sub>. Bill is translating that book<sub>2</sub>.’

Finally, Japanese allows relativization of idioms (Inoue, 1981; Ishii, 1991; Miyamoto, 2017; Whitman, 2013), which is another argument for movement within relative clauses often given (Schachter 1973; Carlson 1977, see also Bhatt 2002, 2015; Sauerland 2003; Cinque 2020). The argument goes that if a noun is relativized and an idiomatic meaning with the modifying verb is retained, these elements must have formed constituents at one point of the derivation. Observe the example below.<sup>45</sup>

- (137) [Karera-ga magarinimo *e*<sub>1</sub> tsuke-ta] ketchaku<sub>1</sub>-wa  
 they-NOM somehow.or.other *e* reach-PST conclusion-TOP  
 yoba-re-na-k-at-ta.  
 be.happy-POT-NEG-PRED-COP-PST  
 ‘They could not be happy about the end [they somehow or other brought it to].’ (*ketchaku-wo tsukeru* = to reach a conclusion = to bring to an end)  
 (Inoue, 1981, 214)

These pieces of data suggest that movement takes place in Japanese relative clauses, in turn motivating the operator derivation. And this derivation only leaves room for an analysis of relative clauses as CPs since the operator needs a landing site.

### 5.3.5.3 Relative clauses and major subjects

There is a class of relative clauses that have been argued to be derived from so-called *major subject constructions* (Kuroda 1992, ch. 9; cf. Kuno 1973; Heycock 1993b; Whitman 2013; Miyamoto 2017). Examples were briefly mentioned in Chapter 3.2.7.2 (and also Chapter 5.3.2.2) and look as follows. Recall that the examples below are cases of relative clauses where the non-verbal modifiers are followed by the monomoraic realizations of MOD, but that forms of the copula can follow the respective modifiers instead.

<sup>45</sup>Idioms are a very difficult case, even in English, and recent discussion has cast doubt upon the validity of the argument that they must be involved in movement derivations (Heycock, 2019; Webelhuth *et al.*, 2019; Pankau, 2020). Furthermore, as mentioned in Bhatt (2002, 48), this argument does not rule out other analyses. See also the appendix in Cinque (2020). In Hoshi (2004), it was argued that the relativization of heads taking part in idiomatic constructions is ruled out in Japanese, but Inoue (1981); Whitman (2013) and Miyamoto (2017) have argued otherwise. Ken Hiraiwa (p.c.) and other speakers have told me that they do not find the idiomatic expression reported in the text entirely natural.

- (138) [Tarō-ga shujinkō-no] monogatari  
 Taro-NOM main.protagonist-MOD story  
 ‘story [in which Taro is the main protagonist]’ (Watanabe, 2010, 66)
- (139) [shūshoku-ga taihen-na] butsurigaku  
 job-NOM hard-MOD physics  
 ‘physics, [where finding a job is difficult]’ (Kuno, 1973, 255)

In (140) and (141), the underlying constructions to (138) and (139) involving a major subject are given. The examples are adapted from the authors mentioned above, (141) is also given in Heycock (1993b, 179).<sup>46</sup>

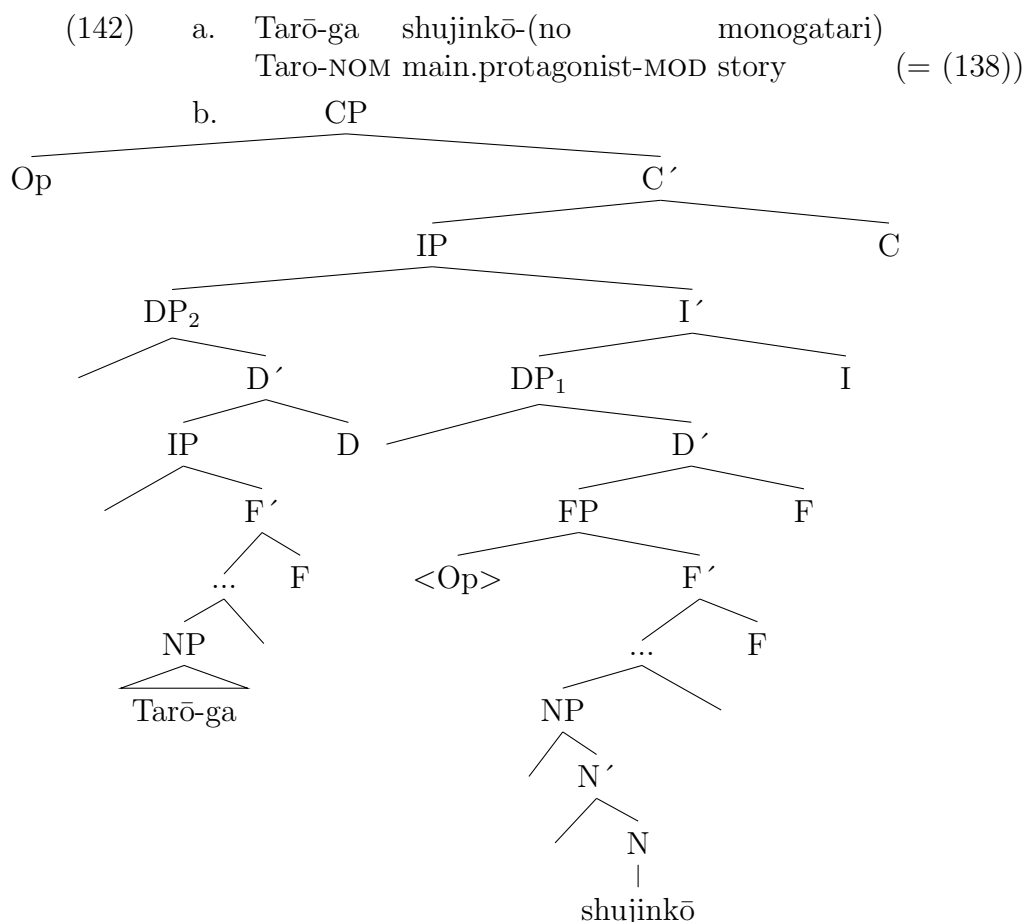
- (140) Sono monogatari-ga Tarō-ga shujinkō-d-a.  
 this story-NOM Taro-NOM main.protagonist-PRED-COP.PRS  
 ‘(As for) this story, Taro is the main protagonist.’
- (141) Butsurigaku-ga shūshoku-ga taihen-d-a.  
 physics-NOM employment-NOM difficult-PRED-COP.PRS  
 ‘Physics is such that finding a job is difficult.’

In major subject constructions, two nouns are bestowed with the nominative particle. This is thus a case where a subject-marked constituent does not bear a theta role (Heycock, 1993b). Rather, the higher GA-marked noun seems to have a focus (or potentially topic) interpretation, as argued in (cf. Vermeulen 2005).

Major subject constructions have been used as an underlying mechanism to account for seemingly gapless relative clauses such as (139) since Kuroda (1992). This suggests that for the relative clause in (138), the following structure is at stake. See Heycock (1993b) and Miyamoto (2017) for the same derivation for (139), on which this is based.

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<sup>46</sup>A reviewer wonders whether (138) might not be derived from a prepositional phrase in which (*sono*) *monogatari* is a locative expression. However, note that this expression could serve as a major subject in a major subject construction as well, since, as pointed out in Chapter 5.3.2.2, this is generally possible for locative (and temporal) expressions (see example (100)). In other words, even if we take the relativized noun to have been part of a prepositional phrase, the relative clause is arguably derived from a major subject construction.



The modifying clause contains two nouns, *Tarō* and *shujinkō* ‘main protagonist’, which, in fact, must both project a DP. The DP *Tarō*, the subject within the relative clause, is situated in Spec, IP of the CP in which *shujinkō* is embedded (although it is questionable whether it originates from somewhere within that DP<sub>1</sub>). Furthermore, the relative clause contains an operator corresponding to the modified noun *monogatari* ‘story’. The major subject construction given above shows that the operator must originate from somewhere within the relative clause, ensuring the correct interpretation. Accounting for its role as a modifier, under the theory adopted here, I assume it is situated in the extended projection of DP<sub>1</sub>.<sup>47</sup>

The derivation is slightly different, and in fact clearer, for relative clauses of the following kind with an adjectival nucleus co-occurring with a subject, which in turn expresses a possession-relation with the modified noun.

- (143) [doresu-ga utsukushi-k-at-ta] jōsei  
 dress-NOM beautiful-PRED-COP-PST woman  
 ‘the woman [whose dress was beautiful]’

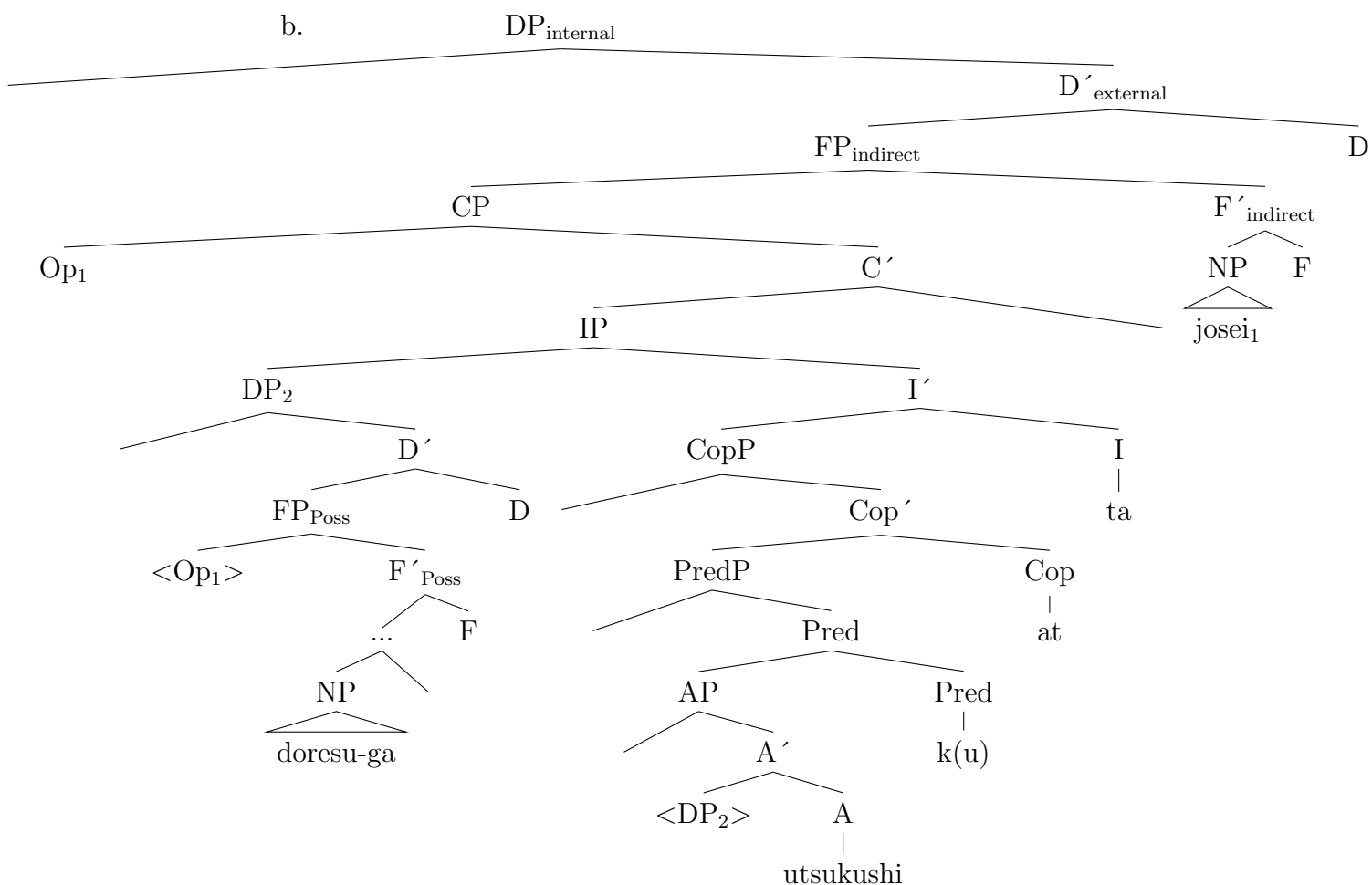
<sup>47</sup>Miyamoto (2017), among others, assumes a subject position, but likely, nothing hinges on this. The point is that the operator originates from somewhere within the relative clause.

In this sentence, the dress belongs to the woman and is described as beautiful. Such sentences, arguably, also have a corresponding major subject construction, namely (144) (and see Chapter 3.2.7.2). Here, *sono josei* ‘that woman’ fulfills several characteristics of a major subject: It has a focus interpretation, it does not bear a theta role, and it seems to be followed by a prosodic break (Vermeulen, 2005, 1346–1348).

- (144) Sono josei-ga        (sono josei-no)        doresu-ga  
      that woman-NOM that woman-MOD dress-NOM  
      utsukushi-k-at-ta.  
      beautiful-PRED-COP-PST  
      ‘As for that woman, her dress was beautiful.’

The relevant difference to (138) is that the operator, representing the noun *josei* ‘woman’, is closely related to the subject DP within the relative clause *doresu* ‘dress’, not to the modifier, which is not a noun, but an adjective. I therefore assume that the operator starts out from the extended projection of that DP (here DP<sub>2</sub>). This explains the structural differences to the derivation of (138) given above in the derivation given below.

- (145) a. [doresu-ga utsukushi-k-at-ta] jousei  
 dress-NOM beautiful-PRED-COP-PST woman (= (143))



Crucially, the DP *doresu* ‘dress’, in which the operator, co-indexed with the head noun, is contained, starts out as the complement of the adjective, ensuring the meaning that it is the dress that is beautiful not the woman, and subsequently raises to Spec, IP of the relative clause. The operator, in turn, raises from the extended projection of DP<sub>2</sub> to Spec, CP. That this must be the case, that some sort of movement takes place, was shown through the availability of reconstruction effects. Here, it is also possible to argue for a precise position of this operator within the extended projection, as it is most likely a position related to possession, ensuring the possessive relationship between ‘dress’ and ‘woman’. I tentatively give a functional projection FP<sub>Poss</sub> (although this might not be relevant, given the broad nature of functional projections in Japanese). Overall, the structure is slightly different than for (138) (and probably even more complex for (139)).

To summarize, the derivation of relative clauses assumed in this book makes use of operator-movement and this gives good piece of evidence for the CP status of Japanese relative clauses. A derivation along these lines is assumed to be universal, thus holds also in seemingly simpler DPs such as (132) mentioned at the beginning

of the chapter. In each case, an operator, which is co-indexed with the head noun, raises from within the relative clause to Spec, CP.

### 5.3.6 Summary

This section showed why the size of relative clauses in Japanese is hard to determine, and eventually presented a unifying analysis as CPs. It was shown that neither the arguments that relative clauses cannot be bigger than IPs, nor that they are of different sizes were convincing. The crucial part to take away is that relative clauses in Japanese only come in one size, which makes the dichotomy *reduced* vs. *full* as well as *CP* vs. *IP* superfluous, and with it the necessity of dedicated projections for these categories.

## 5.4 On the position of relative clauses

Assuming there are several functional projections in the indirect domain, above as well as below numerals, and that relative clauses are all CPs suggests that placement is entirely arbitrary. The question is if this is really the case or if there are factors influencing the position of relative clauses. In this section, it will be argued that the higher positions are reached via independent movement and that a factor influencing this movement is complexity.

### 5.4.1 Introduction

Recall that cross-linguistically the following structural hierarchy is at stake: *demonstrative* > *full relative clause* > *numeral* > *adjective/reduced relative clause*. Translated into terms of head proximity, full relative clauses are further away from the noun than numerals, which in turn are further away than reduced relative clauses (cf. Cinque 2020, ch. 3.5; Cinque 2017; Greenberg 1963; Hawkins 1983; Rijkhoff 2002). This is especially easy to visualize in languages in which all modifiers strictly appear on one side of the noun.

While this applies to Japanese, being strictly head-final, this language does not differentiate between full and reduced relative clauses. This predicts that relative clauses can productively appear to the left or to the right of numerals, which are also situated in the indirect domain. And truly, speakers generally do not have any preference regarding the order of these elements.

- (146) a. [pātī-de at-ta [futari-no josei]]  
           party-LOC meet-PST two-MOD woman  
       b. [futari-no [pātī-de at-ta josei]]  
           two-MOD party-LOC meet-PST woman  
           ‘the two women who I met at a party’

In (146-a), the relative clause in linear order precedes the numeral, while in (146-b) the relative clause linearly follows the numeral. Taking linear order as an indication of structural position suggests that in (146-a) the relative clause is situated in a functional projection above  $FP_{Num}$ , whereas the relative clause in (146-b) is in a functional projection below it. Whether the latter case is the result of movement, or if the relative clause was base-generated in a higher position can at first glance not be deduced. Crucially, the two positions are not the result of the relative clauses being of a different size in each (146-a) and (146-b).

The question now is whether the two options, and ordering of these constituents in general, are simply random. Importantly, speakers find both orders in (146) perfectly fine and do not have a preference for one order over the other.<sup>48</sup> To answer this question, one will first have to investigate if there is a way to make one of the options more or less natural, or more or less acceptable, in contrast to the alternative.

These considerations naturally also apply to adjectival relative clauses, those containing an adjectival nucleus. Considering the findings from the previous chapter, bare adjectives in non-past are ambiguous between a direct and an indirect status. Unambiguous indirect adjectival modifiers can come in several types, but a clear indication is inflection into past tense as well as co-occurrence with subjects.<sup>49</sup> Starting with inflected adjectives, speakers again find both orders – *adjectival relative clause*  $\succ$  *numeral* as well as *numeral*  $\succ$  *adjectival relative clause* – in the ordering pairs presented below grammatical and, more importantly, do not have a preference for one option. The following examples are all taken and adapted from findings in the corpus study reported further below.<sup>50</sup>

<sup>48</sup>Recall that this is not the case for direct modifiers which are significantly degraded to the left of numerals.

<sup>49</sup>Recall that some modifiers analyzed as direct do allow subjects as discussed in Chapter 4.4.3.2.

<sup>50</sup>Although it is a fine line between grammaticality on the one hand and preference on the other, this difference is crucial to be aware of. When discussing such minimal pairs of examples, I checked always for both parameters, although in most cases only a difference in preference emerged. I mark the non-preferred version in the following via the symbol #.



- (147) a. [kanashi-k-at-ta [mittsu-no dorama]]  
sad-PRED-COP-PST three-MOD soap.opera  
b. [mittsu-no [kanashi-k-at-ta dorama]]  
three-MOD sad-PRED-COP-PST soap.opera  
'three soap operas that were sad'
- (148) a. [utsukushi-k-at-ta [futari-no josei]]  
beautiful-PRED-COP-PST two-MOD woman  
b. [futari-no [utsukushi-k-at-ta josei]]  
two-MOD beautiful-PRED-COP-PST woman  
'the two women who were beautiful'

Corresponding examples in other languages do show ordering restrictions. For instance in German, when a participial – that is reduced – relative clause and a numeral co-occur, placing the numeral to the left of the reduced relative clause is strongly preferred. In (149-b), I translated the examples of (147-b) into German, and, although of course there might be subtle differences in information structure depending on the context, according to my judgment, the version in which the reduced relative clause follows the numeral is not only much more natural than the reversed order, but I find the reversed order quite ungrammatical (see also the example in Chapter 2.4).<sup>51</sup>

- (149) a. die drei traurig gewesenenen Seifenoperen (German)  
the three sad be.PST.PTCP.PL soap.opera  
b. ??die traurig gewesenenen drei Seifenoperen  
the sad be.PST.PTCP.PL three soap.opera  
'the three soap operas that were sad'

In fact, there are ways to achieve a similar effect in Japanese, a preference for one order over the other. Significantly, this order is the reversed to the German case. One way to achieve this is to add degree modifiers such as *totemo* 'very' to the relative clause. Almost all speakers I consulted expressed a higher preference for the version with the relative clause in first position, contrary to the minimal pair without the degree adverb, where both options are fine. Given below is the relevant altered version of example (147) with the degree adverb.

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<sup>51</sup>Japanese also contrasts with Mandarin, another strict head-final language. Hsu (2017) reports on many more occurrences of relative clauses in a lower position than in a higher position above numerals and demonstratives in a corpus study he performed and judges the latter case as marked and focused.

- (150) a. [totemo kanashi-k-at-ta [mittsu-no dorama]]  
           very      sad-PRED-COP-PST three-MOD soap.opera  
       b. #[mittsu-no [totemo kanashi-k-at-ta      dorama]]  
           three-MOD very      sad-PRED-COP-PST soap.opera  
           ‘three soap operas that were very sad’

The same effect is achieved when an overt subject appears. Again, speakers showed a stronger preference for the relative clause in first position, now tested for the corresponding example to (148).

- (151) a. [doresu-ga utsukushi-k-at-ta [futari-no josei]]  
           dress-NOM beautiful-PRED-COP-PST two-MOD woman  
       b. #[futari-no [doresu-ga utsukushi-k-at-ta      josei]]  
           two-MOD dress-NOM beautiful-PRED-COP-PST woman  
           ‘the two women whose dresses were beautiful’

Importantly, if a certain order is preferred, it is always the one in which the relative clause precedes the numeral, never the other. But why do speakers prefer the position of the relative clause to the left of the numeral, once a degree modifier or a subject are present?

These relative clauses have been made more *complex* or *heavier*. Although *complexity* and *heaviness* are rather evasive terms and hard to define, let alone measure, I will attempt a descriptive definition below.

For *syntactic complexity*, I adopt the following definition of Hawkins (2007), who relies on an idea going back to Miller and Chomsky (1963).

- (152) Complexity is a function of the amount of structure that is associated with the terminal elements, or words, of a sentence. More structure means, in effect, that more linguistic properties have to be processed in addition to recognizing or producing the words themselves. (Hawkins, 2007, 8)

This means we can speak of higher complexity when a phrase contains additional constituents, especially functional ones. This is definitely the case when an overt subject appears or when an adverbial phrase is merged in the CP domain of the relative clause.

*Prosodic complexity* or *heaviness* on the other hand can be described as equal to length. In other words, the more morae a relative clause consists of, the longer, and in turn the more complex it is.<sup>52</sup>

<sup>52</sup>Some speakers have a higher tolerance than others in this respect. For example Shōta Momma (p.c.) has pointed out to me that for him, the more complex a relative clause to the right of a numeral is made, the less natural he finds it. However, he would still not willingly shift the two units around.

In the following, I will present more empirical evidence to the observation that higher complexity correlates with higher preference of relative clauses to the left of numerals based on a corpus study and subsequent discussion with native speakers on this topic.

### 5.4.2 Complex relative clauses in higher positions: Corpus study

In order to get a clearer picture of the phenomenon I have performed a search query for all occurrences of all adjectival relative clauses containing an adjective inflected for past and a numeral within the same DP, thus modifying the same noun, in the Japanese web corpus jaTenTen11 (Jakubíček *et al.*, 2013).<sup>53</sup> The goal was to see which order of the two constituents is generally more attested and in what way additional constituents inside the relative clause have influence on its position.

There are several reasons why I have chosen to investigate (solely) adjectival relative clauses. From a practical perspective, increasing complexity of this type of relative clause is a straightforward task since degree adverbs are usually freely compatible with adjectives – which is not the case for most verbs and by extension verbal relative clauses where such combinations are often semantically impossible.<sup>54</sup> From a theoretical perspective, I expected this study to complete the picture of adjectives in Japanese, and to give a nice contrast to direct modifiers, out of which adjectives are argued to be the prototypical representatives. On the other hand, one must be aware of the fact that adjectives inflected for past and numerals modifying the same noun might not be entirely natural (see below).

Concretely, I performed searches for numerals,<sup>55</sup> either followed or preceded by an *i*- or a *na*-adjective inflected for past, then followed by a noun. This means in total four different search scenarios were tested. Results were looked through and

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<sup>53</sup>This corpus is accessible via Sketch Engine, <https://auth.sketchengine.eu> (Kilgariff *et al.*, 2014).

<sup>54</sup>A related point is that adjectives were considerably easier to test in the corpus at hand, mainly because I could focus on inflections into past tense. Not only were those clear signs of indirect modification, there were no other tense or aspect forms to test in attributive position. For verbs, I would have needed to include search operations for non-past and other tense and aspect forms, such as progressive, since these all mark instances of indirect modification. Furthermore, syntactic dependencies are not annotated in the interface at hand, which means that getting results for verbal relative clauses as nominal modifiers is quite cumbersome. In fact, one would need to look for all occurrences of a verb and a noun at an arbitrary distance within the same sentence and then to annotate manually whether there are dependencies and what constituents intervene, if any. This aspect is therefore left for future research and potentially a more fitting manner of elicitation.

<sup>55</sup>Concretely, numerals in Japanese consist of a cardinal number morpheme and a numeral classifier reflecting the physical appearance (among other characteristics) of the head noun. There is also an underspecified option in which no classifier appears. In each case, numerals are within the DP obligatorily licensed via MOD spelled out as /no/.

false positives removed in the process. The final results after cleaning the data look as follows.<sup>56</sup>

- (153) *Order of Numerals and Adjectives in Past in Japanese Attested in jaTen-Ten11 (false positives removed, 2022/09/26)*
- a. *i*-adjective in past  $\succ$  Numeral: 149 results
  - b. Numeral  $\succ$  *i*-adjective in past: 26 results
  - c. *na*-adjective in past  $\succ$  Numeral: 74 results
  - d. Numeral  $\succ$  *na*-adjective in past: 7 results

It is striking, especially for a web corpus, that only so few results were obtained. This suggests that the combination of inflected adjectives and numerals is rare in naturally occurring speech. In practically all cases, an appropriate context is needed to justify an inflection of the adjective into past over non-past, as also mentioned by the consulted native speakers (see below).

The first obvious observation is that a higher number of occurrences for the order *adjectival relative clause*  $\succ$  *numeral* is found than for the reversed order. Importantly, the type of adjective, whether the adjective is an *i*- or a *na*-adjective, is not an influencing factor.

In the next step, I annotated every relative clause manually according to whether additional constituents appear or not. The figure below displays all relevant numbers. It indicates how many relative clauses contained additional constituents for each ordering pair and both adjective groups.

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<sup>56</sup>Needless to say, since extraction of false positives was performed manually and this process is rather error-prone, the results must be interpreted with caution.

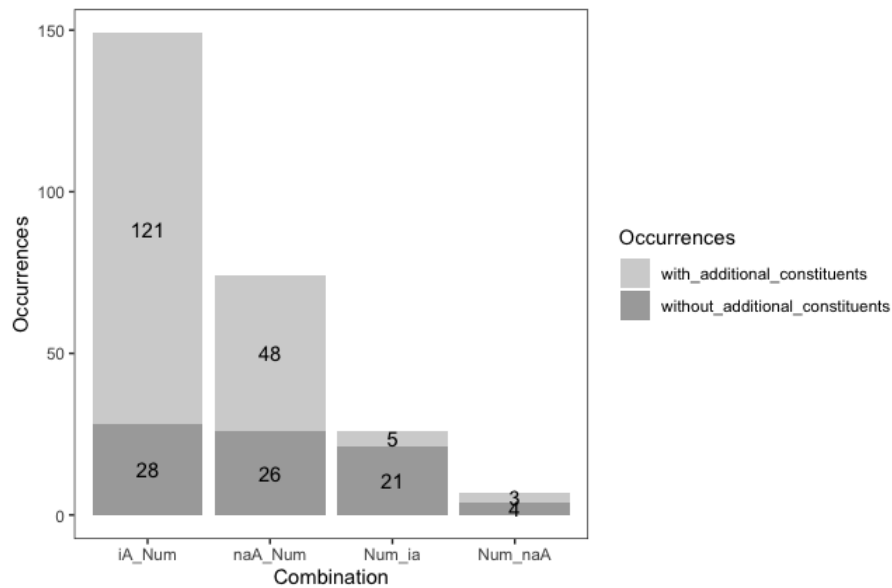


Figure 5.2: Co-Occurrences of Adjectival Relative Clauses and Numerals in Sketch Engine

Taking additional constituents as an indicator of higher complexity, these results confirm the initial observation: More complex adjectival relative clauses are favored to the left of numerals. 81% (121 out of 149) of relative clauses containing *i*-adjectives and 64% (48 out of 74) of those containing *na*-adjectives which appear to the left of the numeral contain at least one additional constituent. For the reversed position, no such trend is detectable: In the position below the numeral, only 20% (5 out of 21) of relative clauses with *i*- and 42% (3 out of 7) of relative clauses with *na*-adjectives contain an additional constituent.

In the final step, I have analyzed which constituents exactly appear in those relative clauses and annotated each relative clause according to the type of constituent it co-occurs with. This allowed a categorization of the relative clauses with constituents into different categories. There are five types of constituents, and, of course, a relative clause can contain more than one additional constituent, although this was uncommon. The most prominent constituents are *degree adverbs* and *subjects*, as exemplified above. Other types of constituents are *adverbial adjuncts*, *verbal clauses* and *compounds*. The relevant numbers are given below for the orders *adjectival relative clause*  $\succ$  *numeral* for both adjective groups.

| Type                                      | Amount |
|-------------------------------------------|--------|
| compound construction                     | 1      |
| subject                                   | 59     |
| degree adverb                             | 18     |
| adverbial adjunct<br>(time, place, other) | 8      |
| verbal clause                             | 18     |
| adjectival clause                         | 0      |
| compound + degree adverb                  | 2      |
| subject + degree adverb                   | 26     |
| verbal clause + degree adverb             | 1      |

Table 5.1: Types of Constituents Inside Adjectival Relative Clauses in the Order *i*-adjective  $\succ$  numeral

| Type                                      | Amount |
|-------------------------------------------|--------|
| compound construction                     | 4      |
| subject                                   | 8      |
| indirect object                           | 1      |
| degree adverb                             | 4      |
| adverbial adjunct<br>(time, place, other) | 18     |
| verbal clause                             | 9      |
| adjectival clause                         | 1      |
| compound + degree<br>Adverb               | 0      |
| subject + degree ad-<br>verb              | 3      |
| verbal clause + degree<br>adverb          | 0      |

Table 5.2: Types of Constituents Inside Adjectival Relative Clauses in the Order *na*-adjective  $\succ$  numeral

To illustrate what is meant by these labels, representative examples are presented just below. For reasons of space and convenience, I give only the DPs.

- (154) [kaiseki-kanō-de-at-ta                      [go-ko-no      hai]]  
analysis-possible-PRED-COP-PST five-CL-MOD embryo  
‘five embryos which were possible to analyze’  
*na*-adjective  $\succ$  numeral; compound<sup>57</sup>
- (155) [inshō-ga              yo-k-at-ta                      [hitotsu-no riyū]]  
impression-NOM good-PRED-COP-PST one-MOD      reason  
‘one reason why the impression was good’  
*i*-adjective  $\succ$  numeral; subject<sup>58</sup>

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<sup>57</sup>castle104.com, 2022/09/26.

<sup>58</sup><http://cocolog-nifty.com>, 2022/09/26.

- (156) [kono mecha kata-k-at-ta [ni-hon-no neji]]  
 this very hard-PRED-COP-PST two-CL-MOD screw  
 ‘those two screws which were very hard’  
*i*-adjective  $\succ$  numeral; degree adverb<sup>59</sup>
- (157) [kodomo-no koro-ni suki-d-at-ta [ni-hiki-no neko]]  
 child-MOD time-LOC fond-PRED-COP-PST two-CL-MOD cat  
 ‘the two cats I was fond of as a child’  
*na*-adjective  $\succ$  numeral; temporal adverbial adjunct<sup>60</sup>
- (158) [kono kōen-de shit-te yo-k-at-ta [futatsu-no kotoba]]  
 this lecture-LOC know-GER good-PRED-COP-PST two-MOD word  
 ‘those two words for which it was good that I learned them in that lecture’  
*i*-adjective  $\succ$  numeral; verbal clause<sup>61</sup>
- (159) [higai-ga mottomo ōki-k-at-ta [itsutsu-no mura]]  
 damage-NOM most big-PRED-COP-PST five-MOD village  
 ‘the five villages where the damage was worst’  
*i*-adjective  $\succ$  numeral; subject + degree adverb<sup>62</sup>

The emerging picture from these findings further confirms the initial hypothesis: Indirect modifiers of a higher complexity are preferred to the left numerals. From a structural perspective, this means they are situated in a functional projection higher than  $FP_{Num}$ .

### 5.4.3 A closer look at the findings

Since the problem remains that only a few relevant examples were attested and because corpora, although being representative samples of a language, are only representative to a certain extent, I have followed up this corpus study with interviews and discussions with native speakers. I had three aims in mind: First, to find out whether representative examples are actually acceptable for speakers. Second, to examine which order of adjectival relative clause and numeral they generally prefer and what role additional constituents play. Third, to find out whether native speakers have intuitions on factors that might influence their preference regarding the order of relative clauses and numerals.<sup>63</sup>

The first thing to note is that the speakers shared the initial impression I had from the corpus analysis. The fact that only a few results are attested is because combinations of inflected adjectives and numerals within the same DP are not en-

<sup>59</sup><https://monote.net>, 2022/09/26.

<sup>60</sup><http://honeypeace.blog11.fc2.com/blog-date-20070808.html>, 2022/09/26.

<sup>61</sup>fc2.com, 2022/09/26.

<sup>62</sup><http://www.ekenzo.jp/shopdetail/006002000016/>, 2022/09/26.

<sup>63</sup>These interviews were conducted over Zoom with three speakers in March 2023. I am incredibly grateful to my consultants.



tirely natural. In terms of the analysis, as mentioned above, pairs of simple relative clauses containing an inflected adjective but no additional constituents in combination with a numeral were by all speakers judged equally fine in both orders, both in terms of grammaticality and acceptability. No speaker had any particular preference for one order over the other. If the adjectival relative clause, however, was presented with an additional constituent, for example a degree modifier such as *totemo* ‘very’, they expressed a clearly higher preference for the position of the relative clause to the left of the numeral. The relevant example is repeated from above (= (150)).

- (160) a. [totemo kanashi-k-at-ta      [mittsu-no dorama]]  
           very      sad-PRED-COP-PST three-MOD soap.opera  
       b. #[mittsu-no [totemo kanashi-k-at-ta      dorama]]  
           three-MOD very      sad-PRED-COP-PST soap.opera  
           ‘three soap operas that were very sad’

When asked for potential reasons for their choice, all consulted speakers noted that for them the sentence becomes easier to parse and that the length of the relative clause influenced their choice to place it DP-initially. One speaker specified that the order in which the numeral is to the left of the relative clause is not only odd, but stylistically awkward with longer relative clauses because the numeral seems too far detached from the noun.

Especially in relative clauses containing subjects, the order *adjectival relative clause*  $\succ$  *numeral* was clearly preferred. Some speakers even noted that for them, the reversed order is decreased not only in preference but in grammaticality. An illustrative example is (161).

- (161) a. [toiawase-ga ō-k-at-ta                      [yattsu-no shitsumon]]  
           inquiry-NOM many-PRED-COP-PST eight-MOD question  
       b. #[yattsu-no [toiawase-ga ō-k-at-ta                      shitsumon]]  
           eight-MOD inquiry-NOM many-PRED-COP-PST question  
           ‘the eight questions with many inquiries’ (Lit. ‘the eight questions  
           where the questions were many’)

A reason for this might be the desire to avoid attachment ambiguities. Such ambiguities can arise when the numeral can logically refer to not only the noun to be modified but also the subject contained in the relative clause. Syntactically, the numeral forms a constituent with the nominal subject within the relative clause, not with the modified head noun. This is indicated below with brackets and in the English translation.

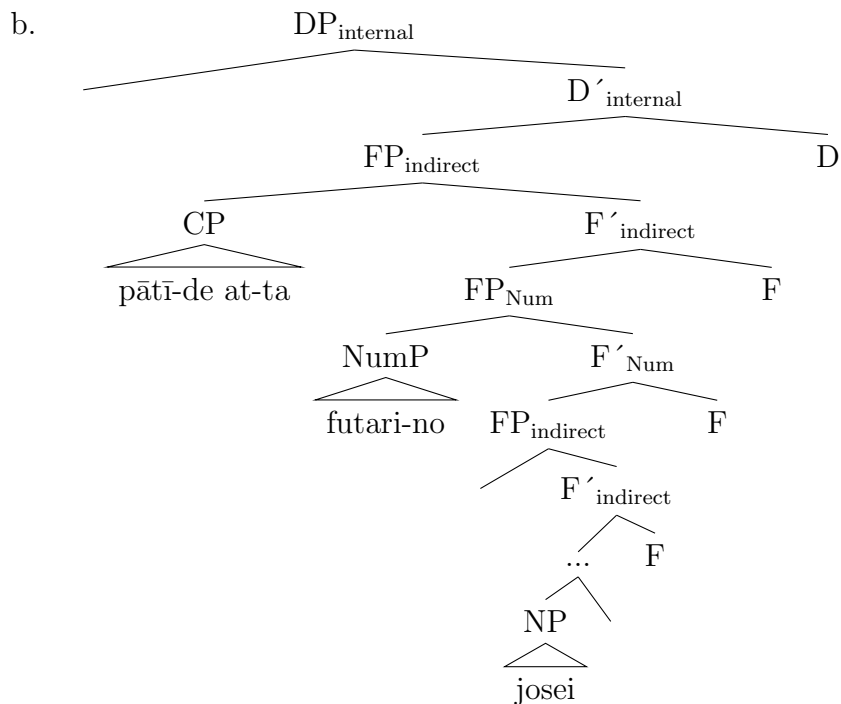
This discussion shows that syntax-external factors have to be considered when it comes to the order of these elements. This is similar to the effect of scope on the order of direct modifiers discussed in Chapter 4.4.3.1 (and see Ramchand and Svenonius 2014). However, the results are not as clear, as also evidenced by the subtle judgments. We cannot determine a certain threshold of *complexity* which bans a relative clause from a lower position. Nevertheless, the observation holds that more complex relative clauses are definitely preferred to the left of numerals. Possible syntactic correlations of this fact will be reviewed in the following.

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languages which also do not mark a difference between the two categories of relative clause morphologically or syntactically.

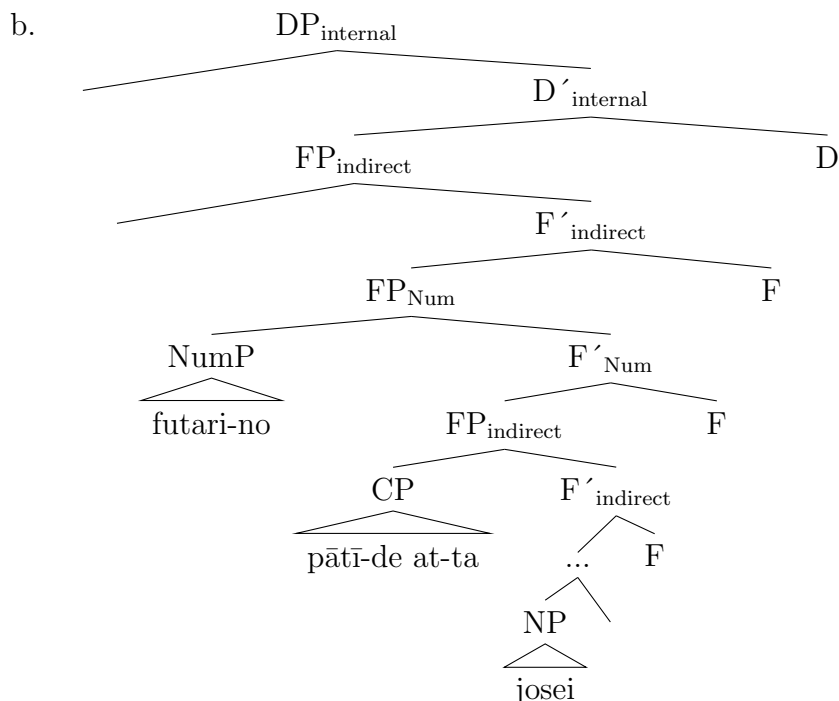
To summarize, the two relative clauses in (146-a) and (146-b) occupy two different positions within the indirect domain and this is shown below.<sup>67</sup>

- (163) a. [pātī-de at-ta [futari-no josei]]  
 party-LOC meet-PST two-MOD woman  
 ‘the two women I met at a party’ (= (146-a))

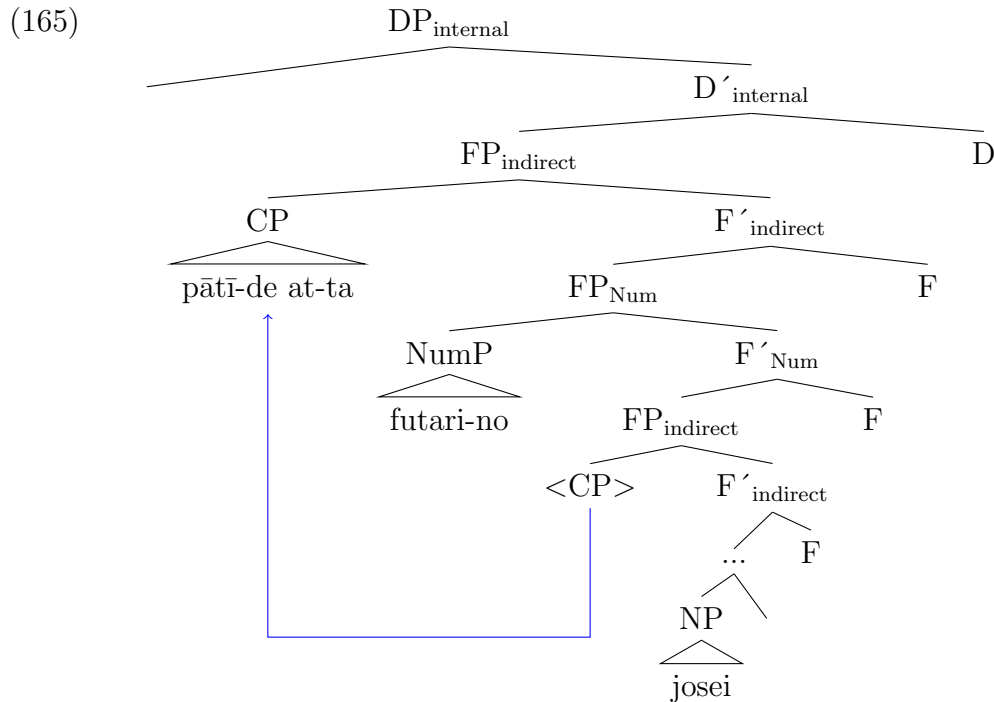


<sup>67</sup>Recall that numerals are selected via MOD as well, but this is simplified here.

- (164) a. [futari-no [pātī-de at-ta josei]]  
 two-MOD party-LOC meet-PST woman (= (146-b))



The question is what the underlying reason for the two ordering possibilities is? Can relative clauses simply be base-generated in different positions? Or is one position a derived position? Based on the fact that complexity is a driving factor that leads speakers to prefer the higher position in a systematic fashion, I opt for the second option: Concretely, relative clauses are base-generated below  $FP_{Num}$  but can optionally move across this projection individually (cf. Laenzlinger 2011; Kim 2019). Furthermore, this movement option is available to all relative clauses, but it correlates with higher complexity. In other words, the more complex a relative clause is, the more prone it is to move up. This means that the derivation of the relative clause in (146-a) is as follows.



The seemingly free order of relative clauses and numerals with potentially only complexity as a driving factor for movement is not straightforwardly explainable in a purely cartographic account, or any purely syntactic account. This holds especially if the only possible movement operations involve (phrasal) movement of the lexical head – either alone or in one of the different pied-piping fashions (Cinque, 2017, 2021, 2023, 2025). However, it becomes clear that the lexical head is not involved in these derivations, hence relative clauses must move individually. Furthermore, adopting the standard assumption that movement is only possible to the left, the lower position must be where relative clauses are base-generated. As a result of left bound movement, relative clauses reach a position above  $FP_{Num}$  and they do this for syntax-external reasons, i.e. complexity. This accounts for the systematic preference for more complex relative clauses in a higher position.<sup>68</sup>

This proposal matches earlier proposals for movement of relative clause (cf. Kim 2019 on individual movement of Korean relative clauses depending on their involvement in the discourse) but also of other modifiers (focus movement, Giusti 1996; Dimitrova-Vulchanova and Giusti 1998; Ihsane and Puskás 2001; Aboh 2004, 2006; Svenonius 2008).

Having established that movement is triggered by syntax-external factors, the movement process observable with complex modifiers seems to be some sort of rescue

<sup>68</sup>A reviewer correctly points out that complexity should not be visible in syntax. However, there is, to my knowledge, no account to better capture the behavior explained here. Note that assuming different areas for base-generation is not an appealing option either. Even if more than one position were accessible for base-generation, the syntax would face the same problem, namely to decide when to merge relative clauses and to decide to merge more complex ones later. It is, at this point, unclear to me how to model this theoretically.

operation. Looking outside of cartography, and generally syntax-based accounts, the fact that longer elements are preferred before shorter elements is in conformity with the so-called *long-before-short principle* that predicts for SOV languages that longer elements are best to the left of shorter elements. This was, for instance, tested empirically for Persian, another SOV language in Faghiri and Samvelian (2020). This principle is the direct opposite of the well-established *Gesetz der wachsenden Glieder* ‘Law of increasing constituents’ of Behaghel (1909), which is not designed for (strictly head-final) SOV languages.

Another interesting parallel concerns compound formation in Japanese in relation to prosodic accentuation.<sup>69</sup> Compounds consisting of three elements can be left- and right-branching. In left-branching compounds, the compound only bears one accent and the three elements are phrased as one unit. In right-branching compounds, on the other hand, there is an additional accent after the first element causing two accentuation phrases (Kubozono 2005, 12–14; cf. Nishiyama 2017). This is shown below adapted from Kubozono (2005, 12–14). Accents mark accented morae, square brackets indicate morphological units, and curly brackets indicate accentual phrases.

- (166) Left-branching compounds  
 [Do'itsu + bu'ngaku]+ kyōkai → {do'itu-bungaku-kyo'okai}  
 ‘Germany + literature + association’ → ‘German Association of literature’
- (167) Right-branching compounds  
 Do'itsu + [bu'ngaku + kyōkai] → {do'itu} {bungaku-kyo'okai}  
 ‘Germany + literature + association’ → ‘Association of German literature’

This means that right-branching compounds are the marked case since they violate one of the accentuation rules of Japanese (Nishiyama, 2017, 153). This is translatable into the long-before-short principle in a straightforward manner. In left-branching compounds, the first two elements form a morphological compound first to which the third is added, meaning a longer sequence (two elements) precedes a shorter sequence (one element). In right-branching compounds, the second and third element have formed a morphological compound to which the first element is added. Therefore, a shorter sequence (one element) precedes a longer sequence (two elements). In accordance with the long-before-short principle, the first option should be favored and this is what happens. Right-branching compounds are more difficult to parse and, hence, involve marked accentuation.

To summarize, relative clauses in Japanese originate in a lower area within the indirect domain and can optionally move upwards to a position above numerals. One driving factor for this movement operation is complexity, a syntax-external fac-

<sup>69</sup>I thank an anonymous reviewer for bringing this parallel to my attention.

tor. These observations fall out nicely under the general long-before-short principle observable in SOV-language. It is desirable to find more instantiations of this principle in Japanese and other SOV-languages, especially pertaining to the DP, and/or the factor complexity.

### 5.4.5 Relative clauses above demonstratives

This remaining section addresses the distribution of *non-restrictive* relative clauses, which are confined to a position above demonstratives, one of the factors that point towards a DP-external domain. Recall the following example from Ishizuka (2008) which shows that those types of relative clause cannot occur below demonstratives but must occur above them.

- (168) \*Ito-san-ni-wa musuko-ga hitori i-ru. [Sono [sakunen isha-ni  
Ito-Mr.-LOC-TOP son-NOM one be-PRS that last.year doctor-LOC  
nat-ta musuko-ga]] kekkon-shi-ta.  
become-PST son-NOM marriage-do-PST  
(‘Mr. Ito has one son. That son, who happened to become a doctor last  
year, got married.’)

\*demonstrative  $\succ$  non-restrictive relative clause (Ishizuka, 2008, 5)

Ishizuka (2008) only considers verbal relative clauses, but the same facts hold for adjectival relative clauses. If they appear higher than demonstratives they can be interpreted restrictively or non-restrictively. A sentence such as (169), where a *na*-adjective precedes the demonstrative and is inflected for past, could be uttered in a context in which Mr./Mrs. Itō has one son and offers additional information about him (non-restrictive), as well as in a context in which s/he has multiple sons and wants to identify that son who got married (namely the shy one).

- (169) [Imamade totemo uchiki-d-at-ta [sono musuko-ga]]  
up.until.now very shy-PRED-COP-PST that son-NOM  
kekkon-shi-ta.  
marriage-do-PST  
1.  $\approx$  ‘That son who was very shy up until now got married.’  
2.  $\approx$  ‘That son, who happened to be very shy up until now, got married.’

On the other hand, it is also impossible for non-restrictive adjectival relative clauses to appear below demonstratives, shown in the adaptation of Ishizuka’s example in (170) where a non-restrictive interpretation is forced.<sup>70</sup>

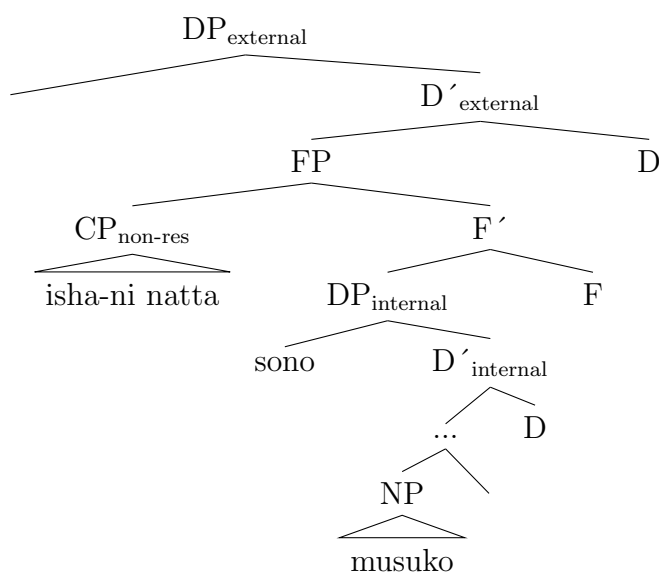
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<sup>70</sup>Again, speakers’ judgment is subtle and some might find such examples acceptable, especially in colloquial speech. However, it seems that when choosing between two options in such a context, speakers might always go for the option in which the non-restrictive relative clause precedes the demonstrative. Future research will shed light on this phenomenon.

- (170) \*Ito-san-ni-wa musuko-ga hitori i-ru. [Sono [totemo  
 Ito-Mr.-LOC-TOP son-NOM one be-PRS that very  
 uchiki-d-at-ta musuko-ga]] kekkon-shi-ta.  
 shy-PRED-COP-PST son-NOM marriage-do-PST  
 ('Mr. Ito has one son. That son, who happened to be very shy, got  
 married.')

From these facts, I draw the conclusion that non-restrictive relative clauses merge in the DP-external domain – the left periphery – and hence are outside the scope of D, predicting their non-restrictive interpretation. This is shown below.

- (171) non-restrictive relative clause > demonstrative

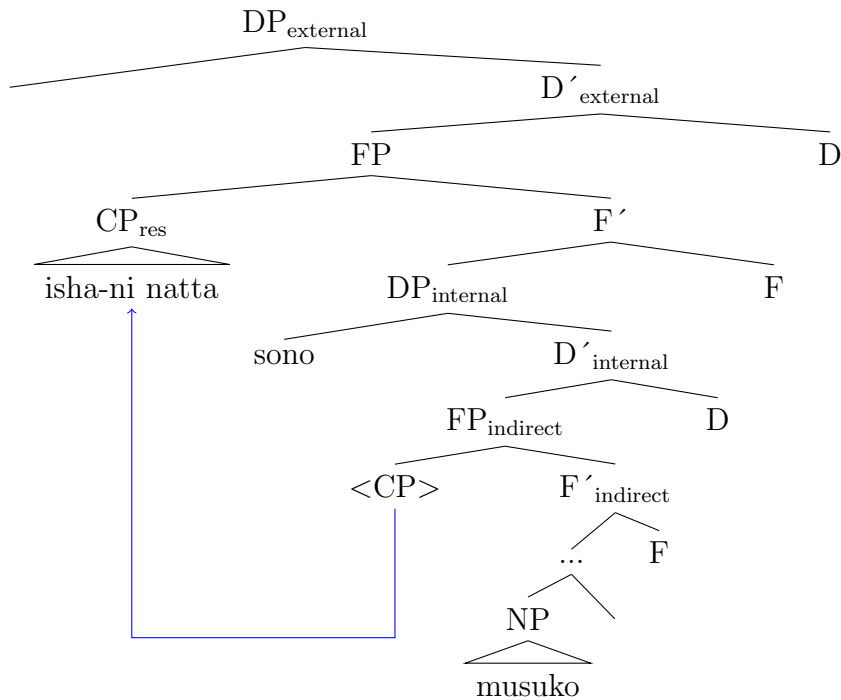


On the other hand, it was shown that restrictive relative clauses merge in the indirect domain below  $FP_{Num}$ . They, hence, have not only the option to move to a functional projection above numerals, but also to the left periphery as shown below, accounting for restrictive relative clauses to the left of demonstratives. Again, they do this independently.<sup>71</sup>

<sup>71</sup>Ishizuka presents an analysis in which two D-projections are assumed in Japanese, a lower one, DemP, where the first part of the demonstrative, namely *ko*- 'speaker-proximate', *so*- 'hearer-proximate' and *a*- 'distant', is generated, and a higher one, DP, headed by *NO*. Having two projections available gives two possible landing sites for relative clauses, which originate very low in her system (Kayne, 1994). In her derivation, movement of the deictic element to Spec, DP yields the correct lexical item. Then the NP moves out of the relative clause to Spec, CP, followed by remnant movement of the IP to Spec, DemP. In this position, it is in the scope of D, therefore can only receive a restrictive interpretation. Additionally, in a further step, the remnant can optionally raise from Spec, DemP to Spec, DP leaving the c-command domain of D and gaining access to a non-restrictive interpretation (see Ishizuka 2008, 8–9). One problem is that the two positions, DP and DemP, lack motivation beyond the derivation proposed here.



(172) restrictive relative clause  $\succ$  demonstrative



Importantly, the two types of relative clauses, restrictive and non-restrictive, occupy different positions, as non-restrictive relative clauses originate DP-externally but restrictive ones move from within the DP-internal domain. The analysis presented here is in line with the analysis of Cinque (2020, 223) and Cinque (2008b), who reduces the differences in interpretation to two different merging positions as well: one outside (= above) demonstratives for non-restrictive and one inside (= below) demonstratives for restrictive relative clauses. As in our system, the additional pattern of Japanese is explained via optional raising of restrictive relative clauses across  $DP_{internal}$  to a position below non-restrictive relative clauses (Cinque, 2008b, 116). Also recall that Kim (2019) proposes the projection  $SpplP$  (*SupplementaryP*) for non-restrictive, but importantly supplementary, relative clauses in the left periphery of Korean, and this is essentially the counterpart to what I simply give as  $FP_{indirect}$ .<sup>72</sup>

Finally, Japanese provides evidence that demonstratives can move from the DP-internal to the DP-external domain. This gives rise to the patterns in which non-restrictive relative clauses appear to the right of demonstratives.<sup>73</sup>

<sup>72</sup>In this respect, note that Japanese offers a parametric difference to Mandarin, where relative clauses in the lower position have access to a restrictive as well as a non-restrictive interpretation, but in the high position are usually interpreted restrictively, although open to a non-restrictive interpretation (Cinque 2020, 235–236; Hsu 2017). Assuming the position in the left periphery is equally reached via independent movement of the relative clause means that the relation of movement and interpretation is different from Japanese and should be investigated. Also recall the German data given in footnote 9 in Chapter 3.1.3.3.

<sup>73</sup>But note that matters are most certainly not as simple as depicted here. Cinque (2008b) proposes a difference between *integrated* and *non-integrated* non-restrictive relative clauses and

A remaining question to be answered is whether there is a driving force for the movement of restrictive relative clauses to the left periphery. I investigated whether a similar effect in terms of complexity can be observed, in other words, whether complexity can also be responsible for an increased preference for relative clauses above demonstratives. As it turns out, this is not the case, exemplified below with an adjectival relative clause containing a subject. While this relative clause would be strongly preferred above numerals within the indirect domain, there does not seem to be any preference for a higher position above a demonstrative. This suggests that, although complexity encourages movement above  $FP_{Num}$ , it does not necessarily do so above the internal D-head. This means then that the driving factor behind the movement of restrictive relative clauses to a position above demonstratives remains undetermined.

- (173) Context: Two friends went to a restaurant yesterday and had several dishes there. Speaker A asks Speaker B about one specific dish.
- a. Kinō it-ta resutoran-no, [ano [sūpu-ga sugoi  
yesterday go-PST restaurant-MOD that soup-NOM very  
kara-k-at-ta ryōri-wa]] nan d-a kke?  
spicy-PRED-COP-PST dish-TOP what PRED-COP.PST Q
  - b. Kinō it-ta resutoran-no, [sūpu-ga sugoi  
yesterday go-PST restaurant-MOD soup-NOM very  
kara-k-at-ta [ano ryōri-wa]] nan d-a kke?  
spicy-PRED-COP-PST that dish-TOP what PRED-COP.PST Q  
'What was [that] dish [in which the soup was very spicy] at the restaurant we went to yesterday?'

To summarize, non-restrictive relative clauses are base-generated DP-externally. Restrictive relative clauses merge in the indirect domain and can optionally move to the left periphery.

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points out that languages such as Italian and other Romance varieties possess both types, but that there might be languages in which only one type exists, among them presumably Japanese (Cinque, 2008b, 121–124). This also concerns the question as to whether non-restrictive relative clauses are base-generated higher than quantificational modifiers in the left periphery or not. Finally, I have assumed that the projections for non-restrictive and restrictive relative clauses in the left periphery are different. This must be the case as stacking of the two types of relative clauses is generally allowed (Cinque, 2008b; Ishizuka, 2008). However, there are different opinions on the ordering principles among non-restrictive and restrictive relative clauses and while Cinque 2008b proposes a hierarchy *non-restrictive relative clause* > *restrictive relative clause* in the left periphery, his judgments are not shared by Ishizuka (2008). For now, I will remain silent on these issues.

## 5.5 Chapter summary and closing remarks

Although indirect modification has been argued to be the only type of modification available in Japanese, many questions have been left open and were addressed in the preceding pages. The following conclusions can be drawn.

- (174)
- a. The complex elements following Japanese modifiers are combinations of the predicative copula, the dummy copula and a tense marker.
  - b. Relative clauses in Japanese must be at least of size IP, but we cannot determine whether they are bigger or not. I have assumed a CP status for several reasons, among them the possibility of focused elements and the better compatibility with the operator-movement approach adopted in this book.
  - c. The difference between full and reduced relative clauses does not exist.
  - d. The indirect domain is comprised of several functional projections reserved for indirect modifiers and a projection reserved for numerals. The projections reserved for indirect modifiers are accordingly not reserved for modifiers of a specific type (full or reduced) or specific syntactic size (IP or CP).
  - e. Indirect modifiers are base-generated below  $FP_{Num}$  but can independently move up to a higher position. One driving factor influencing this movement operation is complexity.
  - f. Non-restrictive relative clauses merge DP-externally.
  - g. Restrictive relative clauses can optionally move to the left periphery.

These findings are similar to the findings in the direct domain and pretty much compatible with previous accounts in the cartographic research field. On the other hand, they emphasize that a certain degree of tolerance is needed when it comes to movement and ordering. Abandoning syntactic size and category as criteria that influence the position of indirect modifiers in Japanese leads to the conclusion that the functional projections inside the indirect domain only correlate to these parameters when they are overtly realized in a given language. In, for example, English or German, relative clauses of different sizes can be distinguished on the surface structure and therefore dedicated functional projections are justified. However, this is not the case in Japanese. From a perspective of language-uniformity, I suggest that the structure of the indirect domain presented by Japanese, and similar languages, is the underlying structure. Languages then show parametric, language-specific, differences as to whether they realize different functional projections dedicated to different types of indirect modifiers or not.

# Chapter 6

## Concluding Remarks

The goal of this book was to present an analysis of the Japanese DP and its internal modifiers and to transfer the findings into cross-linguistic perspective. In this final chapter, I present the main findings, discuss some additional points and raise questions for future research.

### 6.1 Main findings

#### 6.1.1 Summary

The following main findings emerged over the course of the discussion.

- (1) a. The Japanese DP is made up of a DP-external and a DP-internal domain. The DP-internal domain is split up into a domain for direct and a domain for indirect modification.
- b. The DP-external domain, the left periphery, hosts functional projections for quantificational modifiers, non-restrictive relative clauses, potentially focused modifiers, as well as unspecific functional projections for (restrictive) indirect modifiers as well as direct modifiers. Those modifiers move to this domain independently.
- c. Modifiers within the DP are licensed by an additional functional head, called MOD, which licenses them as attributive modifiers. It licenses all modifiers, irrespective of word class and modification type. I argued that this head is universal. It is realized as /i/ and /na/ following the two adjective groups and as /no/ elsewhere. It is not realized when the tense head is overt.
- d. On the other hand, the complex elements following Japanese adjectives and *no*-modifiers in indirect structures are combinations of the predicative copula, the dummy copula and a tense marker.

- e. Direct modifiers exist in Japanese. However, Japanese suggests that the direct domain consists of a possible infinite number of functional projections which are not necessarily dedicated to fine-grained semantic categories. This explains the lack of ordering principles observable in this language. The direct domain in languages such as English, on the other hand, does contain fine-grained functional projections. The reason for this can be sought in parametric differences. In other words, Japanese lacks the language-specific inventory (Units of Languages) to realize specific functional projections. There are also in-between languages, e.g. Persian and Modern Greek, in which a subpart of the functional projections are realized, and it seems that if functional projections are realized, it is always those reserved for direct-only modifiers.
- f. Indirect modifiers only come in one size, they are CPs. The indirect domain is higher than the direct domain and encompasses several functional projections above and below a projection hosting numerals. Indirect modifiers are base-generated below this projection but can independently move to any projection above  $FP_{Num}$ . More complex indirect modifiers are more likely to be subject to such movement operations.

I argued that the structure of the DP is uniform across languages, i.e. all languages share the same universal spine. Importantly, I left room for cross-linguistic variation as to how this spine is filled. Japanese presents (up to now) the most underspecified case in which functional projections are unspecific and not dedicated to modifiers of different types, word classes, semantic categories or sizes. Other languages are more precise in their structure of the DP, and for each relevant domain respectively. Below, I repeat the DP-structure, based on Japanese, proposed in this book.



the dichotomy between direct and indirect modification and their connection to the different DP-domains.

### 6.1.3 What Japanese can offer for cartography

However, it is important to acknowledge that this transfer becomes only possible if one makes the correct amendments and is open to draw critical conclusions, which brings me to the second important insight of this study: The contribution Japanese can offer to cartography.

Perhaps the most important contribution is that the notorious lack of ordering restrictions, which has for decades been the main argument to deny direct (attributive) modification in Japanese, is not problematic at all; rather the presence or absence of ordering restrictions and type of modification are two orthogonal issues. Japanese suggests that all languages start out with unspecified functional projections and that only some languages can realize specific functional projections. The fine-grained hierarchy of functional projections is, on the one hand, one of the ground pillars of cartography; on the other hand, it has constantly been the fuel for criticism put forward by various scholars (cf. Larson 2021).

By integrating the ideas that some languages exhibit more specific functional projections than others together with the concept of paradigmatic variability into the research field, I proposed that we can expand our knowledge of the DP and take this as a starting point for further investigation.

At this point, I would like to emphasize that the account proposed in this book, and built mostly on Japanese data, does in no way undermine the core predictions and achievements of cartography. This concerns, most of all, the perhaps biggest achievements: To reduce left-right asymmetries and different orders of modifiers in relation to the lexical head, while at the same time ruling out unattestable (Cinque, 2025) orders, to different movement operations the lexical head can undergo (see Chapter 2.1.2).<sup>1</sup>

The system proposed here shares with standard cartography the assumption of distinct merging domains for demonstratives, numerals and adjectives. In fact, Japanese presents a language where this order is reflected in linear order to a maximal extent. Assuming demonstratives are merged in DP (here: in Spec, DP), numerals within the indirect domain, and adjectives within the direct domain, if direct, or the indirect domain, if indirect, but then in a lower position than NumP, the relative order is unaffected by the broad direct and indirect domains. Applying this, for instance, to a language such as Persian, which reflects the basis order Dem

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<sup>1</sup>I am grateful to an anonymous reviewer for also raising this point.

$\succ \text{Num} \succ \text{N} \succ \text{A}$ , the noun moves across each modifying adjective. This means that these general assumptions are unaffected by the idea of unspecific functional heads.

The only caveat exists with regard to (direct) adjectives, or more broadly *modifiers*, where, as I argued, language variation comes into play. In Japanese, (direct) modifiers do not merge in a specific order, hence there is no fixed underlying order. However, since the noun does not move in this language anyway, this observation, only with regard to Japanese, does not interfere with U20-effects. Should a counterpart language to Japanese be found, and Persian at least shows certain indications, then free order of adjectives (or: modifiers) in postnominal position will hold.

On the other hand, in languages that do have the language-specific entities to form specific functional projections in the direct domain, the relevant ordering principles outlined in Chapter 2.1.2 will hold regardless. Specifically, only those orders are possible that are the result of the various movement operations accessible to the noun, and combinations thereof. Unattestable orders are ruled out by virtue of not being derivable by these operations. It is a fact that when we looking at the variety of languages investigated through a cartographic lens, astounding parallels in terms of ordering emerge (Cinque, 2025, sec. 2) giving evidence that the principles of ordering and of realizing functional projections are uniform.

In other words, the core assumptions, and achievements, of cartography are unaffected by the theory and the empirical data points presented throughout this book, however, certain assumptions, specifically the nature of the functional projections within the direct and the indirect domain and their strict mapping to order, are challenged by these aspects, which eventually has led to an evolution of cartographic theory.

Another crucial amendment, that enriches the DP structure, is the integration of the syntactic head MOD as a universal mediator of modification as it offers a reference to the internal structure of modifiers. Tightening the connection between the structure of the DP and the internal structure of its modifiers can only be beneficial for future research, as it will allow us to draw conclusions from the macro- to the micro-perspective and vice versa.

## 6.2 Open questions and directions for future research

This study probably opened as many new questions as it has answered, which I consider a good thing. Below, I will outline some immediate prospects, topics and concerns.



### 6.2.1 Cartography: Where do we go from here?

Assuming the DP structure I proposed here is universal and Japanese comes close to the baseline language in which almost no specific, or dedicated, functional heads exist within the direct, and indirect, domain begs the question whether similar languages exist. If they do, what could such languages have in common? Can we derive areal, genealogical or syntactic similarities and draw typological generalizations? In the general spirit of cartography, precise typological considerations ensure the system to be as theoretically and empirically sound as possible. A precise typology of languages according to the degree they adhere to ordering restrictions or, more importantly, allow ordering violations is therefore imperative for further progression in cartographic theory.

Besides the rather lenient ordering observable in Japanese and the rather strict ordering observable in many other languages, what about further in-between languages besides the ones investigated here (Chapter 4.4.2)?

The same applies for the indirect domain. Contrary to the situation in the direct domain, I expect that we find a large variety of languages of the Japanese type, given the fact that many (not only prenominal relative clause) languages seem to abide by the same principles, namely to only show one type of relative clause overtly. Possible questions thus are: Do we find similar patterns with regard to ordering in such languages? Is complexity a driving factor in these languages as well?

Another clear direction for future research only alluded to here is the precise structure of the left periphery. I suggested, in line with authors such as Grohmann (2003), Laenzlinger (2005, 2011) and Kim (2019), that the DP-external domain is universal. I showed that we need to assume this area in order to account for rather high direct and indirect modifiers, but also non-restrictive relative clauses, which originate there in Japanese. As for focused modifiers, they can productively appear in the left periphery; but they do not need to, and sometimes are forbidden from doing so. Future research should point out the precise nature of this area. Questions that come to mind are: What other projections do we find in the left periphery? To what degree and according to which factors are the categories in this domain language-specific? Are there ordering principles in the left periphery, both from a cross-linguistic and language-specific perspective?

### 6.2.2 Internal structure

The final research desideratum is the internal structure of modifiers. The functional head MOD as well as the complex structure of indirect modifiers proposed here are not only compatible with the universal DP structure, they also expand it significantly. Assuming the existence of MOD in Japanese must mean this head exists in every

language and languages differ in if and how they realize it. There are many similar elements which arguably are instances of MOD, Persian *ezāfe* or Mandarin DE – or generally all elements called *linkers* (Richards, 2023) – but of course in the majority of languages we do not see such elements. Even in Japanese, NO is never present when overt tense is realized. How do we account for these observations? And can we draw generalizations from different languages that exhibit such elements? Taking the internal structure of modifiers as important for research on the DP as the establishment of ordering restrictions and new categories of modifiers itself means this is another endeavor we should tackle in the future.

### 6.3 A final suggestion

I will close this book with a suggestion: I think it is desirable that syntactic research keeps informing the field of Japanese Studies, or in fact any field of Area Studies, but importantly this also holds the other way around. The point is that these two disciplines can learn so much from one another, and so much potential remains untapped if researchers do not make use of it. Many Japanese Studies scholars will dismiss cartographic assumptions, also the ones defended here, immediately, as they seem too specific and rigid. Nevertheless, as I hope to have demonstrated, they make descriptive generalizations graspable by offering explananda thereof. On the other hand, the cartographic research field is dependent on specialists in the relevant fields, and only experts can grant them the deep understanding of relevant phenomena that is needed to derive aforementioned explananda. My deep hope is to have inspired more studies in the future making use of both these research fields, Japanese studies and theoretical syntax, to enhance our knowledge in these two disciplines and beyond.

# Appendices

In this appendix, I discuss two issues: The nature of so-called *bound markers*, suffixes which might appear with relational modifiers in Japanese and have influence on their syntactic behavior, and the word class status of *no*-modifiers.

## A: Bound markers and expanded modifiers

This section gives an overview over *bound markers* – also called *relational nouns* – those suffixes that form so-called *expanded modifiers* out of Japanese relational modifiers discussed in Chapter 4.3.2 (cf. Nagano 2016; Shimada and Nagano 2018). First of all, bound markers specify the meaning of a relational modifier. In the DP below, the bound marker -IRI ‘put in(-side)’ is added to the noun/relational modifier *chīzu* ‘cheese’ and specifies the relationship between *cheese* and the modified noun *pan* ‘bread’. The expanded modifier formed out of this combination denotes precisely *bread with cheese put inside*, whereas the DP *chīzu-no pan* can mean anything from ‘bread made of cheese’, ‘bread containing cheese’ to ‘bread made for cheese’. Like relational modifiers, expanded modifiers appear exclusively with NO in attributive position.

- (1)    *chīzu-iri-no*                *pan*  
         cheese-put.in-MOD bread  
         ‘bread with cheese put inside’

The array of bound markers roughly corresponds to the list of categories for relational modifiers presented in Chapter 4.3.2, but includes some more categories such as TASTE, BELONGING, STATE, PURPOSE, LEVEL and COLOR. The following is a (possible non-exhaustive) list, also taken from Nagano (2016, 56–57).<sup>2</sup>

- (2)    *Nagano’s list of bound markers*  
         MATERIAL: *sei* ‘made by’, *iri* ‘added’  
         ORIGIN: *sei* ‘made in’, *kei* ‘descended from’, *shushin* ‘coming from’, *umare*

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<sup>2</sup>That *no*-modifiers are often formed, or often contain, elements for word formation, preferably suffixes, is also noted in Muraki (2012, 195–205) who gives, besides his list of non-predicative modifiers, also a list of possible prefixes or suffixes often co-occurring with *no*-modifiers.

‘born in’  
 SHAPE/SIZE: *kei* ‘shape’, *kata/gata* ‘shape, size’  
 TASTE: *aji/mi* ‘taste’, *fūmi* ‘taste’  
 TYPE: *sei* ‘type, nature’, *gata* ‘type’, *kei* ‘type’  
 STATE: *jō* ‘state’, *jōtai* ‘state’, *sugata* ‘wearing’  
 BELONGING: *kumi* ‘group’, *ha* ‘group, school’, *shugi* ‘-ism’, *shozoku* ‘belong-  
 ing to’  
 SIMILARITY: *fū* ‘like’, *ryū* ‘like, in the style of’  
 POSSESSION/INGREDIENT: *tsuki* ‘with’, *mochi* ‘with’, *iri* ‘added’, *ari* ‘with’,  
*nashi* ‘-less, without’, *darake* ‘-ful, -ridden’  
 PURPOSE/TARGET: *yō* ‘for’, *muke* ‘meant for’, *senyō* ‘exclusively for’  
 LOCATION: *mae/zen* ‘front’, *shita/ka* ‘under’, *naka* ‘in, inside’, *ue/jō* ‘on,  
 above’, *chū* ‘inside’, *kan* ‘between’, *iki* ‘bound for’, *hatsu* ‘departing from’,  
*muki* ‘toward, faced to’, *kake* ‘hanged on’  
 TIME: *mae* ‘before’, *go* ‘after’, *chū* ‘during’  
 STATUS/POSSESSION: *jin* ‘nationality’, *shi* ‘specialist’, *fu* ‘female’, *kan* ‘offi-  
 cial’, *ko* ‘worker’, *jo* ‘female’, *toshite* ‘as’  
 LEVEL: *kyū* ‘level’, *reberu* ‘level’, *do* ‘degree’, *i* ‘level’  
 TOPIC: *jō* ‘on, about’, *garami* ‘about, related to’, *kanren* ‘related to’  
 COLOR: *shoku* ‘color’, *iro* ‘color’

Some of these markers are derived from nouns which can be used on their own, such as *jōtai* ‘state’, while other bound markers cannot be used independently, for example *-darake* ‘-ful, -ridden’.

These markers come with interesting syntactic peculiarities. When they are added to a, formerly non-predicative, relational modifier, the resulting expanded modifier becomes possible in predicative position as shown below (Nagano, 2016; Shimada and Nagano, 2018).<sup>3</sup>

- (3) Kono pan-wa chīzu-iri-d-a.  
 this bread-TOP cheese-put.in-PRED-COP.PRS  
 ‘This bread is with cheese inside.’
- (4) Kono kuruma-wa Nihon-sei-d-a.  
 this car-TOP Japan-produce PRED-COP.PRS  
 ‘This car is Japanese.’ (= was produced in Japan)

Following the analysis of DE used for coordination as the predicative copula (cf. Chapter 5.2), we expect expanded modifiers to be also able in coordinative structures. This is indeed the case.

<sup>3</sup>The following data points were also part of the interview conducted for relational modifiers and reported on in Chapter 4.

- (5) a. chīzu-iri-de herushi-na pan  
 cheese-put.inside-PRED healthy-MOD bread  
 ‘bread that is cheesy (= cheese put inside) and healthy’  
 b. Nihon-sei-de jōbu-na kuruma  
 Japan-produce-PRED robust-MOD car  
 ‘car that is Japanese (= produced in Japan) and robust’

Finally, while paraphrasing NO with the canonical copula DE-AR-U was never possible with relational modifiers, this operation is possible with expanded modifiers.

- (6) Nihon-sei-de-ar-u tokei  
 Japan-produce-PRED-COP-PRS watch  
 ‘a Japanese watch’ (Shimada and Nagano, 2018, 75)

On the other hand, in contrast to canonical relational modifiers, these extended modifiers do not exhibit nominal use.

- (7) \*Chūgoku-sei-ga kawat-ta  
 China-produce-NOM change-PST  
 ‘(produced in) China changed’

This suggests that they either do not have a nominal status anymore, or that they have become lexicalized. Adjectival status, at first glance, seems equally doubtful. They still remain non-gradable, in combination with the relative degree adverb *totemo* ‘very’ and in comparative clauses (8), and cannot be nominalized via the suffix -SA ‘ness’ (9).

- (8) a. \*totemo Nihon-san-no kuruma  
 very Japanese-production-MOD car  
 ‘(very Japanese car)’  
 b. \*Watashi-ga okusan-no kuruma-to kurabe-reba Nihon-san-no  
 I-NOM wife-MOD car-COM compare-if Japan-production-MOD  
 kuruma-wo kat-ta.  
 car-ACC buy-PST  
 ‘(Compared to my wife, I bought a more Japanese car.)’  
 (9) a. \*Chūgoku-sei-sa  
 China-produce-ness  
 b. \*Nihon-san-sa  
 Japan-produce-ness

However, at least for some suffixes, especially those derived from lexical verbs, gradability in comparative constructions with the particle *yorī* ‘than’ is possible. (10), in which the quantity of cheese is compared, was judged acceptable by three out of five consultants.

- (10) [LAWSON-no pan-to kurabe-reba] motto chīzu-iri-no]  
 LAWSON-MOD bread-COM compare-if more cheese-put.inside-MOD  
 pan-wo kat-ta.  
 bread-ACC buy-PST  
 ‘I bought bread with more cheese inside compared to bread from LAWSON.’

That *chīzu-iri* is inherently verbal is visible when paraphrasing this modifier with a verbal relative clause, either intransitive or transitive. In these constructions, comparison is equally possible.

- (11) a. [LAWSON-no pan-yori chīzu-ga hait-te-i-ru] pan  
 LAWSON-MOD bread-than cheese-NOM be.inside-GER-be-PRS bread  
 ‘bread with more cheese inside than bread from LAWSON’  
 b. [LAWSON-no pan-yori [chīzu-wo ire-ta] pan]  
 LAWSON-MOD bread-than cheese-ACC put.inside-PST bread  
 ‘bread in which *pro* has put more cheese than in bread from LAWSON’

In order to give a syntactic explanation for predicative use, Shimada and Nagano (2018) adhere to a KIND-head following Adger (2013). This KIND-head, which is more of a variable, is present in predicative position and is what is really modified by the relational modifier, meaning that essentially we are dealing with a modification structure (see especially Adger 2013, ch. 4 for the underlying analysis of relational nouns). This proposal is in spirit very similar to the semantic *kind-reading* in McNally and Boleda (2004). Recall the following example from Catalan.

- (12) La tuberculosi pot ser pulmonar. (Catalan)  
 the tuberculosis can be pulmonary  
 ‘The tuberculosis could be (of the) pulmonary (kind/type).’  
 (McNally and Boleda, 2004, 189)

Also note the following example in English from Bauer *et al.* (2013) containing the relational adjective *nuclear* in predicative position. Predicative use is further ameliorated by a measure phrase containing an endpoint modifier (Fábregas 2014, see Shimada and Nagano 2018, 80).

- (13) 75% of French electricity is nuclear. (Bauer *et al.*, 2013, 318)

In this example, a noun such as *kind* or *type* seems to be understood. Therefore, the sentence can be paraphrased accordingly either with the overt noun repeated or a generic *kind*-noun.

- (14) a. 75% of French energy is nuclear energy.  
 b. 75% of French energy is of the nuclear kind/type.

Japanese seems to overtly express such a KIND-head in the form of bound markers. This essentially suggests the following structure in which what really is selected by ModP is a DP whose lexical head is the bound marker and in whose extended projection the modifier *chīzu* resides (repeated from (1) above).

- (15) a. *chīzu-iri-no pan*  
 cheese-put.in-MOD bread  
 ‘bread with cheese put inside’
- b.
- 
- ```

graph TD
    DP1[DPinternal] --- D1[D']
    DP1 --- D2[D]
    D1 --- FP1[FP]
    D1 --- D3[D]
    FP1 --- ModP[ModP]
    FP1 --- F1[F']
    ModP --- Mod'1[Mod']
    Mod'1 --- DP2[DPinternal]
    Mod'1 --- Mod1[Mod]
    DP2 --- D4[D']
    D4 --- FP2[FP]
    D4 --- D5[D]
    FP2 --- XP[XP]
    FP2 --- F2[F']
    XP --- chīzu[chīzu]
    F2 --- NP1[NP]
    F2 --- F3[F]
    NP1 --- iri[iri]
    Mod1 --- no[no]
    F1 --- NP2[NP]
    F1 --- F4[F]
    NP2 --- pan[pan]
    
```

The only question is, whether *chīzu* is not selected by MOD or whether this head is covert again. Presumably, we see some kind of lexicalization here and therefore this syntactic head might not be necessary. However, a direct counterargument against a status of *chīzu-iri* as lexical compound comes from structures as in (16).

- (16) *[[oishi-i chīzu-iri-no] pan]*
 tasty-MOD cheese-put.in-MOD bread
 ‘bread with tasty cheese put inside’

While the *i*-adjective *oishi-i* ‘tasty’ can modify *pan* ‘bread’, it can also modify *cheese* as indicated by the brackets. If we were dealing with a lexical compound the first part should not be able to be modified.

There are thus many issues left open for the research on bound markers and expanded modifiers to be addressed in the future.

B: A taxonomy of *no*-modifiers

In this section, I present an overview over the taxonomy of *no*-modifiers I established. More details can be found in the data paper contained in the repository under <https://osf.io/u9txp/>.

I have tested every lexeme in the data set for the following five variables.

- (17)
 - a. occurrences in attributive use
 - b. element in attributive position: NA or NO
 - c. predicative use with copula
 - d. referential use with case particles for subject and object
 - e. nominalization

Based on these variables I have established the following taxonomy of *no*-modifiers.

- (18) Taxonomy of *no*-modifiers
 - a. *no*-adjectives (Group 1)
 - b. Dual membership: Adjectival use with NO and NA but not as noun (Group 2)
 - c. Dual membership: Use as noun and adjectival use as *na*-adjective, in some instances as *no*-modifiers (Group 3)
 - d. Pure nouns (Group 4)
 - e. Direct modifiers (Group 5)

There are several important things to note: Group 3 can be further divided into lexemes where attributive use is only possible with NA, and those where it is also possible with NO. An instance of the former would be *kenkō* ‘healthy, health’ an instance of the latter *byōdō* ‘equal, equality’. Since this is a semantic factor, I have not established a different group, but sorted these modifiers into two subgroups. Second, since modification status and word class are treated as orthogonal issues, for any given lexeme nominal or adjectival status is independent of its role as direct modifier. For some lexemes, one could argue that they should rather be put in another group than group 5. However, since direct modification is such an important part of this study, I have dedicated a whole group to direct modifiers. Finally, note that group 5 can be subdivided into those direct modifiers without a nominal status (group 5a) and those with a nominal status (group 5b).

Taking all these factors into account, an overview of the groups and relevant characteristics looks as follows.

Group	1	2	3a	3b	4	5a	5b
Example	mumei 'un-known'	tokubetsu 'special'	kenkō 'health(-y)'	byōdō 'equal(-ity)'	byōki 'sickness'	seirai 'innate'	tagai 'each other'
MOD	no	no/na	no/na	no/na	no	no	no
Predicative Use	✓	✓	✓	✓	✓	×	×
Referential Use	×	×	✓	✓	✓	×	✓
Nominalization	✓	✓	marginal	×	×	marginal	×

Table 6.1: Taxonomy of *no*-Modifiers

The emerging picture shows a spectrum from more adjectival to nominal status. Group 1 fulfill the most adjectival criteria, while the other extreme, group 4, consists of true nouns that fulfill none of the criteria defined for adjectives. Below, tables with relevant examples of each group are given.

Lexeme	-no	-na	Copula	Case	-sa
Mumei 'unknown'	217	11	46	4	0
Hadashi 'barefoot'	101	0	58	8	0
Kogata 'small-size'	522	8	125	10	0

Table 6.2: Group 1: *no*-Adjectives, 215 lexemes

Lexeme	-no	-na	Copula	Case	-sa
Pittari 'fitting perfectly'	301	69	338	0	0
Muyō 'un- necessary'	196	115	182	4	2
Kō'un 'good luck'	106	134	209	0	0

Table 6.3: Group 2, *no/na*-Modifiers, 80

Lexeme	-no	-na	Copula	Case	-sa
Kurashikku 'classic'	102	93	37	50	1
Kyū 'ur- gent, urgency'	21	605	138	118	1
Manzoku 'pleased'	218	138	345	248	0

Table 6.4: Group 3: Adjectival/Nominal, 105

Lexeme	-no	-na	Copula	Case	-sa
Hantai 'opposite, contrary'	1140	23	656	510	0
Gūzen 'accident(- al)'	327	20	371	124	0
Erīto 'elite'	77	1	63	51	0

Table 6.5: Group 4, Nouns: 147

Examples of direct modifiers without nominal status are the following.

Lexeme	-no	-na	Copula	Case	-sa
Kōzen 'publicly known'	40	0	0	0	0
Seirai 'in- nate'	92	0	0	5	0
Eien 'eter- nity, eter- nal'	793	12	38	31	0

Table 6.6: Group 5: Direct Modifiers, 511

Finally, the subgroup of *direct nominals* contained in group 5 looks as follows. A direct nominal is a direct modifier with at least 5 occurrences with case particles or a ratio higher than 1>5 for nominal use. This applies to a total of 90 lexemes. Good examples are most relational modifiers, including modifiers of material such as *tetsu* 'iron', but also *ōkuchi* 'large (amount)' or *shōsai* 'detailed, details'.

Lexeme	-no	-na	Copula	Case	-sa
Tetsu 'iron'	1117	2	39	348	0
Ōkuchi 'large (amount)'	64	1	2	47	0
Shōsai 'de- tailed, de- tails'	32	787	46	472	2

Table 6.7: Group 5b: Direct Nominal Modifiers, 90

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